

Service Manual

LG-V500

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1. INTRODUCTION

1.1 Purpose

LG-V500 manual provides the information necessary to repair, calibration, description and download the features of this model.

1.2 Regulatory Information

A. Security

To all fraud, the unauthorized use of telecommunications system by an unauthorized part (for example, persons other than your company's employees, agents, subcontractors, or person working on your company's behalf) can result in substantial additional charges for your telecommunications services.

System users are responsible for the security of own system. There are may be risks of toll fraud associated with your telecommunications system. System users are responsible for programming and configuring the equipment to prevent unauthorized use. The manufacturer does not warrant that this product is immune from the above case but will prevent unauthorized use of common carrier telecommunication service of facilities accessed through or connected to it. The manufacturer will not be responsible for any charges that result from such unauthorized use.

B. Incidence of Harm

If a telephone company determines that the equipment provided to customer is faulty and possibly causing harm or interruption in service to the telephone network, it should disconnect telephone service until repair can be done. A telephone company may temporarily disconnect service as long as repair is not done.

C. Changes in Service

A local telephone company may make changes in its communications facilities or procedure. If these changes could reasonably be expected to affect the use of the phones or compatibility with the net work, the telephone company is required to give advanced written notice to the user, allowing the user to take appropriate steps to maintain telephone service.

D. Maintenance Limitations

Maintenance limitations on the phones must be performed only by the manufacturer or its authorized agent. The user may not make any changes and/or repairs expect as specifically noted in this manual. Therefore, note that unauthorized alternations or repair may affect the regulatory status of the system and may void any remaining warranty.

E. Notice of Radiated Emissions

This model complies with rules regarding radiation and radio frequency emission as defined by local regulatory agencies. In accordance with these agencies, you may be required to provide information such as the following to the end user.

F. Pictures

The pictures in this manual are for illustrative purposes only; your actual hardware may look slightly different.

G. Interference and Attenuation

A phone may interfere with sensitive laboratory equipment, medical equipment, etc. Interference from unsuppressed engines or electric motors may cause problems.

H. Electrostatic Sensitive Devices

ATTENTION

Boards, which contain Electrostatic Sensitive Device (ESD), are indicated by the  sign.

Following information is ESD handling:

- Service personnel should ground themselves by using a wrist strap when exchange system boards.
- When repairs are made to a system board, they should spread the floor with anti-static mat which is also grounded.
- Use a suitable, grounded soldering iron.
- Keep sensitive parts in these protective packages until these are used.
- When returning system boards or parts like EEPROM to the factory, use the protective package as described.

2. PERFORMANCE

2.1 Product Name

LG-V500: GPS, Bluetooth 4.0, WIFI 802.11 a/b/g/n

2.2 Supporting Standard

Item	Feature	Comment
Supporting Standard	GPS Bluetooth 4.0 WIFI 802.11 a/b/g/n	
Frequency Range	GPS: 1574.42 MHz ~ 1576.42 MHz Bluetooth : 2400 MHz ~ 2483 MHz WIFI: 2400 MHz ~ 2483.5 MHz 5150 MHz ~ 5825 MHz	
Application Standard		

2.3 Main Parts

Item	Part Name	Comment
AP Chip	APQ8064	

2.4 HW Features

Item	Feature	Comment
Form Factor	DOP type	
Battery	Standard : Li-Ion , Capacity : 4,600mAh	
Size	216.8x 126.5 x 8.3 (mm)	
Weight	349.38 g (TBD)	With Battery
PCB	L1F2, 10 Layers , 0.8t	
Stand by time	460 hr	
Charging time	5 hr under.	
Display	Main LCD TFT Main LCD (8.3", 1920X1200)	
Built-in Camera	5M Camera, 1.3M secondary	
ANT	WiFi : SUS + Carrier Type GPS : SUS + Carrier Type	
System connector	5 Pin Micro USB	USB 2.0 HS
Ear Phone Jack	3.5Phi, 4 Pole, Stereo	
PC synchronization	Yes	
Memory(AP)	eMMC : 16GB LP-DDR2 : 8Gb	
Data & Fax	Built in Data & Fax support	
BlueTooth	V4.0 BLE (BT Low Energy)	
MIDI(for Buzzer Function)	72 Poly, MP3 Ringtone	
Music Player	MP3/ WMA/AAC/AAC+/WAV/AC3	
Video Player	MPEG4, H.263, H.264 @ 1080p 30fps/ Divx HD	
Camcorder	MPEG4, H.264, H.263 @ 1080p 30pfs	
Voice Recording	Yes	
Speaker Phone mode Support	Yes	
Travel Adapter	Yes	
Stereo Headset	No	
Data Cable	Yes	
T-Flash (External Memory)	Yes	

2.5 SW Features

Item	Feature	Comment
Battery Charging	0 ~ 21 Levels	Android 4.0 (ICS)
Key Volume	0 ~ 7 Level	
Audio Volume	1 ~ 15 Level	
Time / Date Display	Yes	
Multi-Language	Yes	English, Espanol, Franch, Korea, Chinese
Quick Access Mode	Phone / Space / Browser / Menu	
PC Sync	No	
CLIP / CLIR	Yes	
Voice Memo	Yes	Support voice recorder
Download	No	
Game	Yes	
Calendar	Yes	
Memo	Yes	There is no limitation on the number of items. It depends on available memory amount.
World Clock	Yes	
Unit Convert	No	
Stop Watch	Yes	
Wall Paper	Yes	
Download Melody / Wallpaper	Yes	Over web browser
MMS	Yes	
EONS	Yes	
CPHS	Yes	V4.2
ENS	Yes	
Camera	Yes	5M AF / 1.3M
IrDa	No	
Bluetooth	Yes	Ver. 4.0 BLE
Memo pad	Yes	Rich Note
TTY	Yes	
SyncML	Yes	
IM	Yes	Gtalk
Email	Yes	

2.6 H/W SPEC.

1) WLAN 802.11a transceiver specification

Item	Specification
Transmit Frequency	5150 MHz ~ 5825 MHz (CH36~CH165)
Tx Power Level	$\leq 23\text{dBm}$ (5150 ~ 5350 MHz), $\leq 30\text{dBm}$ (5470 ~ 5725 MHz)
Frequency Tolerance	within ± 20 PPM
Chip clock Frequency Tolerance	within ± 20 PPM
Spectrum Mask	≤ -20 @ $\pm 11\text{MHz}$ offset (9MHz ~ 11MHz) ≤ -28 @ $\pm 20\text{MHz}$ offset (11MHz ~ 20MHz) ≤ -40 @ $\pm 30\text{MHz}$ offset (20MHz ~ 30MHz)
Transmitter constellation error (rms EVM)	$\leq -5\text{dB@6Mbps}$, $\leq -8\text{dB@9Mbps}$, $\leq -10\text{dB@12Mbps}$, $\leq -13\text{dB@18Mbps}$, $\leq -16\text{dB@24Mbps}$, $\leq -19\text{dB@36Mbps}$, $\leq -22\text{dB@48Mbps}$, $\leq -25\text{dB@54Mbps}$
Spurious Emissions	$< -36\text{ dBm @ } 30\text{ MHz to } 47\text{ MHz}$ $< -54\text{ dBm @ } 47\text{ MHz to } 74\text{ MHz}$ $< -36\text{ dBm @ } 74\text{ MHz to } 87,5\text{ MHz}$ $< -54\text{ dBm @ } 87,5\text{ MHz to } 118\text{ MHz}$ $< -36\text{ dBm @ } 118\text{ MHz to } 174\text{ MHz}$ $< -54\text{ dBm @ } 174\text{ MHz to } 230\text{ MHz}$ $< -36\text{ dBm @ } 230\text{ MHz to } 470\text{ MHz}$ $< -54\text{ dBm @ } 470\text{ MHz to } 862\text{ MHz}$ $< -36\text{ dBm @ } 862\text{ MHz to } 1\text{ GHz}$ $< -30\text{ dBm @ } 1\text{ GHz to } 5.15\text{ GHz}$ $< -30\text{ dBm @ } 5.35\text{ GHz to } 5.47\text{ GHz}$ $< -30\text{ dBm @ } 5.725\text{ GHz to } 26.5\text{ GHz}$
Rx Min input Sensitivity	PER $\leq 10\%$ -82dBm@6Mbps , -81dBm@9Mbps , -79dBm@12Mbps -77dBm@18Mbps , -74dBm@24Mbps , -70dBm@36Mbps -66dBm@48Mbps , -65dBm@54Mbps
Rx Max input Sensitivity	$\geq -30\text{dBm}$ (6,9,12,18,24,36,48,54Mbps) @ PER $\leq 10\%$

2. PERFORMANCE

Rx Adjacent Channel Rejection	PER ≤ 10%, ACR ≥ 16dB@6Mbps, ACR ≥ 15dB@9Mbps, ACR ≥ 13dB@12Mbps, ACR ≥ 11dB@18Mbps, ACR ≥ 8dB@24Mbps, ACR ≥ 4dB@36Mbps ACR ≥ 0dB@48Mbps, ACR ≥ -1dB@54Mbps ※ACR shall be measured by setting the desired signal's strength 3 dB above the rate-dependent sensitivity specified in min input sensitivity
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2) WLAN 802.11b transceiver specification

Item	Specification
Transmit Frequency	2400 MHz ~ 2483.5 MHz (CH1~CH13)
Tx Power Level	≤ 20dBm
Frequency Tolerance	within ±25 PPM
Chip clock Frequency Tolerance	within ±25 PPM
Spectrum Mask	≤ -30 @ $f_c - 22\text{MHz} < f < f_c - 11\text{MHz}$ and $f_c + 11\text{MHz} < f < f_c + 22\text{MHz}$ ≤ -50 @ $f < f_c - 22\text{MHz}$ and $f > f_c + 22\text{MHz}$
Power ramp on/off time	≤ 2us
Carrier Suppression	≤ -15dB
Modulation Accuracy(Peak EVM)	≤ 35%
Spurious Emissions	< -36 dBm @ 30MHz ~ 1GHz < -30 dBm above @ 1GHz ~ 12.75GHz < -47 dBm @ 1.8GHz ~ 1.9GHz < -47 dBm @ 5.15GHz ~ 5.3GHz
Rx Min input Sensitivity	≤ -76dBm(1Mbps,2Mbps,5.5Mbps,11Mbps) @ FER ≤ 8%
Rx Max input Sensitivity	≥ -10dBm(1Mbps,2Mbps,5.5Mbps,11Mbps) @ FER ≤ 8%
Rx Adjacent Channel Rejection	≥ 35dB @FER ≤ 8%, interference input signal -70dBm@ $f_c \pm 25\text{MHz}$ (11Mbps)

3) WLAN 802.11g transceiver specification

Item	Specification
Transmit Frequency	2400 MHz ~ 2483.5 MHz (CH1~CH13)
Tx Power Level	≤ 20dBm
Frequency Tolerance	within ±25 PPM
Chip clock Frequency Tolerance	within ±25 PPM
Spectrum Mask	≤ -20 @ ±11MHz offset (9Mhz ~ 11MHz) ≤ -28 @ ±20MHz offset (11MHz ~ 20Mhz) ≤ -40 @ ±30MHz offset (20MHz ~ 30Mhz)
Transmitter constellation error (rms EVM)	≤ -5dB@6Mbps, ≤ -8dB@9Mbps, ≤ -10dB@12Mbps, ≤ -13dB@18Mbps, ≤ -16dB@24Mbps, ≤ -19dB@36Mbps, ≤ -22dB@48Mbps, ≤ -25dB@54Mbps
Spurious Emissions	< -36 dBm @ 30MHz ~ 1GHz < -30 dBm above @ 1GHz ~ 12.75GHz < -47 dBm @ 1.8GHz ~ 1.9GHz < -47 dBm @ 5.15GHz ~ 5.3GHz
Rx Min input Sensitivity	PER ≤ 10% -82dBm@6Mbps, -81dBm@9Mbps, -79dBm@12Mbps -77dBm@18Mbps, -74dBm@24Mbps, -70dBm@36Mbps -66dBm@48Mbps, -65dBm@54Mbps
Rx Max input Sensitivity	≥ -20dBm(6,9,12,18,24,36,48,54Mbps) @ PER ≤ 10%
Rx Adjacent Channel Rejection	PER ≤ 10%, ACR ≥ 16dB@6Mbps, ACR ≥ 15dB@9Mbps, ACR ≥ 13dB@12Mbps, ACR ≥ 11dB@18Mbps, ACR ≥ 8dB@24Mbps, ACR ≥ 4dB@36Mbps ACR ≥ 0dB@48Mbps, ACR ≥ -1dB@54Mbps ※ ACR shall be measured by setting the desired signal's strength 3 dB above the rate-dependent sensitivity specified in min input sensitivity

4) WLAN 802.11n transceiver specification

Item	Specification
Transmit Frequency	2400 MHz ~ 2483.5 MHz (CH1~CH13) 5150 MHz ~ 5825 MHz (CH36~CH165)
Tx Power Level	$\leq 20\text{dBm}$ (2400 ~ 2483.5 MHz) $\leq 23\text{dBm}$ (5150 ~ 5350 MHz), $\leq 30\text{dBm}$ (5470 ~ 5725 MHz)
Frequency Tolerance	within ± 25 PPM(2400 ~ 2483.5 MHz) within ± 20 PPM(5150 ~ 5725 MHz)
Chip clock Frequency Tolerance	within ± 25 PPM(2400 ~ 2483.5 MHz) within ± 20 PPM(5150 ~ 5725 MHz)
Spectrum Mask	≤ -20 @ $\pm 11\text{MHz}$ offset (9MHz ~ 11MHz) ≤ -28 @ $\pm 20\text{MHz}$ offset (11MHz ~ 20MHz) ≤ -45 @ $\pm 30\text{MHz}$ offset (20MHz ~ 30MHz)
Transmitter constellation error (rms EVM)	$\leq -5\text{dB}@6.5\text{Mbps}$, $\leq -10\text{dB}@13\text{Mbps}$, $\leq -13\text{dB}@19.5\text{Mbps}$, $\leq -16\text{dB}@26\text{Mbps}$, $\leq -19\text{dB}@39\text{Mbps}$, $\leq -22\text{dB}@52\text{Mbps}$, $\leq -25\text{dB}@58.5\text{Mbps}$, $\leq -28\text{dB}@65\text{Mbps}$
Spurious Emissions	[2400 MHz ~ 2483.5 MHz (CH1~CH13)] $< -36\text{ dBm}$ @ 30MHz ~ 1GHz $< -30\text{ dBm}$ above @ 1GHz ~ 12.75GHz $< -47\text{ dBm}$ @ 1.8GHz ~ 1.9GHz $< -47\text{ dBm}$ @ 5.15GHz ~ 5.3GHz [5150 MHz ~ 5725 MHz (CH36~CH140)] $< -36\text{ dBm}$ @ 30 MHz to 47 MHz $< -54\text{ dBm}$ @ 47 MHz to 74 MHz $< -36\text{ dBm}$ @ 74 MHz to 87,5 MHz $< -54\text{ dBm}$ @ 87,5 MHz to 118 MHz $< -36\text{ dBm}$ @ 118 MHz to 174 MHz $< -54\text{ dBm}$ @ 174 MHz to 230 MHz $< -36\text{ dBm}$ @ 230 MHz to 470 MHz $< -54\text{ dBm}$ @ 470 MHz to 862 MHz $< -36\text{ dBm}$ @ 862 MHz to 1 GHz $< -30\text{ dBm}$ @ 1 GHz to 5.15 GHz $< -30\text{ dBm}$ @ 5.35 GHz to 5.47 GHz $< -30\text{ dBm}$ @ 5.725 GHz to 26.5 GHz

2. PERFORMANCE

Rx Min input Sensitivity	$PER \leq 10\%$ -82dBm@6.5Mbps, -79dBm@13Mbps, -77dBm@19.5Mbps -74dBm@26Mbps, -70dBm@39Mbps, -66dBm@52Mbps -65dBm@58.5Mbps, -64dBm@65Mbps
Rx Max input Sensitivity	$\geq -20\text{dBm}$ @ $PER \leq 10\%$ (2400 ~ 2483.5 MHz) $\geq -30\text{dBm}$ @ $PER \leq 10\%$ (5150 ~ 5725 MHz)
Rx Adjacent Channel Rejection	$PER \leq 10\%$, $ACR \geq 16\text{dB}$ @6.5Mbps, $ACR \geq 13\text{dB}$ @13Mbps, $ACR \geq 11\text{dB}$ @19.5Mbps, $ACR \geq 8\text{dB}$ @26Mbps, $ACR \geq 4\text{dB}$ @39Mbps, $ACR \geq 0\text{dB}$ @52Mbps $ACR \geq -1\text{dB}$ @58.5Mbps, $ACR \geq -2\text{dB}$ @65Mbps ※ACR shall be measured by setting the desired signal's strength 3 dB above the rate-dependent Sensitivity specified in min input sensitivity

5) GPS receiver specification

Item	Specification
Receive Frequency	1574.42 MHz ~ 1576.42 MHz
Minimum Sensitivity	Success rate(1 satellite) $\geq 60\%$ @-154dBm

6) Current consumption

	Stand by
WiFi Only(Sleep Mode)	10 mA under

** Test condition for Standby current consumption should be as below.

: Measurement time 1hr, with Agilent & CMT JIG

7) Battery life time

	Stand by
WiFi Only(Sleep Mode)	460 hours.

8) Charging hour

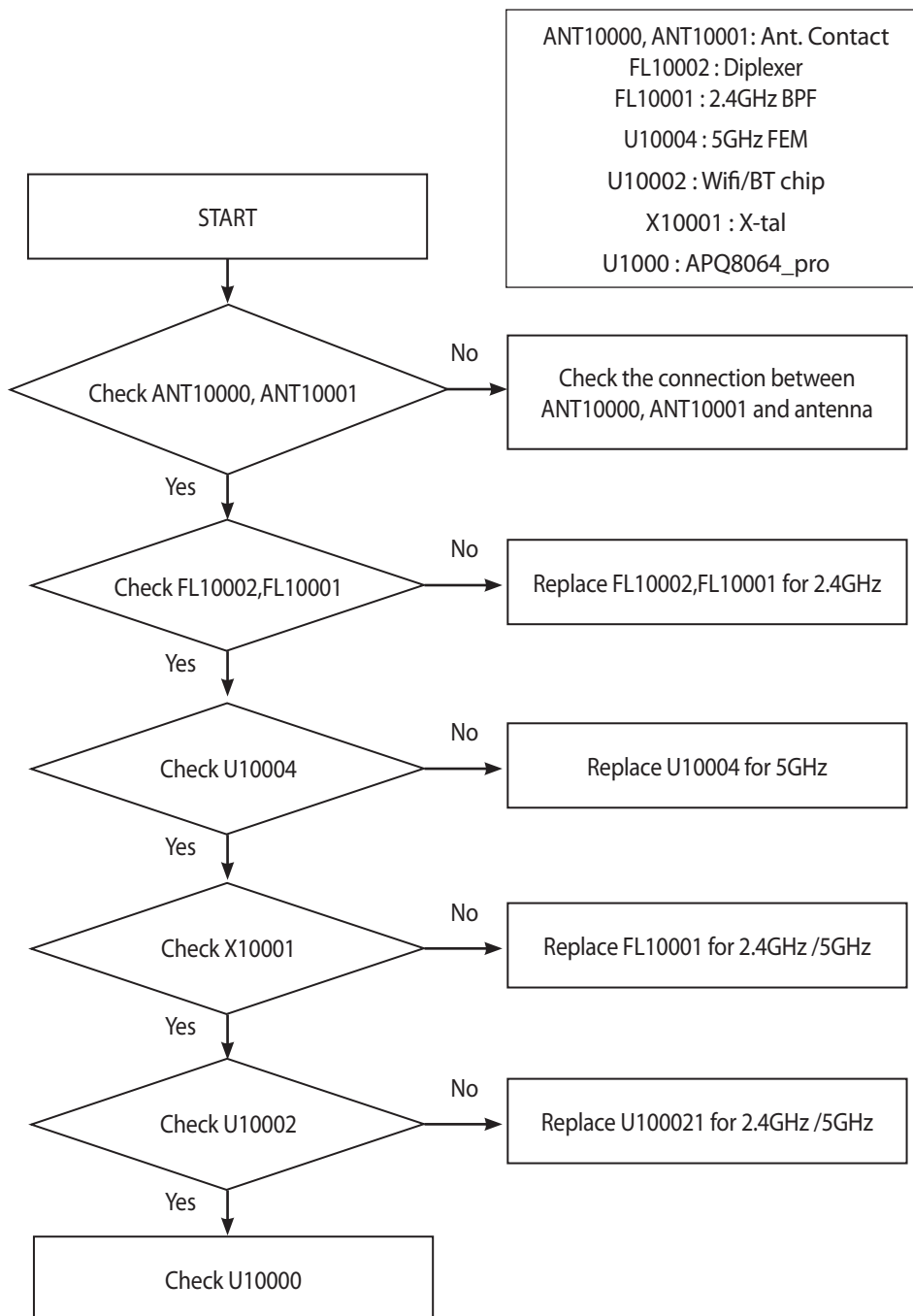
5hour. under (4,600mAh battery, 1.8A TA)

9) Battery indicator

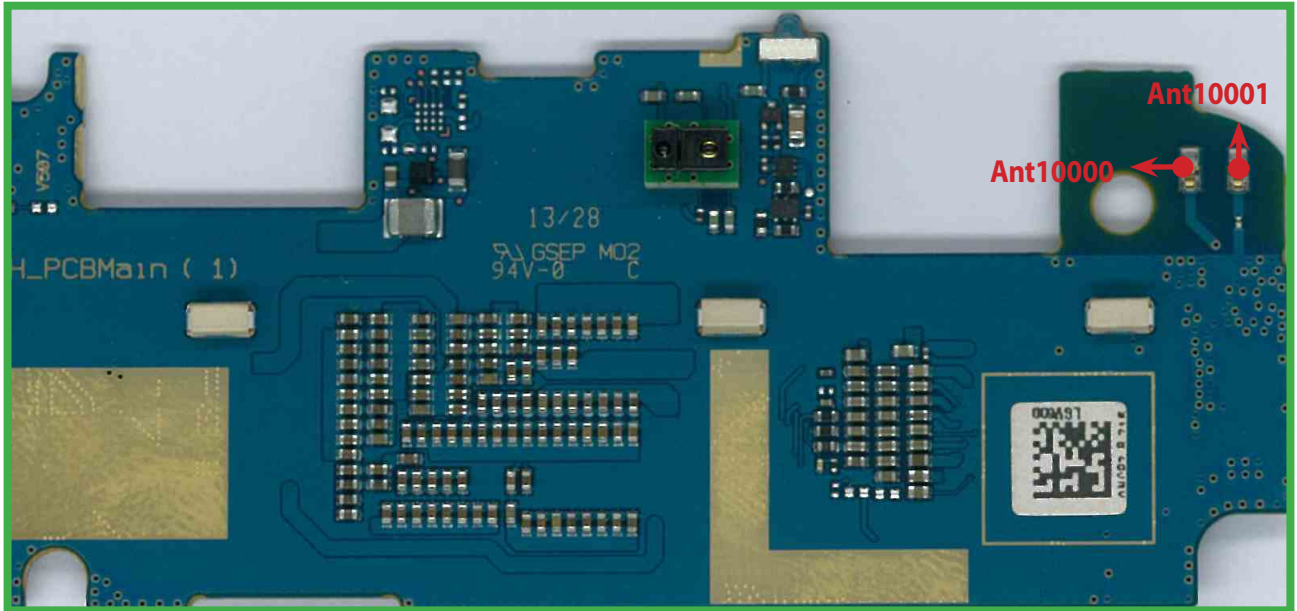
Battery Bar	Specification	Unit
BAR 20 (Full)	98% over	Remain%
BAR 20 -> 19	98% → 97%	
BAR 19 -> 18	93% → 92%	
BAR 18 -> 17	88% → 87%	
BAR 17 -> 16	83% → 82%	
BAR 16 -> 15	78% → 77%	
BAR 15 -> 14	73% → 72%	
BAR 14 -> 13	68% → 67%	
BAR 13 -> 12	63% → 62%	
BAR 12 -> 11	58% → 57%	
BAR 11 -> 10	53% → 52%	
BAR 10 -> 9	48% → 47%	
BAR 9 -> 8	43% → 42%	
BAR 8 -> 7	38% → 37%	
BAR 7 -> 6	33% → 32%	
BAR 6 -> 5	28% → 27%	
BAR 5 -> 4	23% → 22%	
BAR 4 -> 3	18% → 17%	
BAR 3 -> 2	13% → 12%	
BAR 2 -> 1	8% → 7%	
BAR 1 -> 0	3% → 2%	
Low Battery Pop-up	15%, 5% Low battery pop-up	
POWER OFF	1% under	

3. TROUBLE SHOOTING

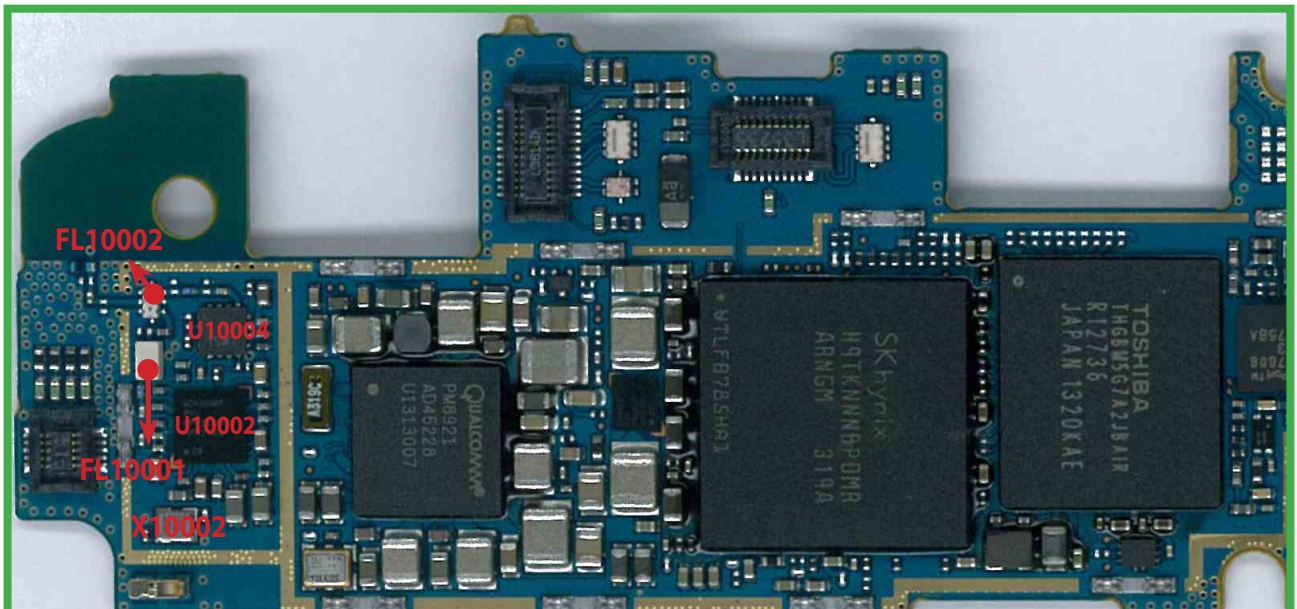
3.1 Wifi Module



3. TROUBLE SHOOTING



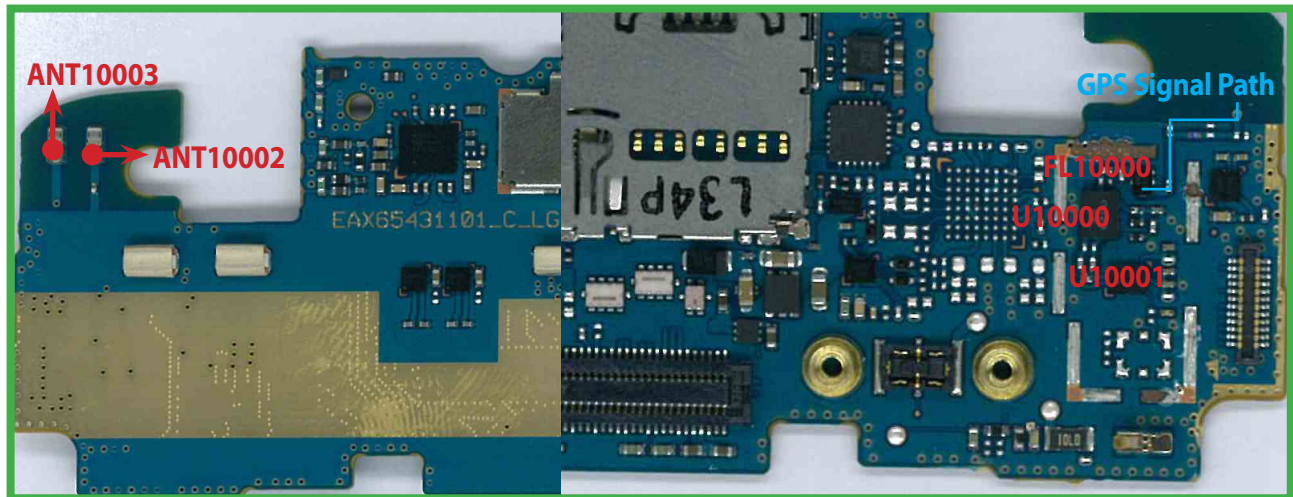
TOP



BOT

3.2 GPS Module

The WGR7640 integrates GPS/GNSS solution.

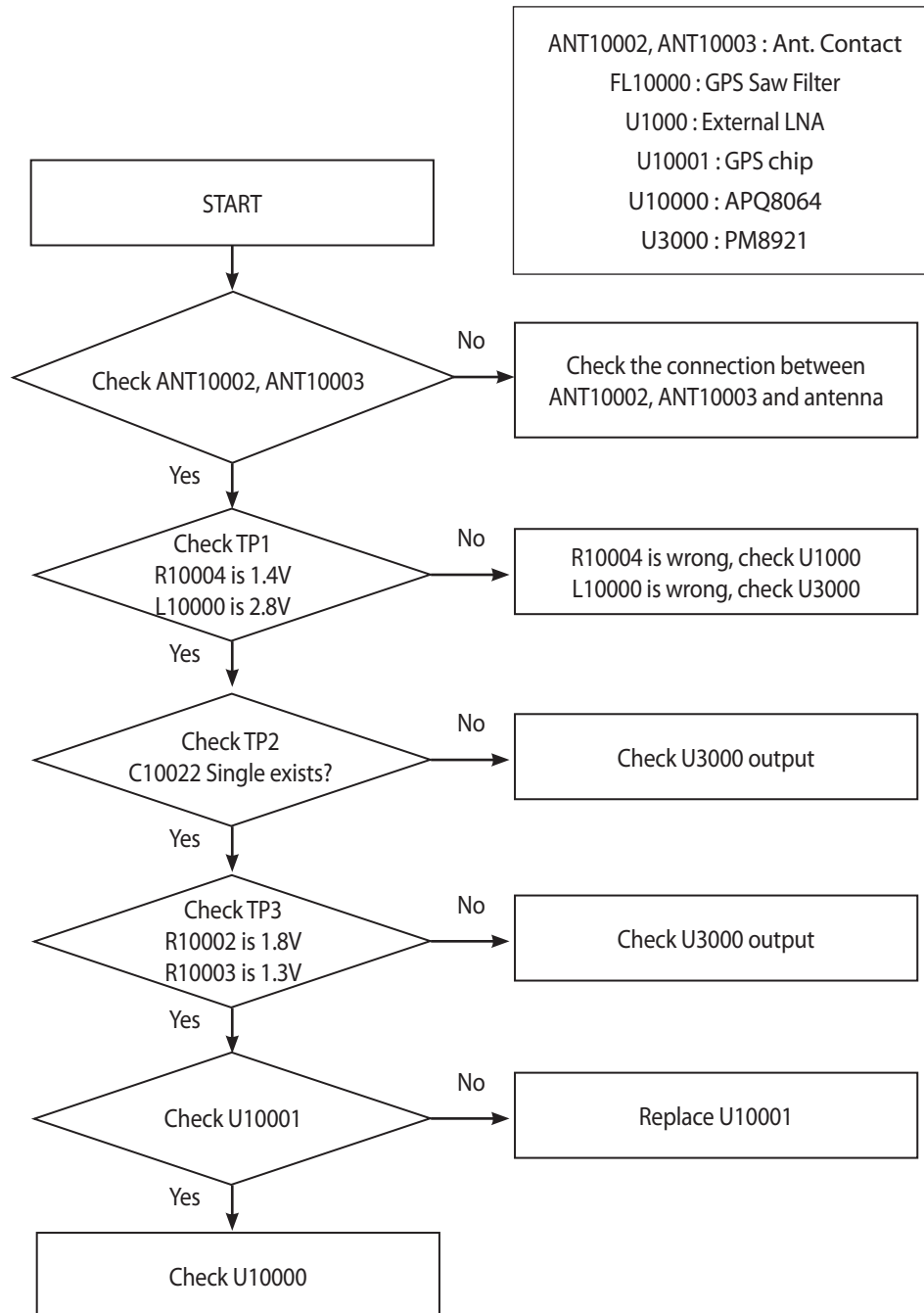


TOP / Left Side

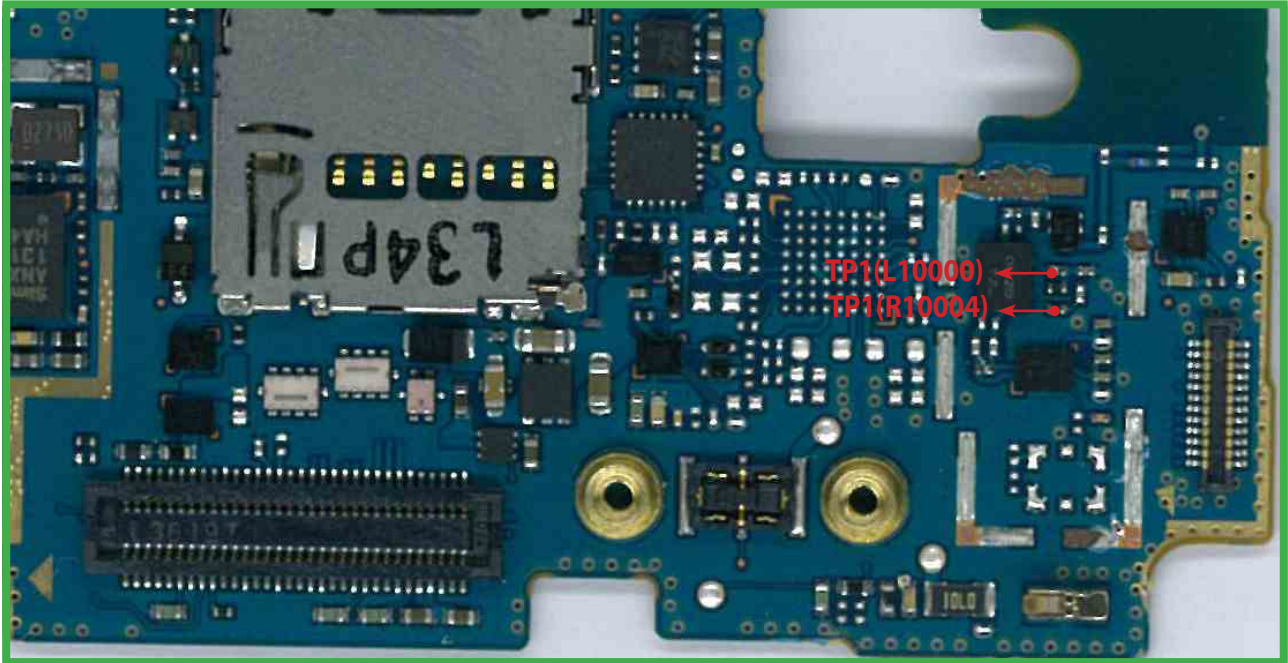
Bottom / Right Side

Reference	Description
ANT10002/ANT10003	Antenna Contact
FL10000	Saw Filter
U10000	External LNA
U10001	GPS Chip(WGR7640

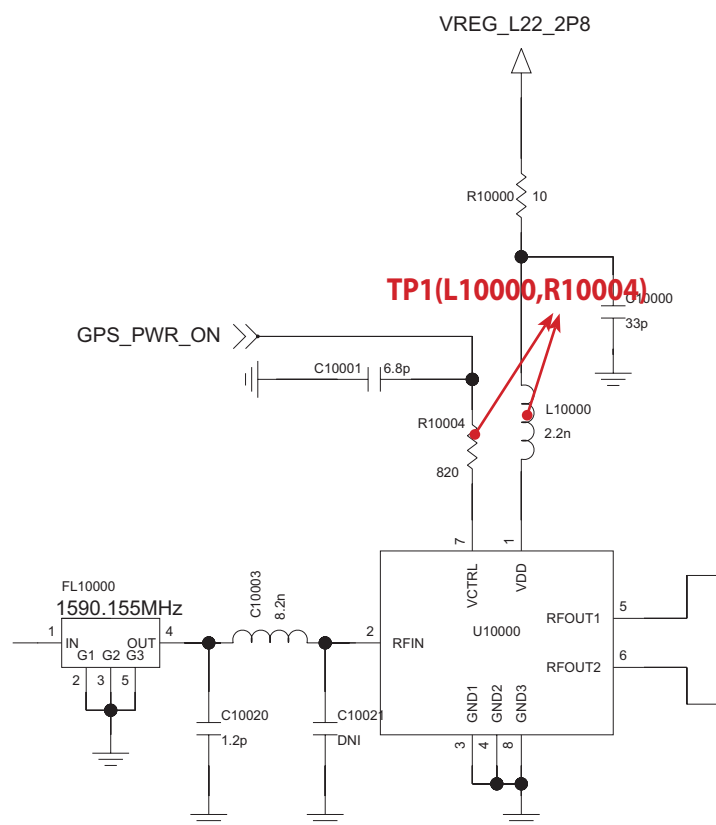
3. TROUBLE SHOOTING



3. TROUBLE SHOOTING

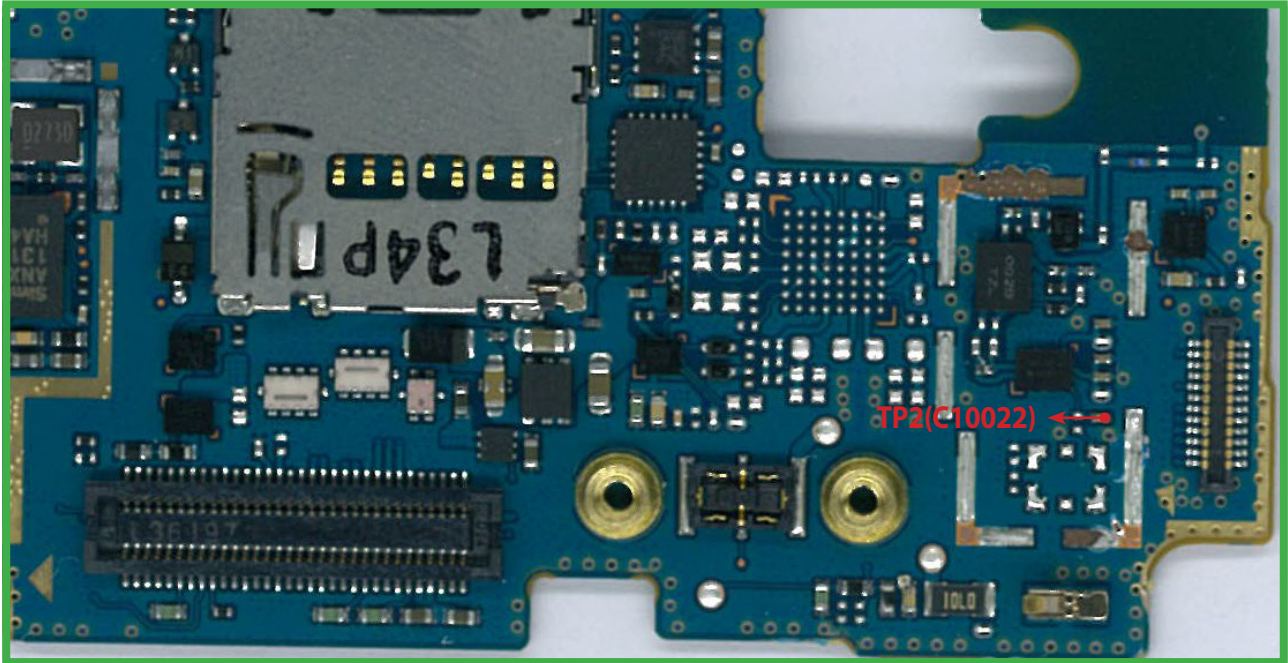


PCB Bottom Side (Inside SMT Type SHIELD CAN)



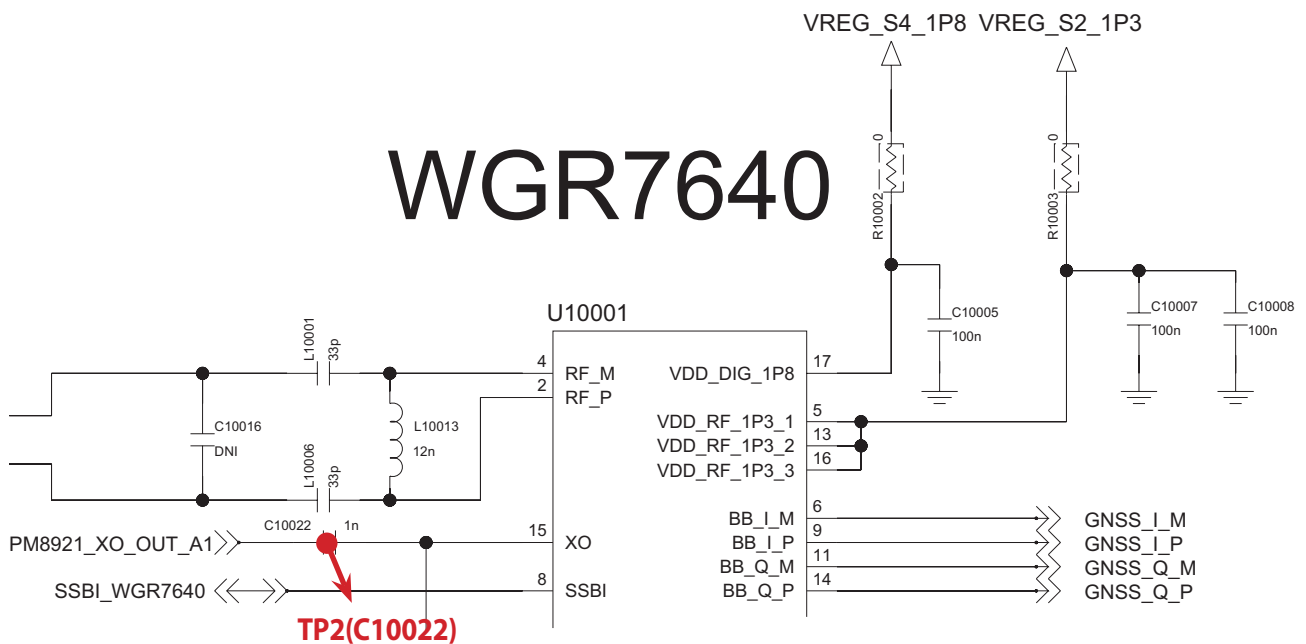
Schematic of External LNA

3. TROUBLE SHOOTING



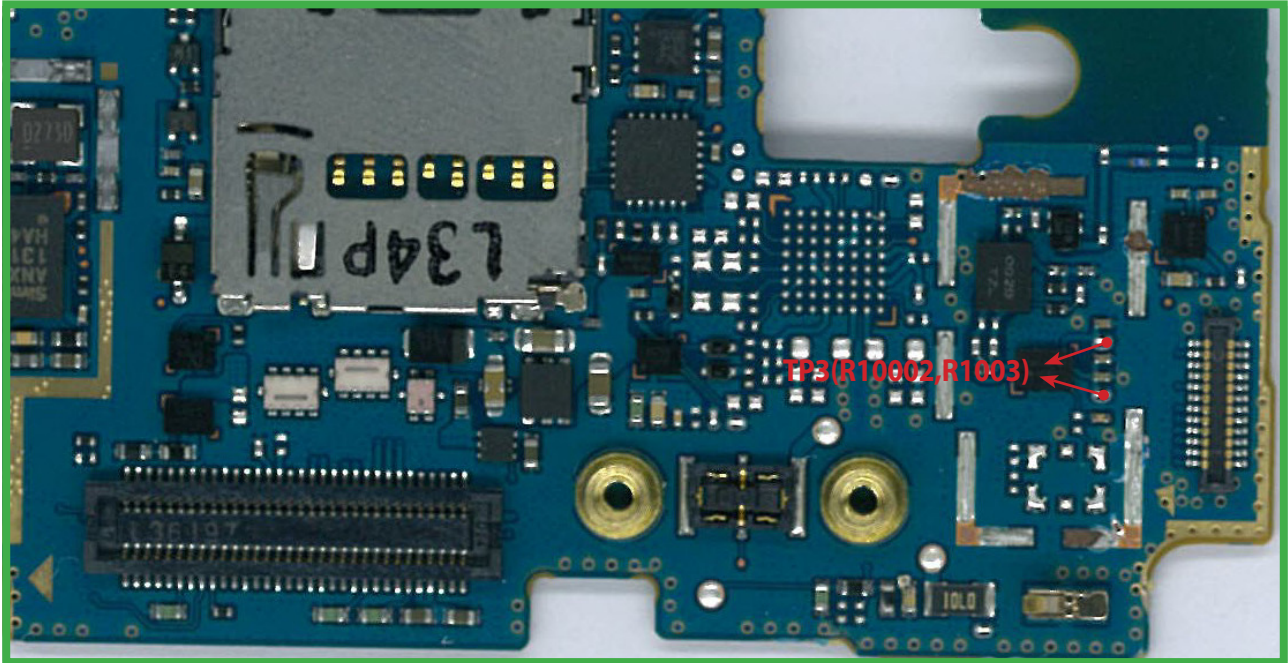
PCB Bottom Side (Inside SMT Type SHIELD CAN)

WGR7640

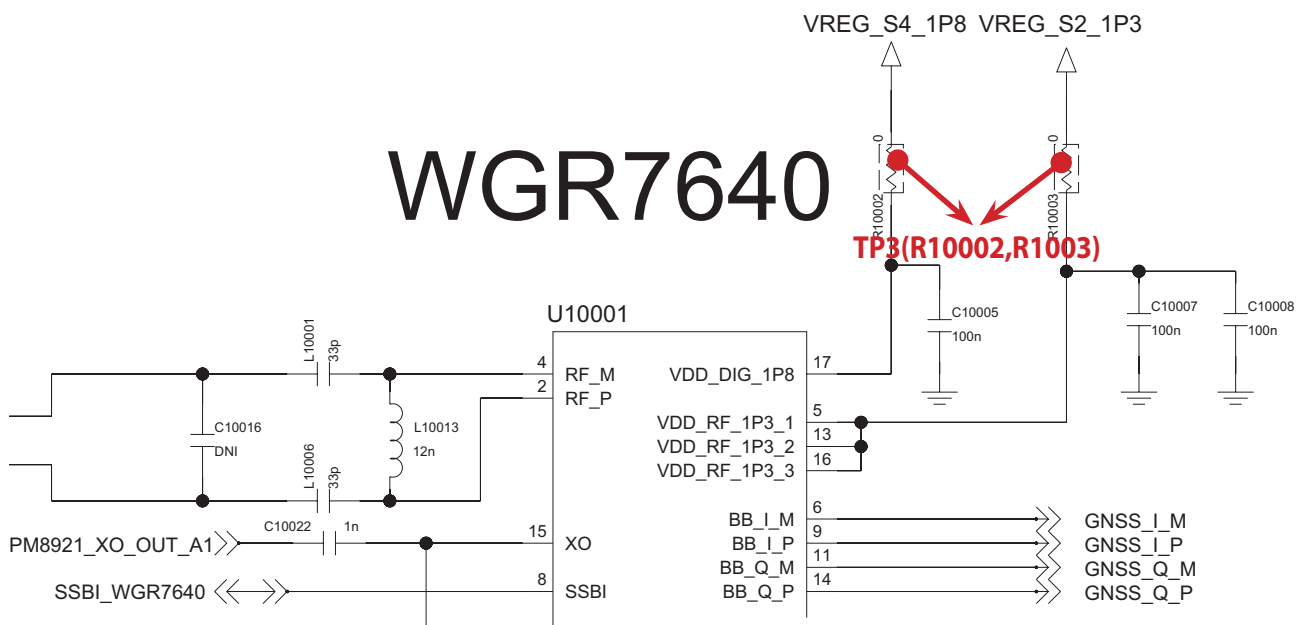


Schematic of GPS Chip(WGR7640)

3. TROUBLE SHOOTING



PCB Bottom Side (Inside SMT Type SHIELD CAN)



Schematic of GPS Chip(WGR7640)

3.3 Charging

LG-V500 uses PMIC (Power Management IC) for charging control. The PMIC (PM8921) circuit includes the charging pass transistor (SSM6J502NU), detects the VDD and VBAT voltages, and its control driver. The LG-V500 charging current limit is set around 1400mA.

Charging Procedure

- Supply the VBUS_USB_IN use a TA or u-USB cable
- PM8921 control the charging current (VPWR)
- Charging current flows into the main battery

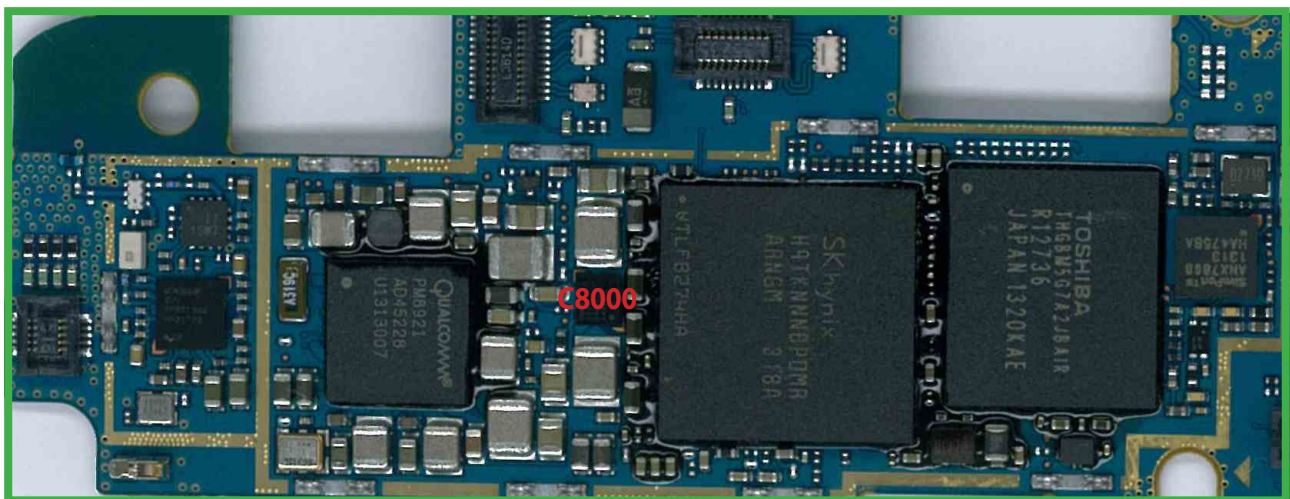
Troubleshooting Check Point

- Connection of the TA or u-USB cable
- Current path from PMIC to FET: C8001, L8000
- Charging current path (Battery FET: SSM6J502NU)

VBAT between FET and Main Battery C8018:

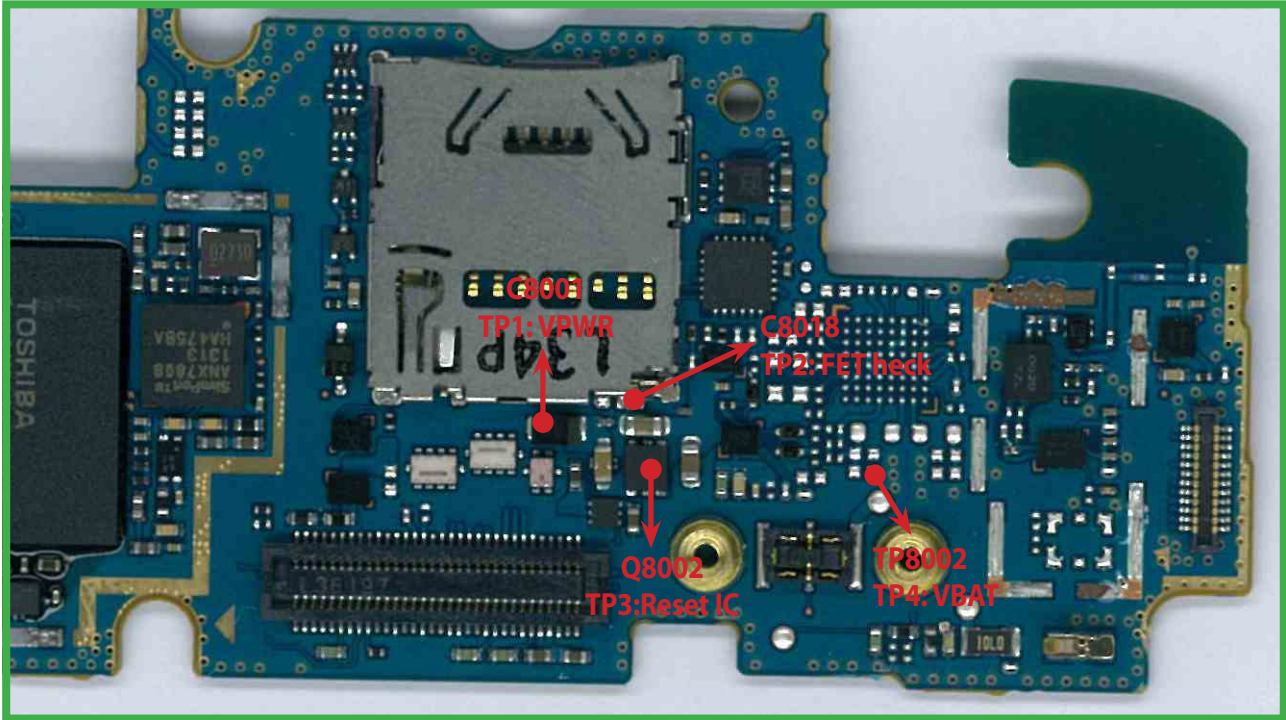
- Main battery

Battery Voltage check (TP8002)

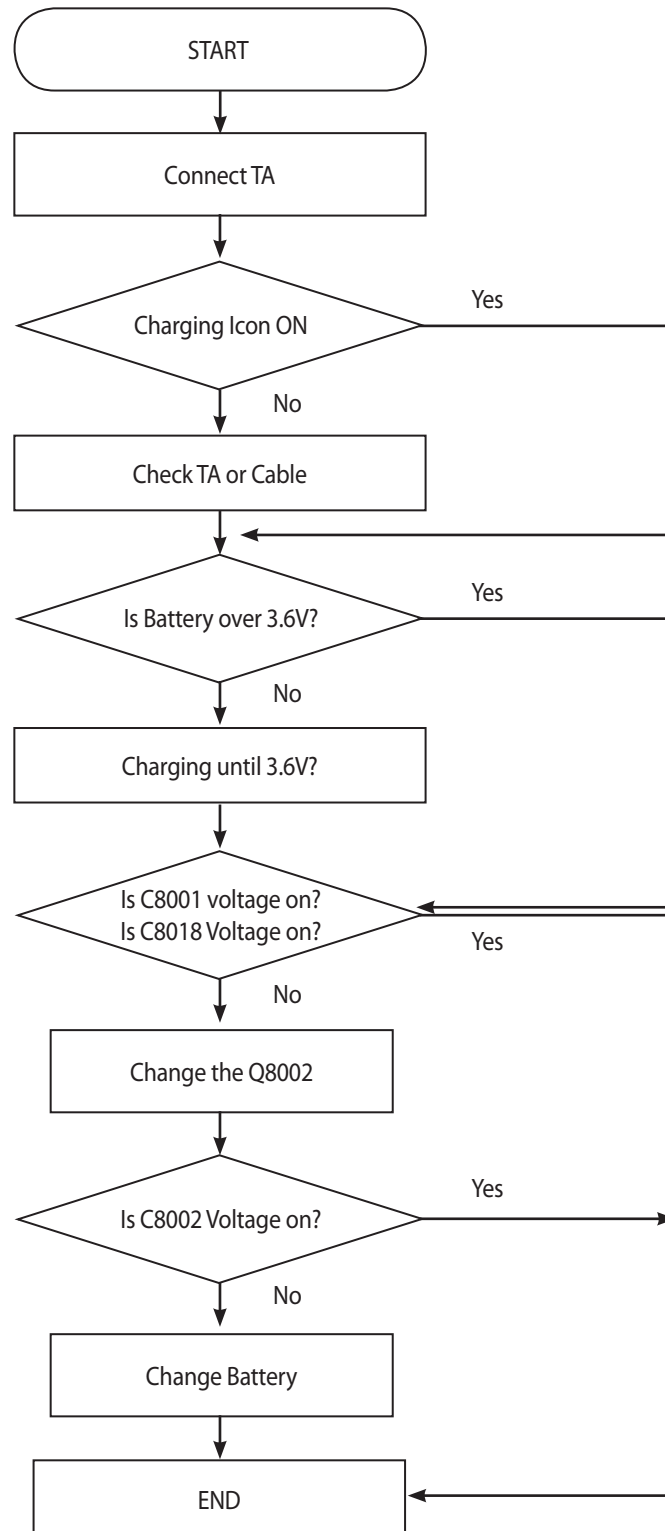


PCB Bottom Side (Inside SMT Type SHIELD CAN)

3. TROUBLE SHOOTING



PCB Bottom Side

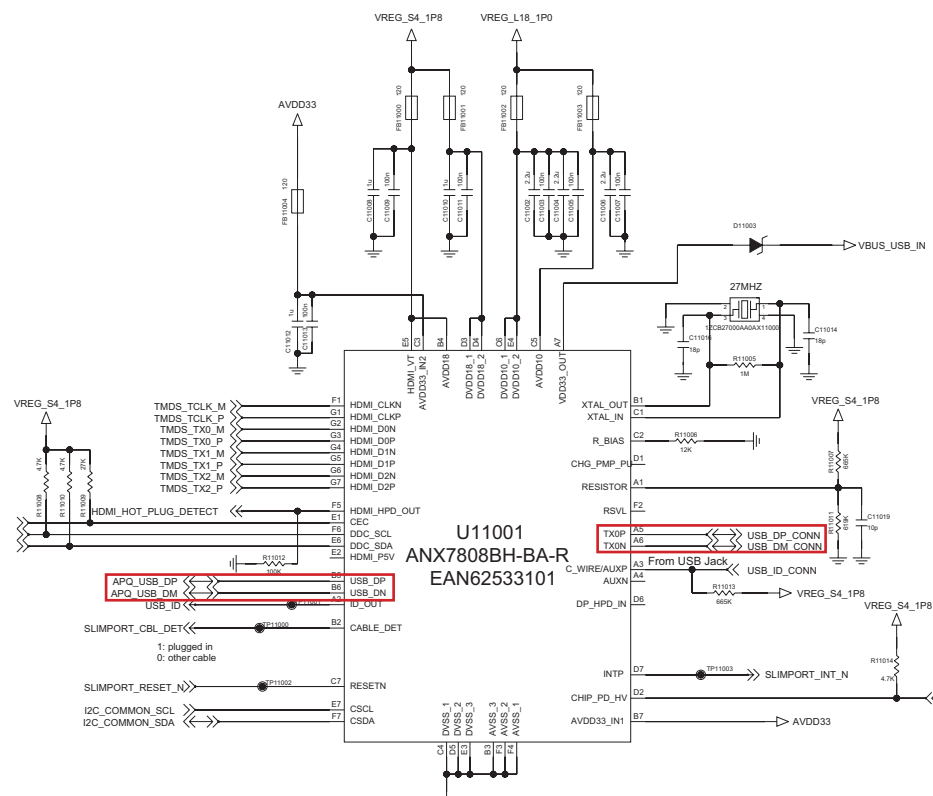
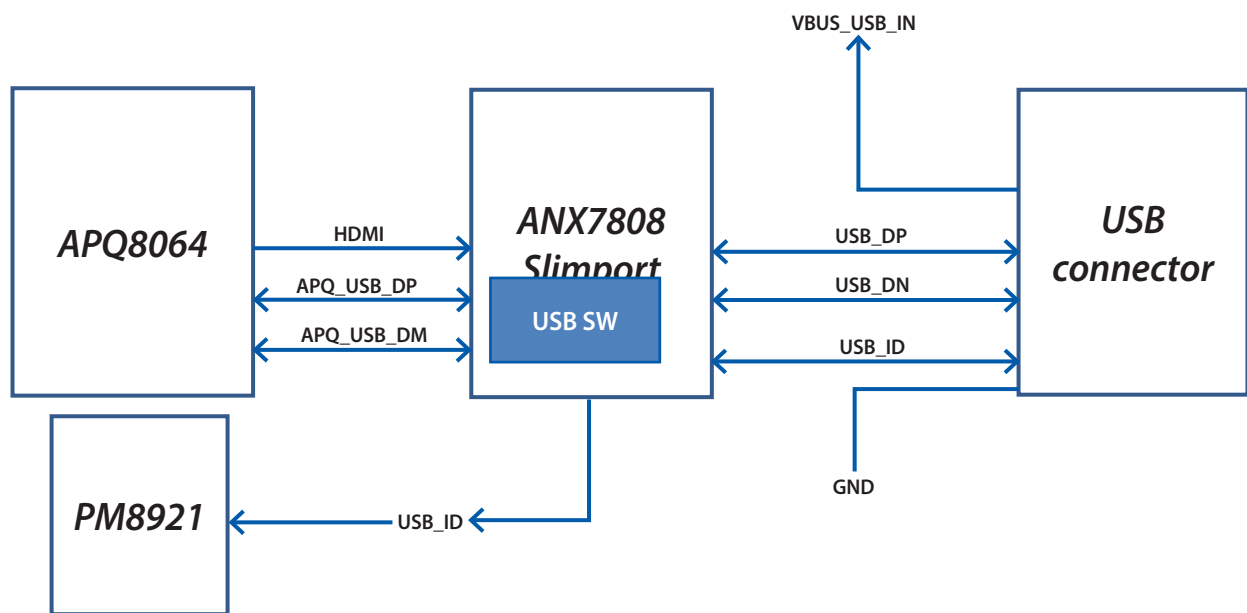


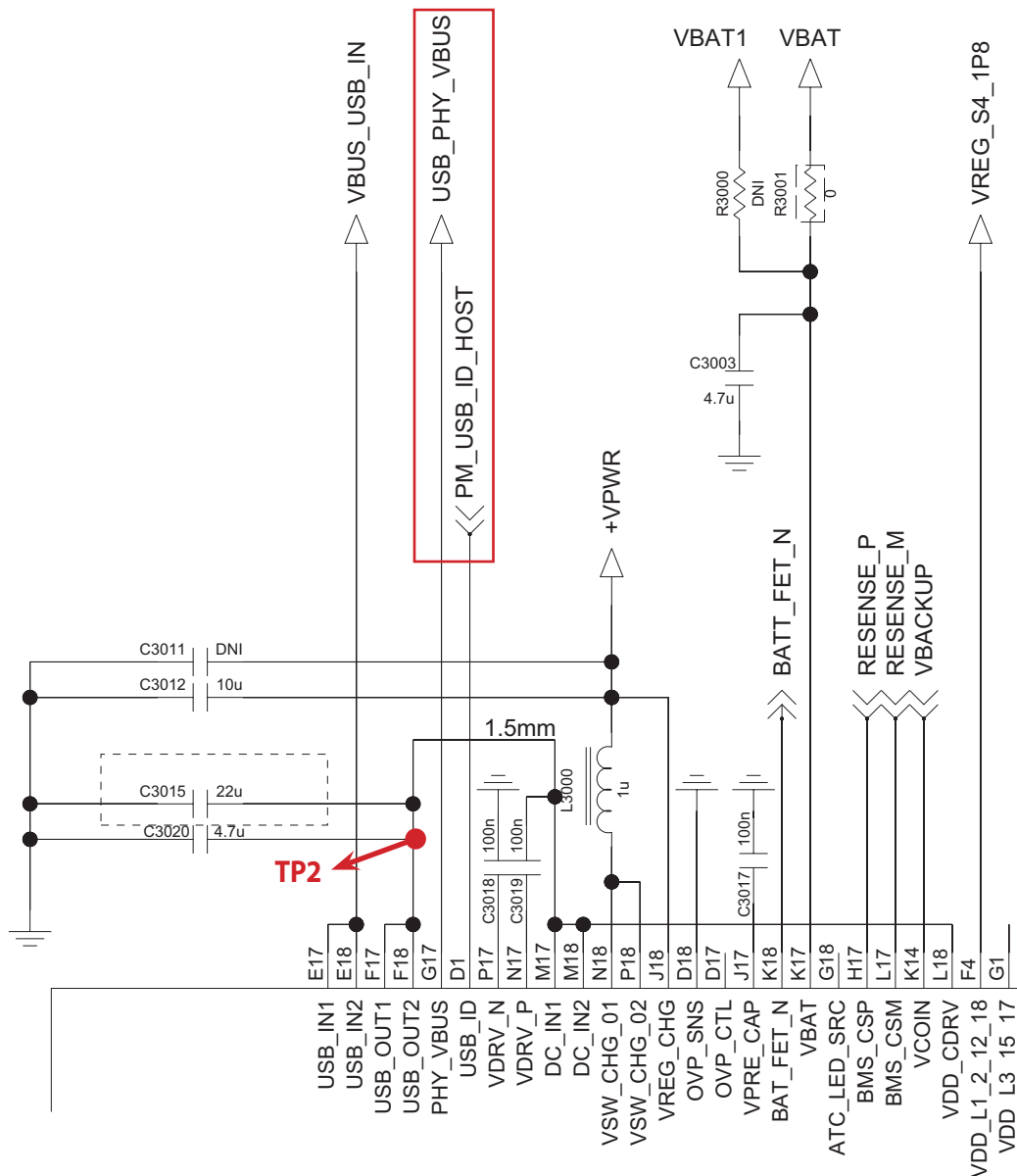
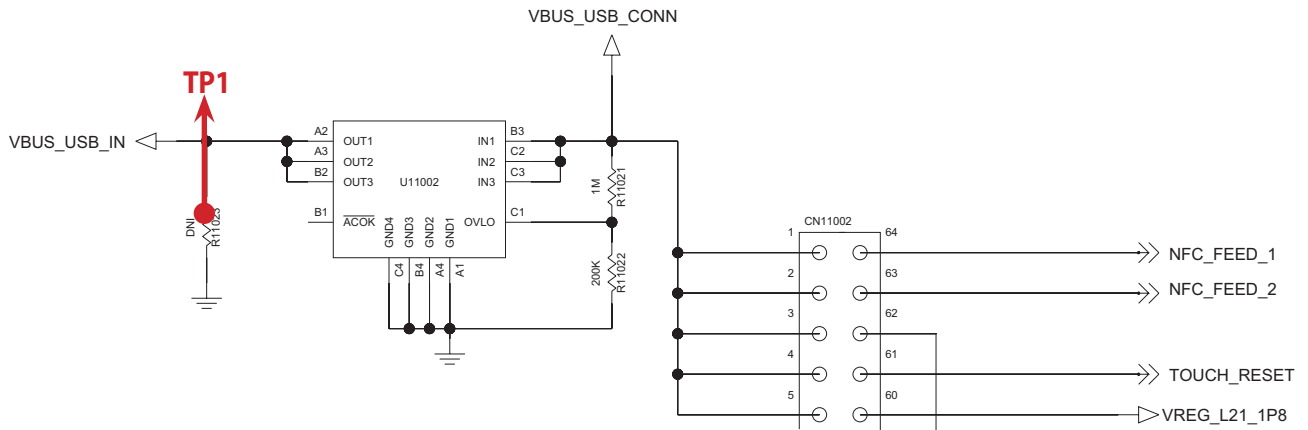
3.4 USB

The sequence of LGV500 USB is,

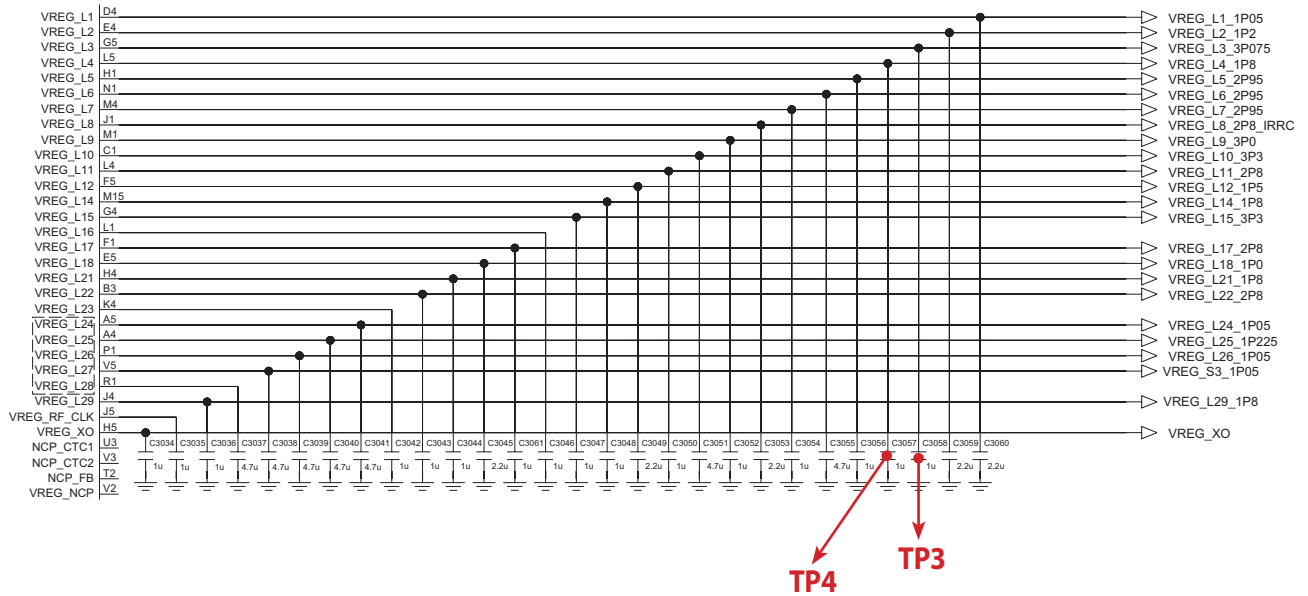
USB connected to LGV500 -> VBUS_USB_IN(R11023) goes to 5V -> VREG_L3_3P075 go to 3.075V -> USB_DATA is triggered -> USB work

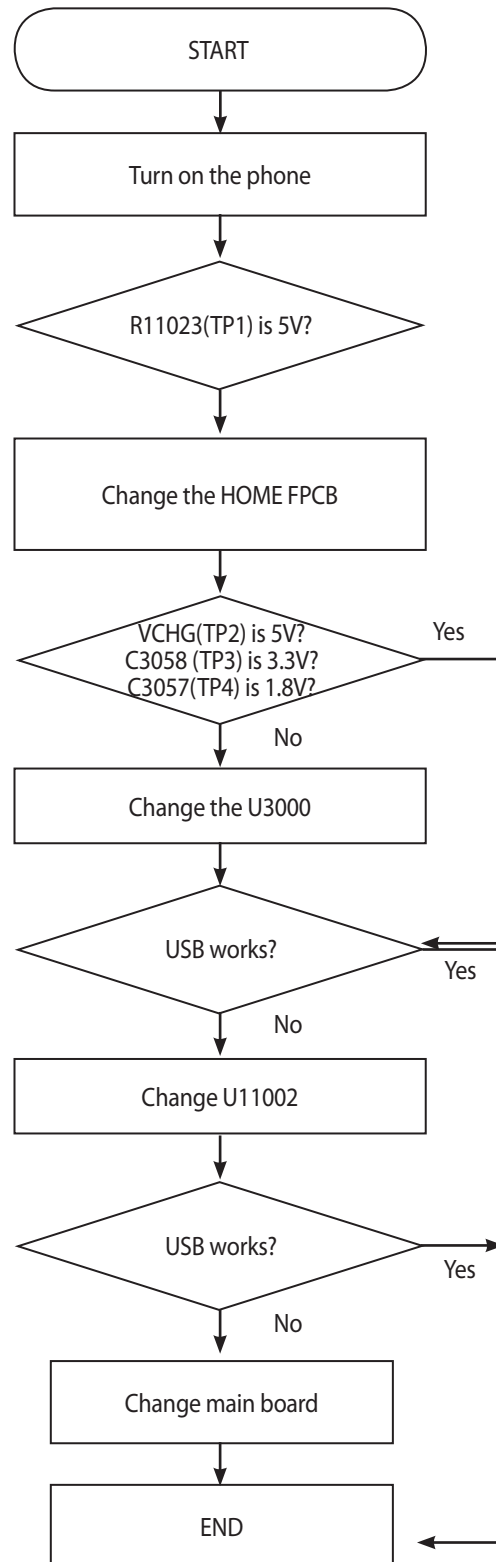
Block Diagram of USB & UART connection is shown below



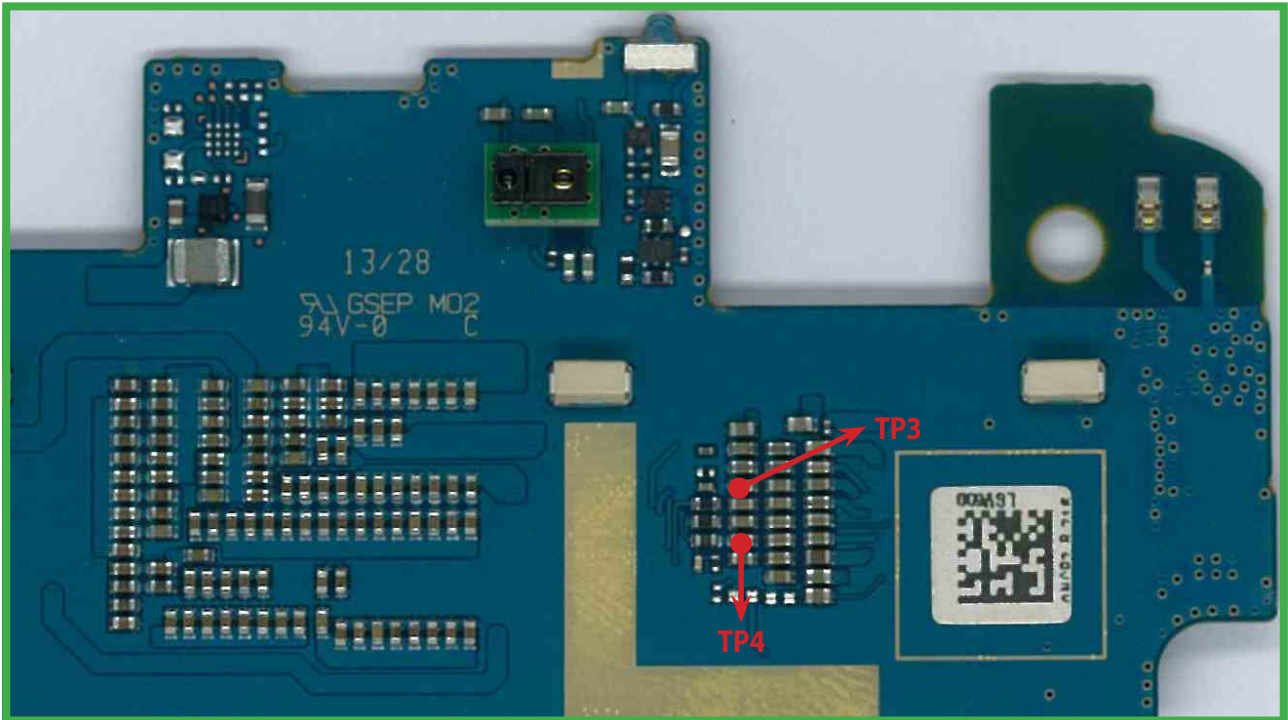


3. TROUBLE SHOOTING

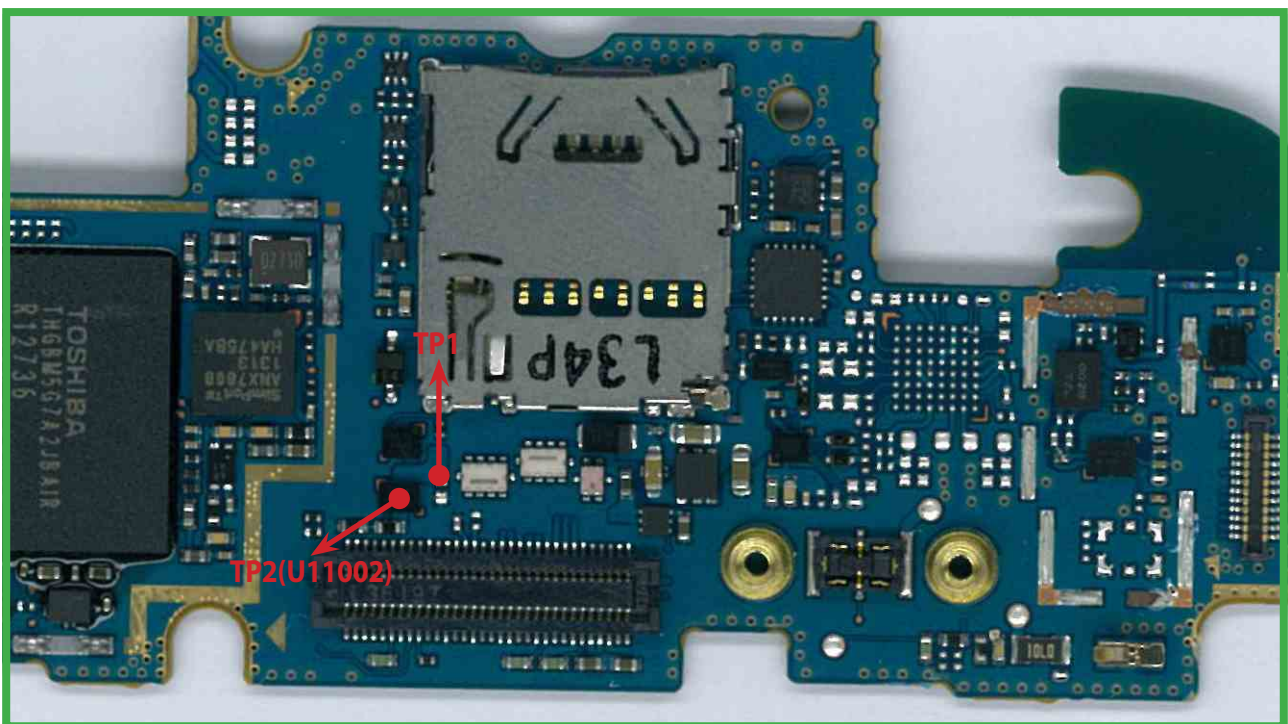




3. TROUBLE SHOOTING



PCB TOP Side

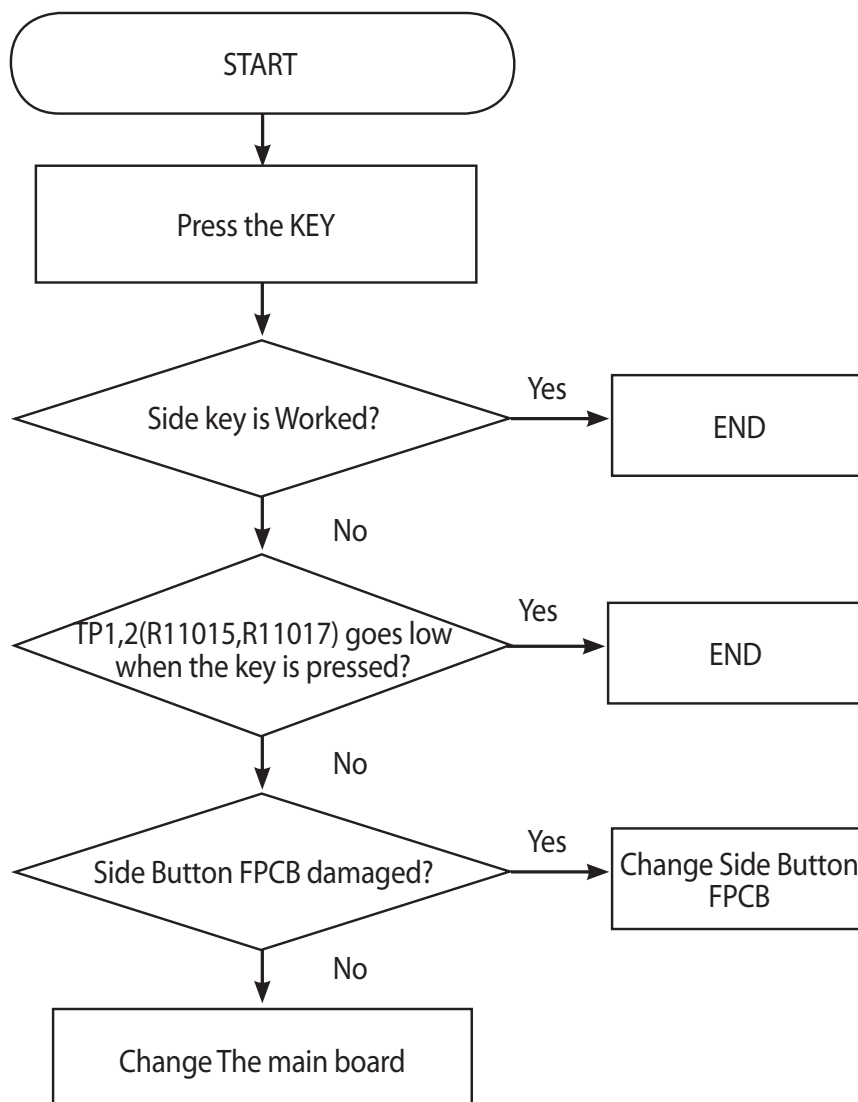
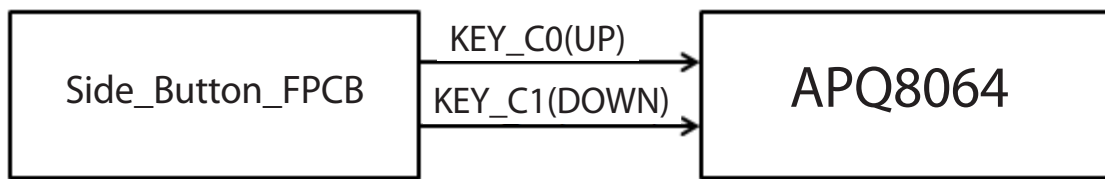


PCB Bottom Side

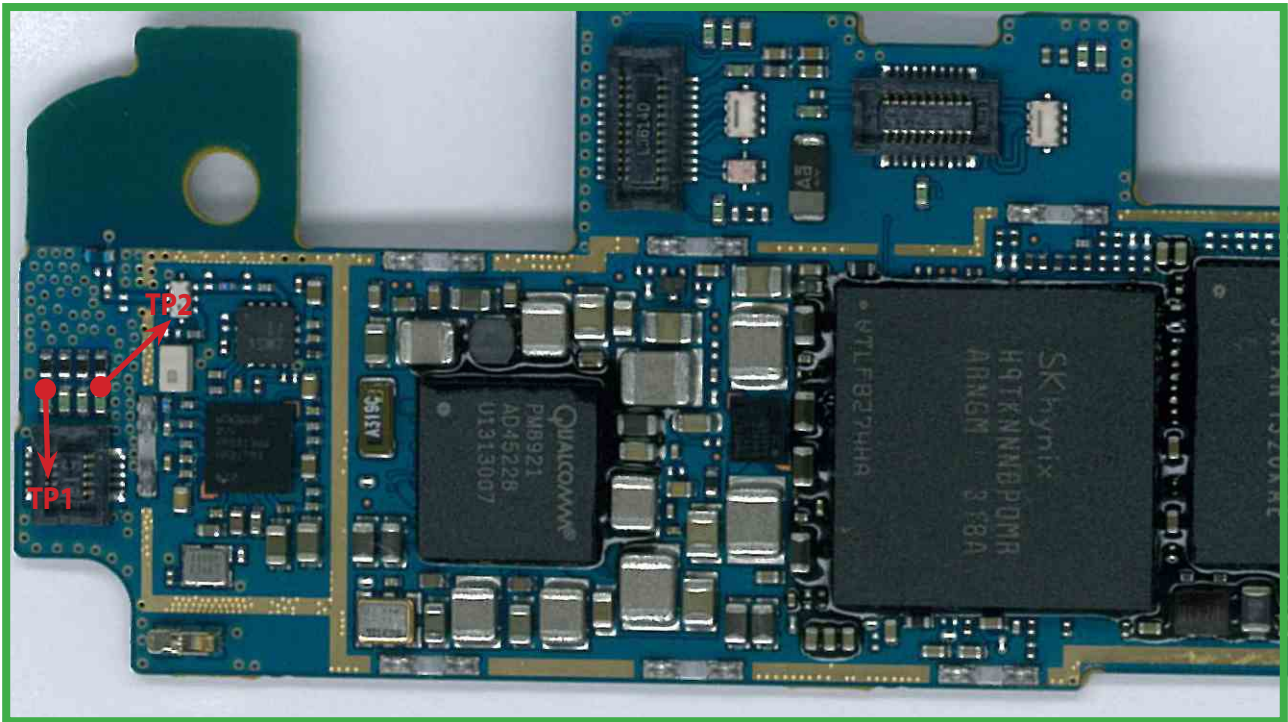
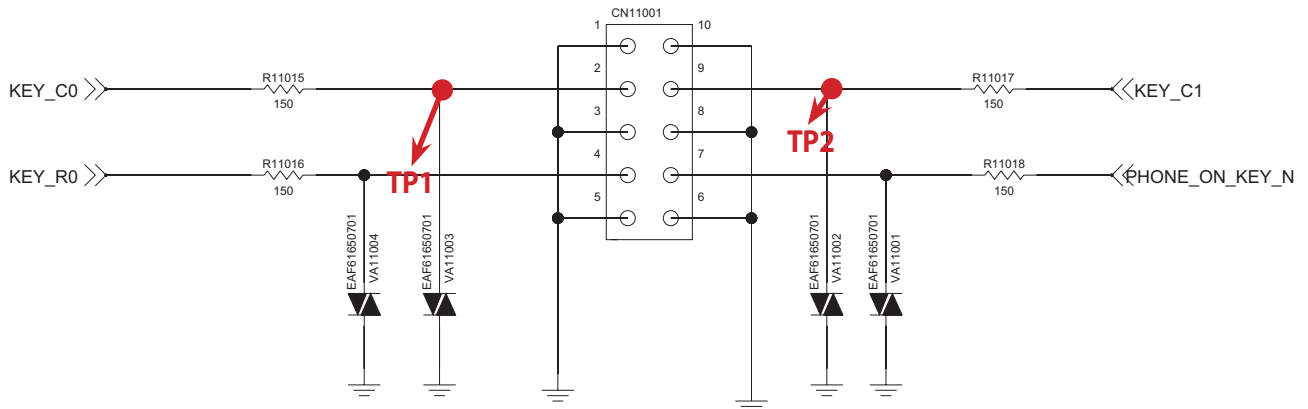
3.5 Side Key

The sequence of detecting LGV500 Side key is,

Side key pressed on LGV500 -> KEY_C0(UP), KEY_C1(DOWN) goes low
Block Diagram of Side key connection is shown below



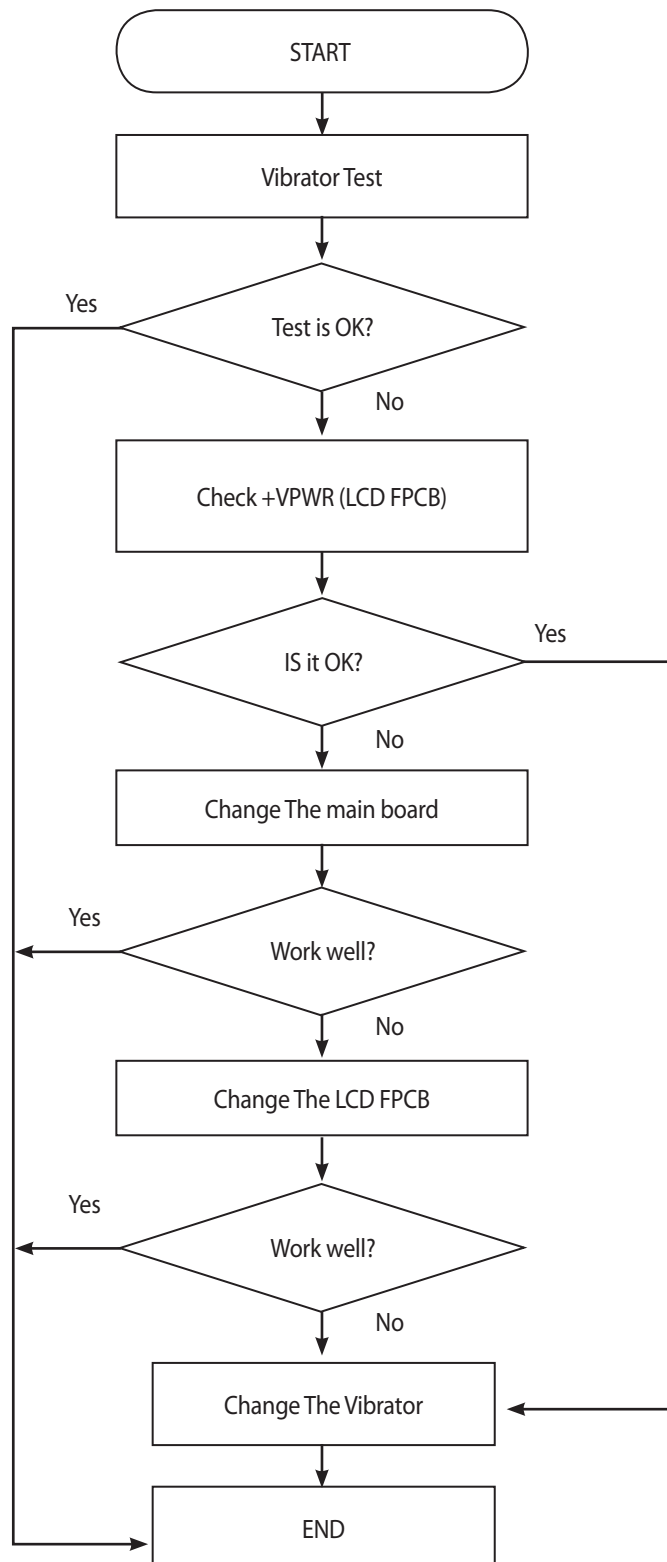
POWER & VOLUME KEY



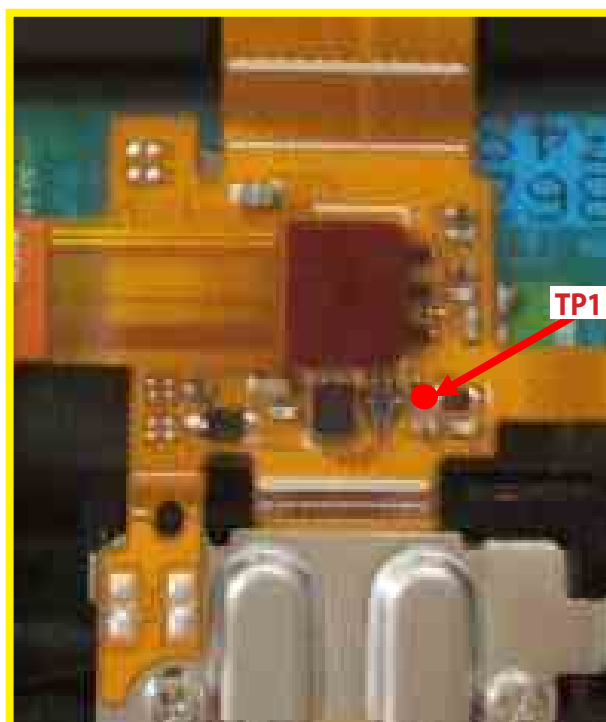
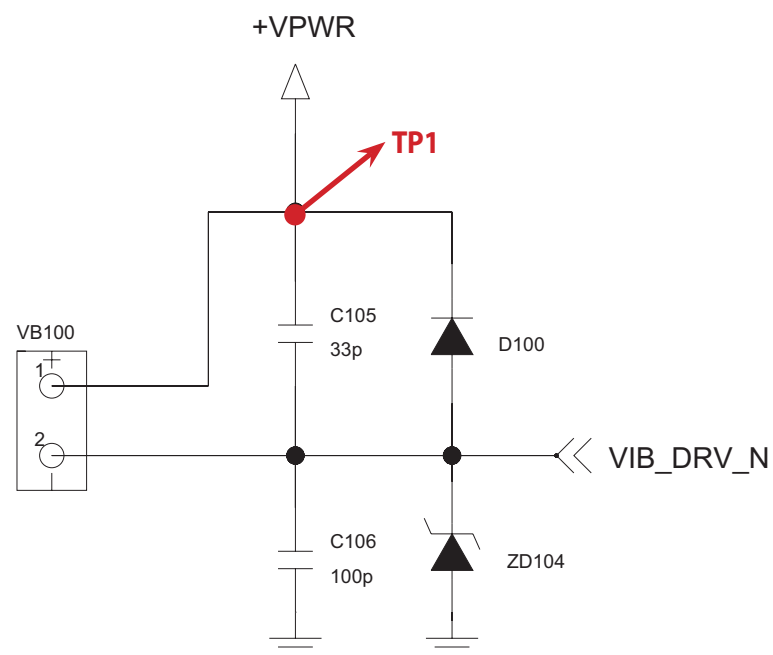
PCB Bottom Side

3.6 Vibrator

Check out the setting menu on the phone. If not, check Test points shown on the pictures.

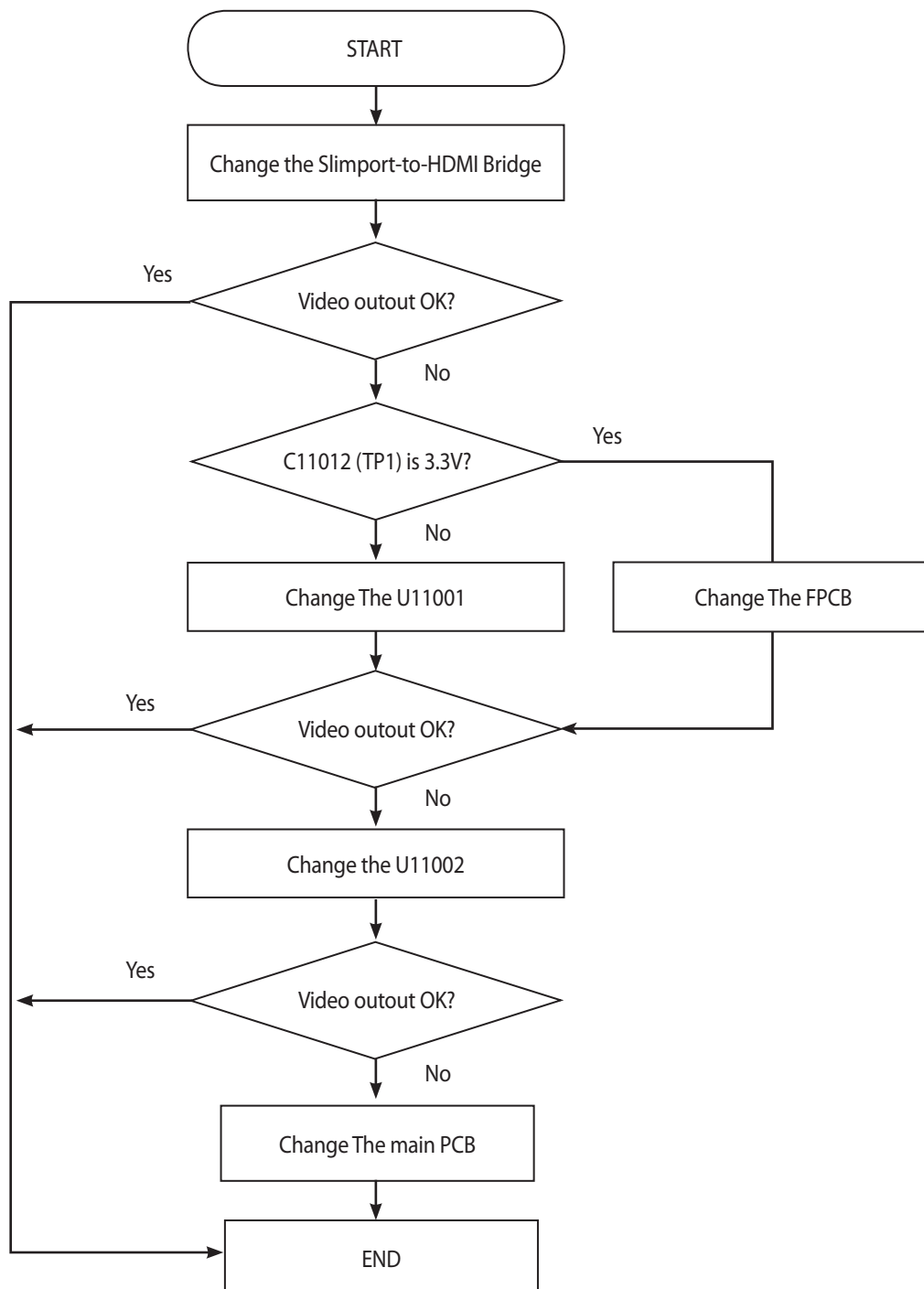


MOTOR

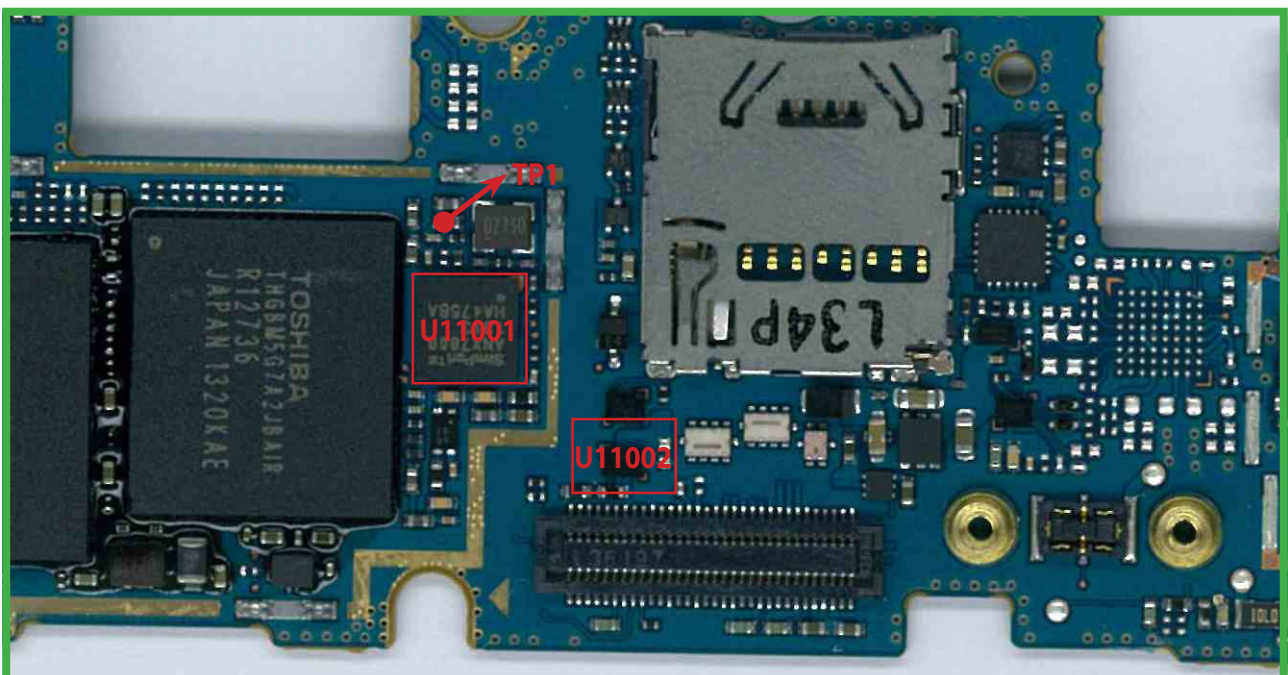
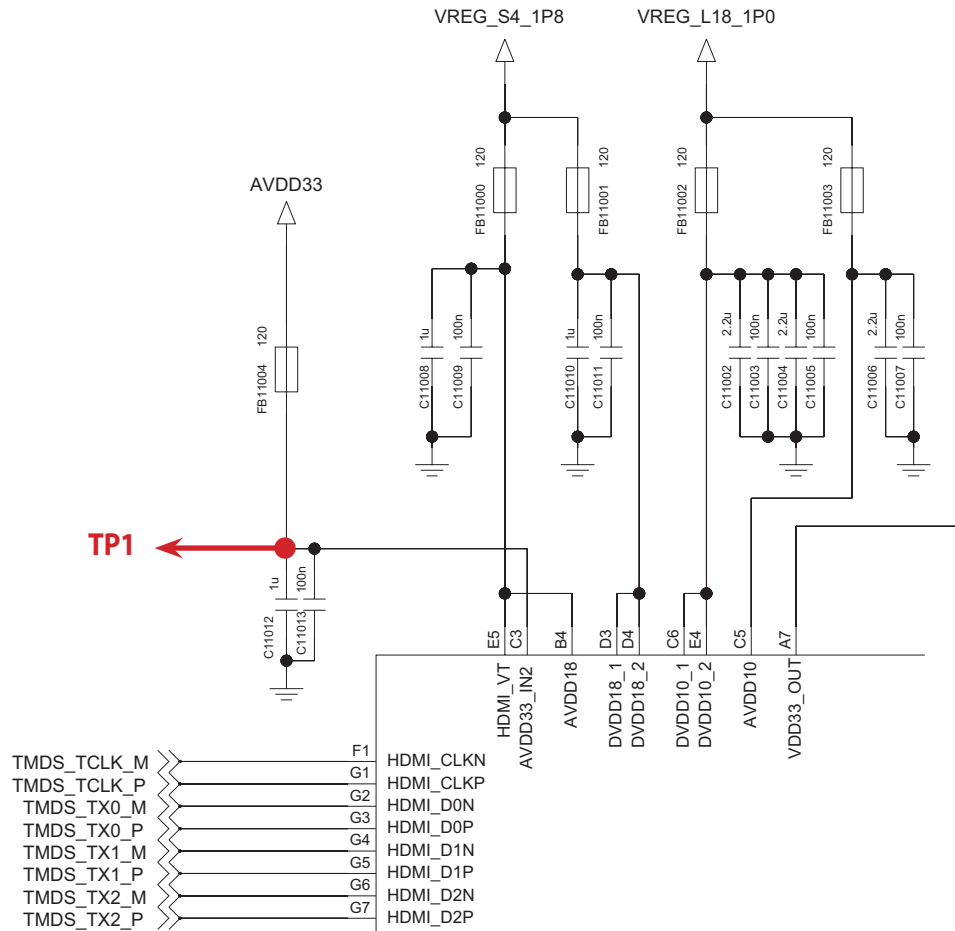


3.7 HDMI

Check out the setting menu on the phone and then set the proper resolution for the outer display as shown below



3. TROUBLE SHOOTING

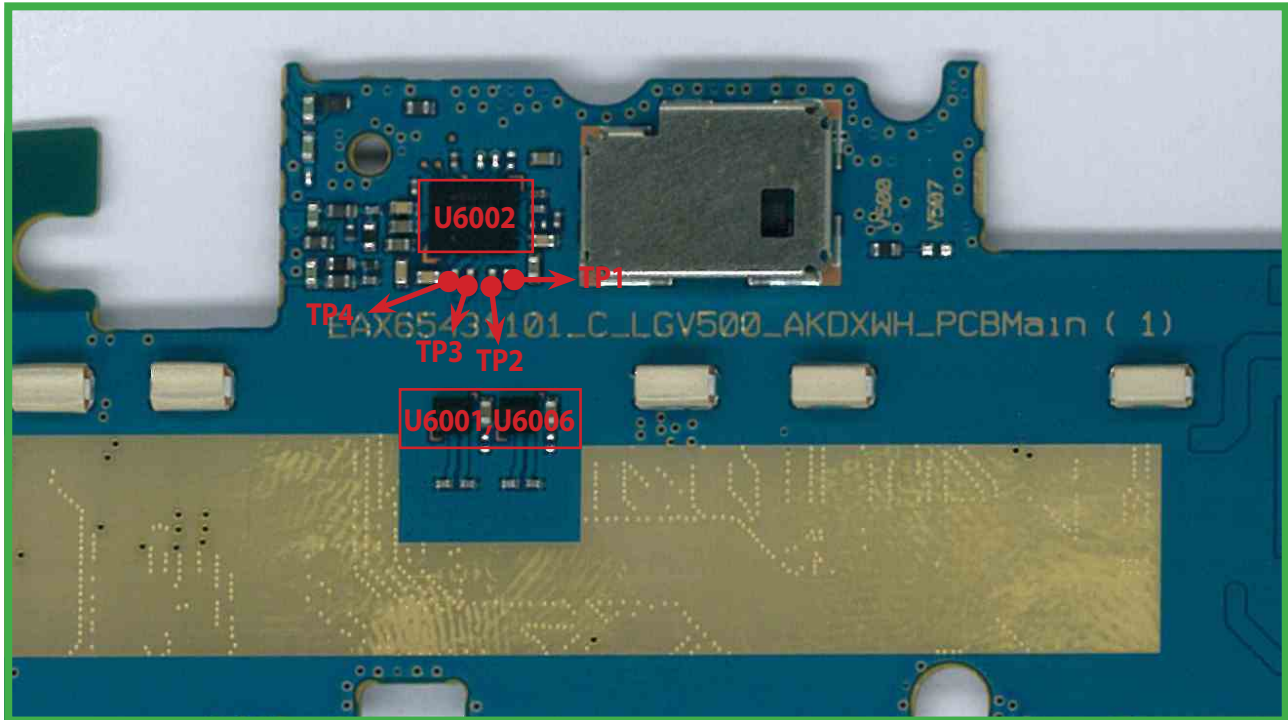


PCB Bottom

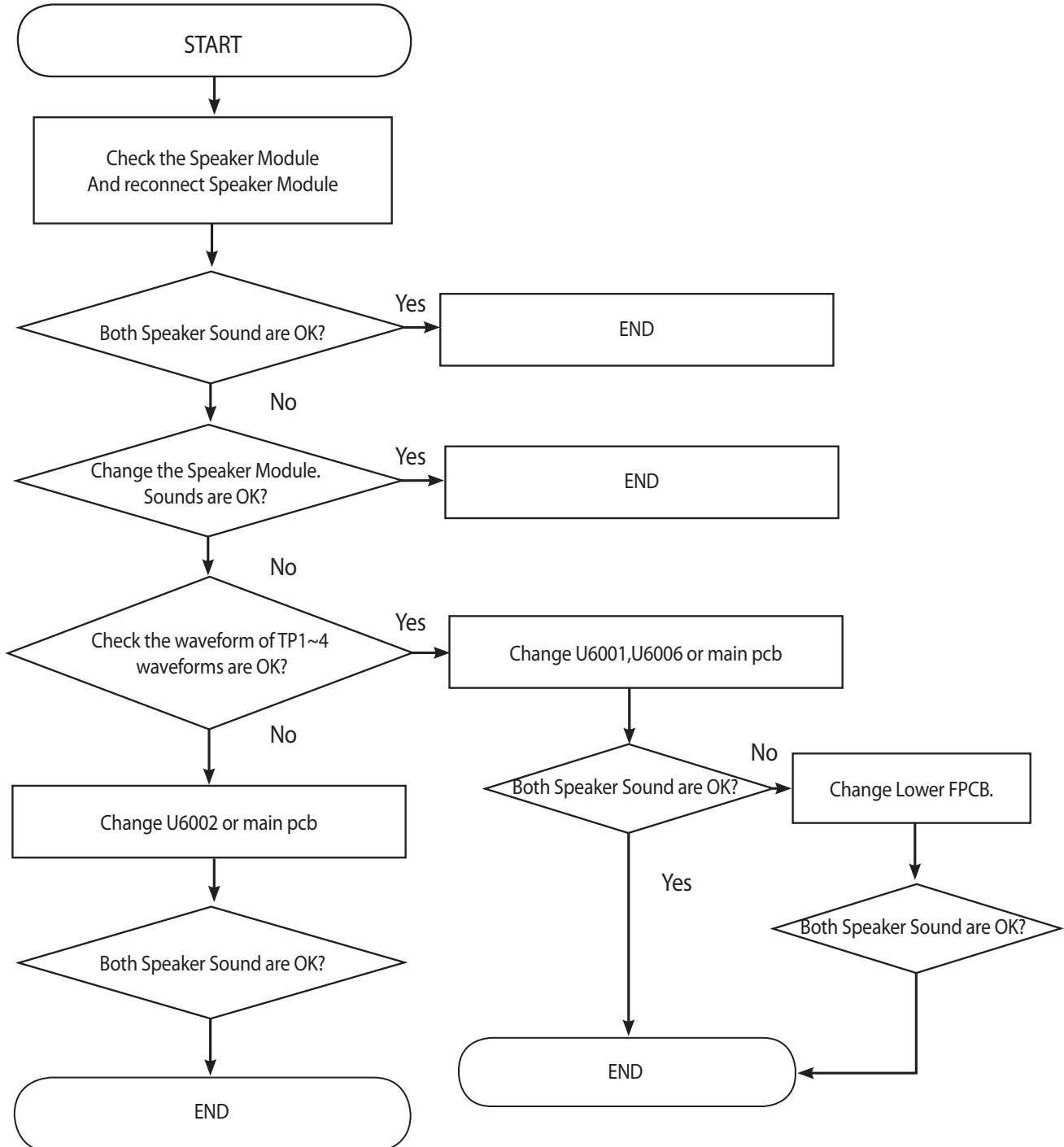
3.8.1 Speaker Troubleshooting

Speaker sound signal Path

3. TROUBLE SHOOTING

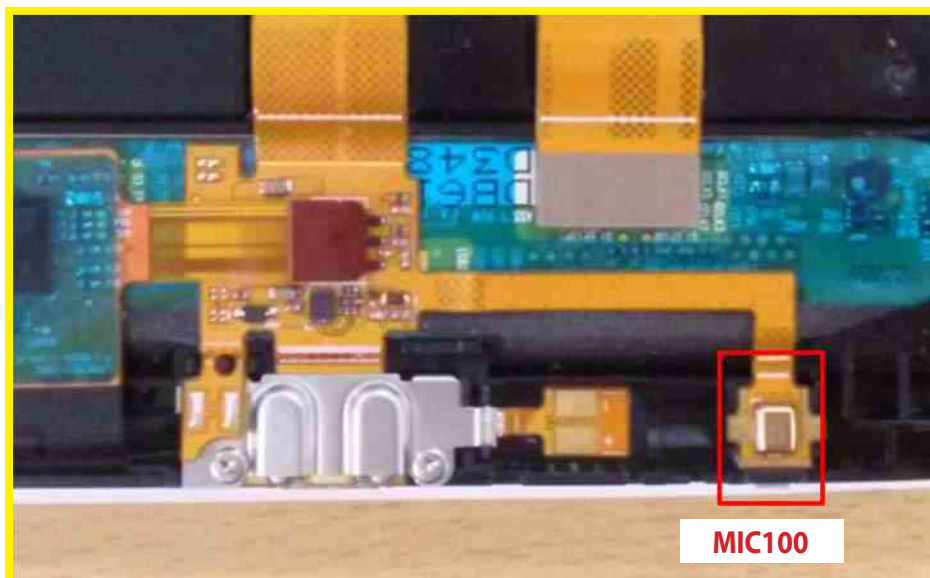
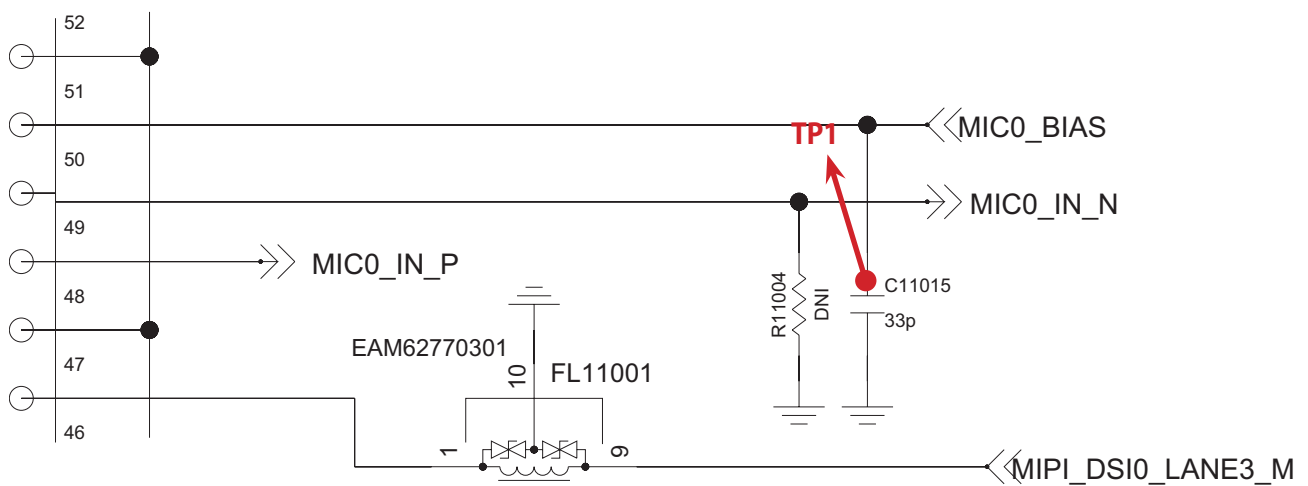


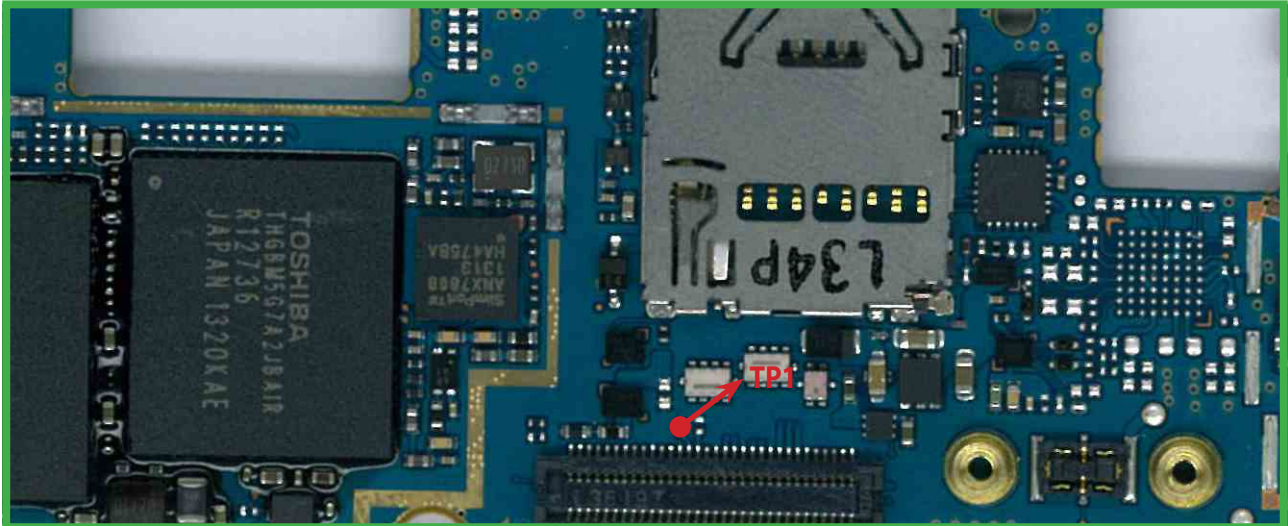
PCB TOP



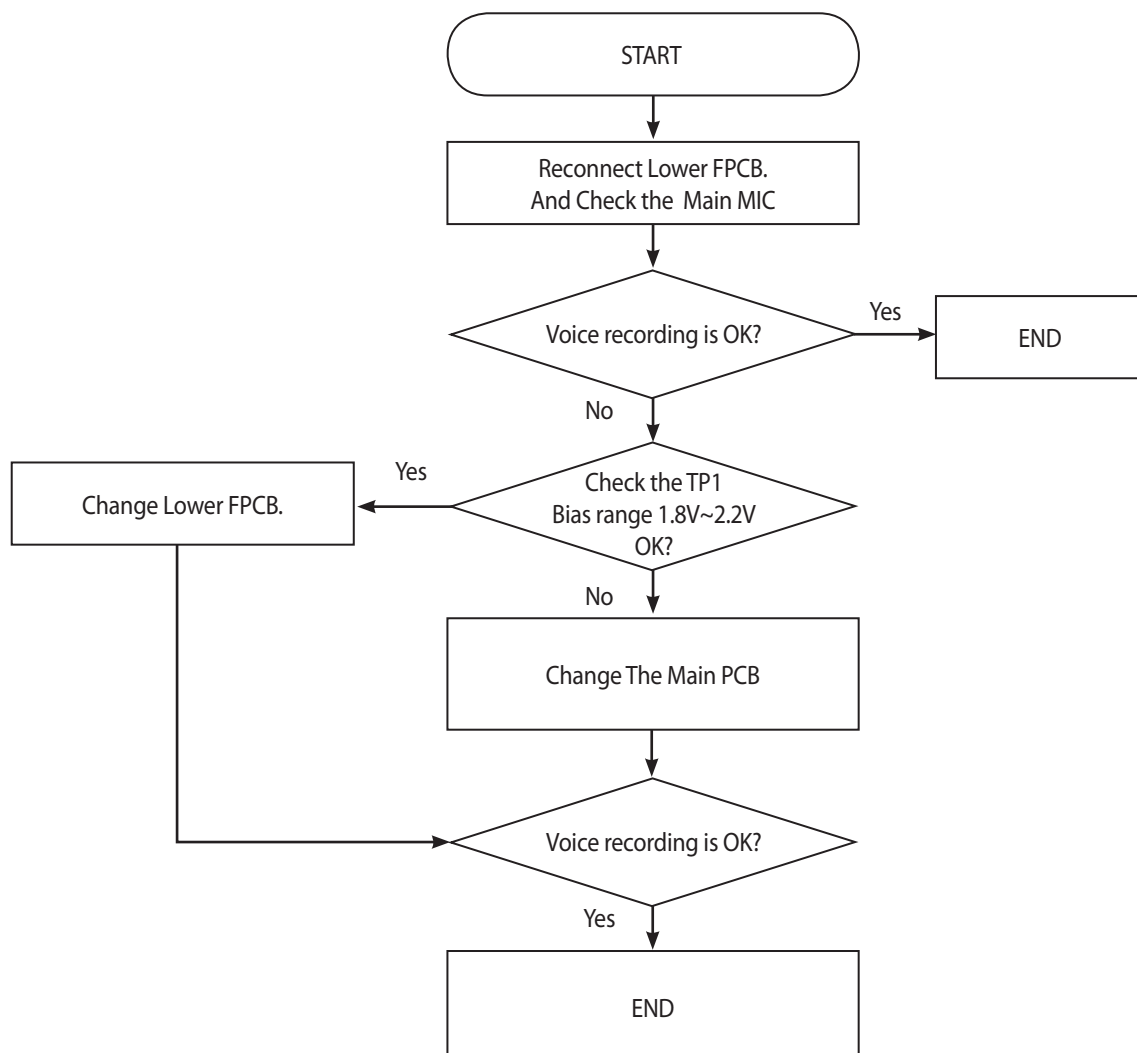
3.8.2 Main Mic. Troubleshooting

Main mic. signal comes from MEMS mic.(MIC100), amplified by WCD9310 (U6002), and Power is supplied by Battery (VBAT).





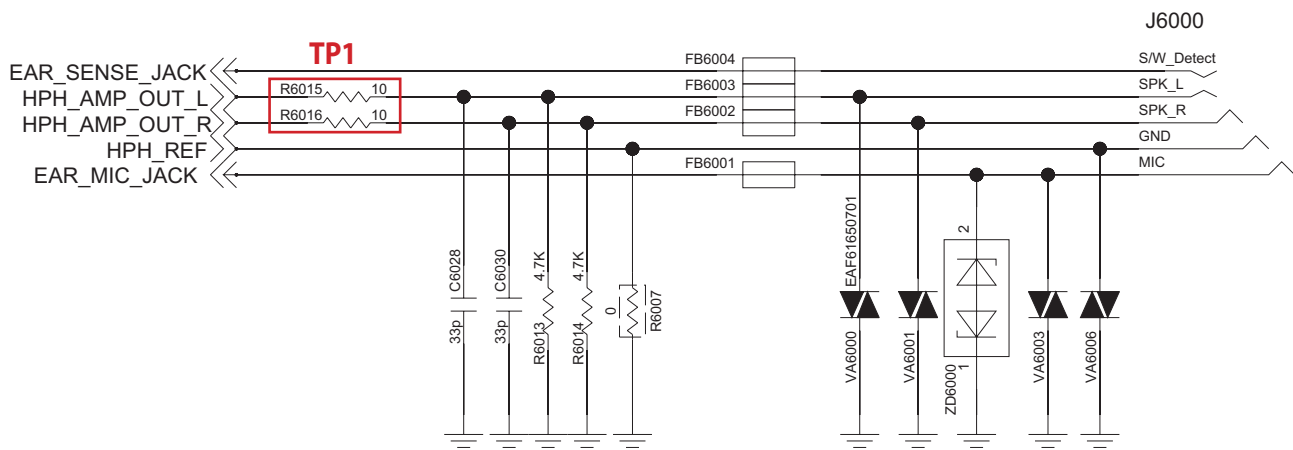
PCB Bottom Side



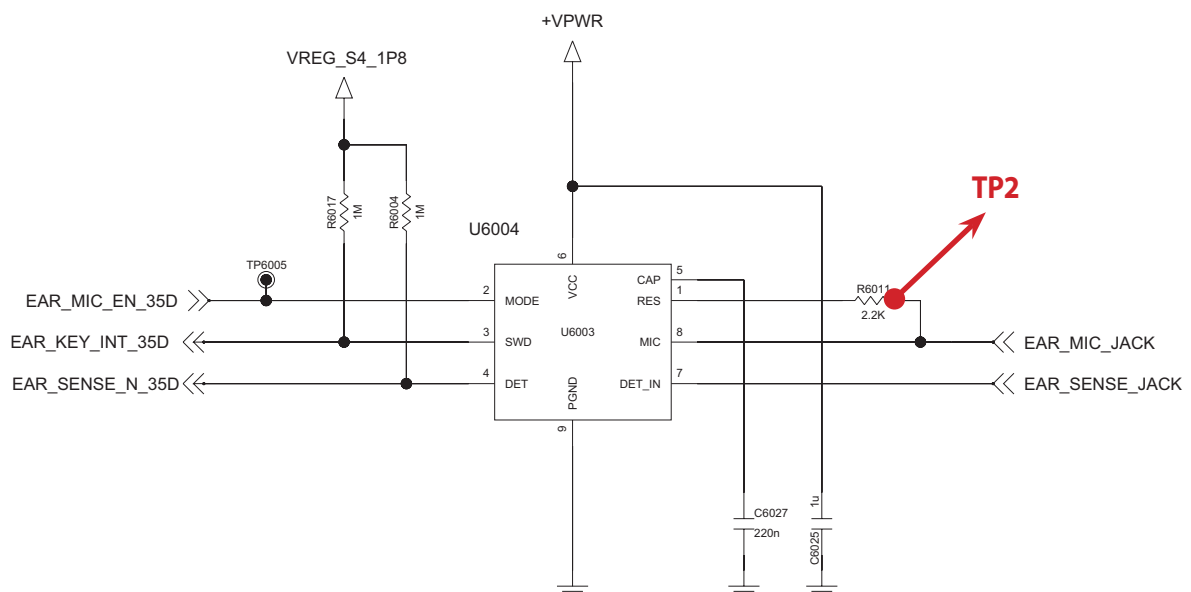
3.8.3 Ear-Mic. Troubleshooting

Ear-Mic. signals are generated by APQ8064 (U1000), amplified by WCD9310 (U6002) and Power is supplied by Battery (VBAT).

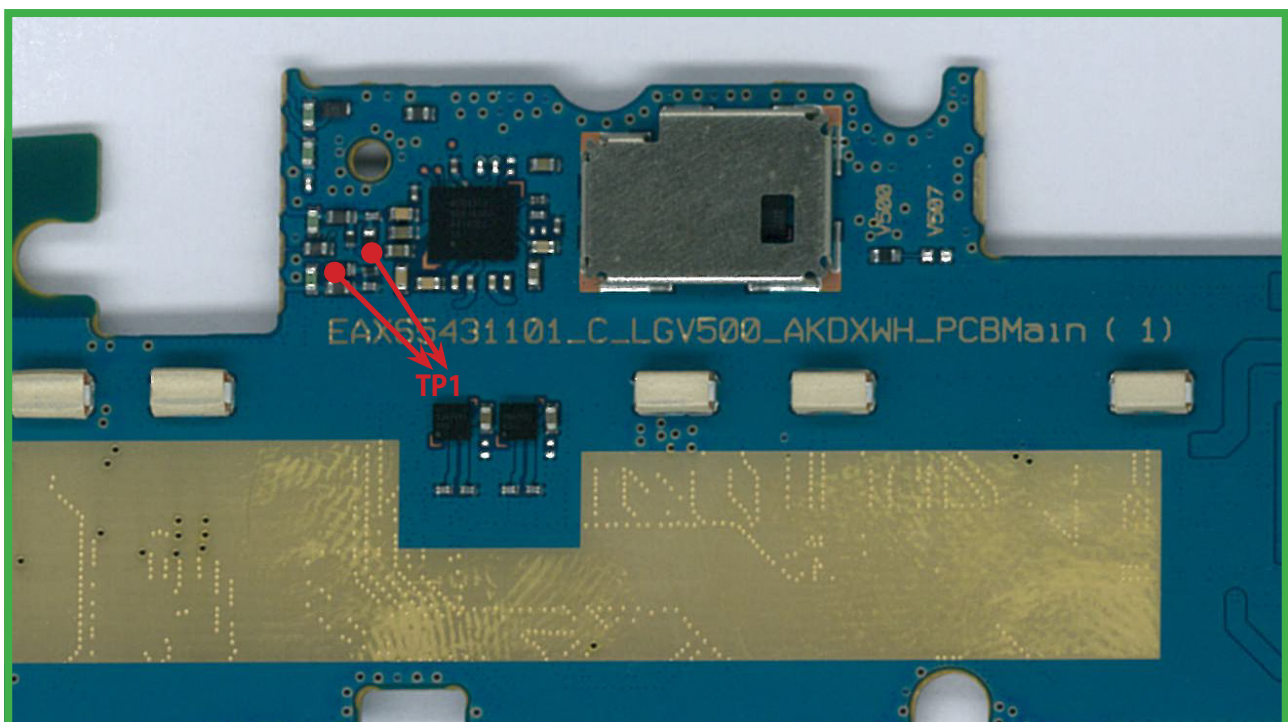
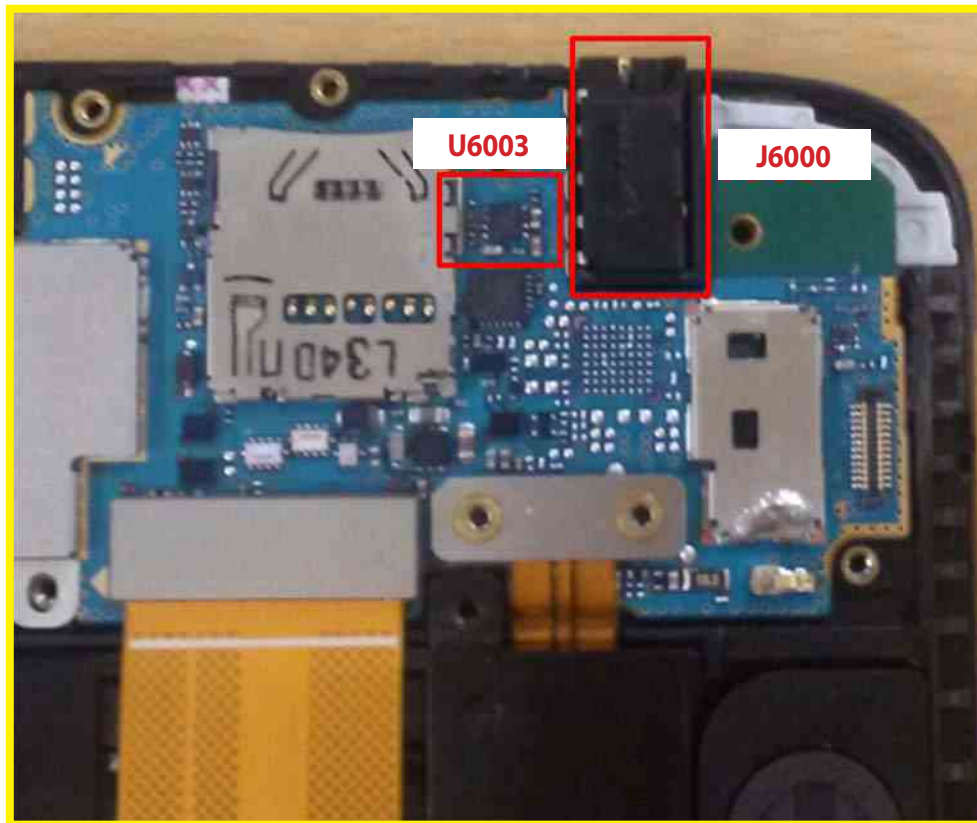
3.5phi Earjack



EJ DETECT

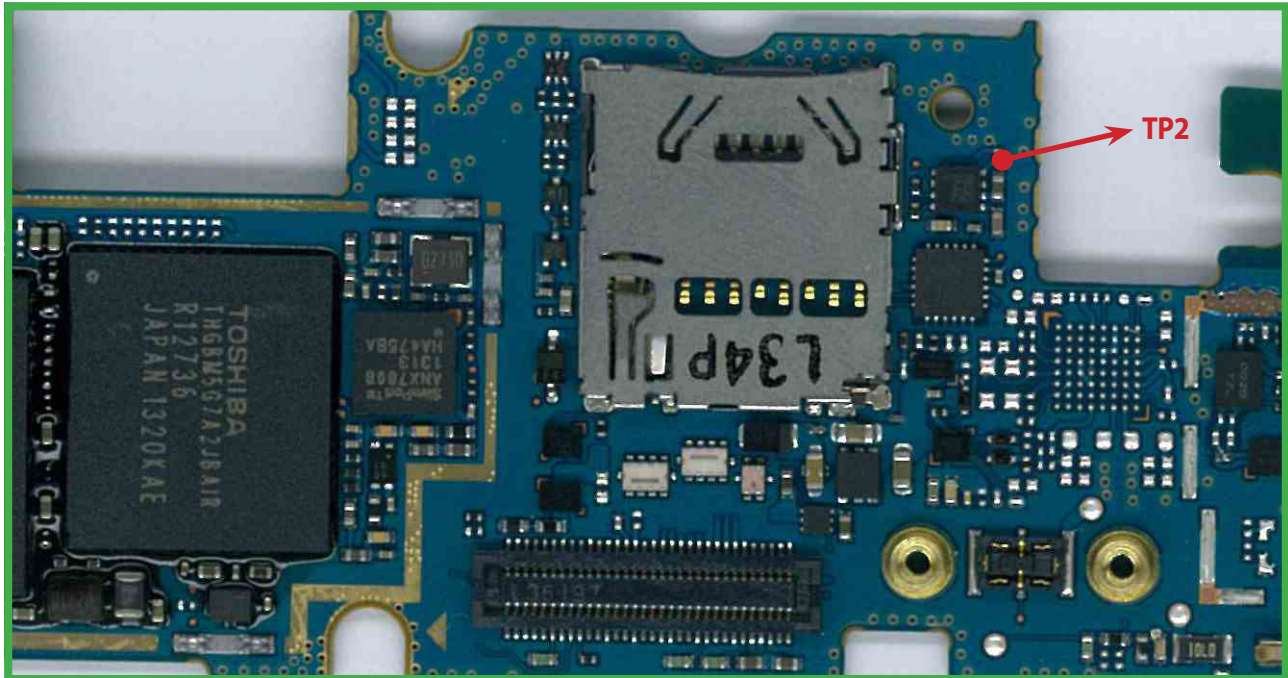


3. TROUBLE SHOOTING



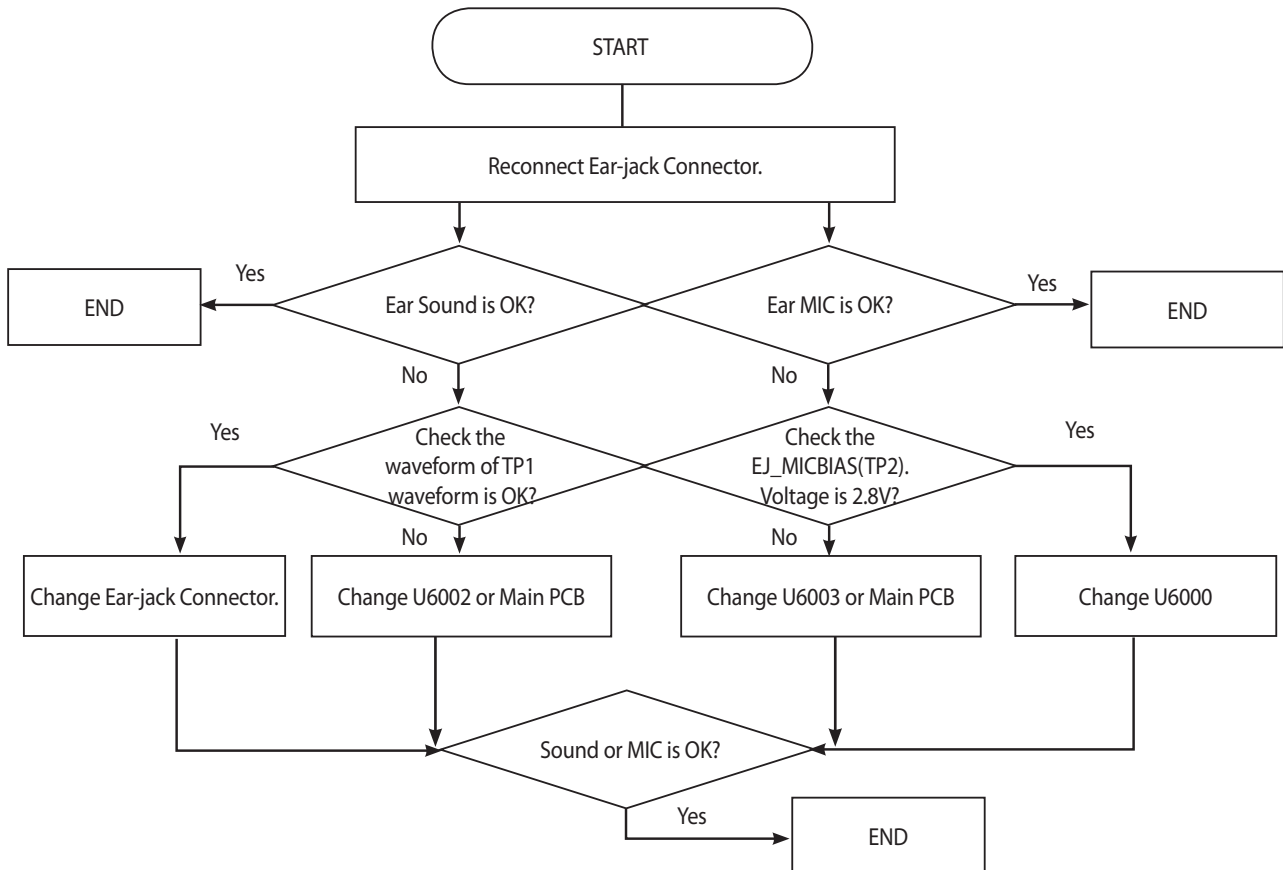
PCB TOP Side

3. TROUBLE SHOOTING



PCB Bottom

3. TROUBLE SHOOTING

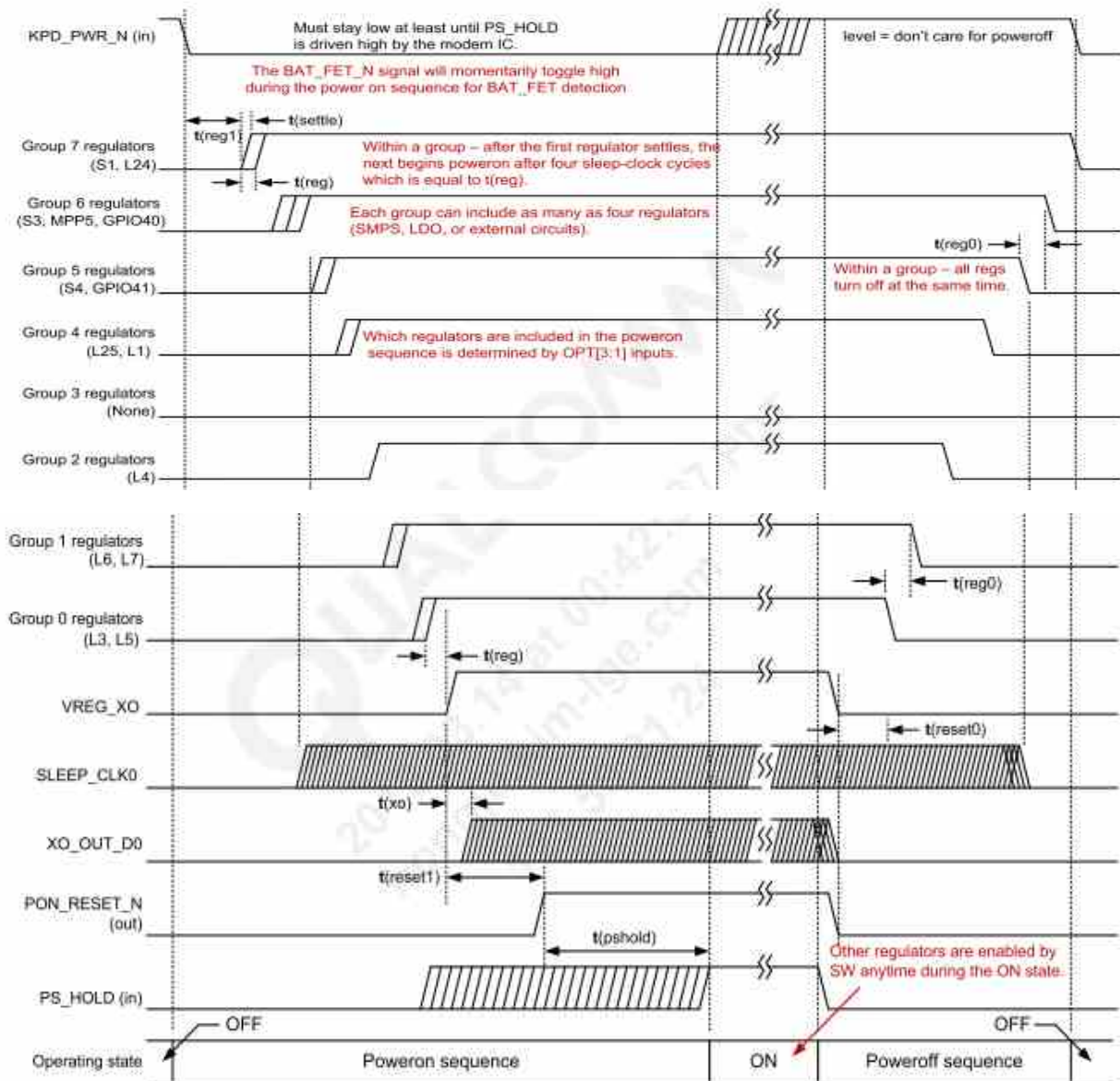


3.9 PMIC

The main power source of MAX8921 is provided by APQ8064.

Power ON sequence of MAX8921 is,

PWR key press -> PWR_ON_KEY go to low (D8001) -> PM8921 Power Up -> VREG_S1_1P225 (C3026) -> VREG_S3_1P05 (C3028) -> VREG_S4_1P8(C3029) -> TCXO(X3000) -> PM8921_PON_RESET_N (R1017) 1.8V CHECK->APQ_PS_HOLD(R3010)High-> APQ_RESOUT_N (R5020) goes to high.

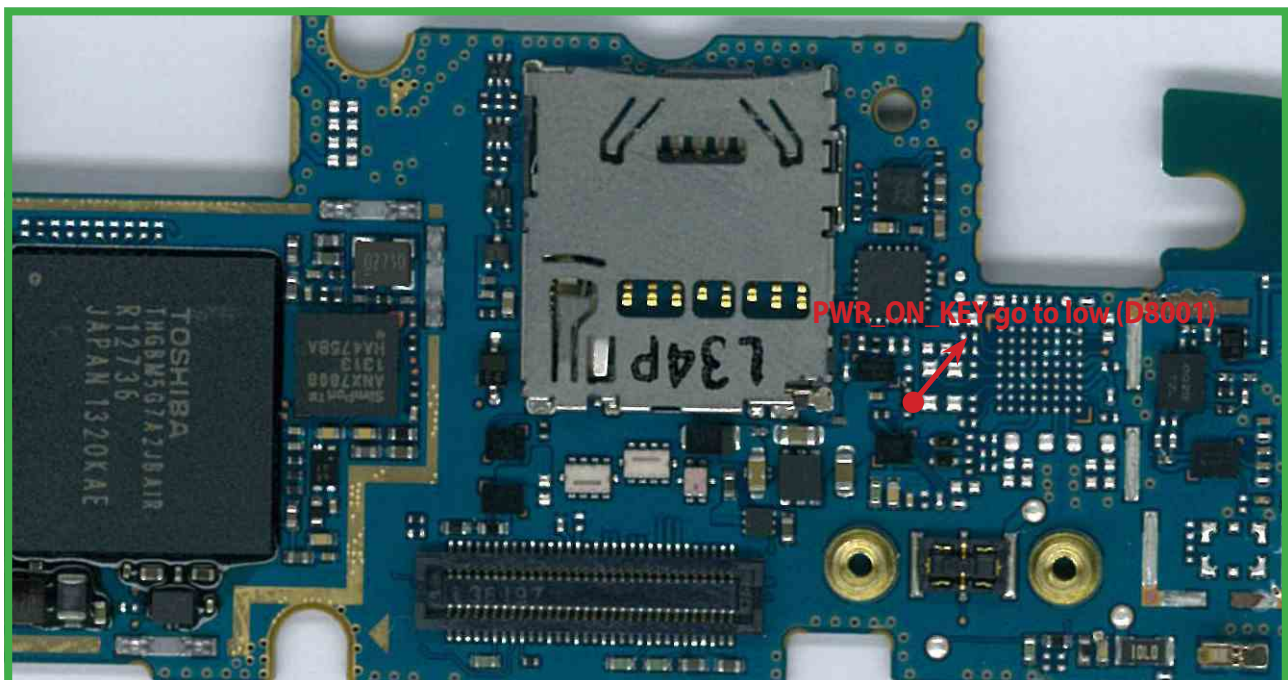
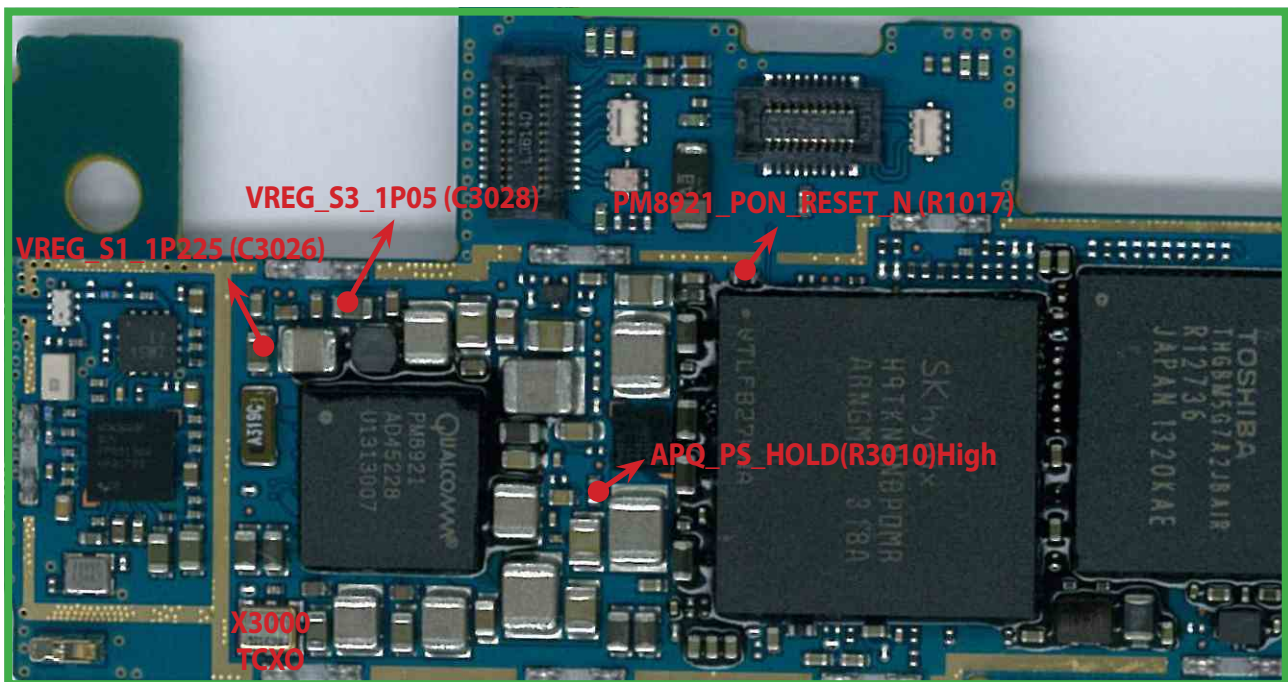


3. TROUBLE SHOOTING

Main PCB Top place

PWR key press -> PWR_ON_KEY go to low (D8001) -> PM8921 Power Up -> VREG_S1_1P225 (C3026) -> VREG_S3_1P05 (C3028)
->VREG_S4_1P8(C3029)-> TCXO(X3000) -> PM8921_PON_RESET_N (R1017)

1.8V CHECK->APQ_PS_HOLD(R3010)High-> APQ_RESOUT_N (R5020) goes to high

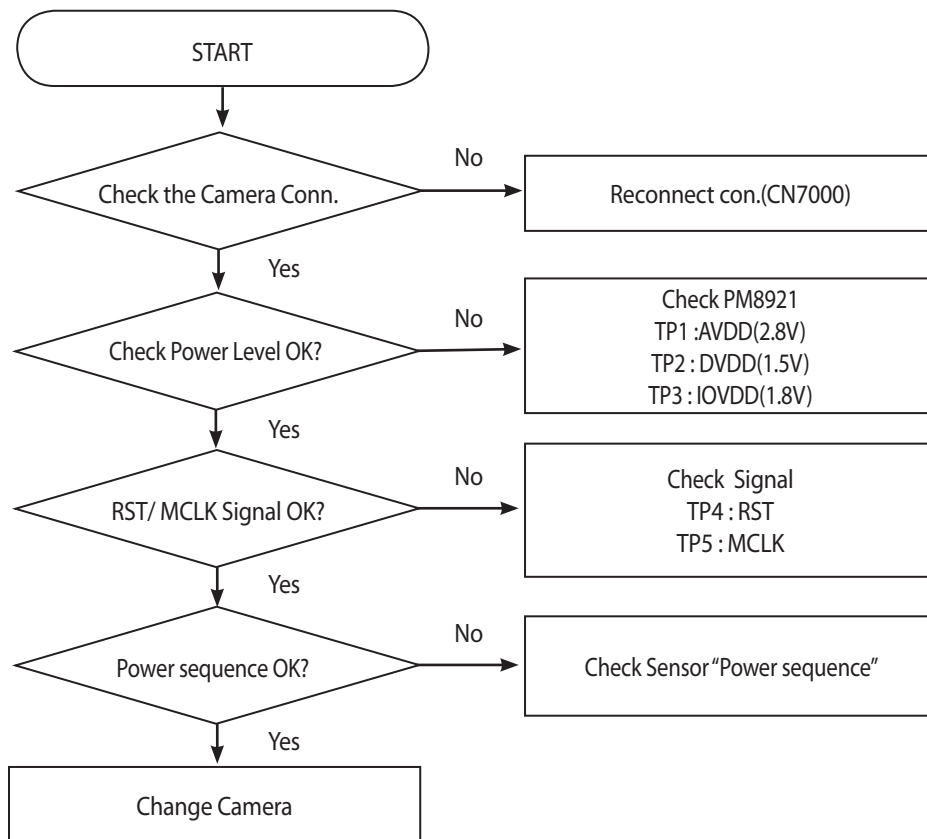


PCB Bottom

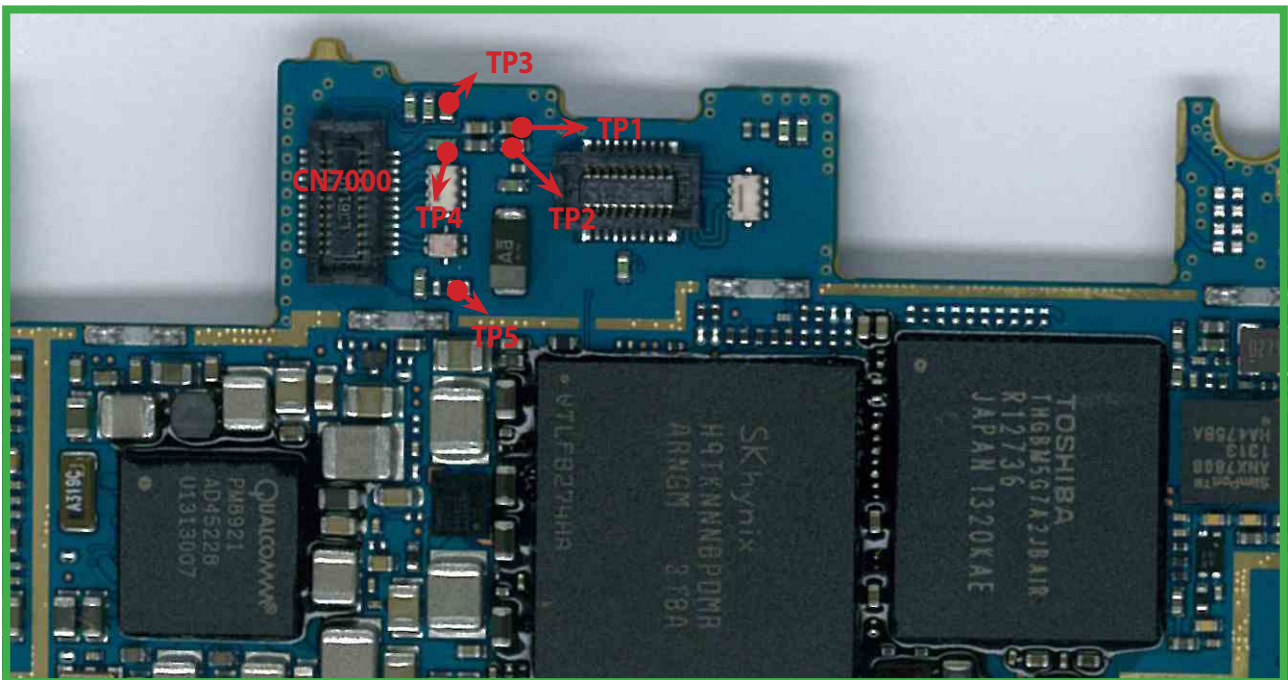
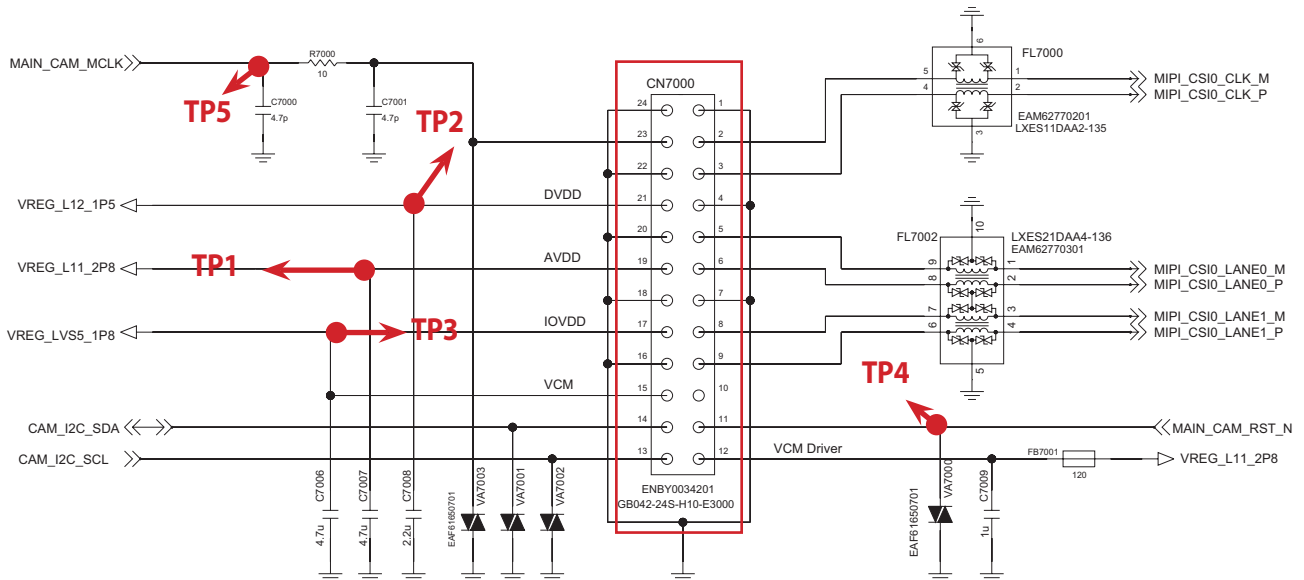
3.10 Camera

3.10.1 5M AF Camera Trouble Shooting

5M camera control signals are generated by AP8064



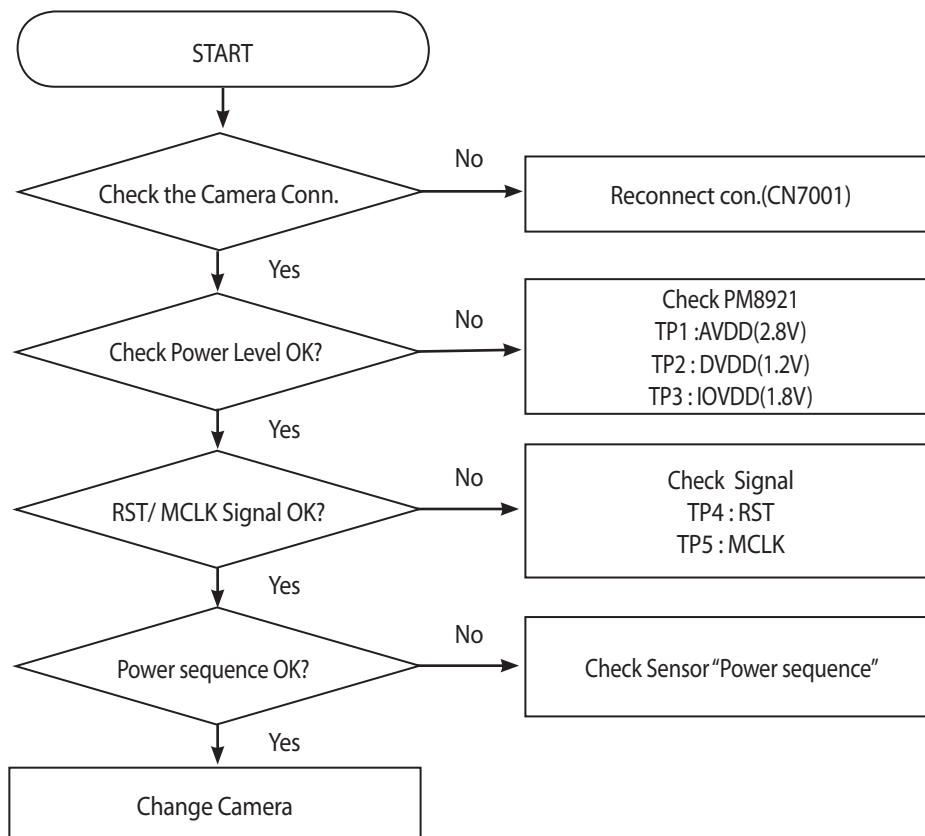
3. TROUBLE SHOOTING



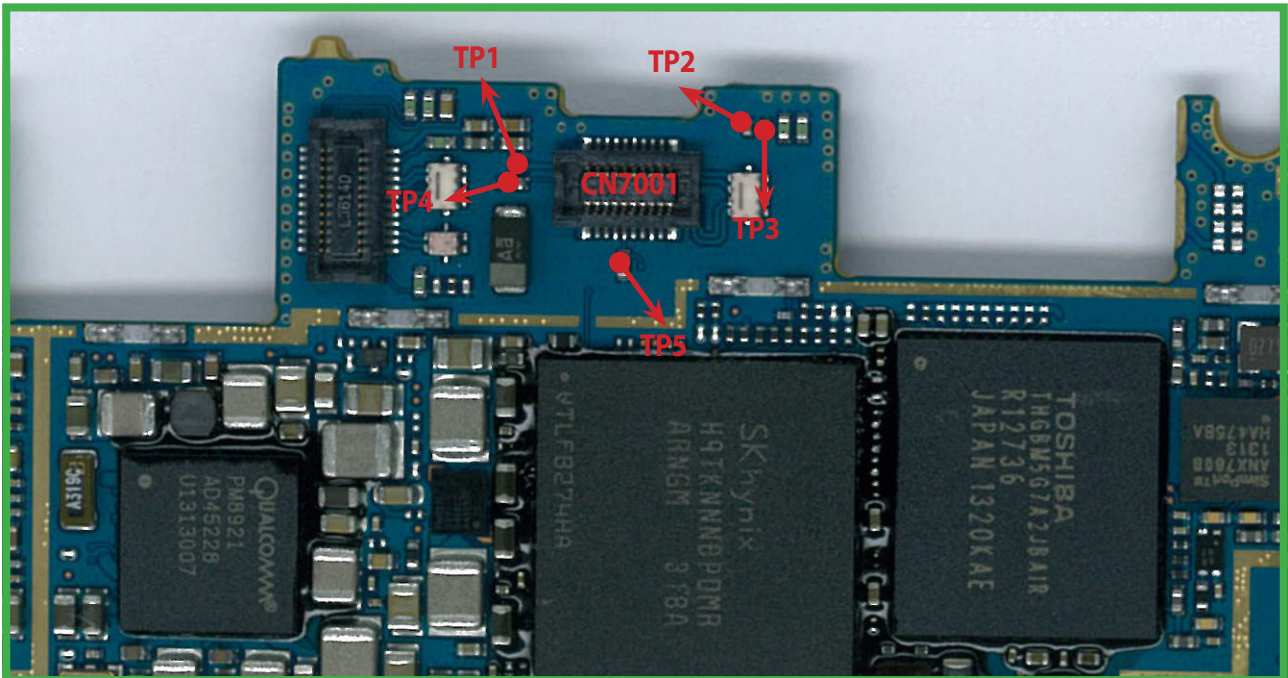
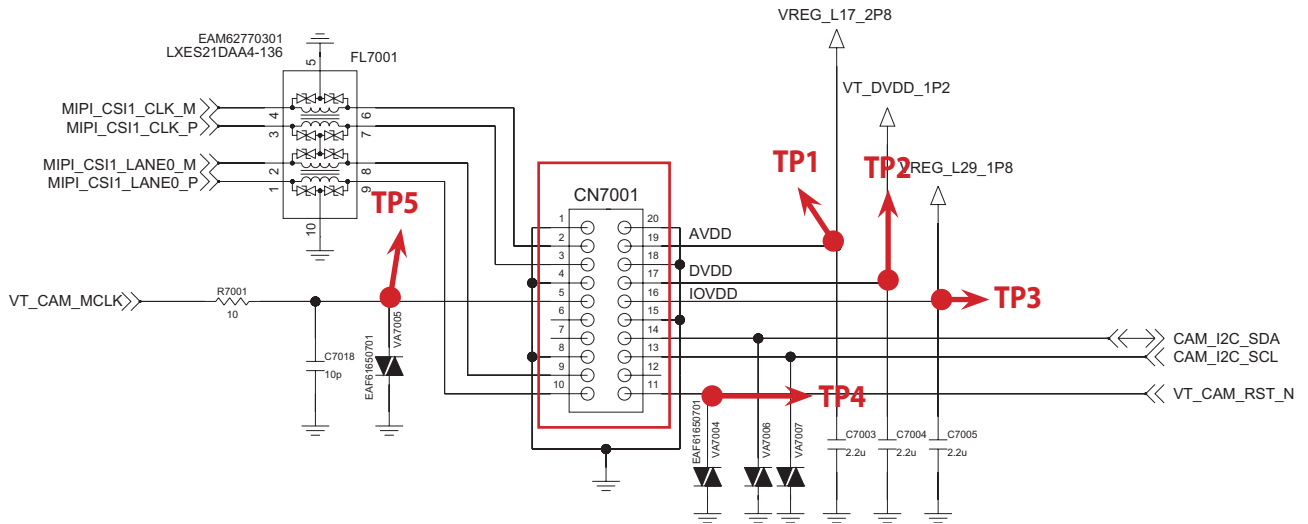
PCB Bottom

3.10.2 1.3M AF Camera Trouble Shooting

1.3M camera control signals are generated by AP8064



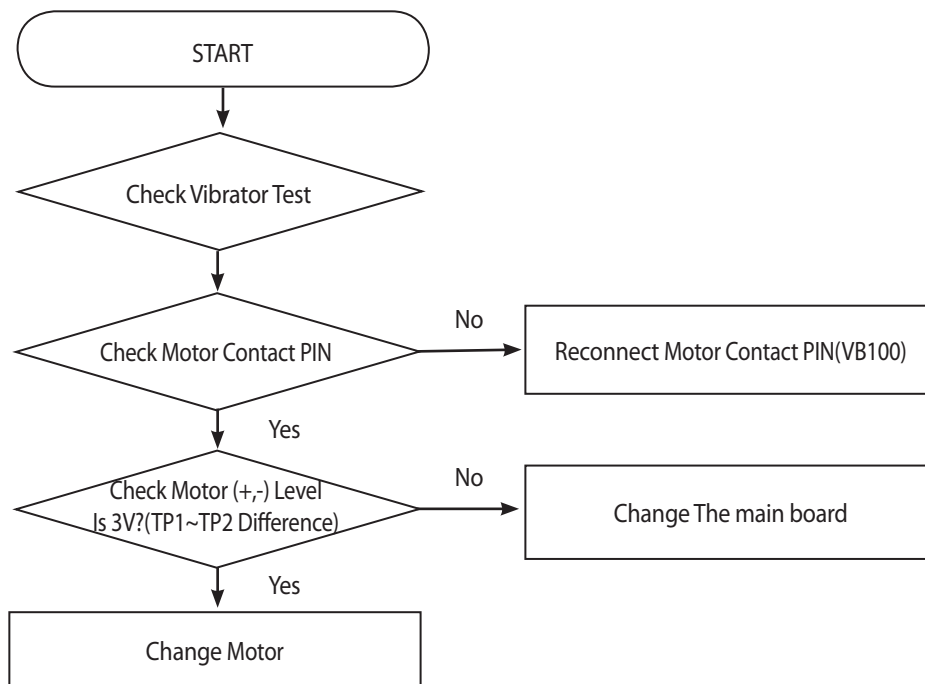
3. TROUBLE SHOOTING



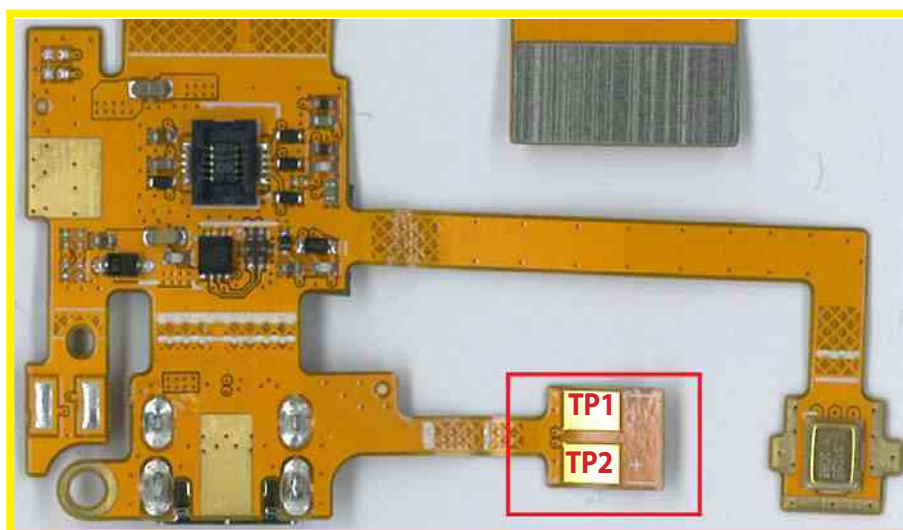
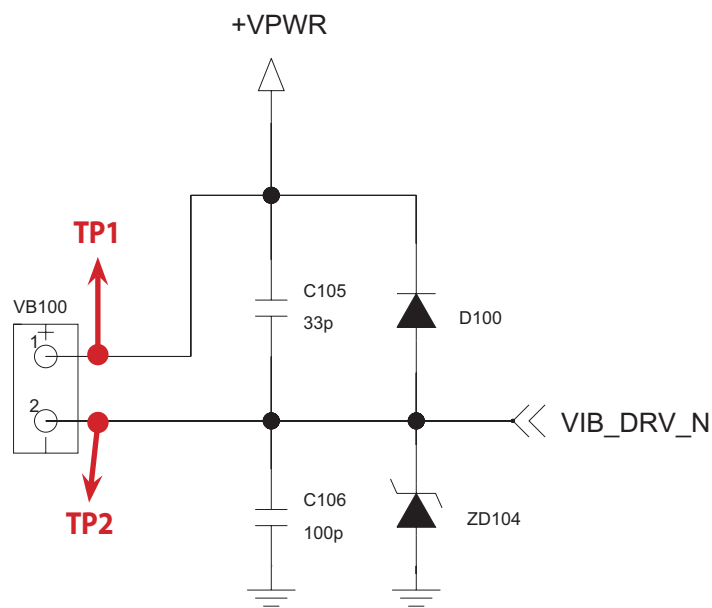
PCB Bottom

3.11 Motor

Check out the setting menu on the phone. If not, check Test points shown on the pictures.



MOTOR

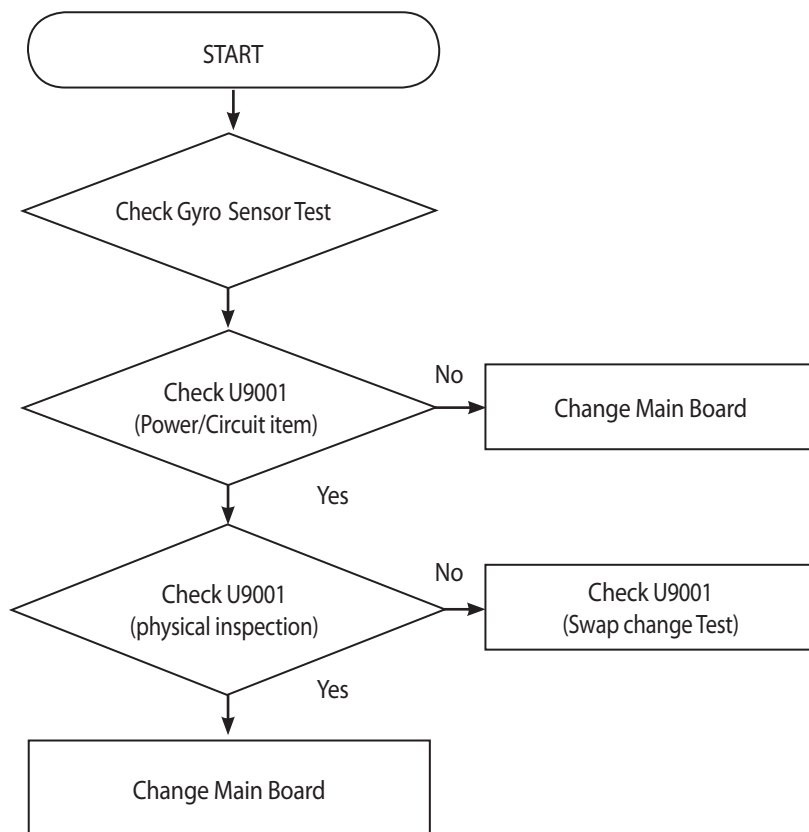


F_LOWER_BOT

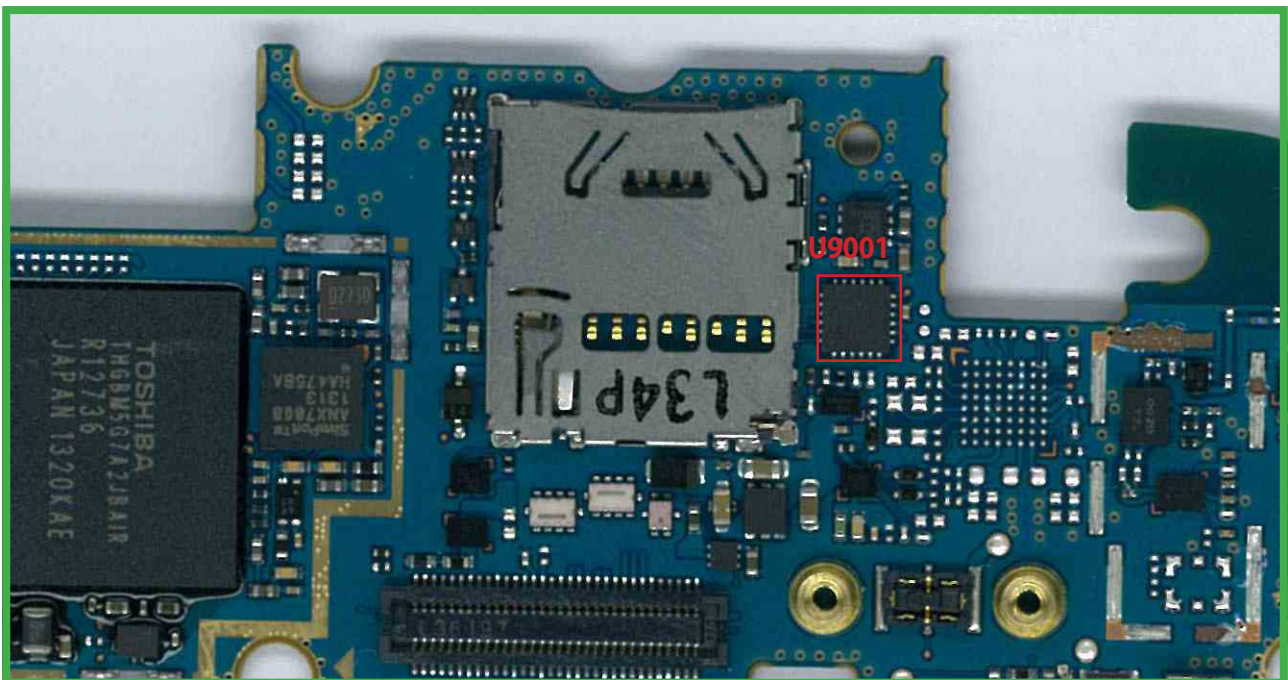
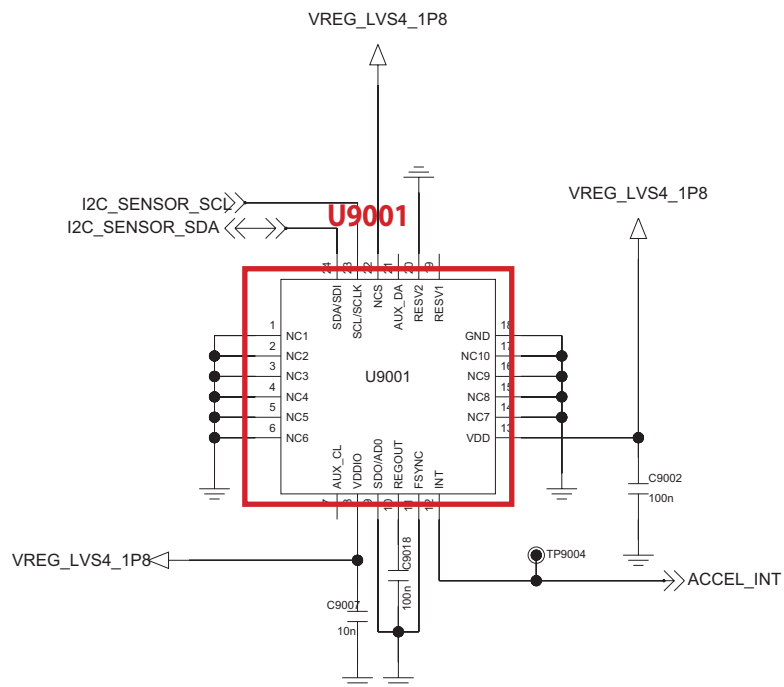
3.12 Sensor

3.12.1 Gyro Sensor troubleshooting

When the Gyro sensor does not work, the reason is chip problem. If the Gyro sensor IC is damaged, it will do not work even though power is supplied and APQ8064 controls normally. Therefore if the Gyro sensor is damaged, change the Gyro sensor IC or main board.



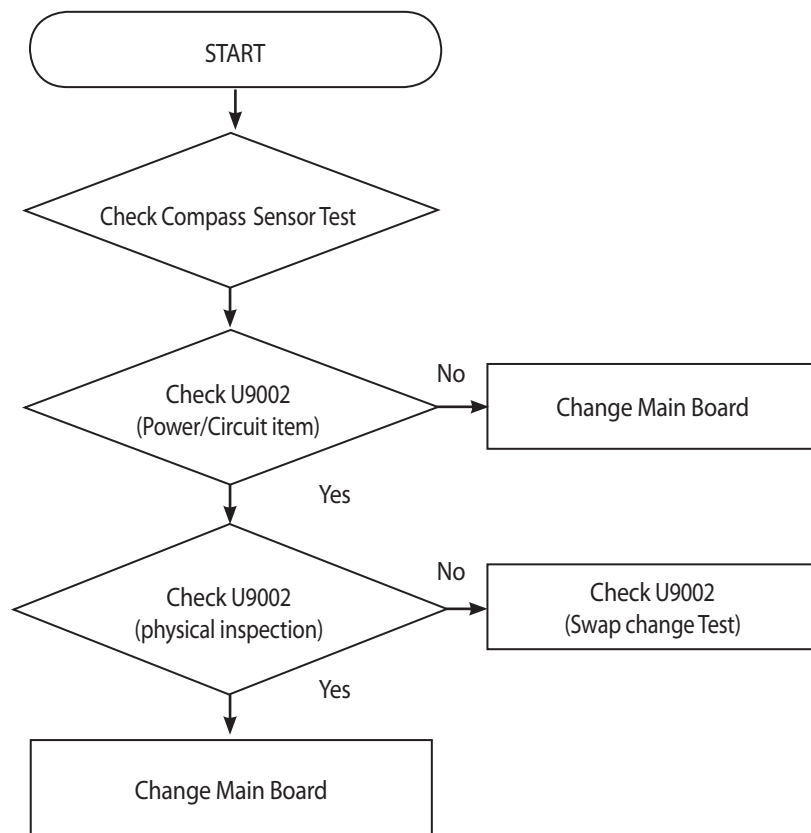
GYRO SENSOR



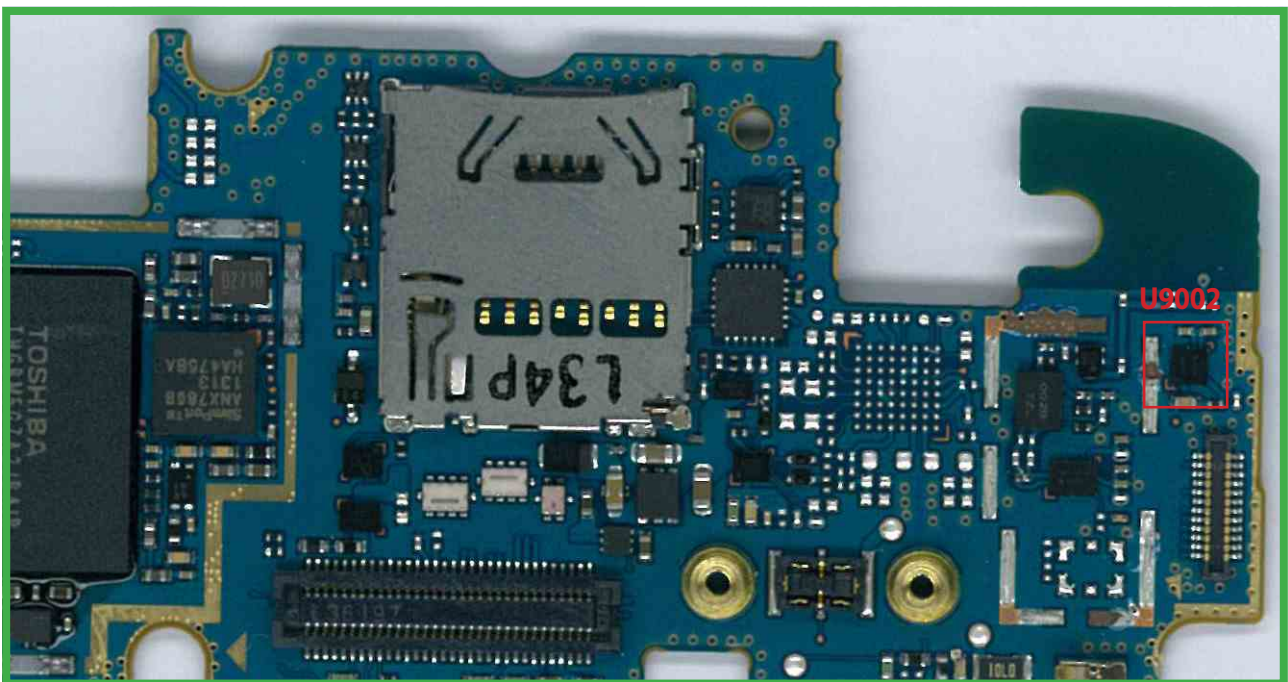
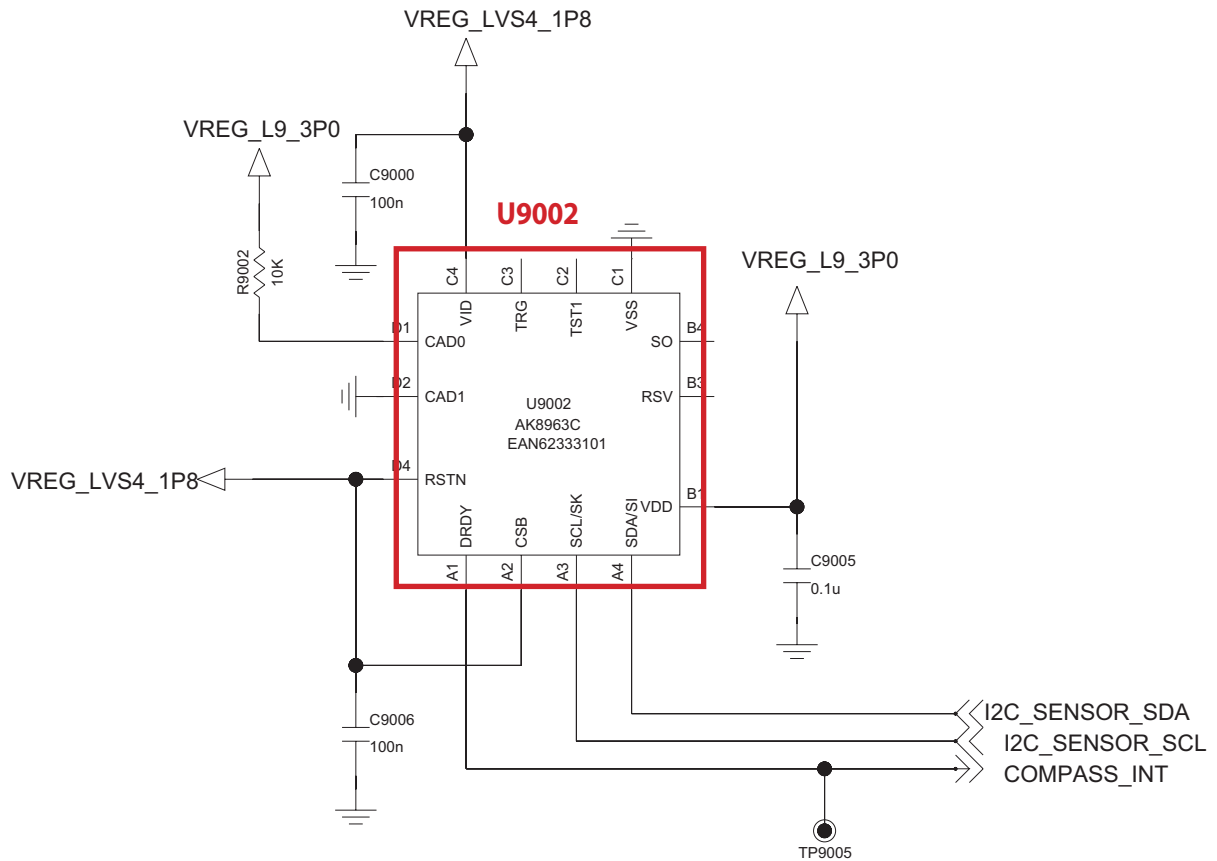
PCB Bottom

3.12.2 Compass sensor troubleshooting

The compass sensor is calibrated by the gyro sensor data using SW algorithm. Therefore Gyro sensor error makes the compass sensor malfunction.



3. TROUBLE SHOOTING

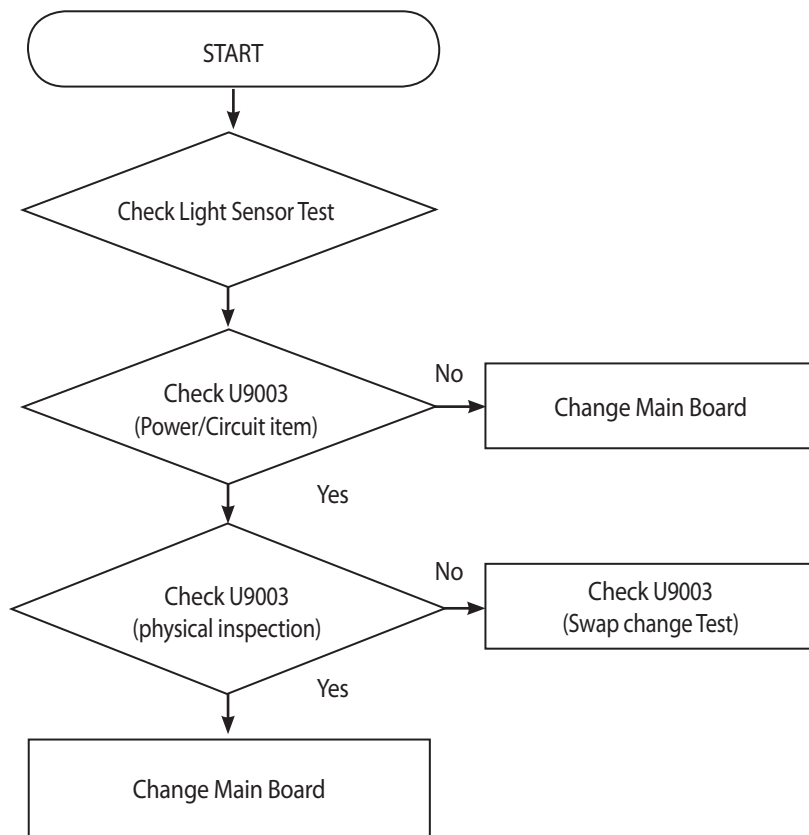


PCB Bottom

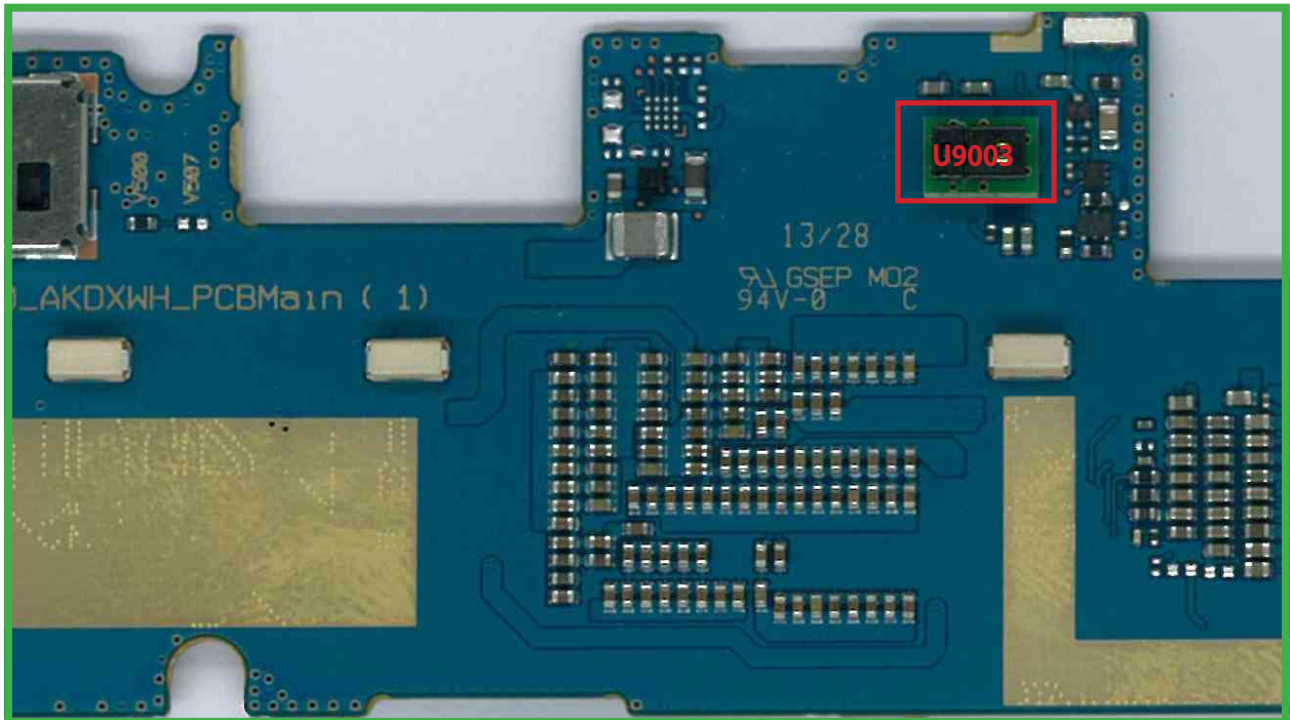
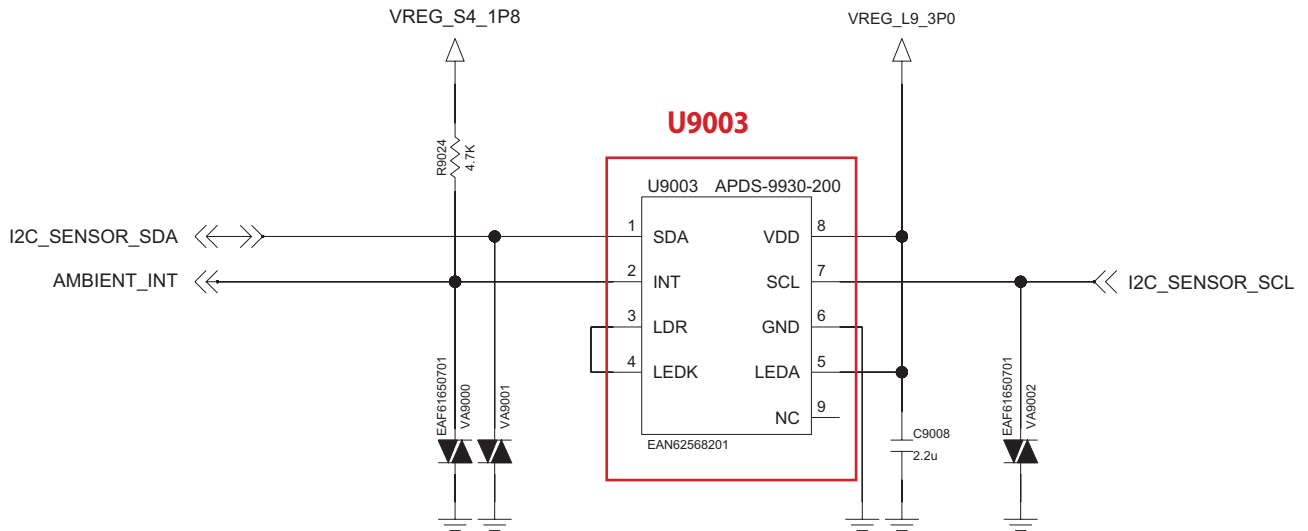
3.12.3 Light Sensor on/off trouble

Light Sensor is worked as below:

Menu Key click -> Settings click -> Display click -> Brightness click -> Automatic brightness click.



3. TROUBLE SHOOTING

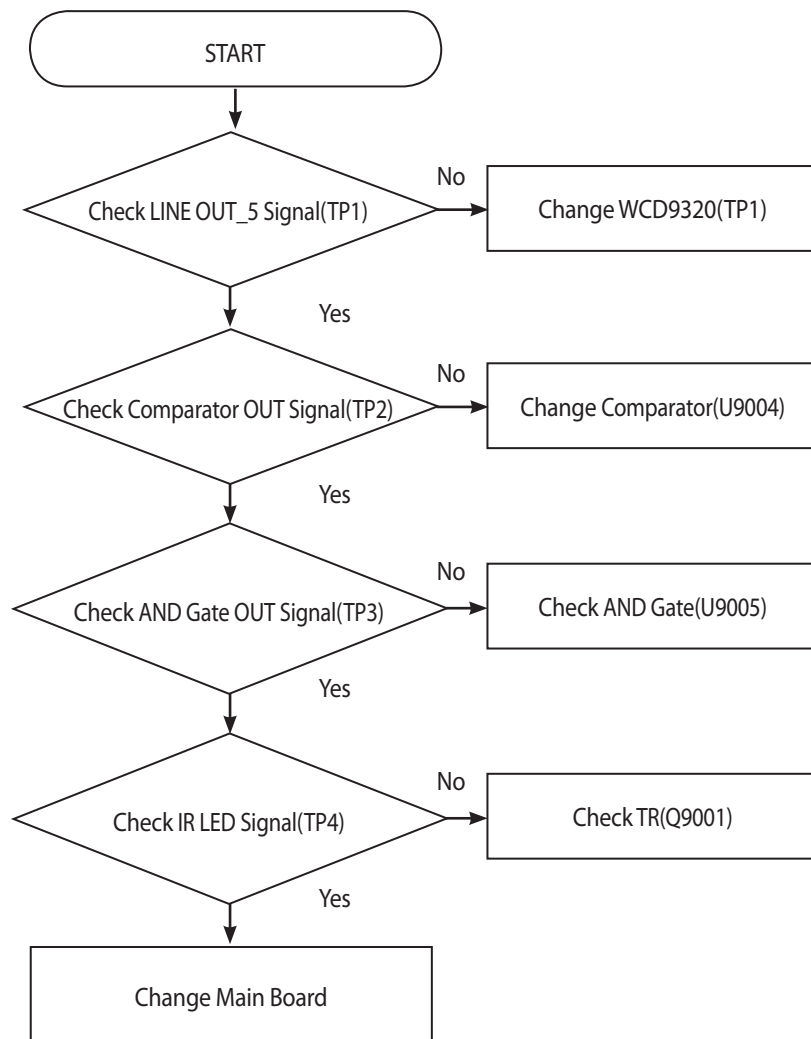


PCB TOP Side

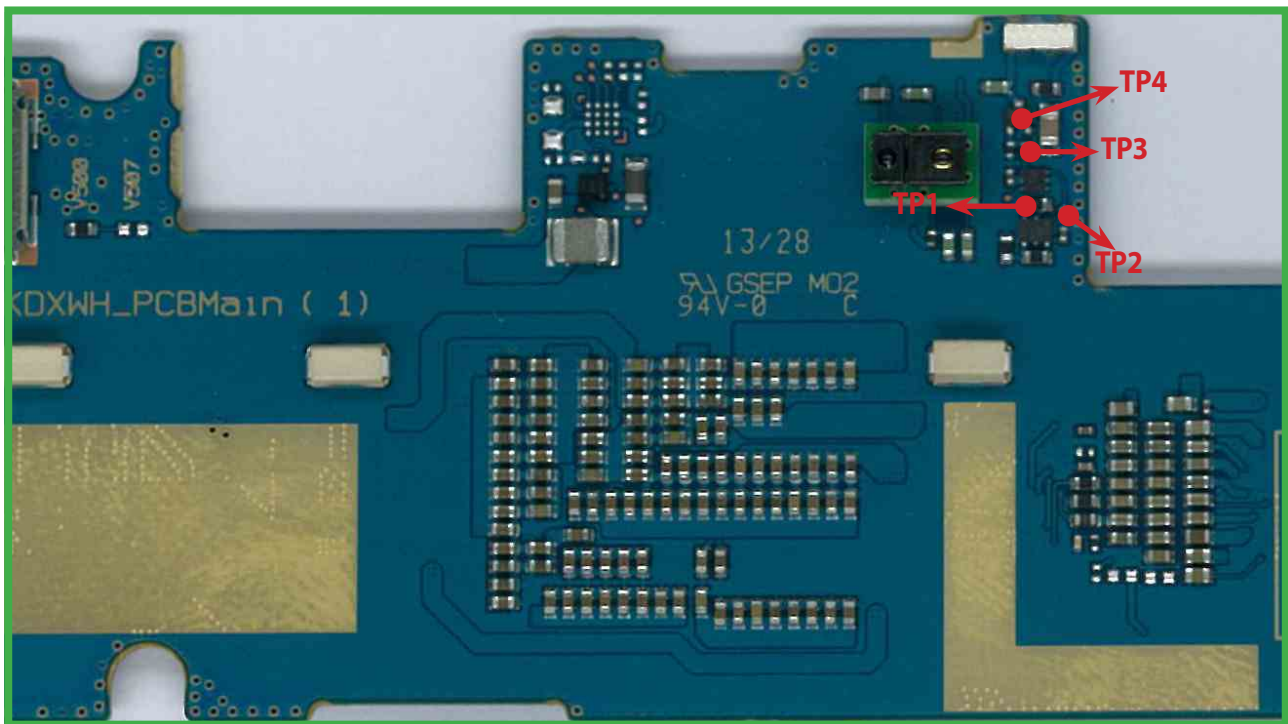
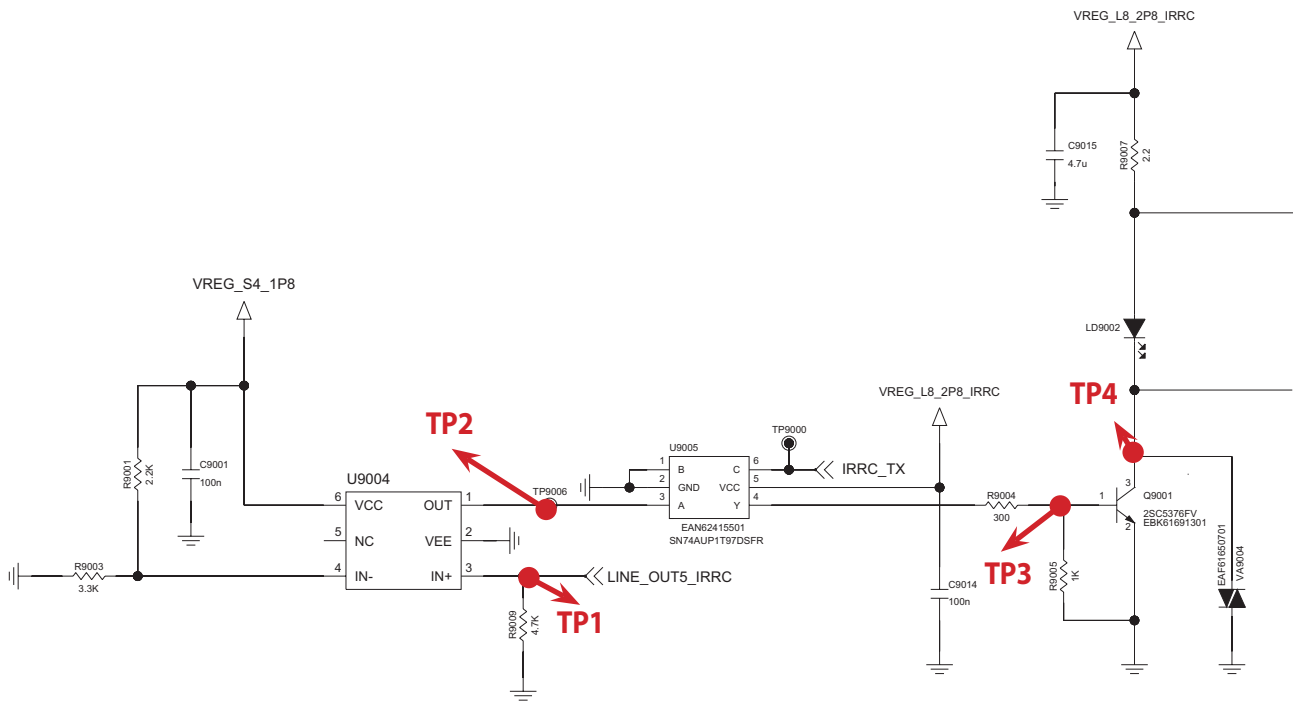
3.13 IIRC

3.13.1 IIRC troubleshooting

The IIRC function is worked by SW (Remote database) algorithm.

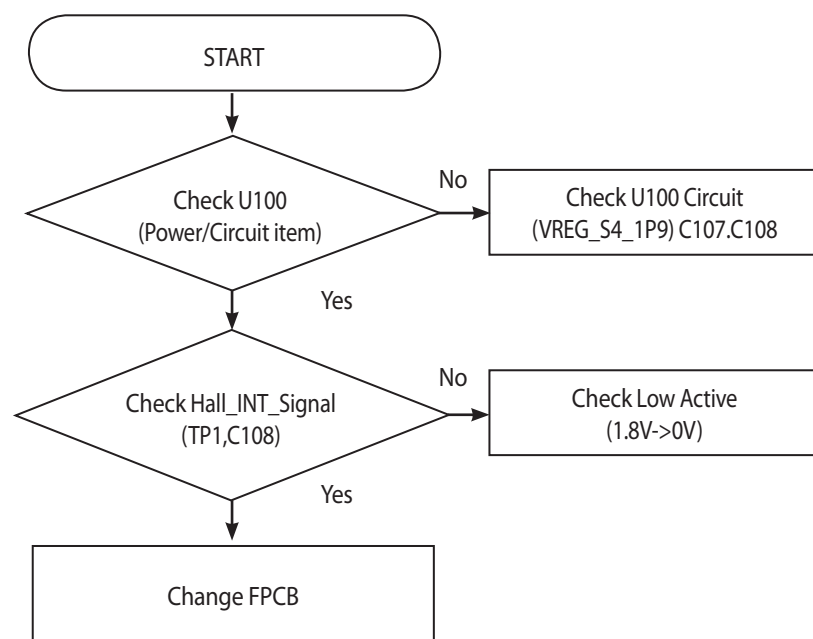


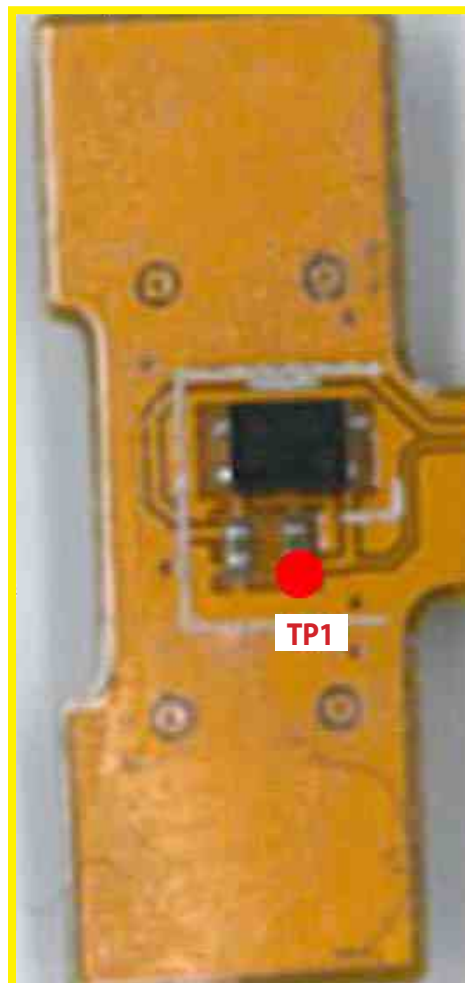
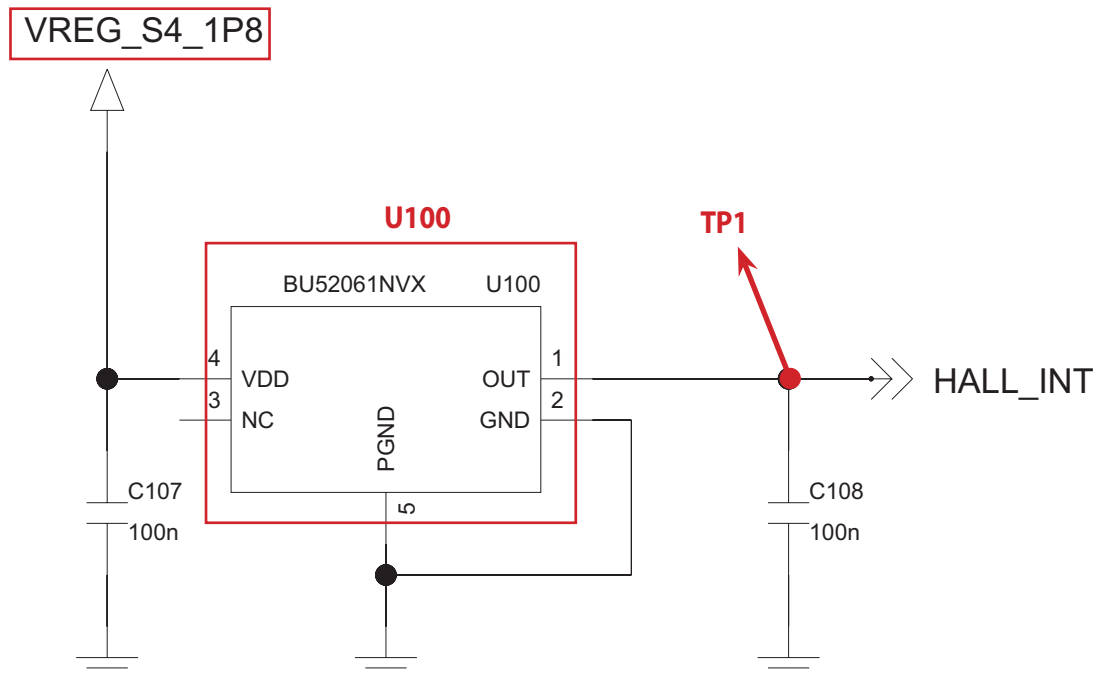
3. TROUBLE SHOOTING



PCB TOP Side

3.13.2 Cover Case(Hall Sensor) troubleshooting

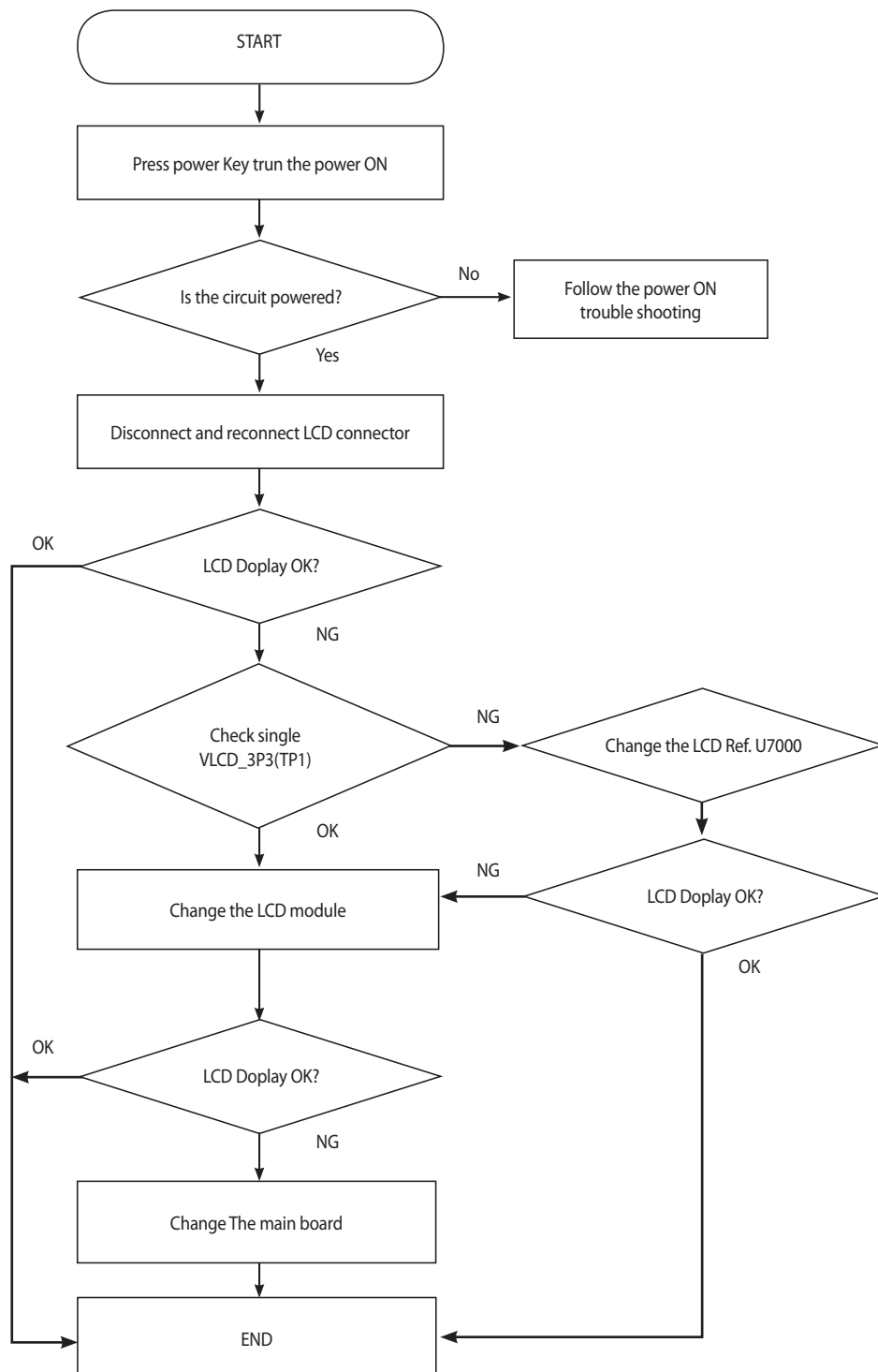




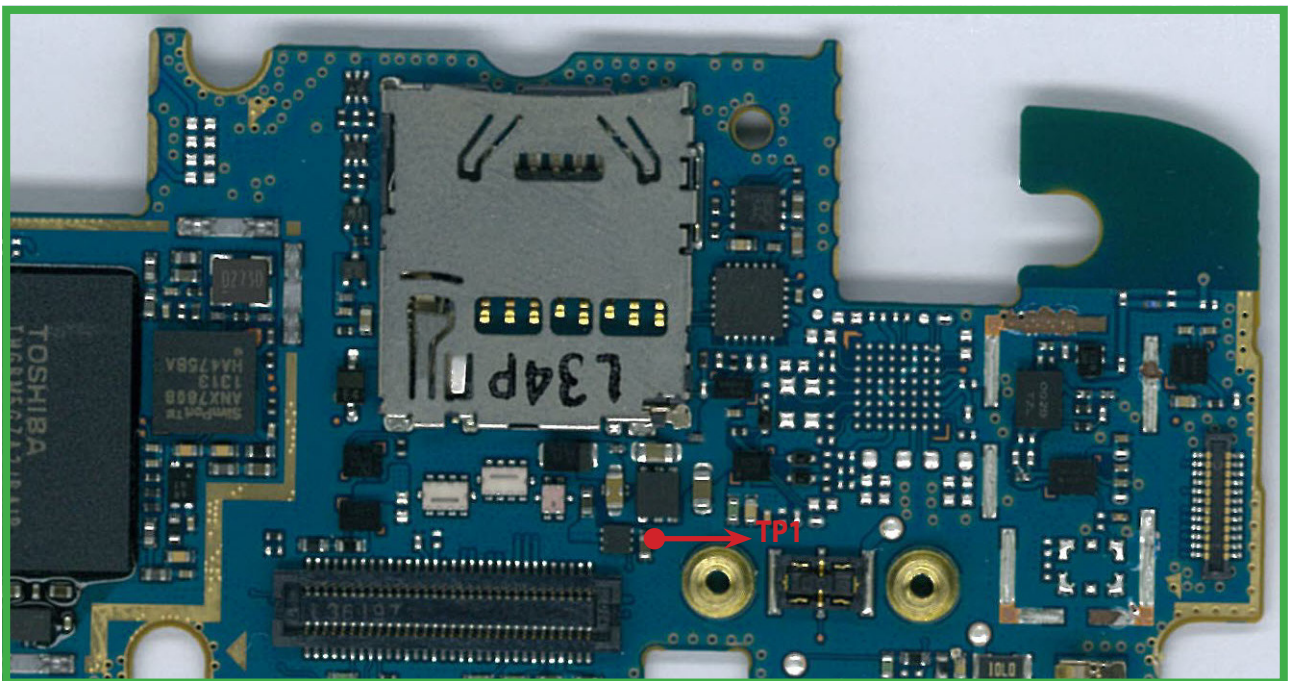
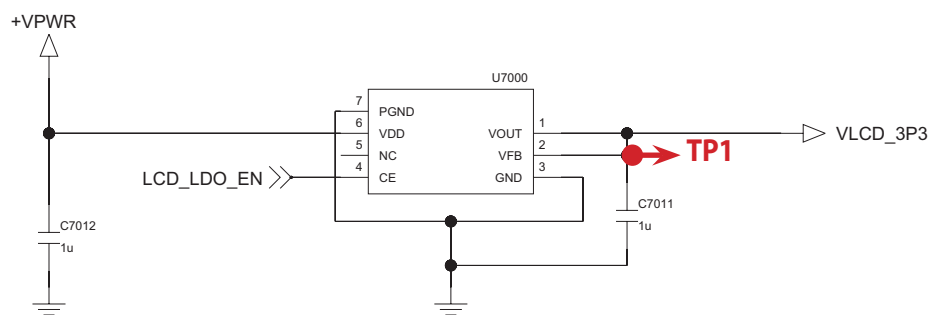
3.14 LCD

Main LCD control signals are generated by APQ8064.

Those signal's path are : APQ8064 -> LCD Module



3.3V 500mA LCD POWER

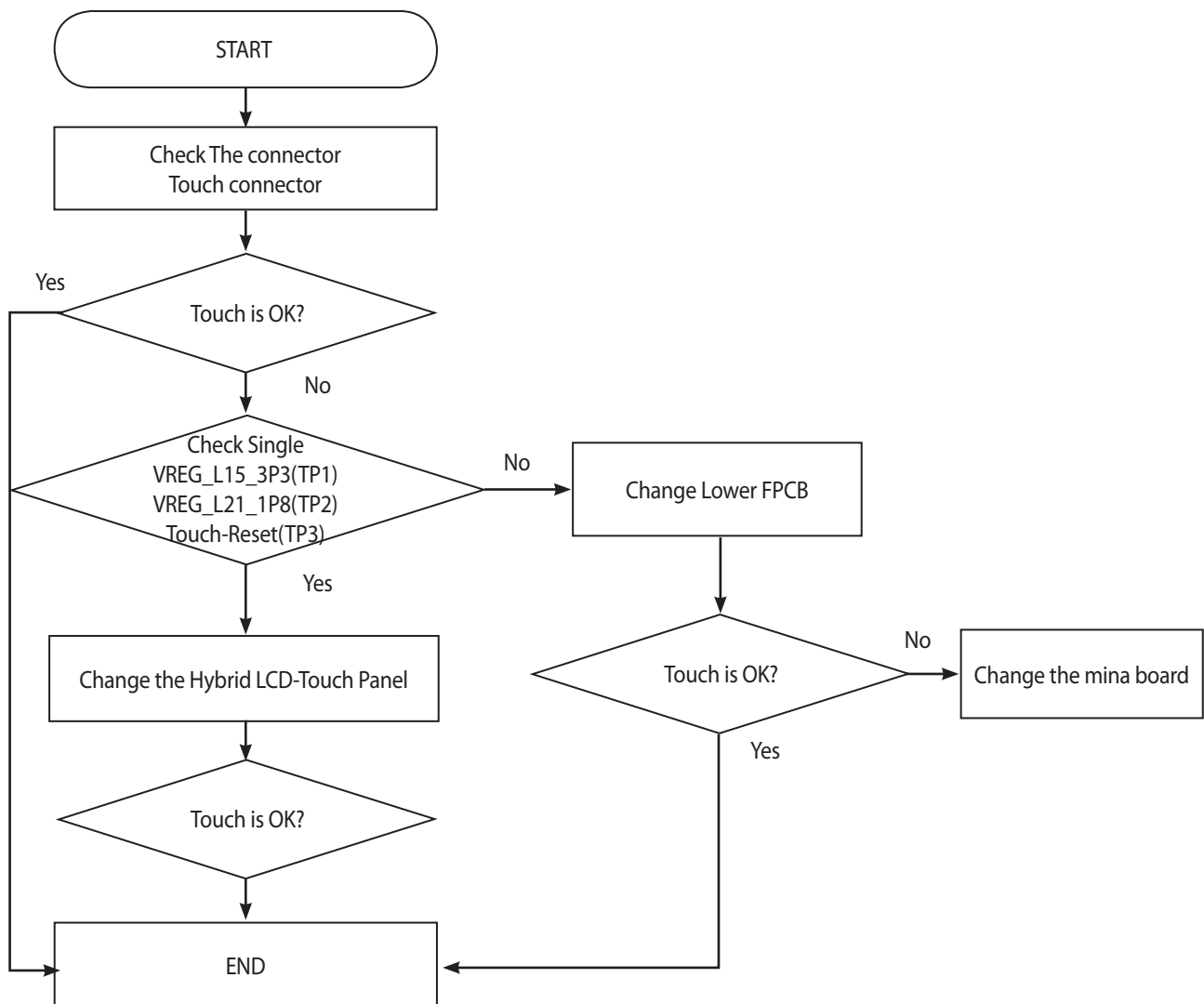


PCB Bottom

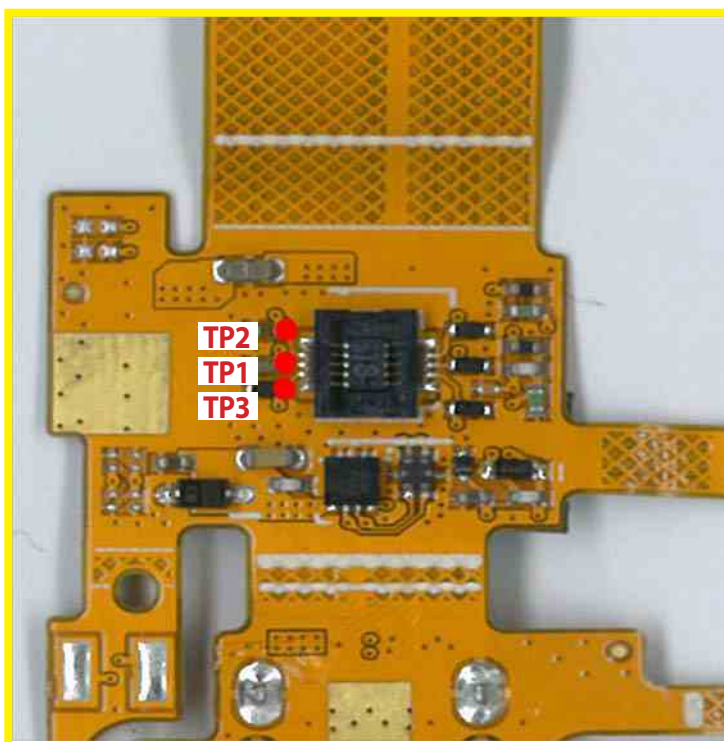
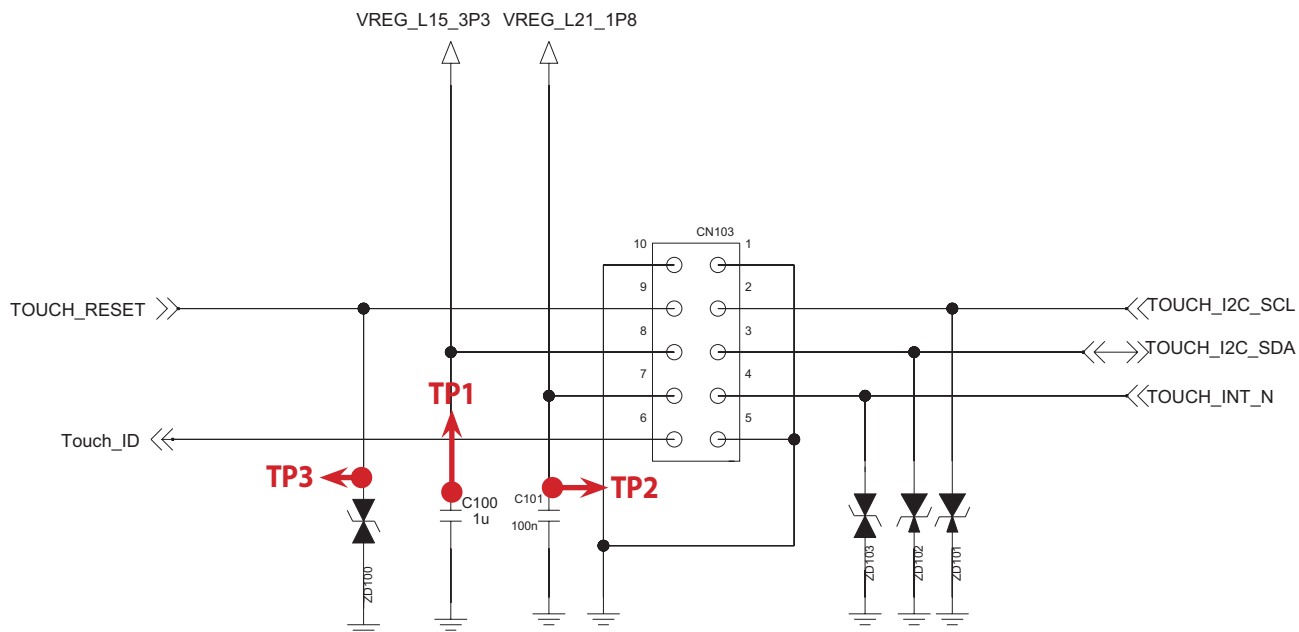
3.15 Touch trouble

Touch control signals are generated by APQ8064. Those signal's path are :

APQ8064 -> Low FPCB -> Touch Module

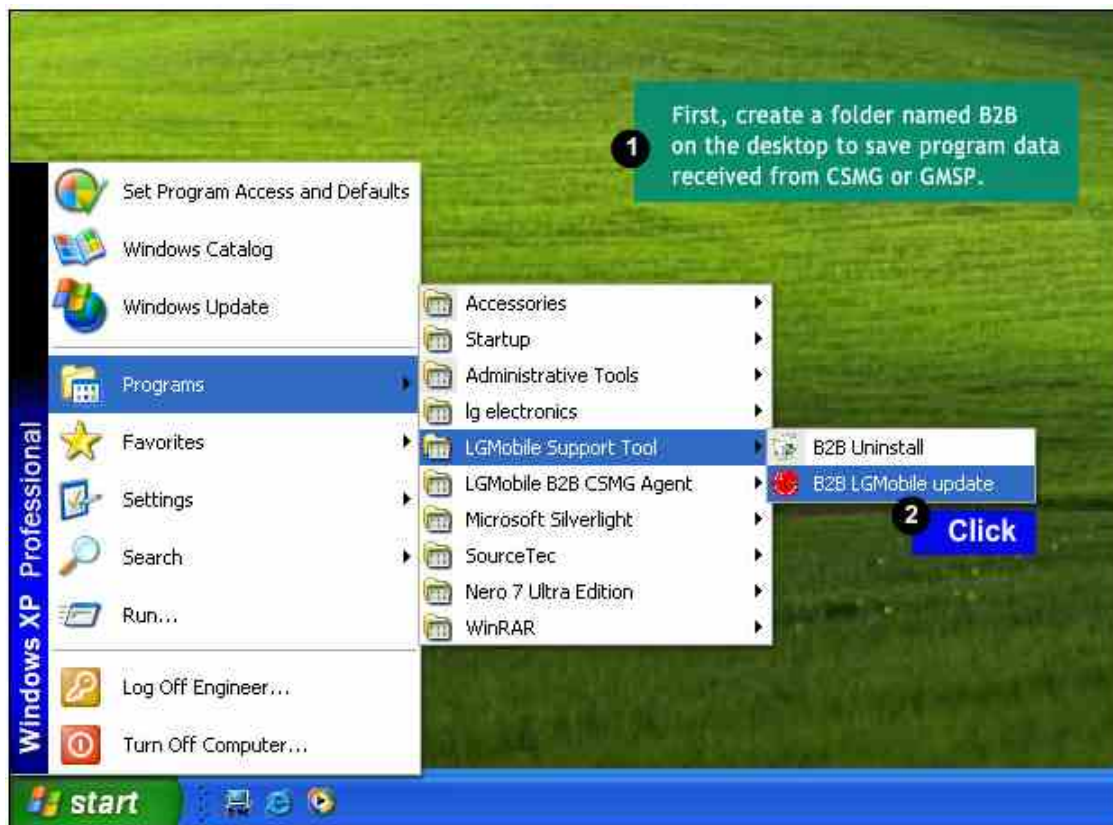


TOUCH CONNECTOR

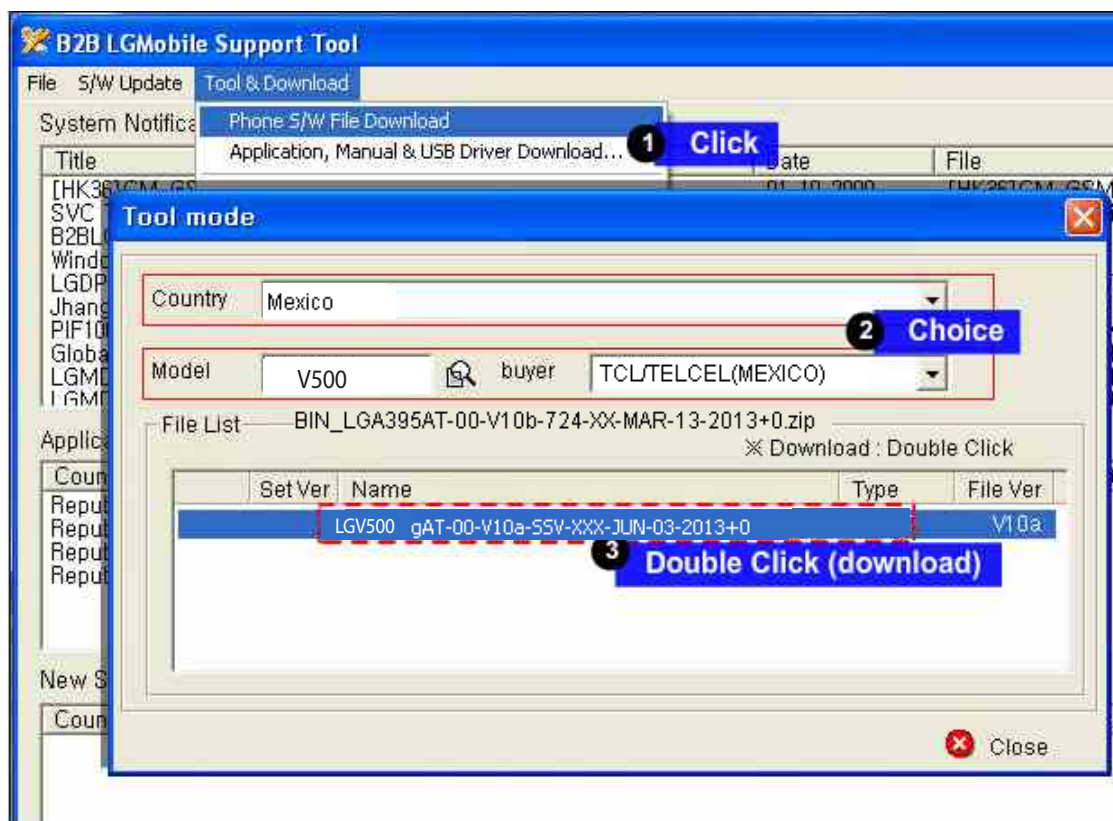
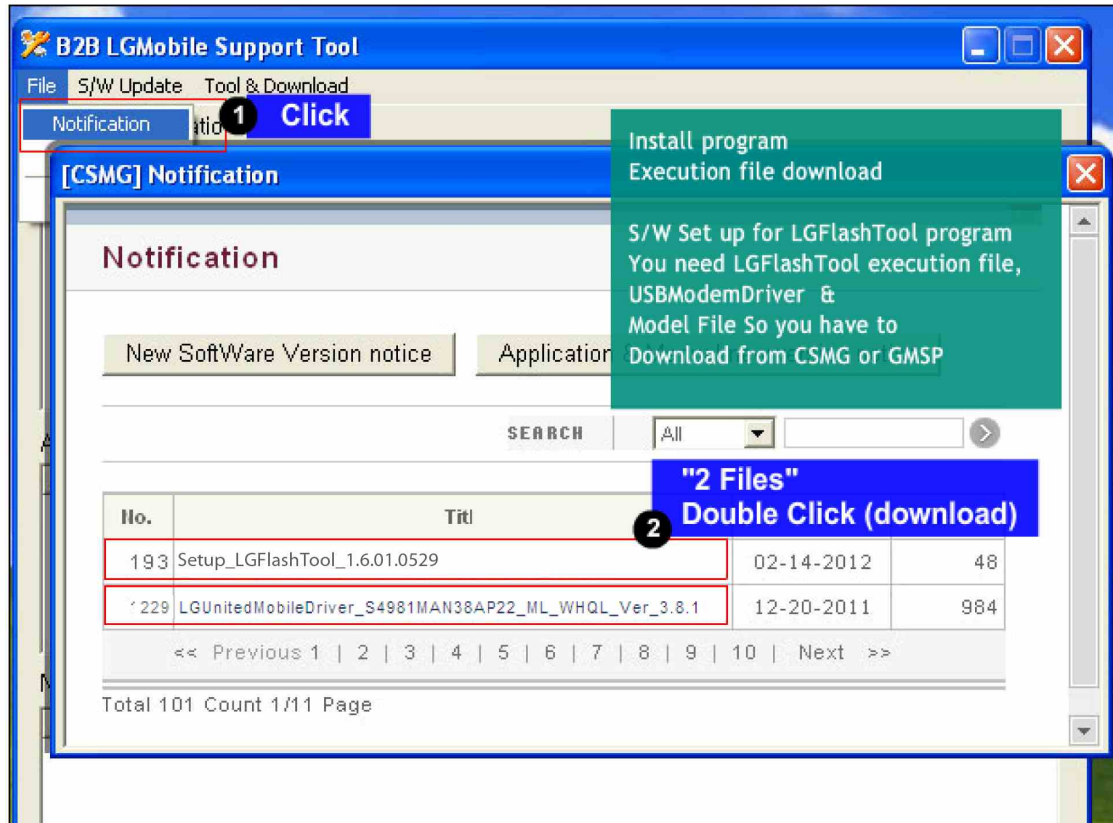


4. DOWNLOAD

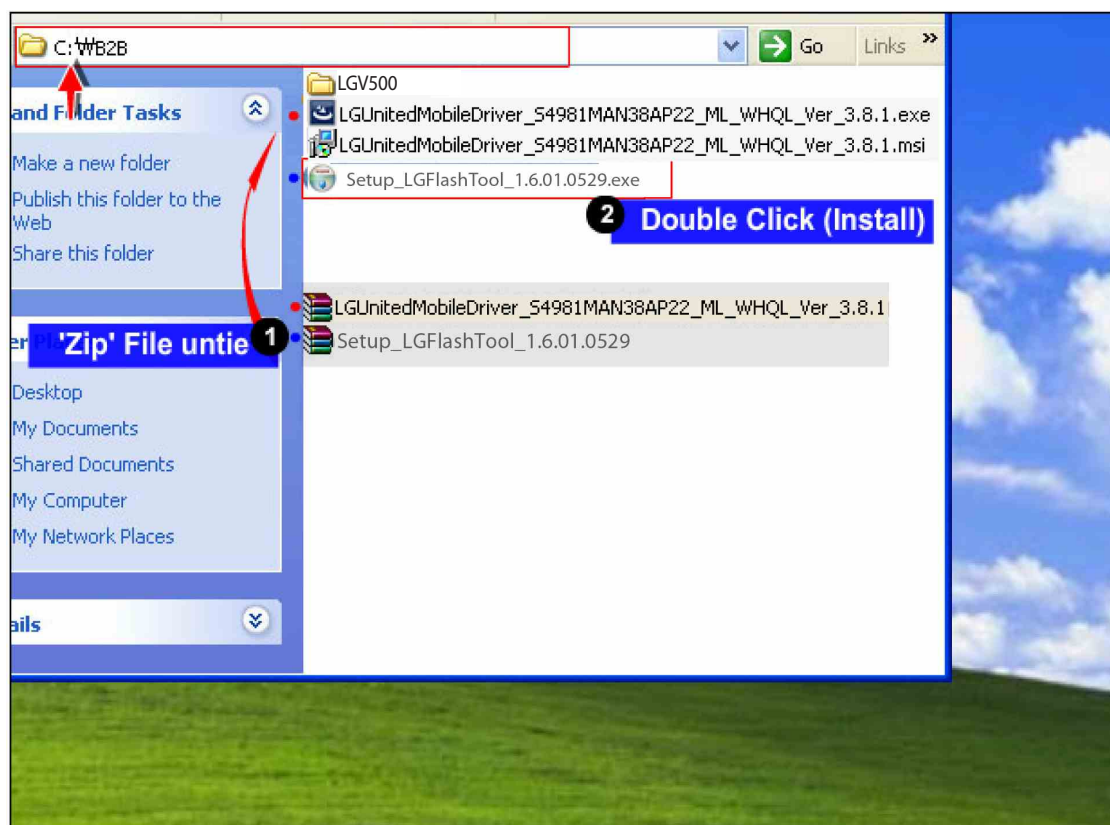
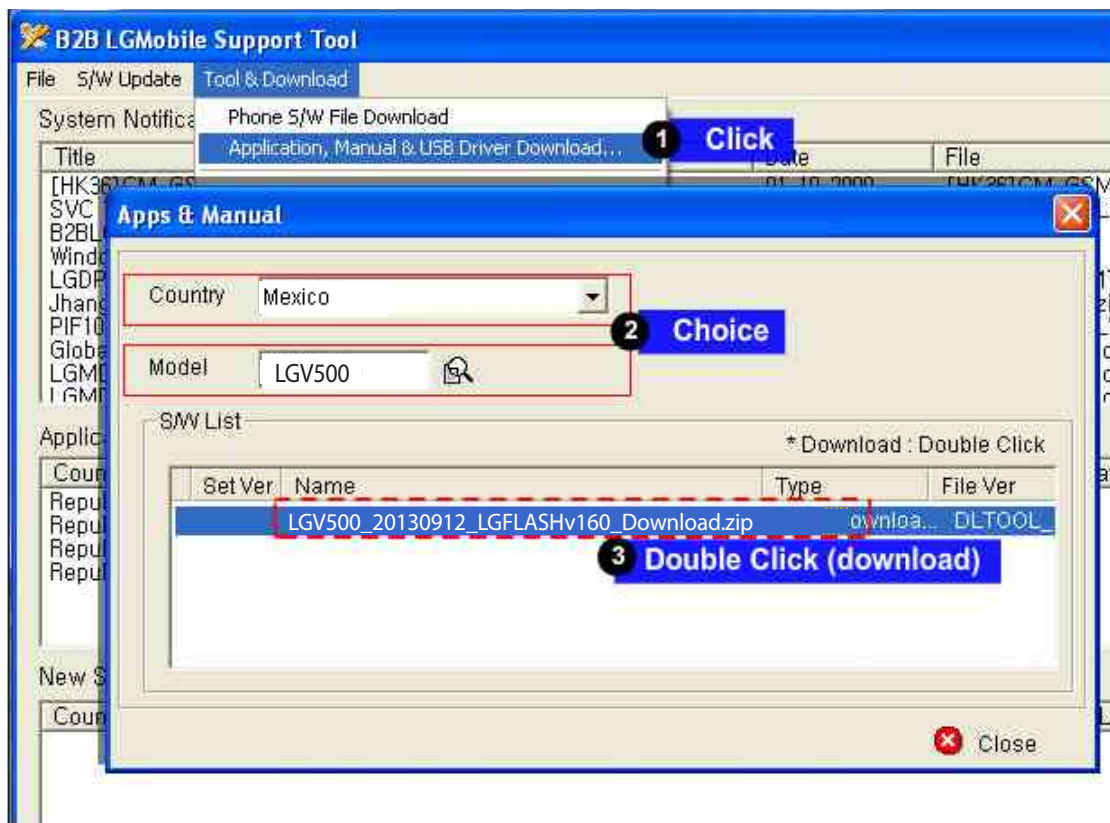
TOOL INFORMATION			
TOOL VERSION	DLL NAME	USB DRIVER	
LGFLASHv160	LGV500_20130912_LGFLASHv160_ Download	LGUnitedMobileDriver_S4981MAN38AP22_ML_WHQL_ Ver_3.8.1	
Please Check the Version to “B2B”			
H/W			
	Name	Part No.	SW
D/L Cable	Micro 5P (56-open-910K) USB DLC	RAD32167835	TOT



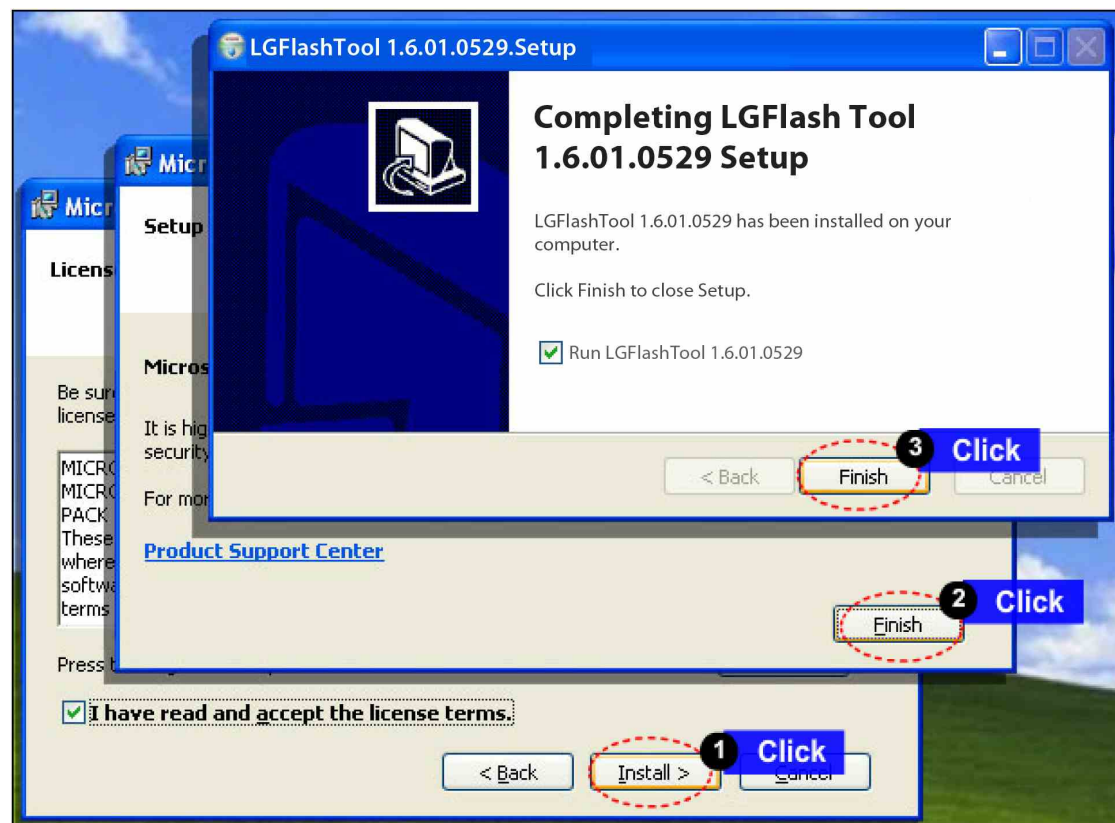
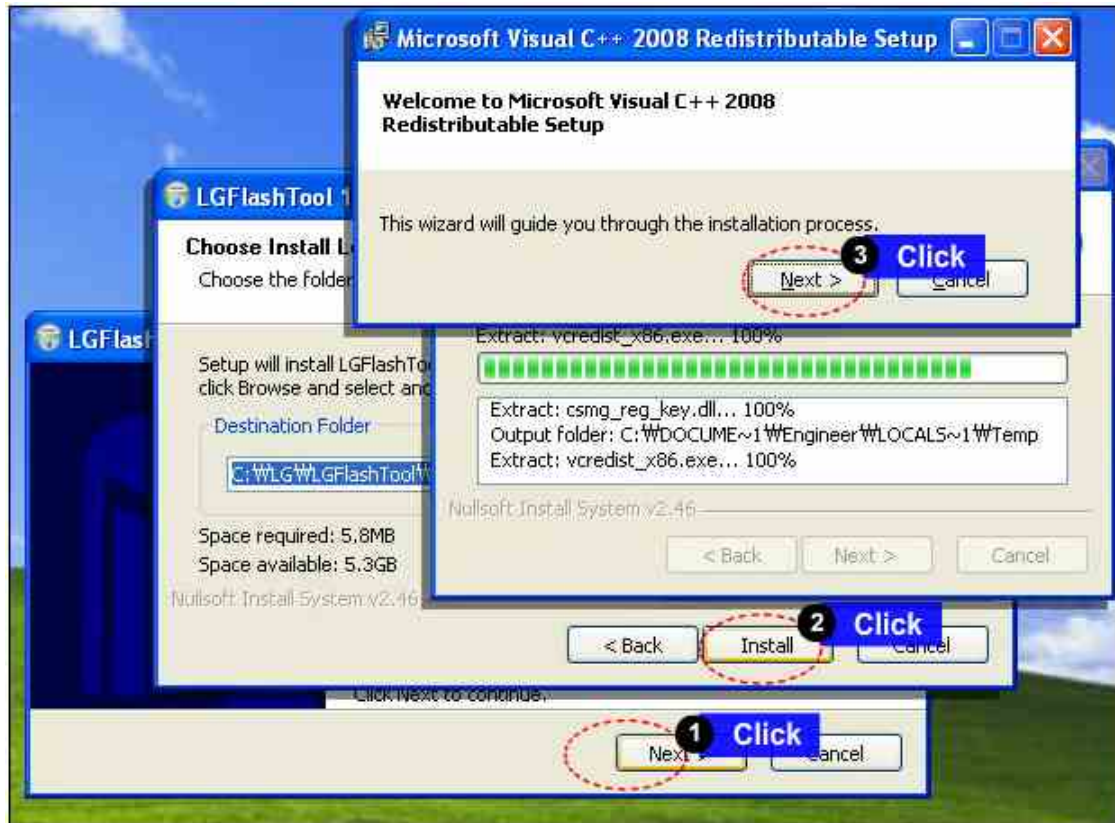
4. DOWNLOAD



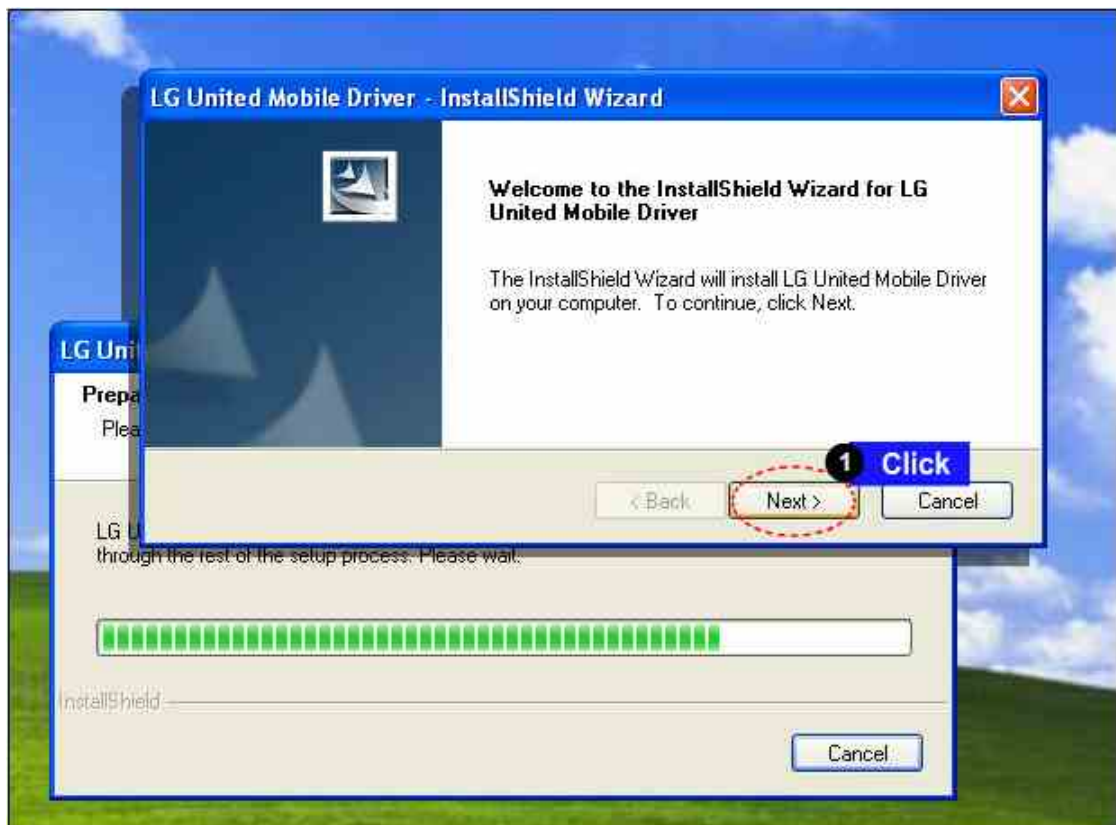
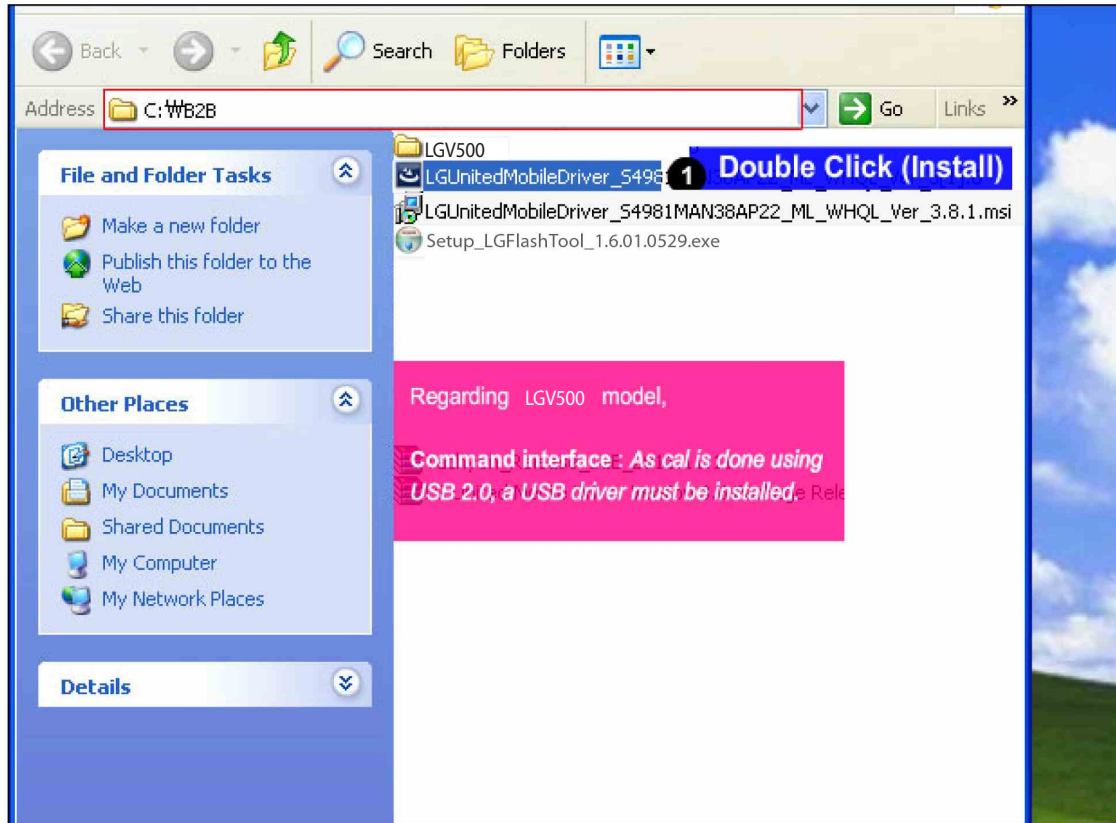
4. DOWNLOAD



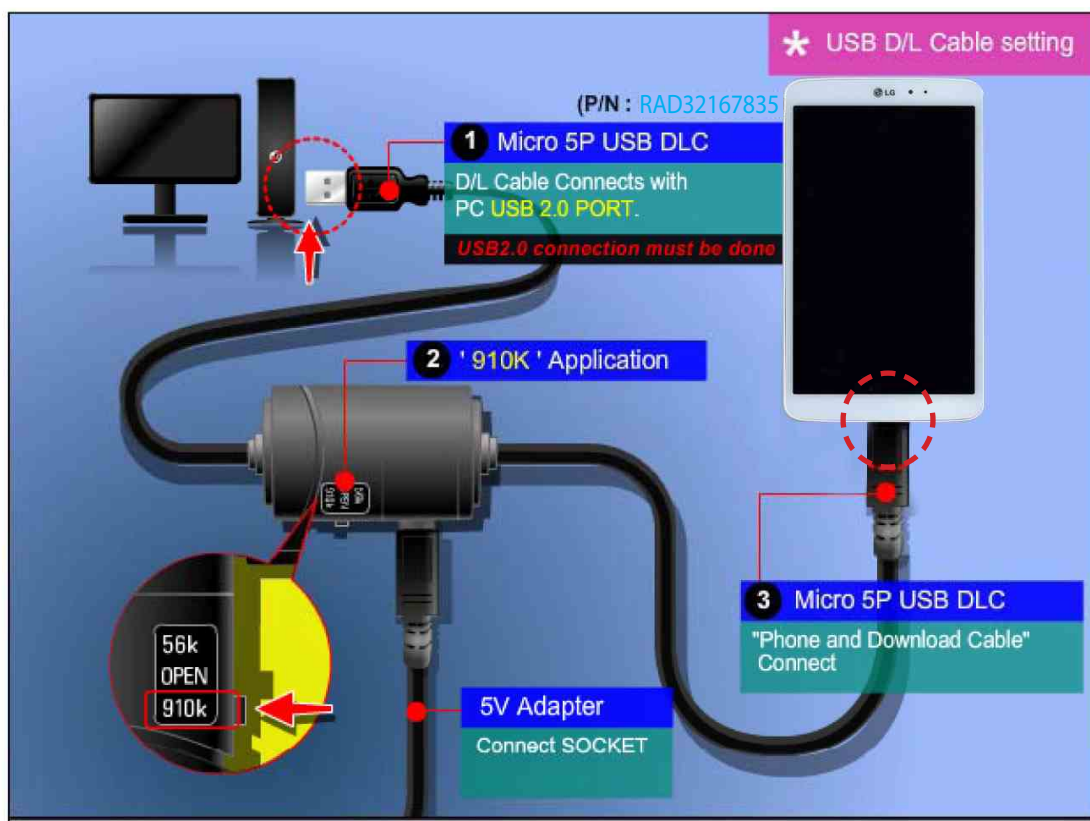
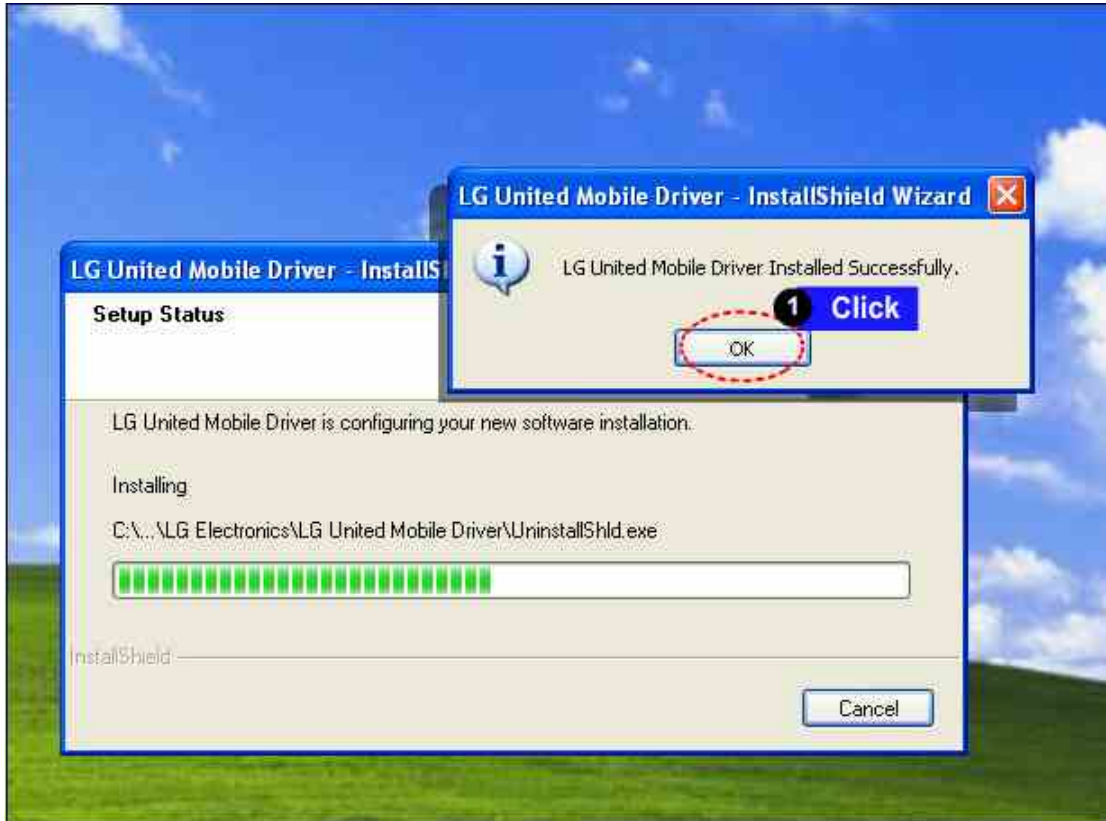
4. DOWNLOAD



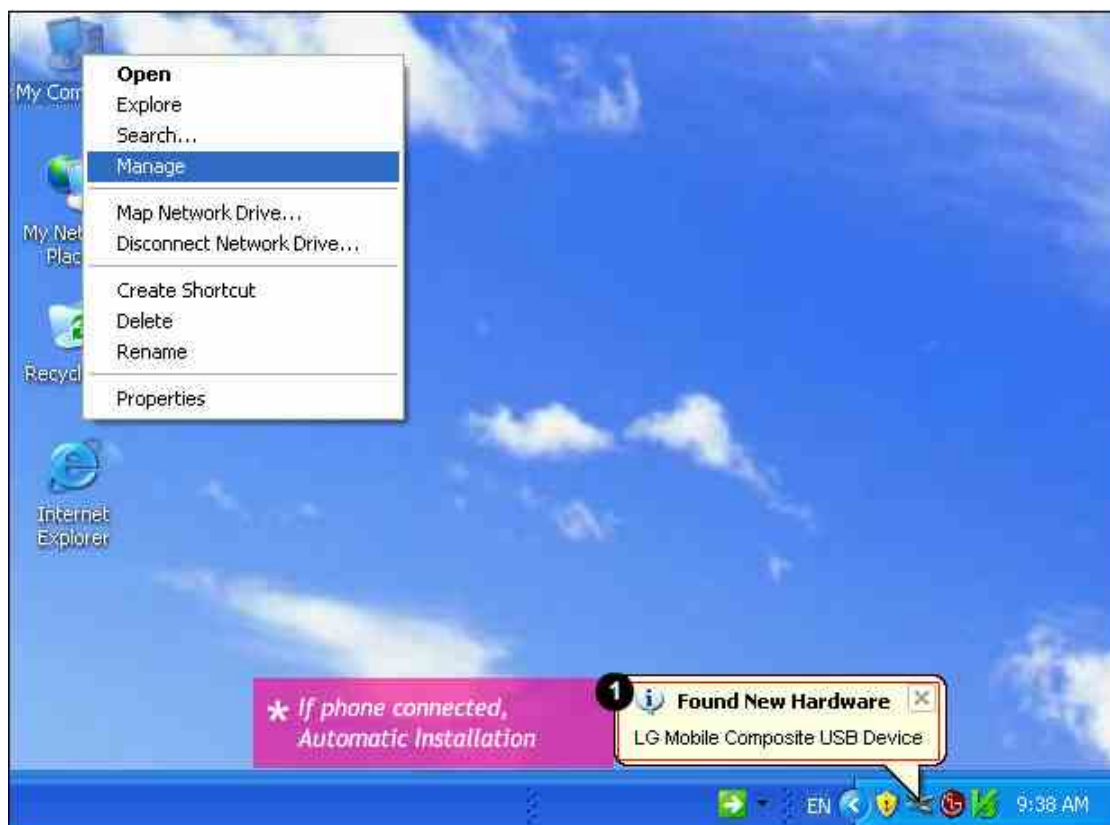
4. DOWNLOAD

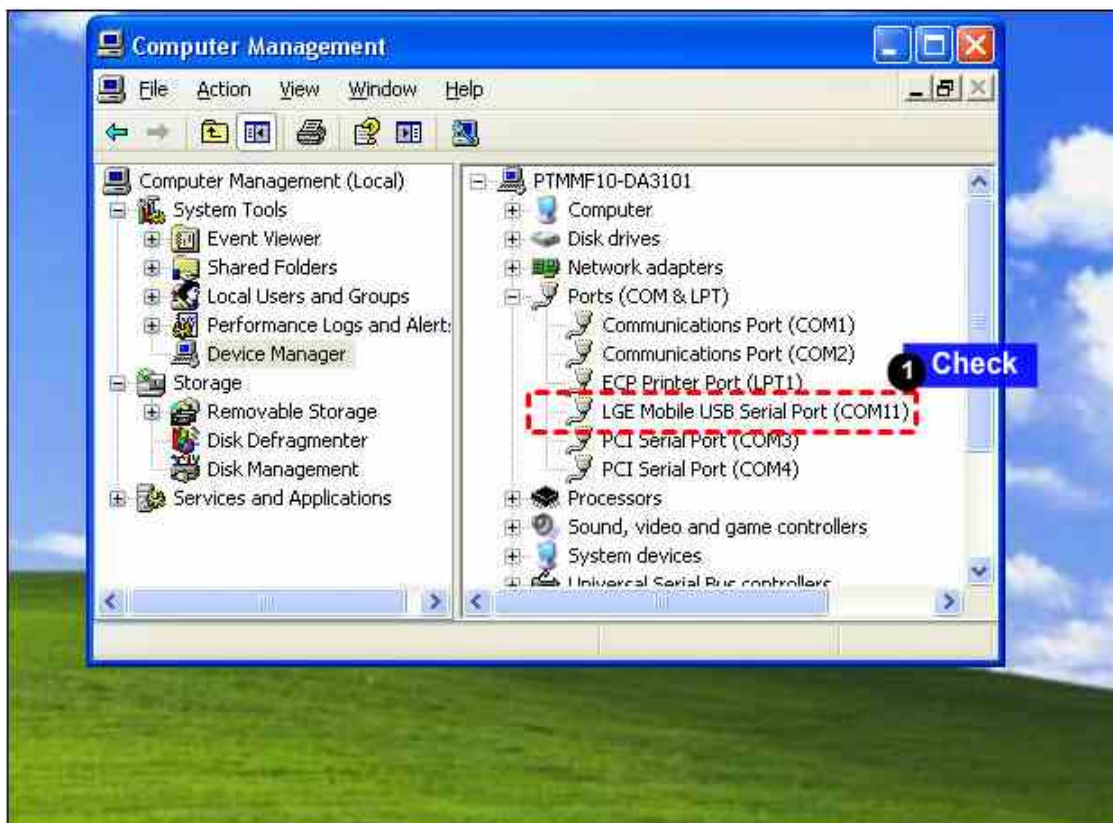
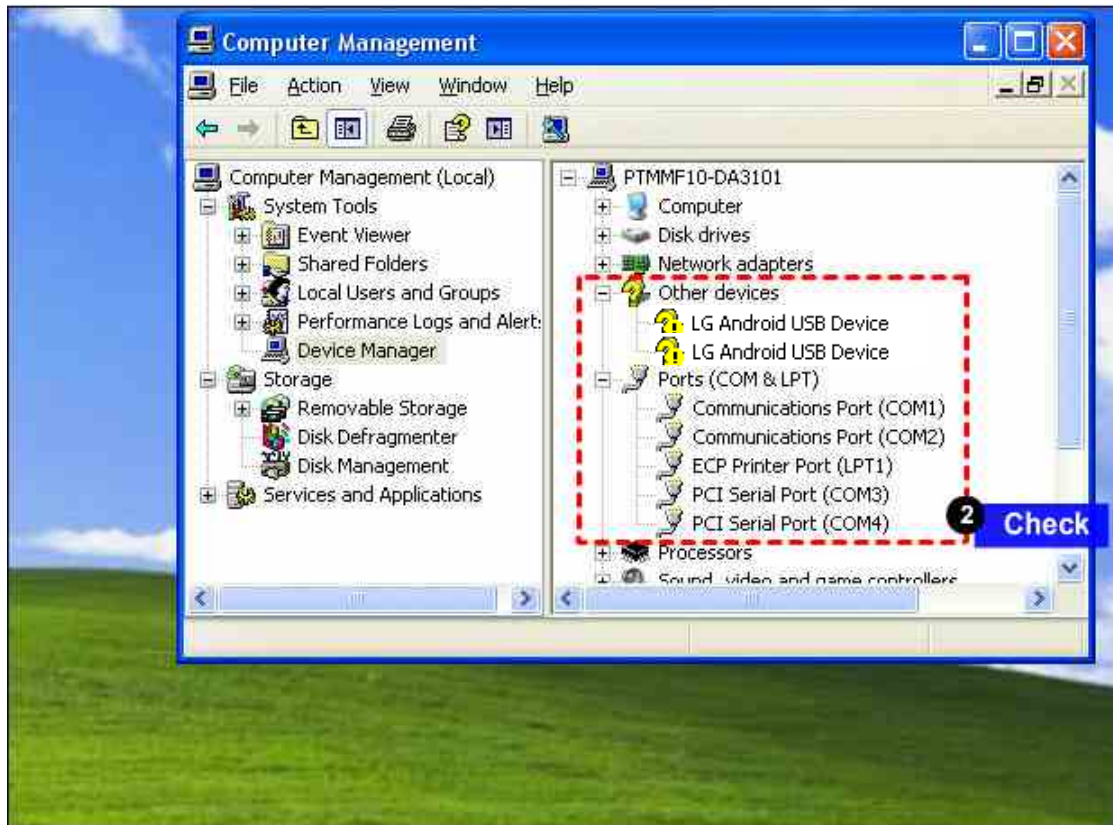


4. DOWNLOAD

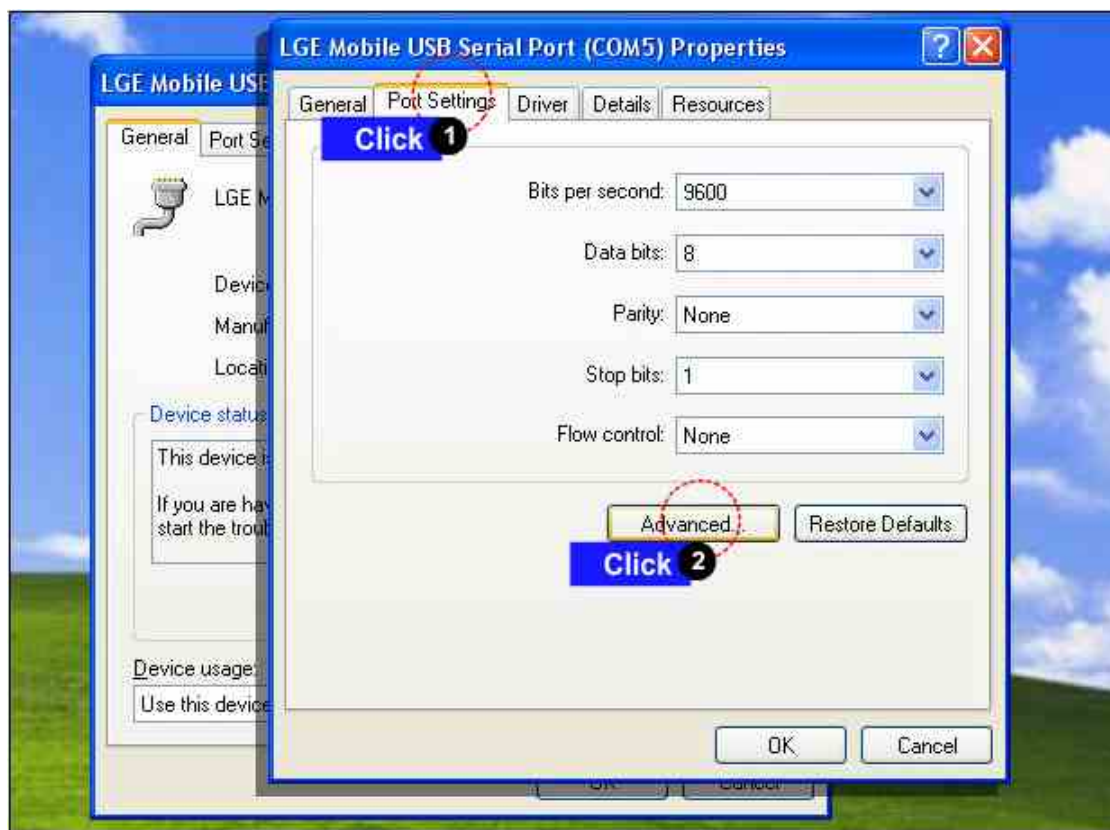
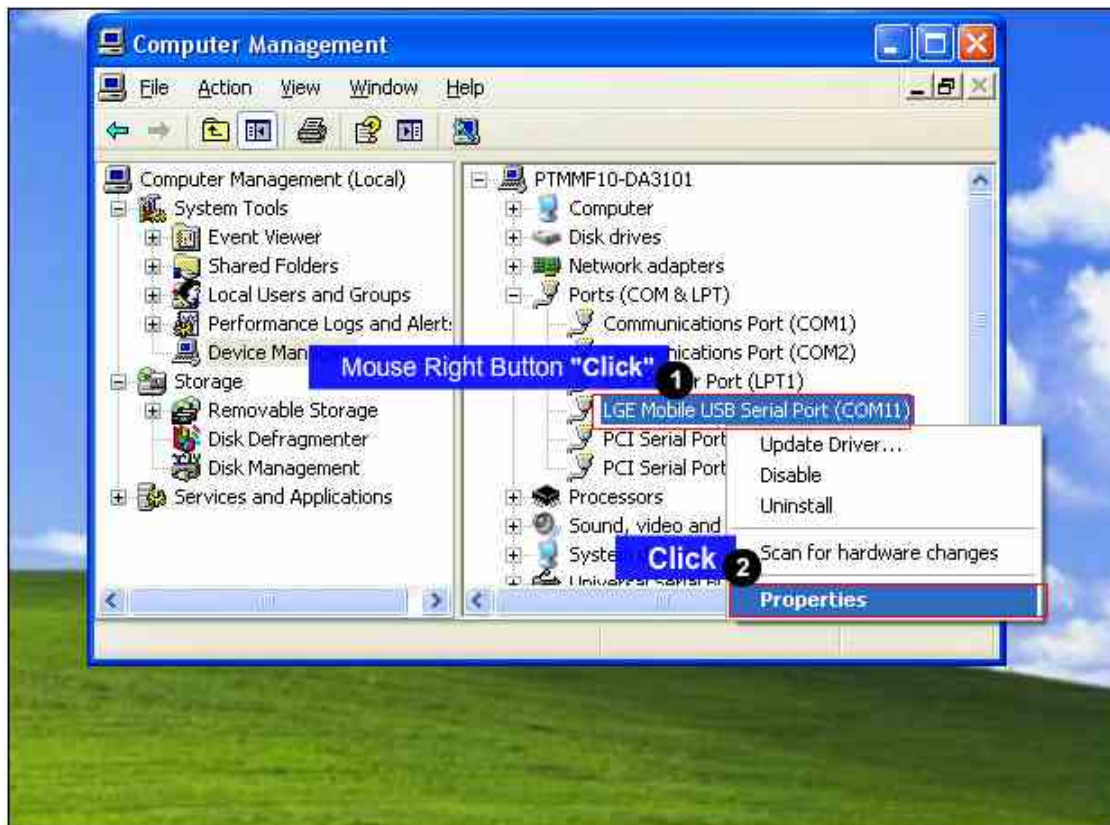


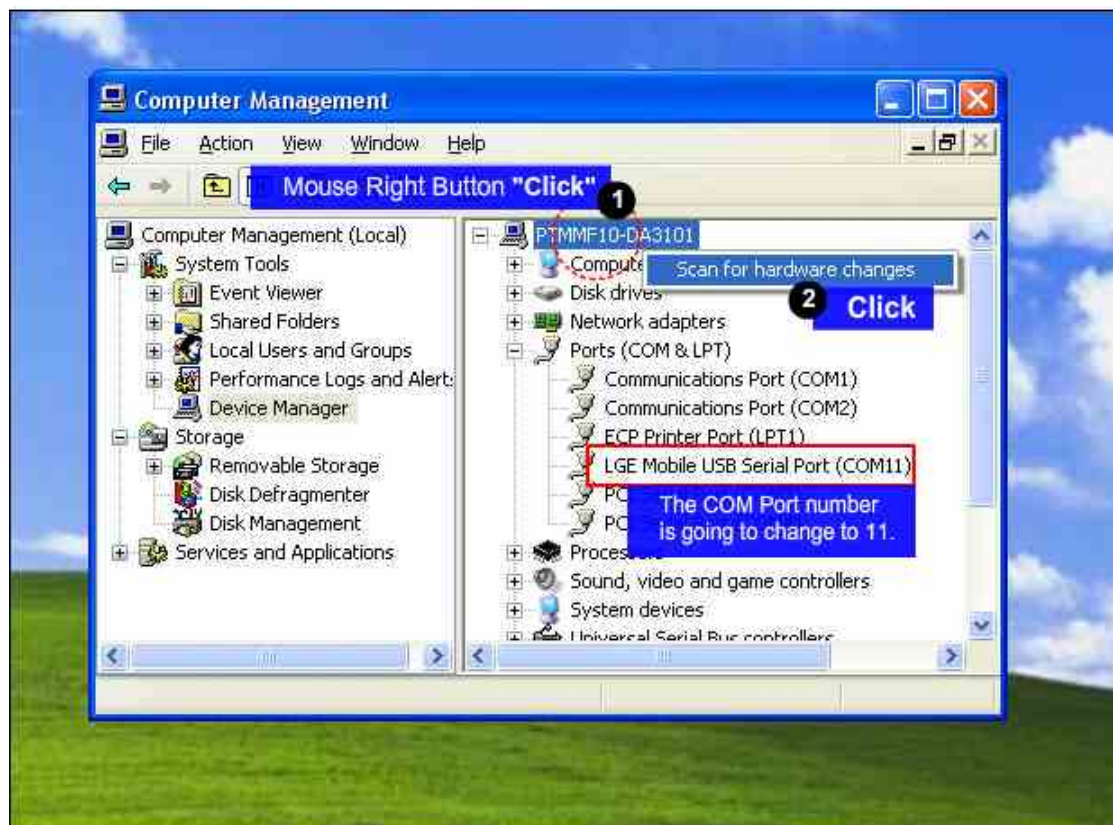
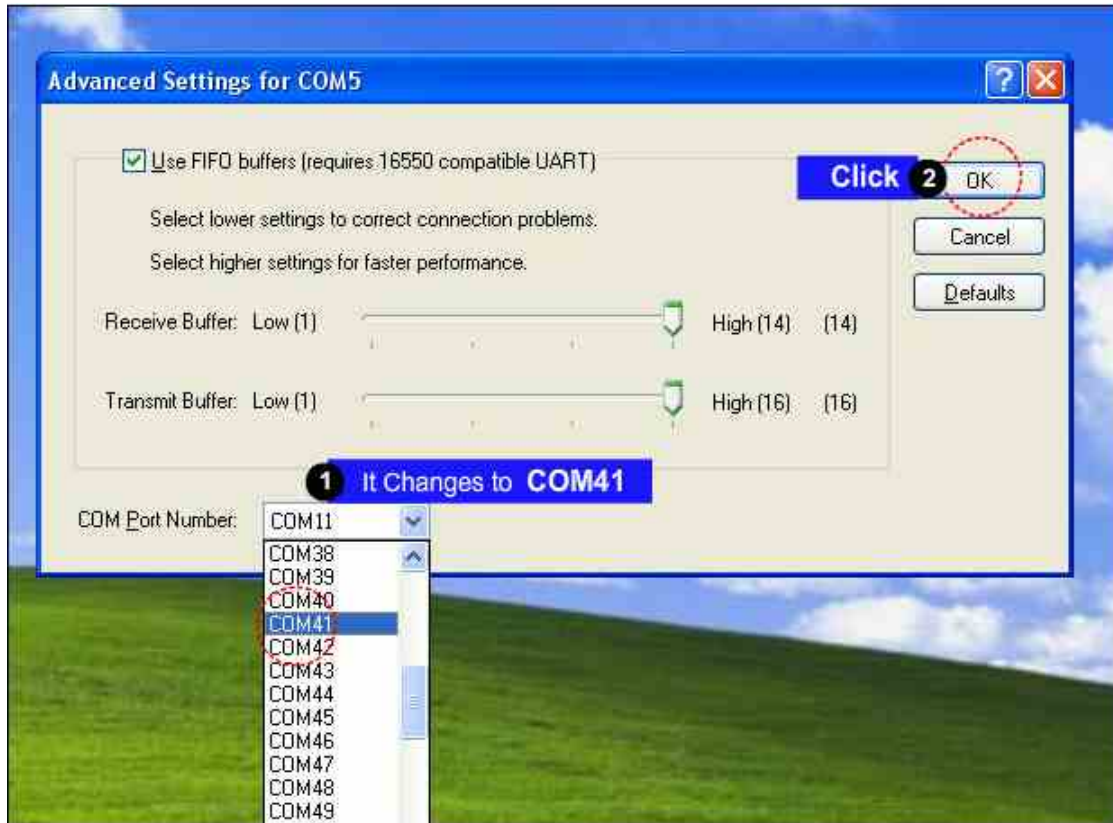
4. DOWNLOAD



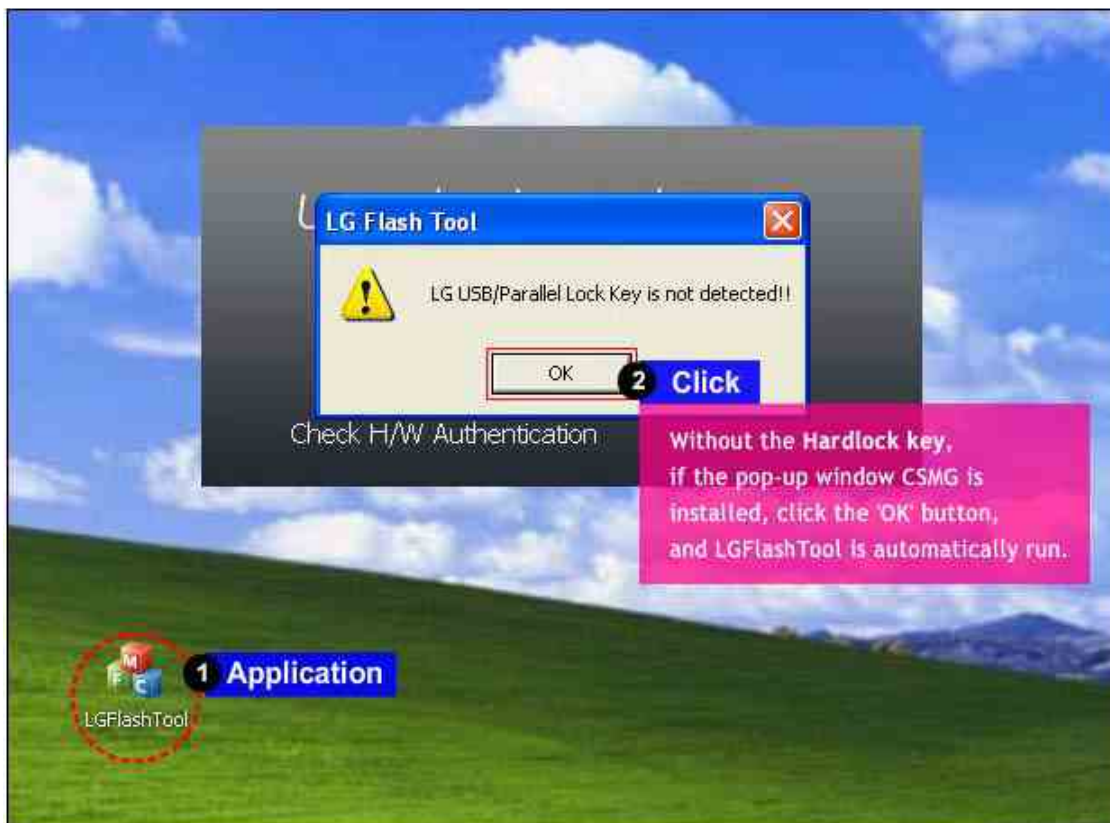
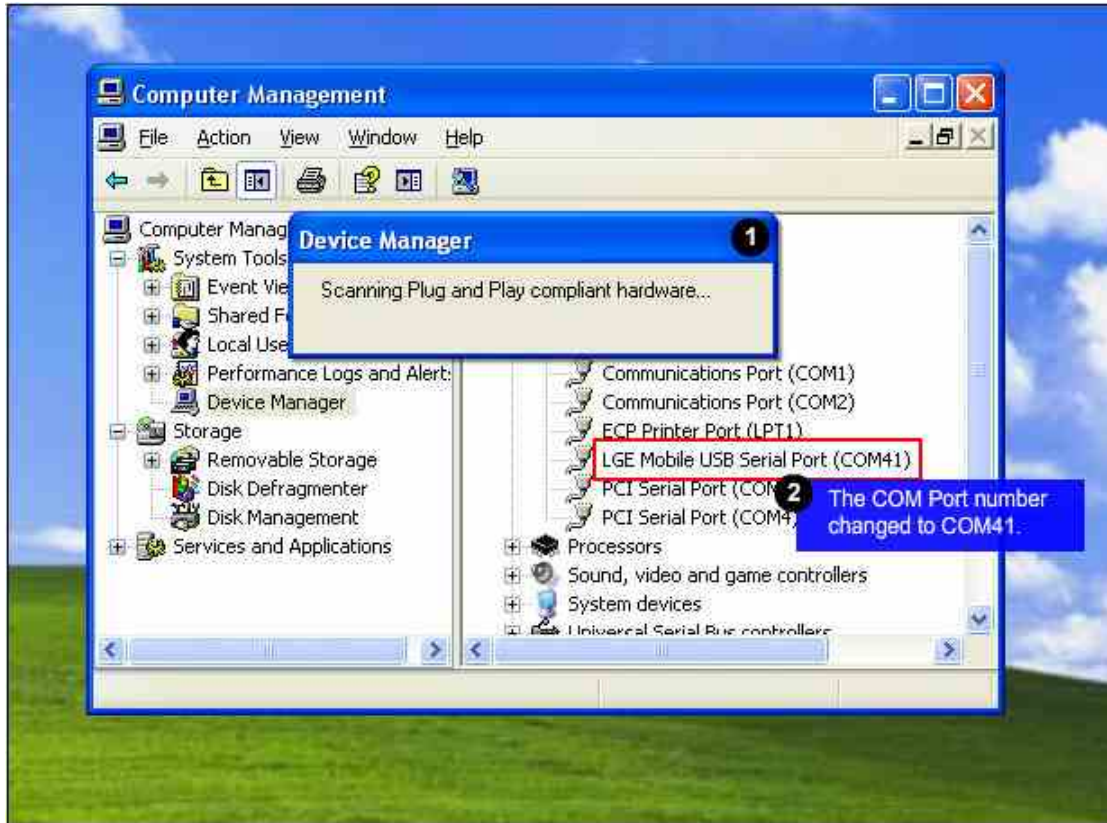


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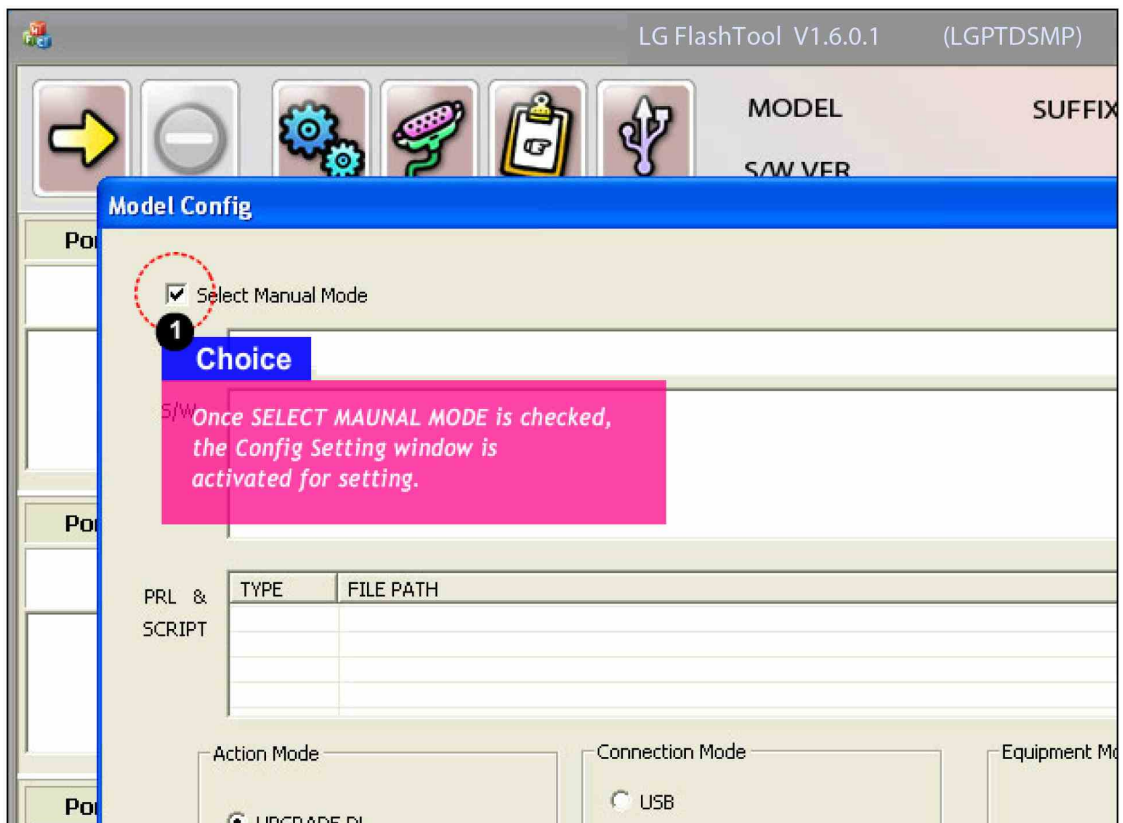




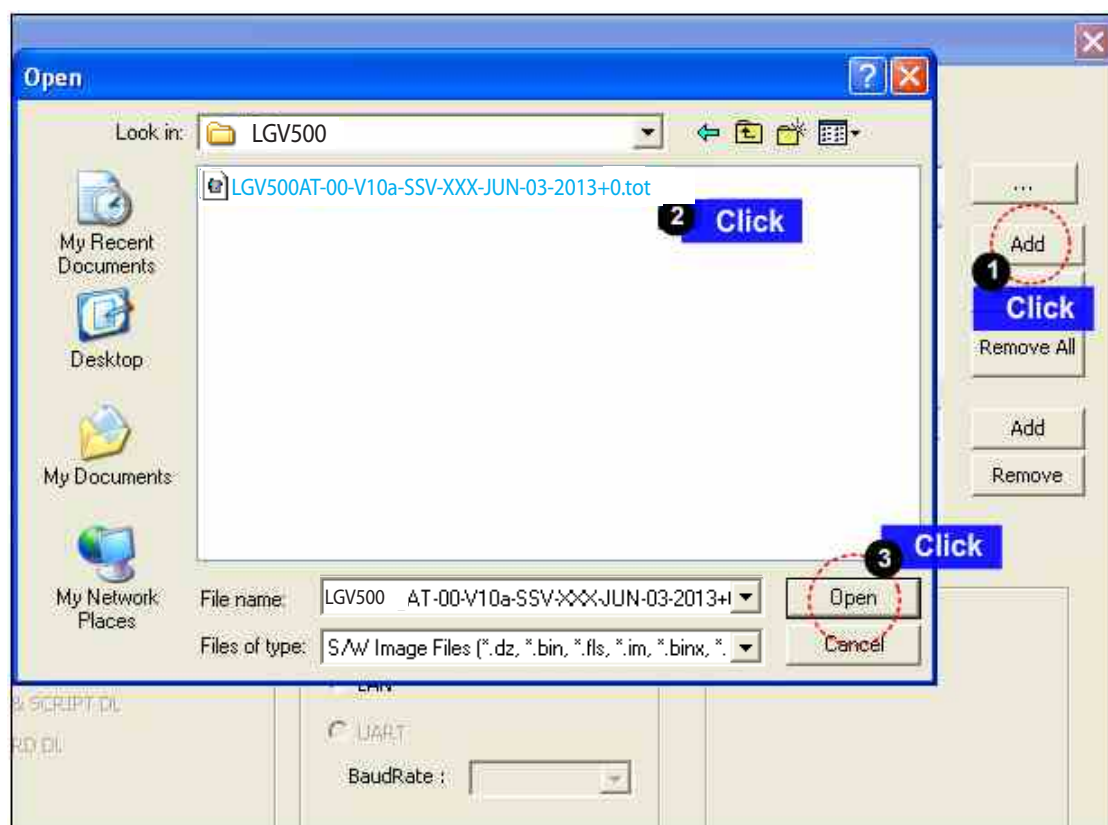
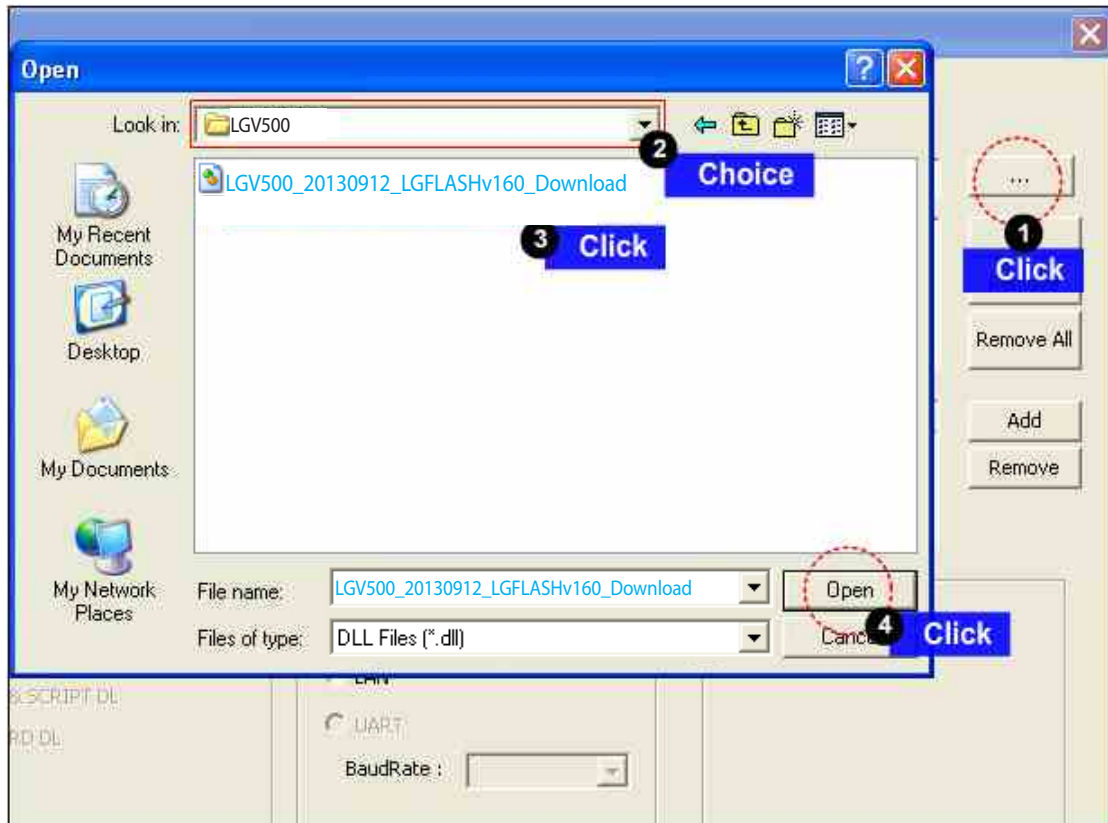
4. DOWNLOAD

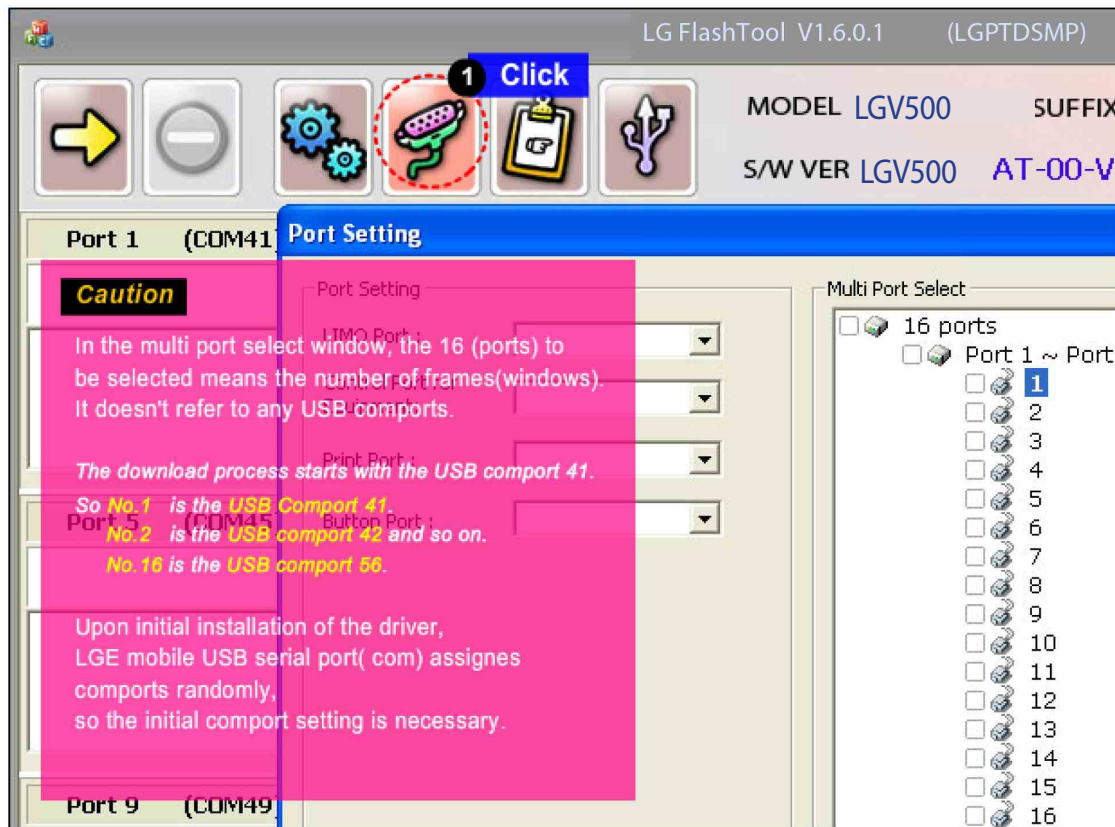


4. DOWNLOAD

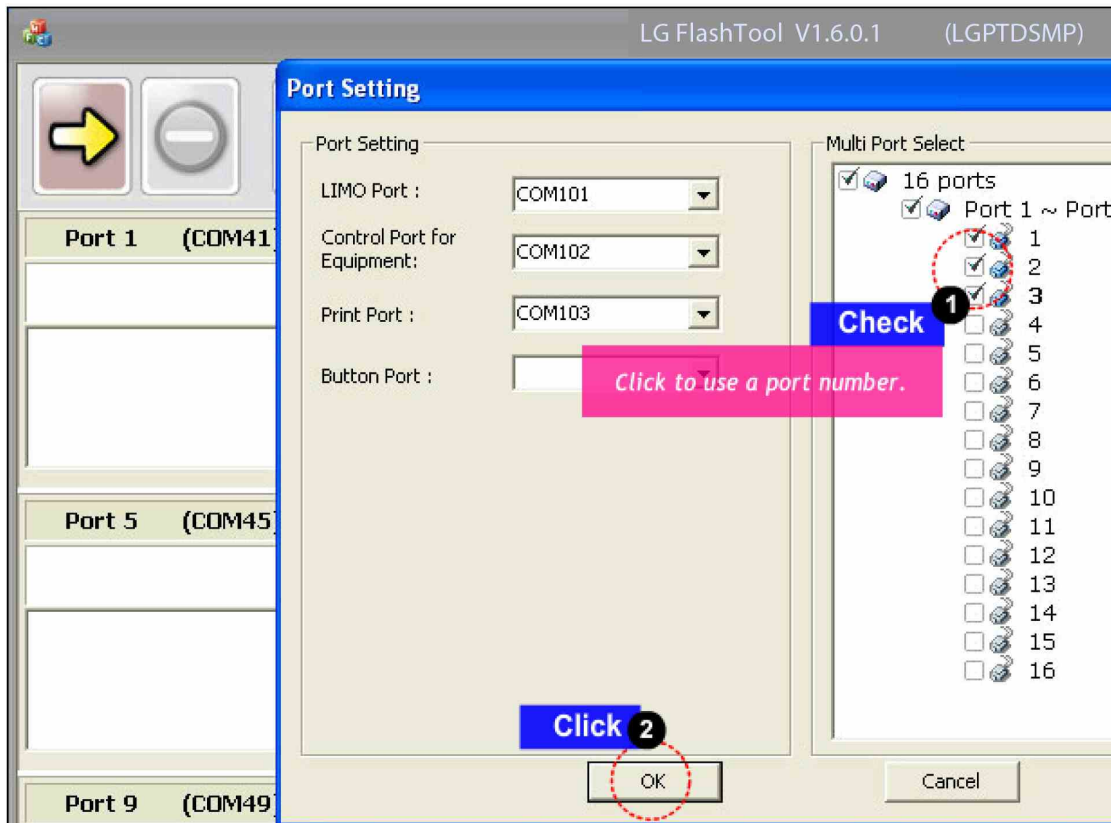


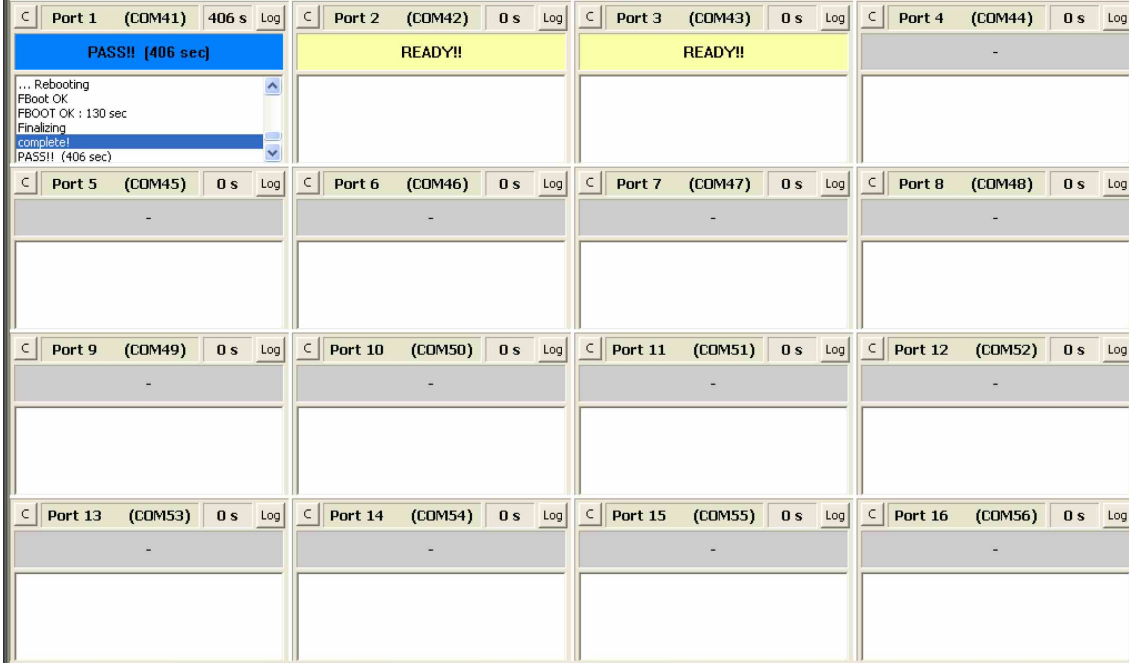
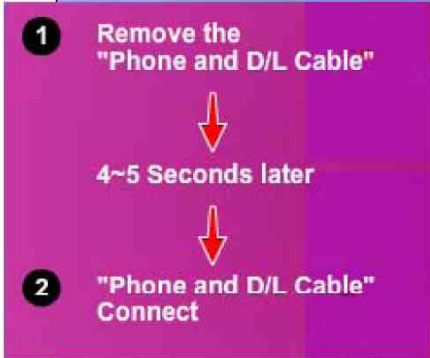
4. DOWNLOAD



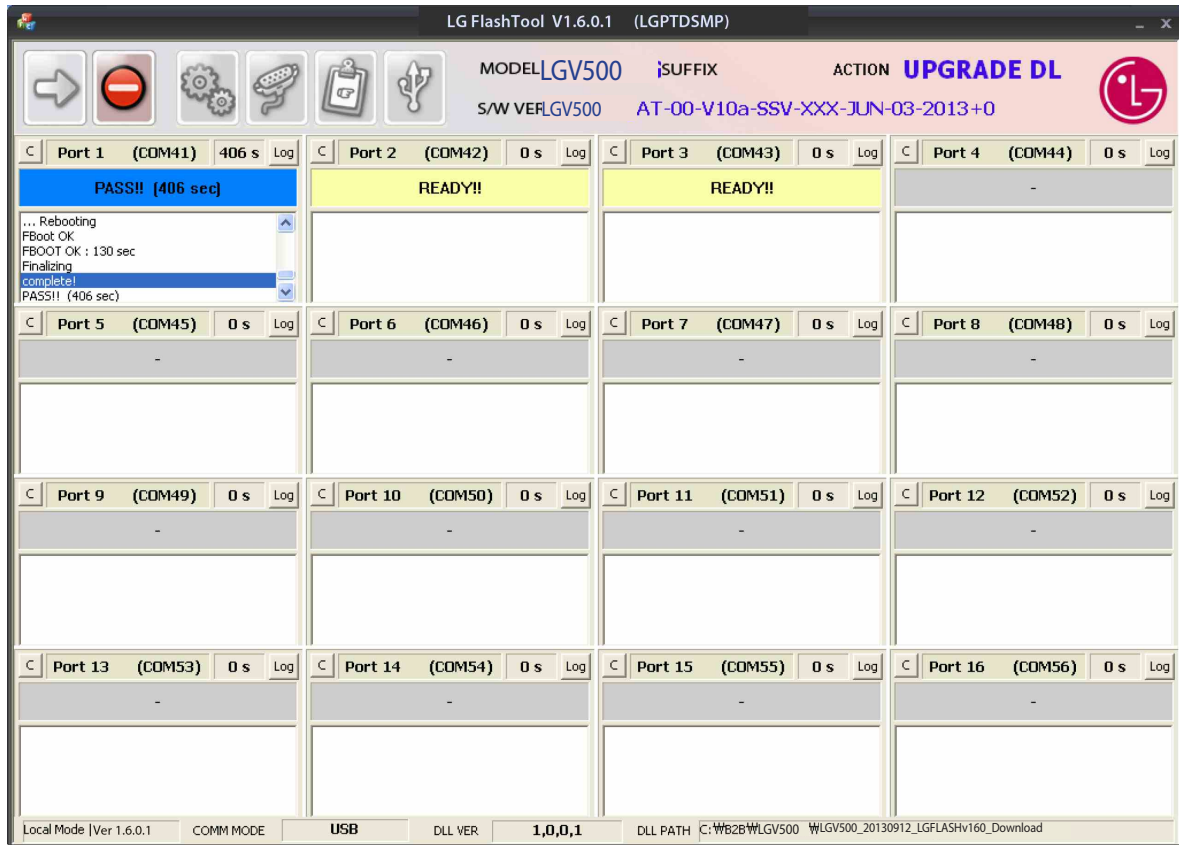


4. DOWNLOAD



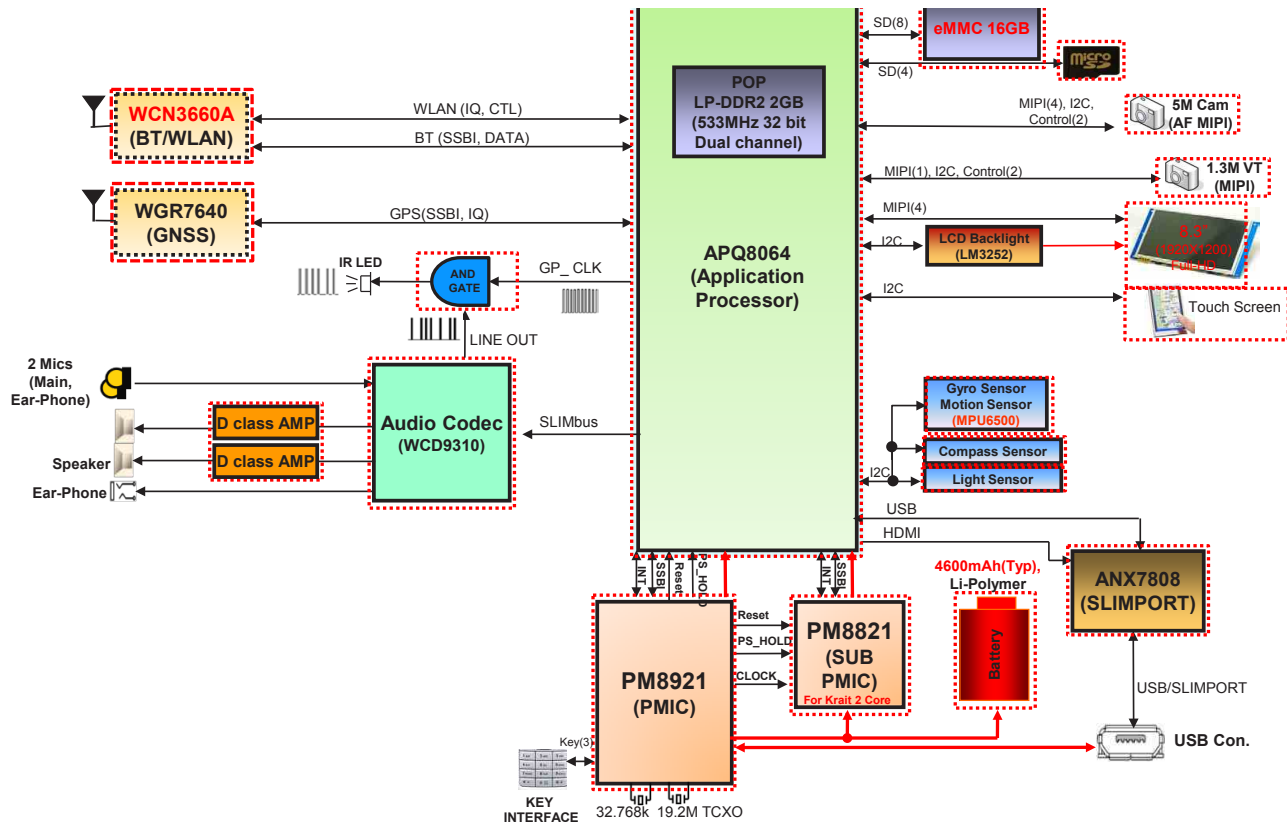


4. DOWNLOAD



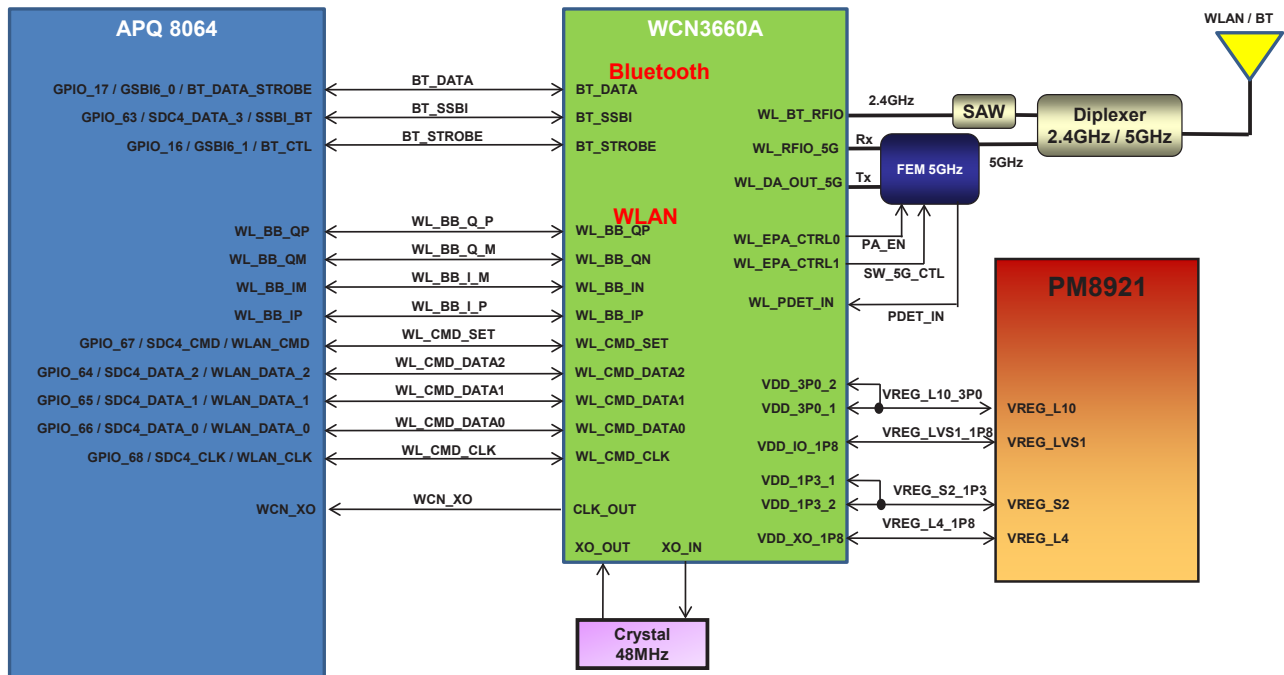
5. BLOCK DIAGRAM

[V500] Total Block Diagram



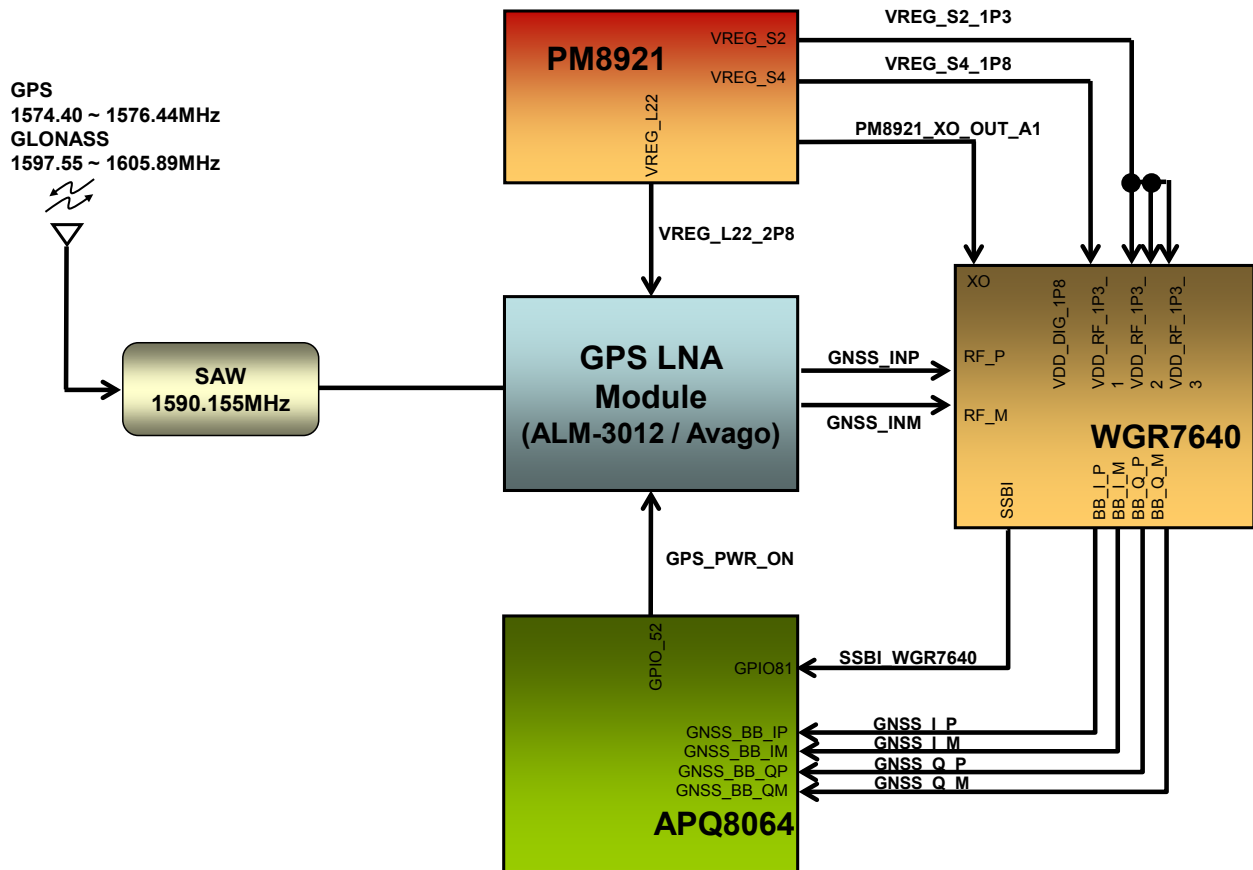
5. BLOCK DIAGRAM

[V500] Block diagram : RF SUB BAND 1 (BT/WI-FI-COB)

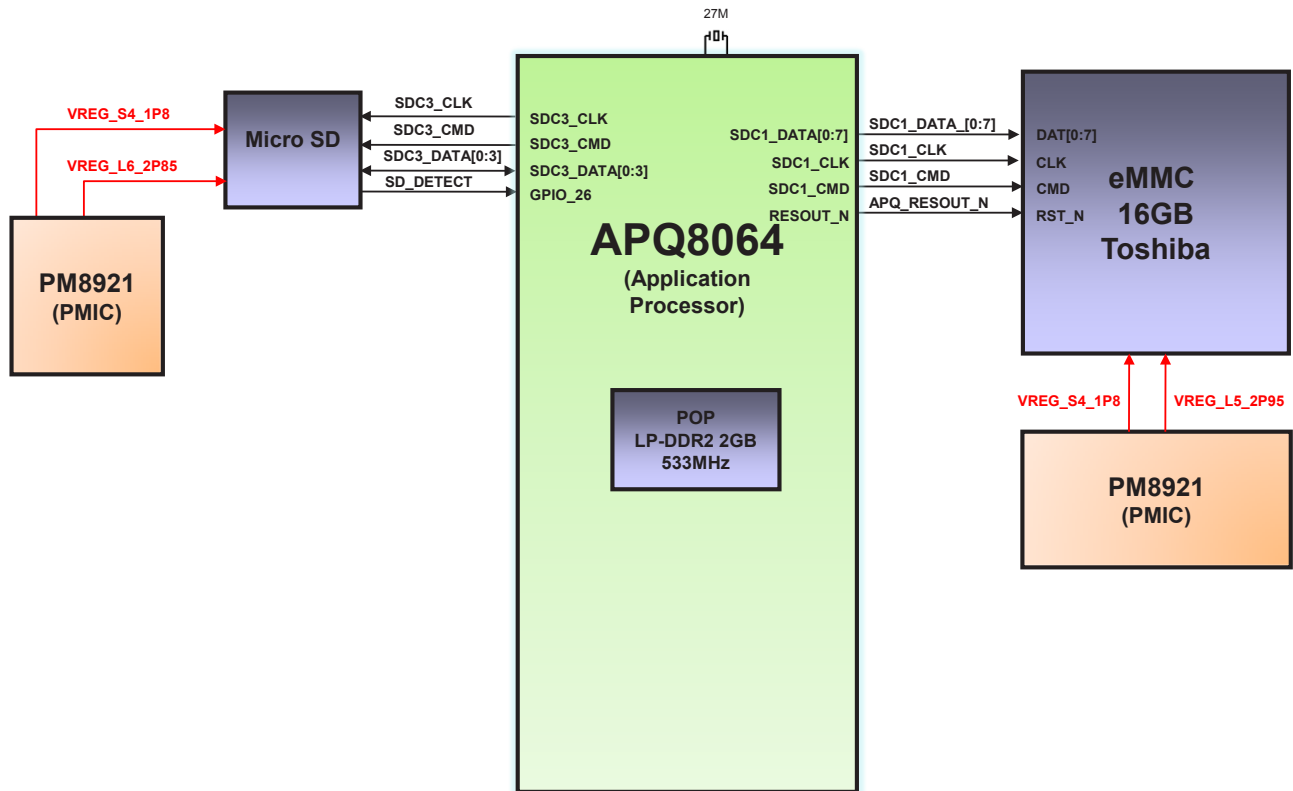


5. BLOCK DIAGRAM

[V500] RF SUB BAND 2 (GNSS)

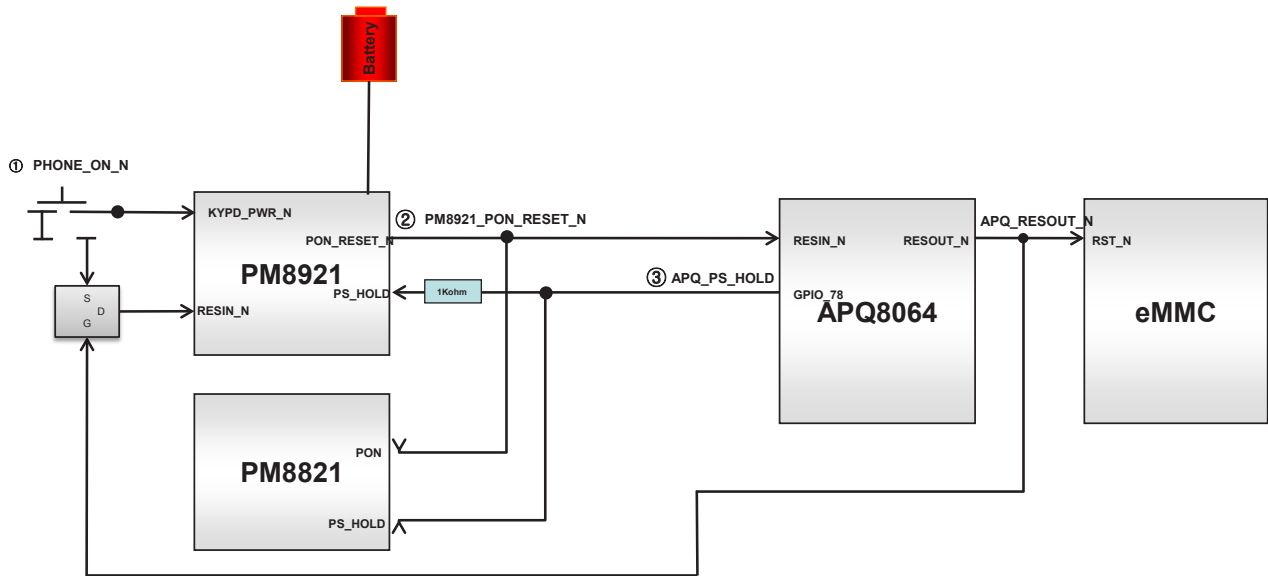


[V500] Block diagram : Memory, Micro-SD

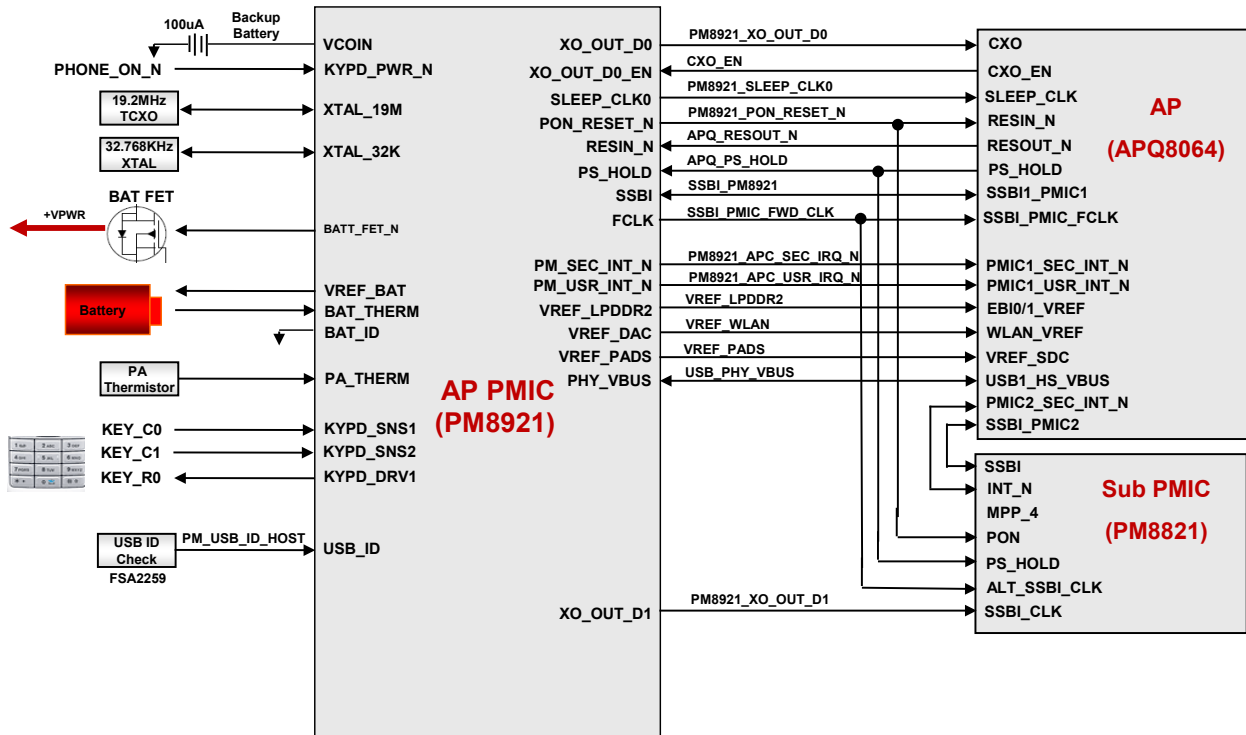


[V500] Block diagram : POWER UP/RESET TREE

<Booting Sequence>

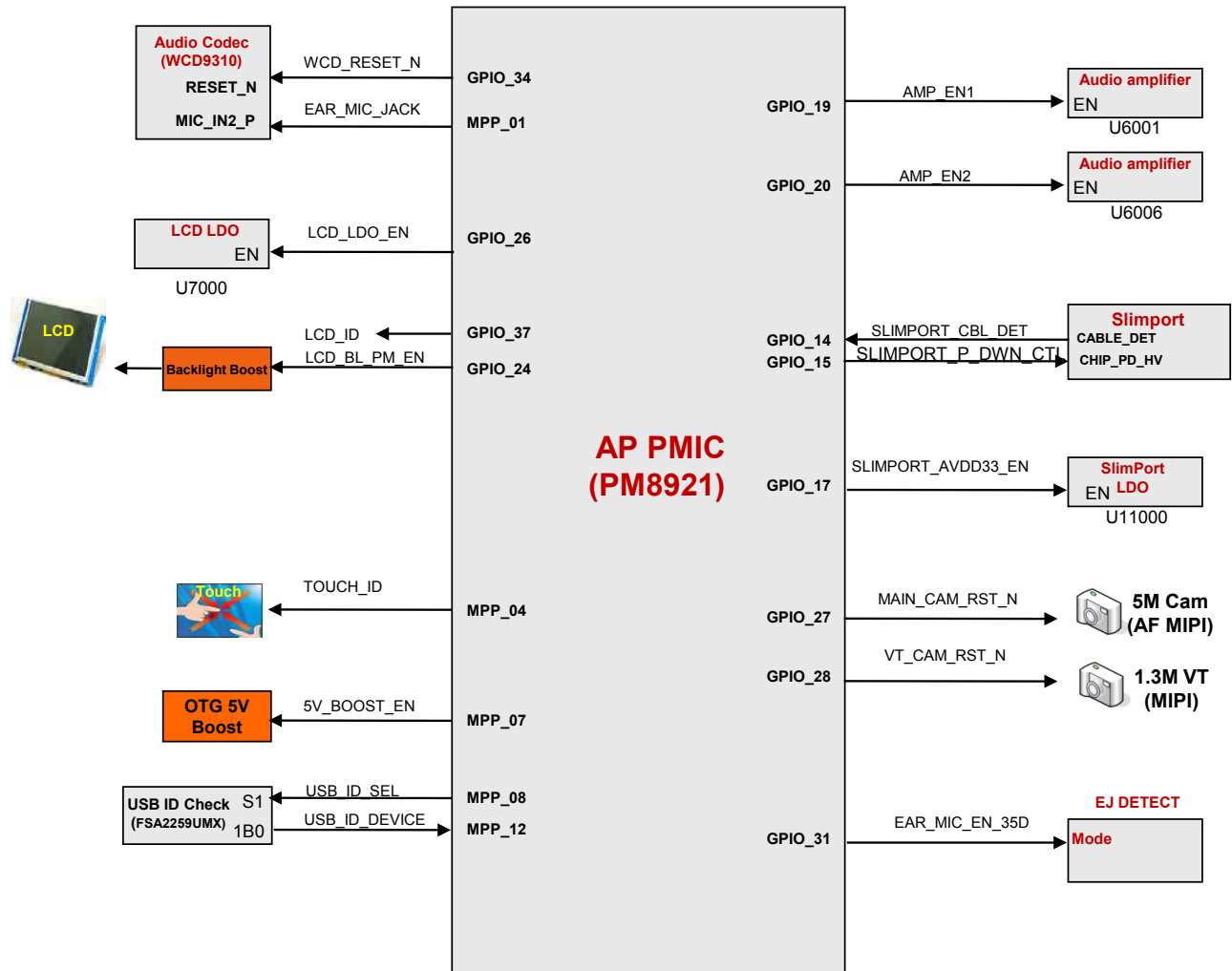


[V500] Block diagram : PM8921 Control INTERFACE



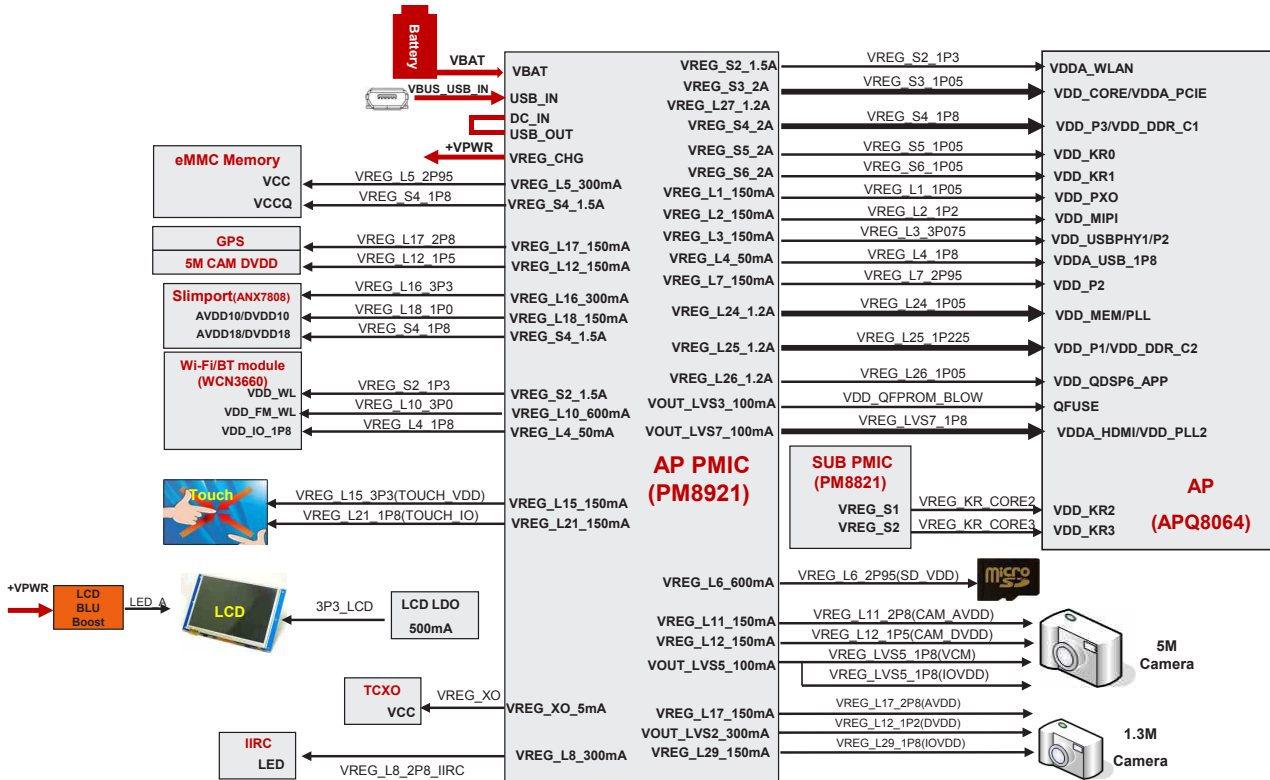
5. BLOCK DIAGRAM

[V500] Block diagram : PM8921 GPIO/MPP INTERFACE



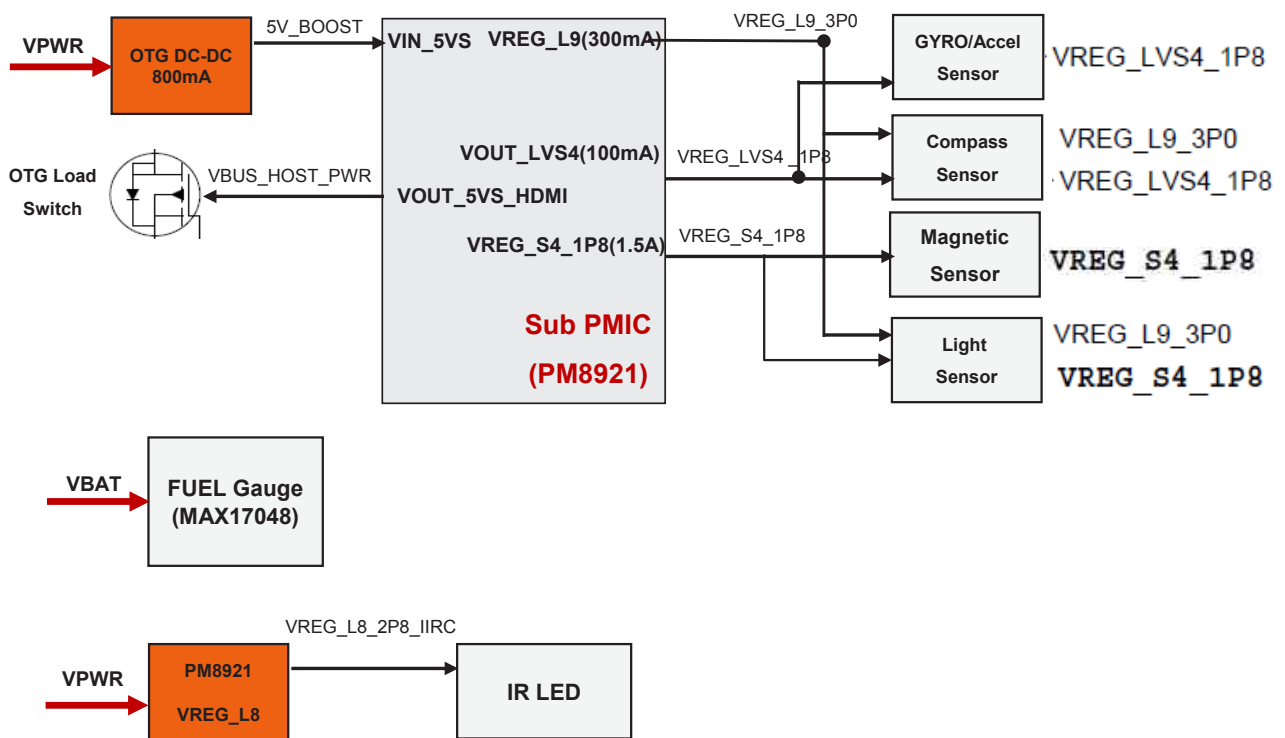
5. BLOCK DIAGRAM

[V500] Block diagram : PM8921 POWER INTERFACE1 _4_4

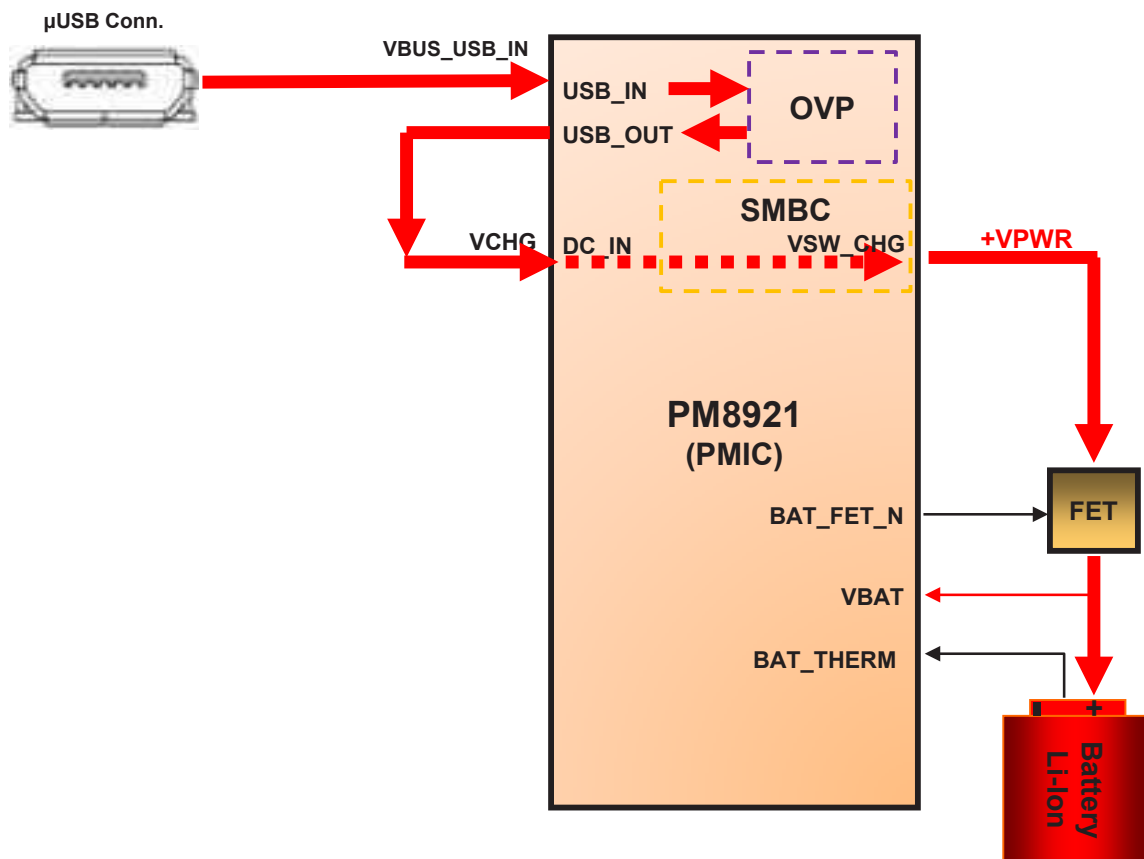


5. BLOCK DIAGRAM

[V500] Block diagram : PM8921 POWER INTERFACE2

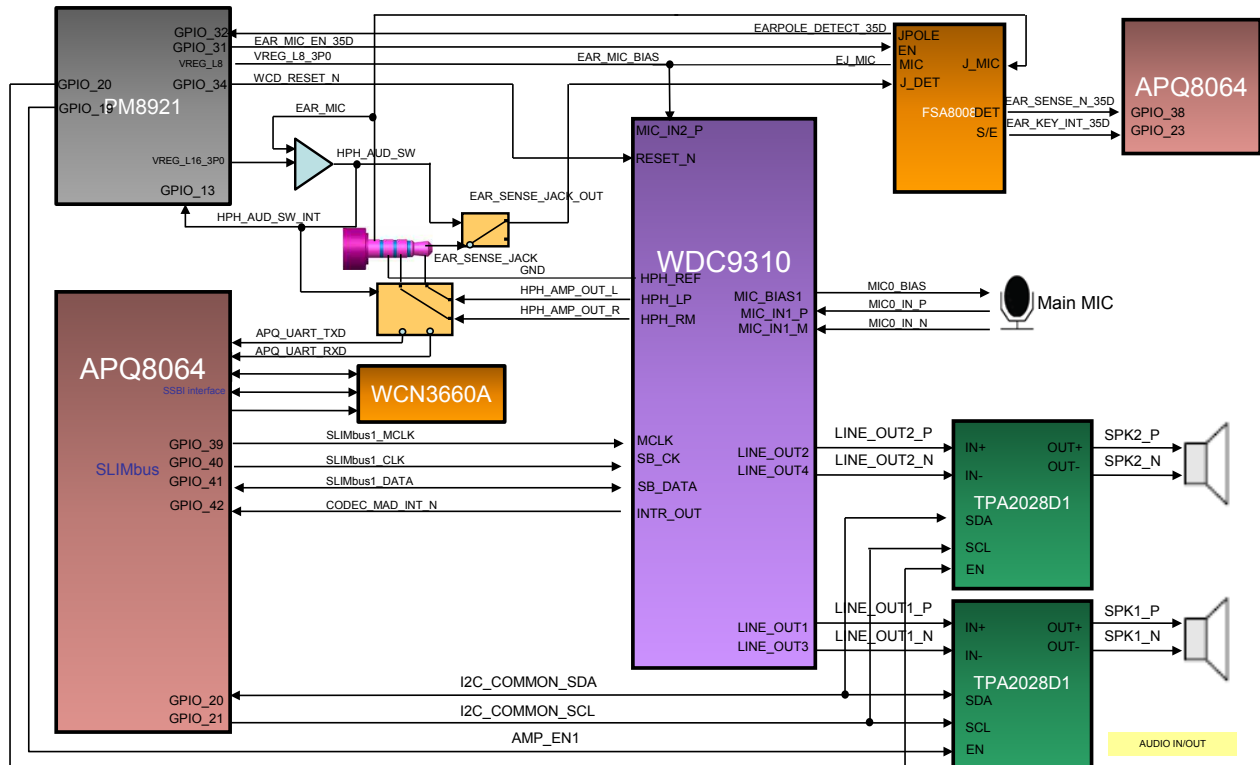


[V500] Block diagram : CHARGING



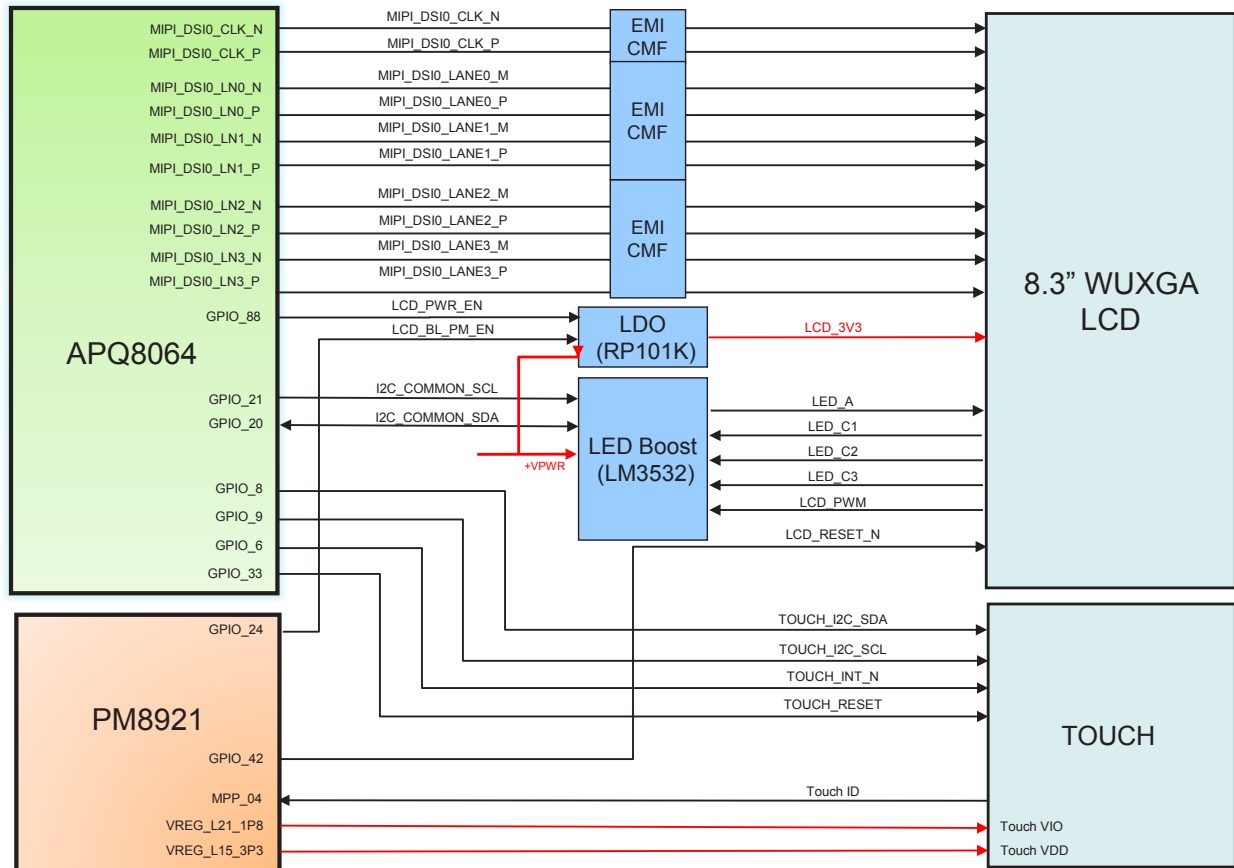
5. BLOCK DIAGRAM

[V500] Block diagram : AUDIO



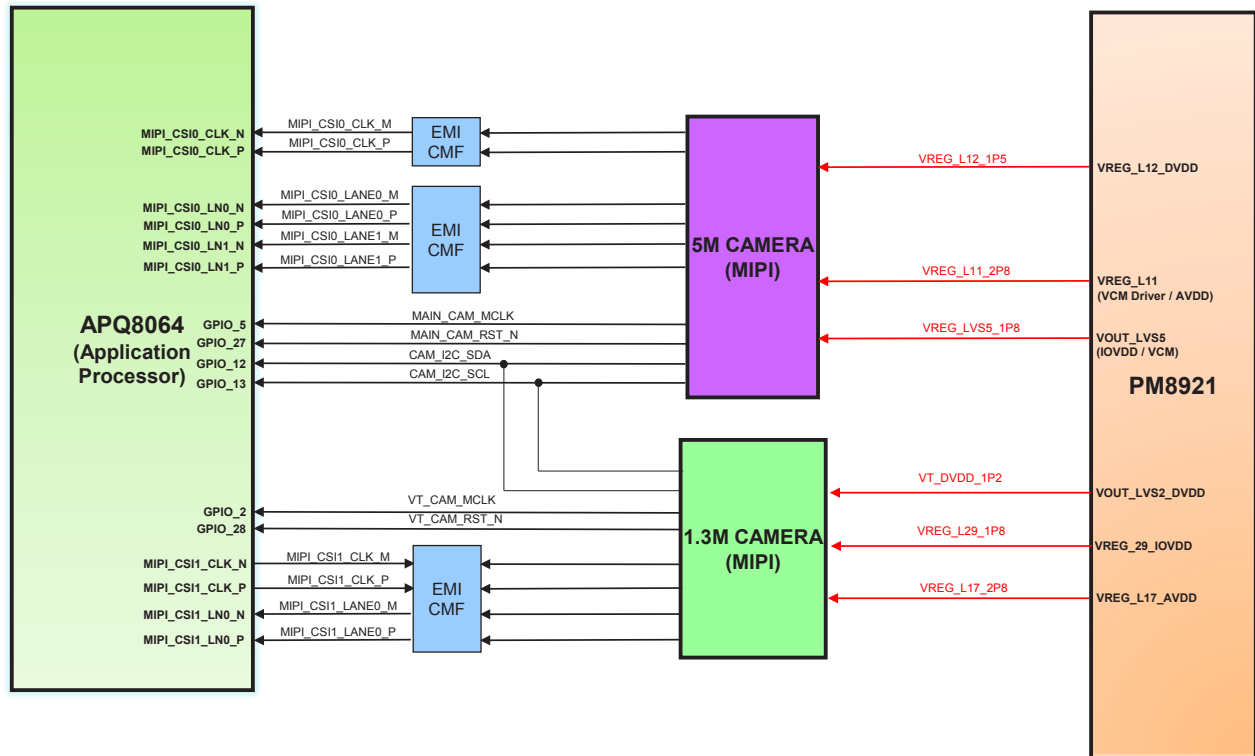
5. BLOCK DIAGRAM

[V500] Block diagram : LCD/TOUCH



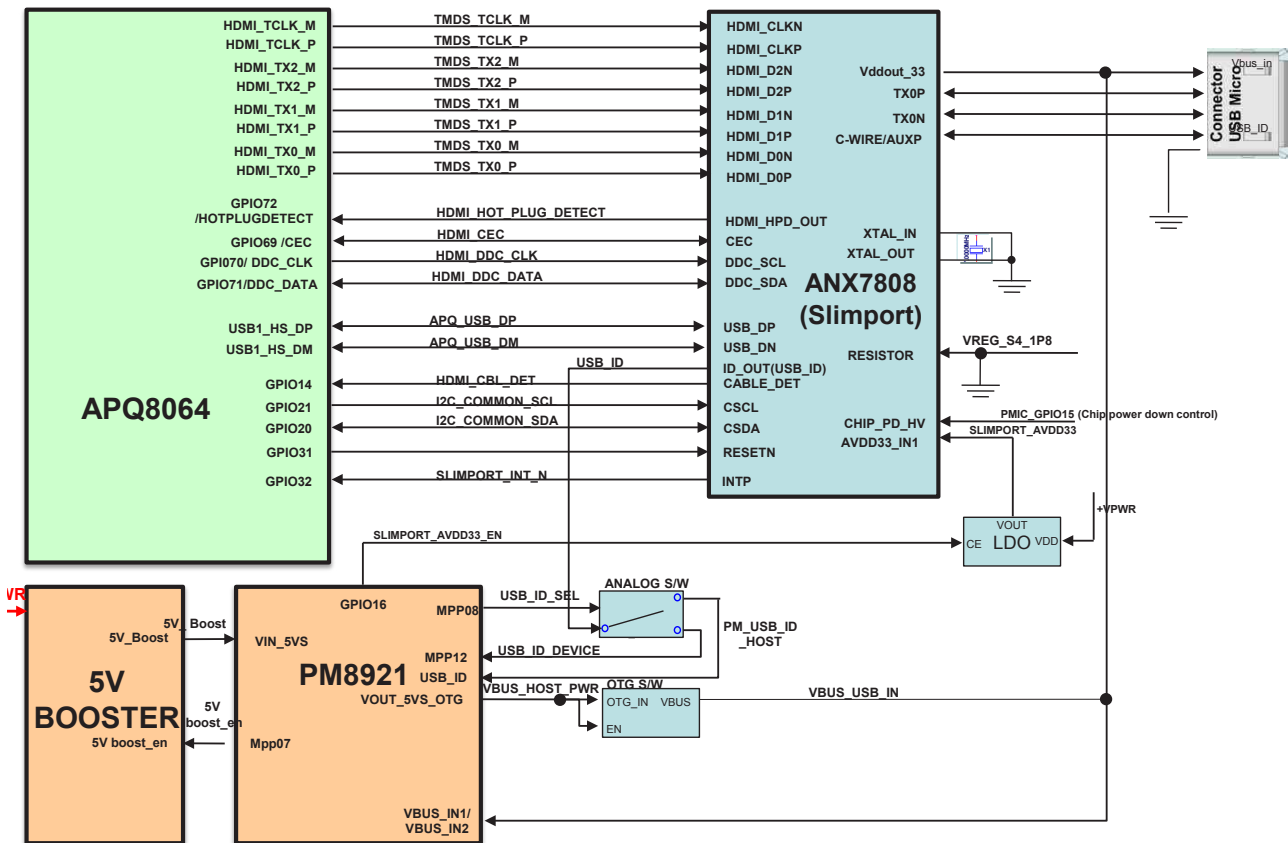
5. BLOCK DIAGRAM

[V500] Block diagram : Camera(5M, 1.3M)



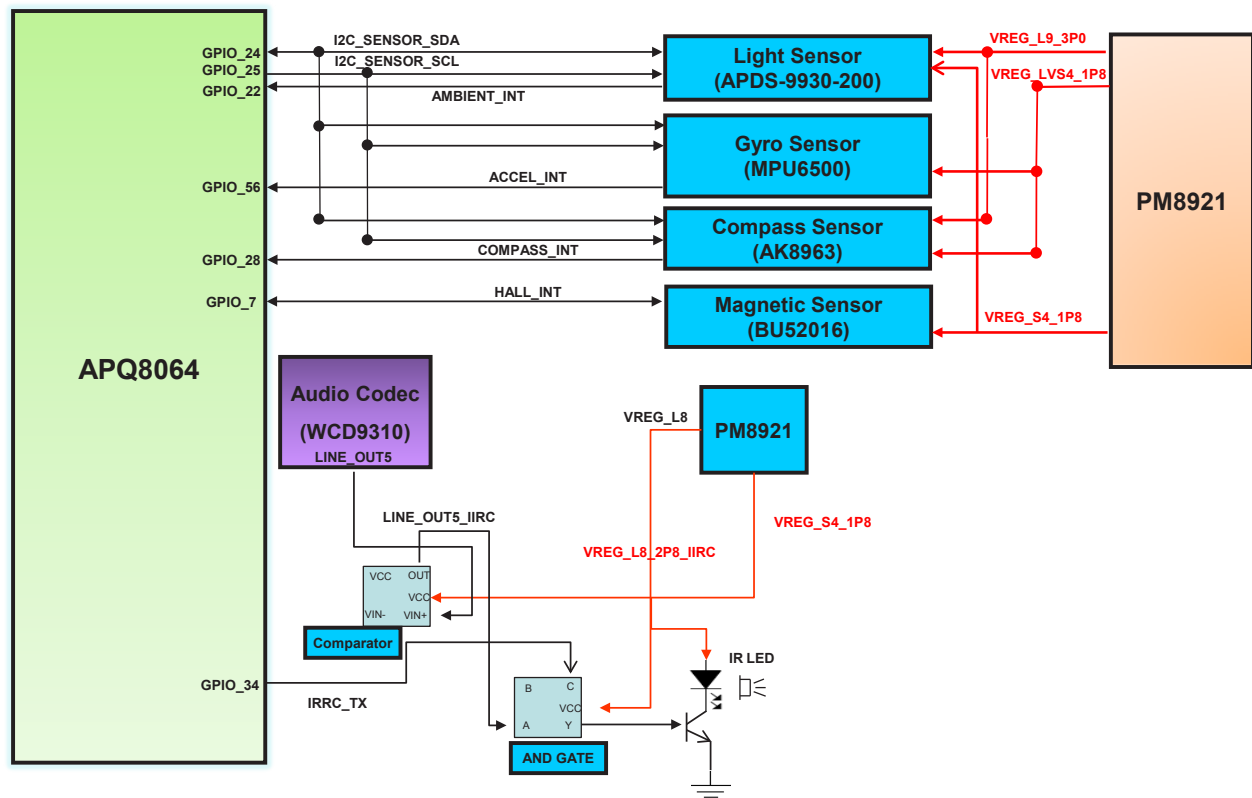
5. BLOCK DIAGRAM

[V500] Block diagram : SLIMPORT/USB OTG



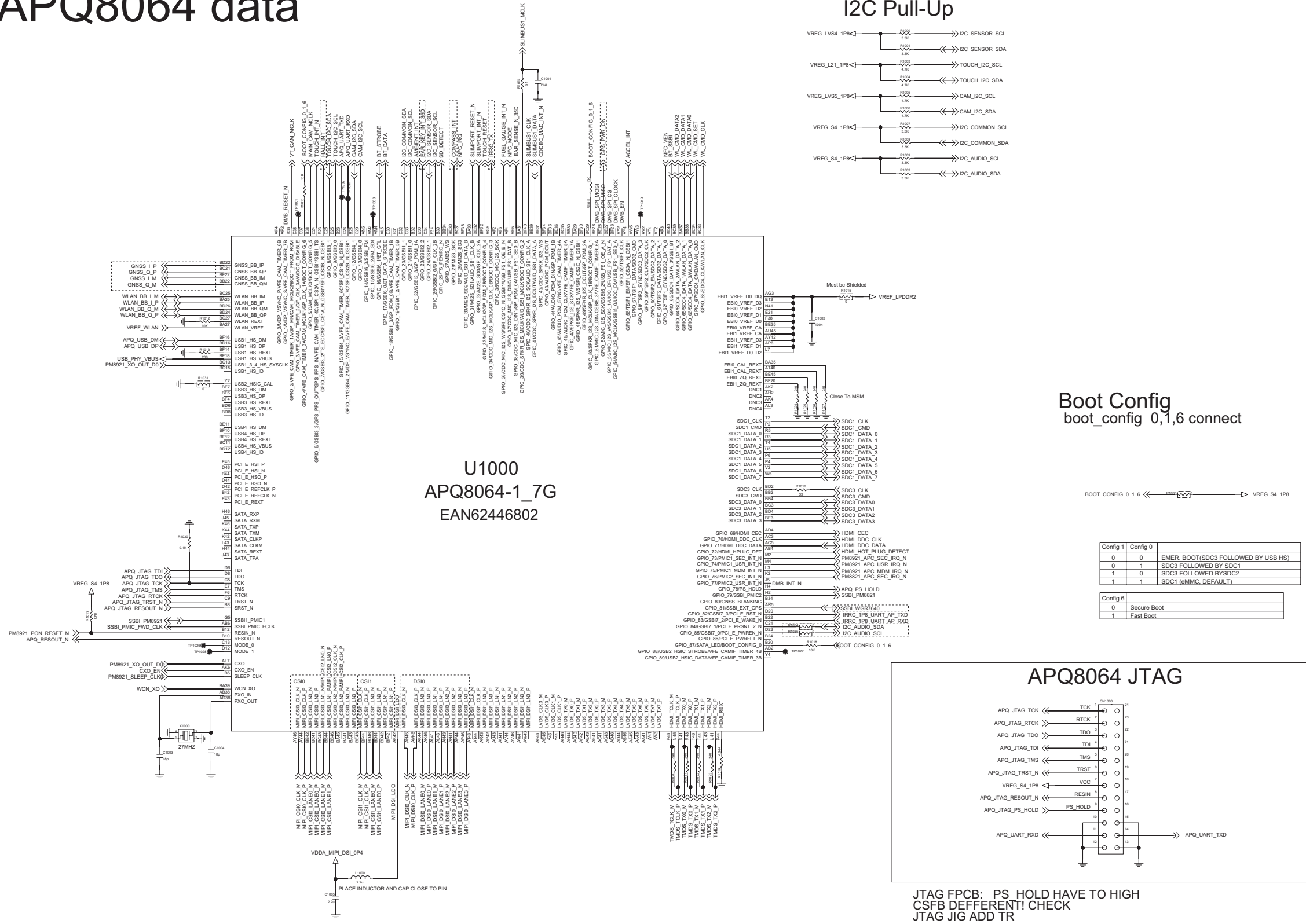
5. BLOCK DIAGRAM

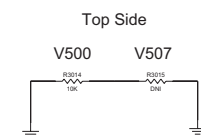
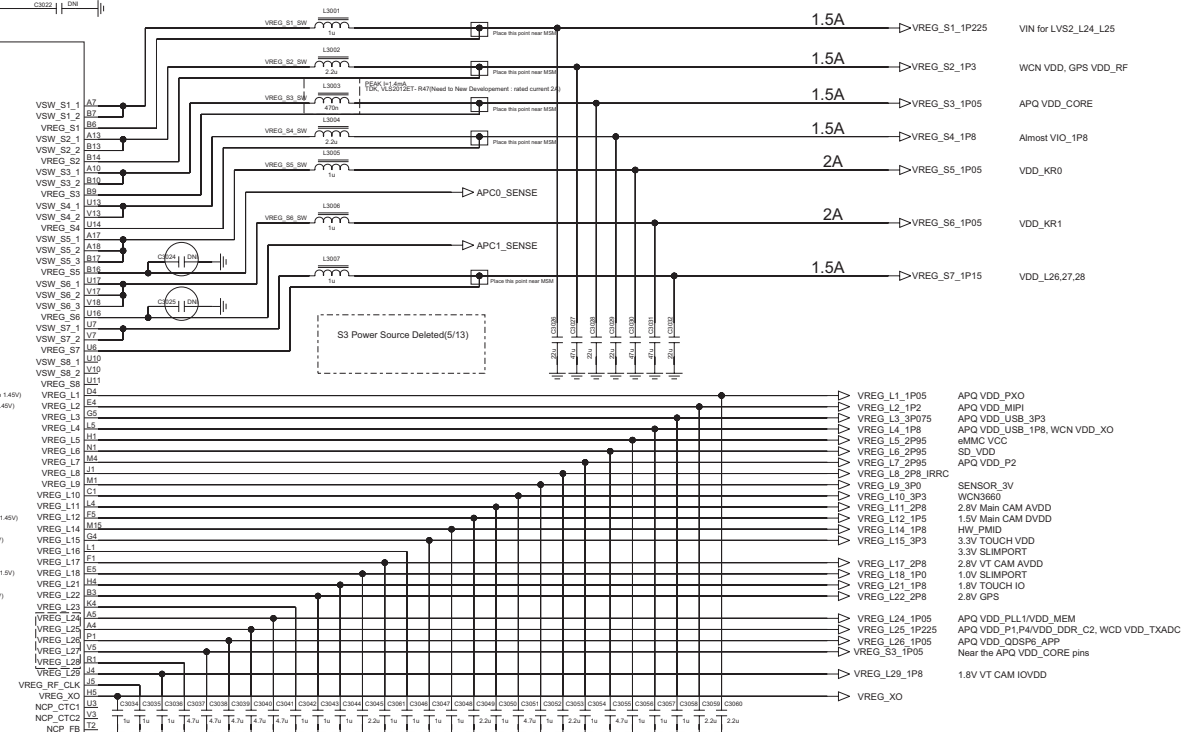
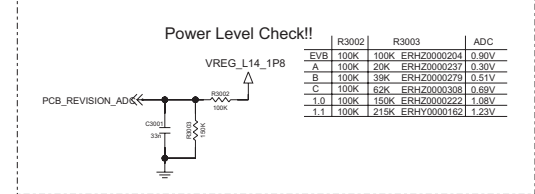
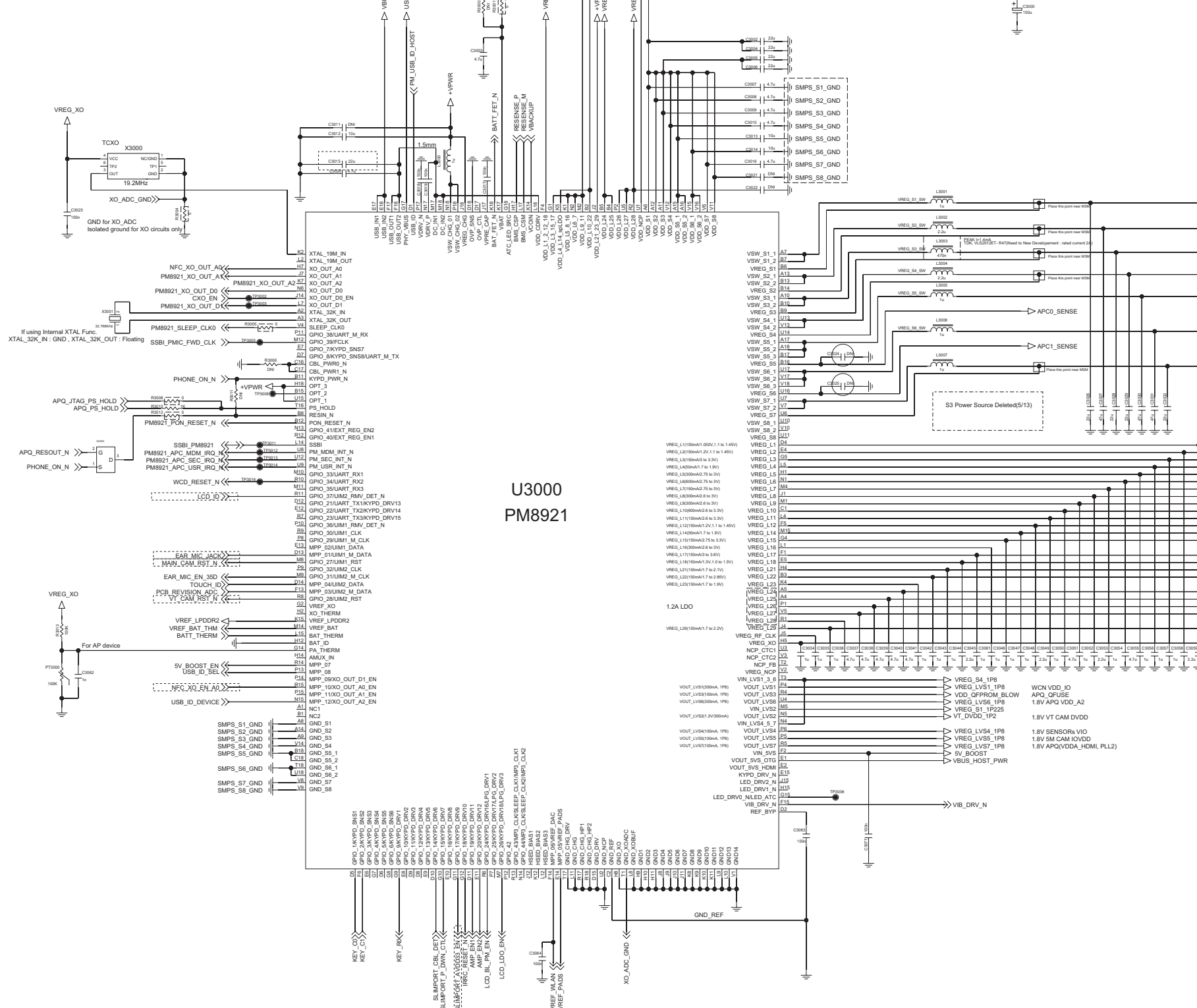
[V500] Block diagram :SENSOR's / IIRC



6. CIRCUIT DIAGRAM

APQ8064 data



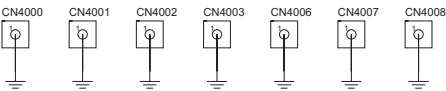
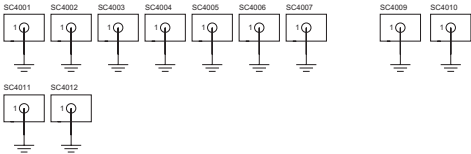


Batt Conn Guide boss

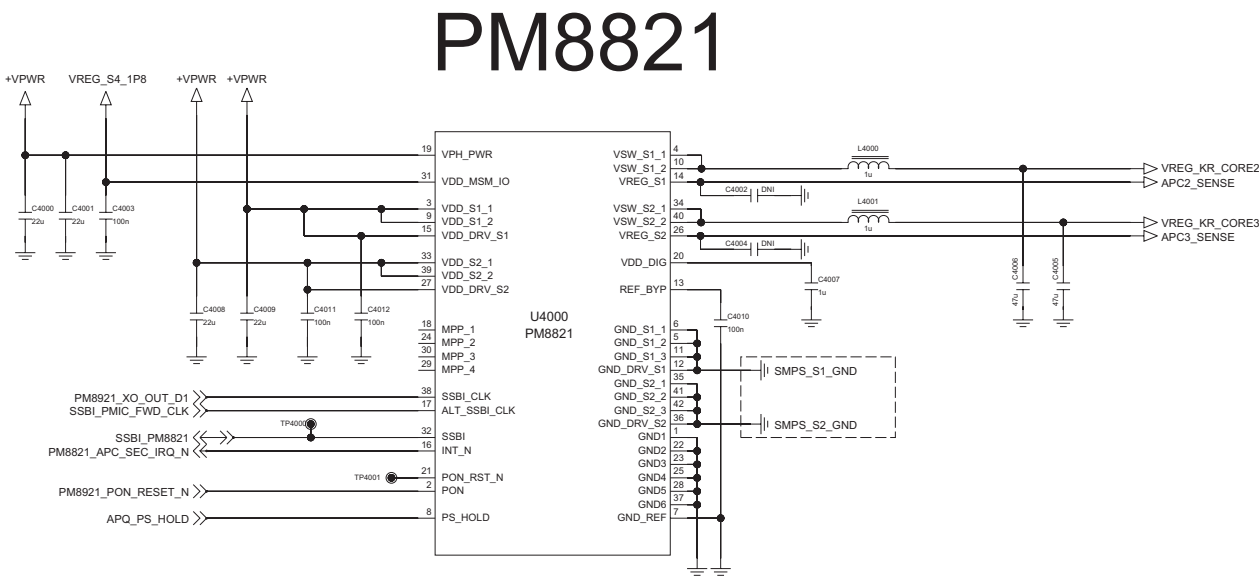


Contact Clip
TOP

Shield Can Clip

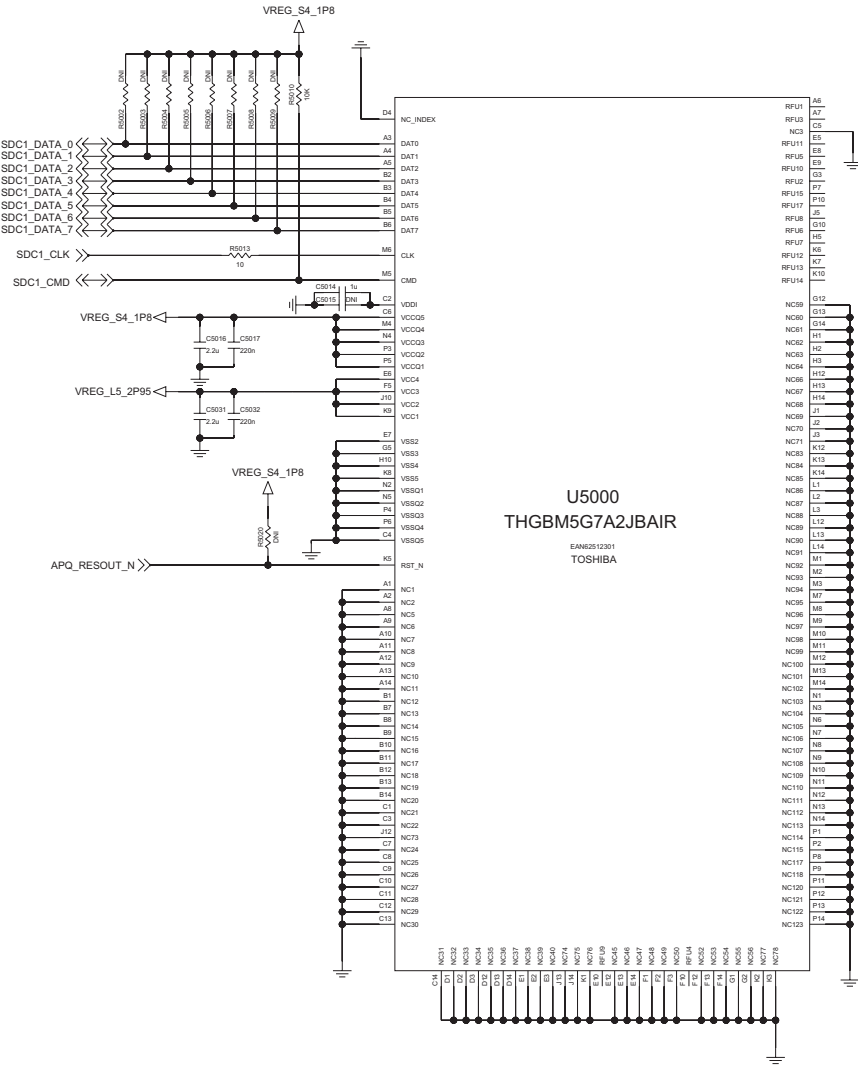


BOTTOM



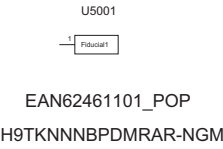
Memory

eMMC 16GB



LPDDR2

16Gb 533MHz(38 nm, 4Gb X 4), SK HYNIX POP

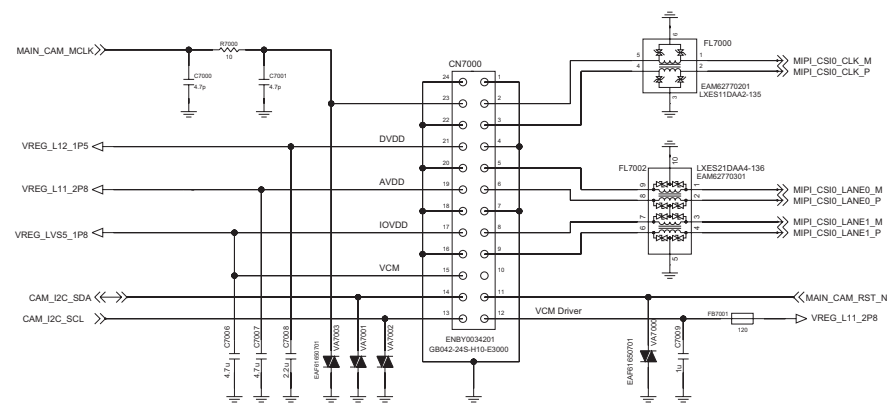




Camera & LCD Backlight

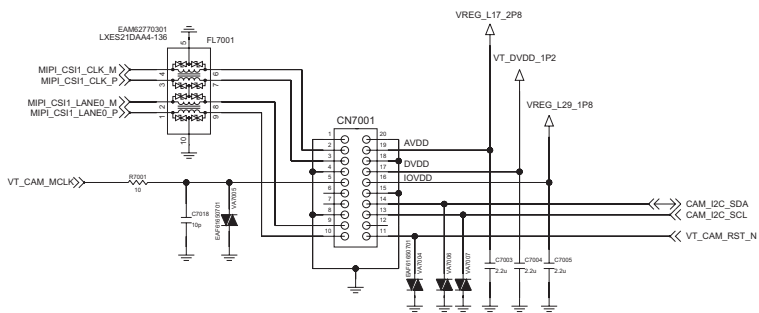
5M_CAM_MIPI

5M BSI, SS, 90 deg (EBP61842101)

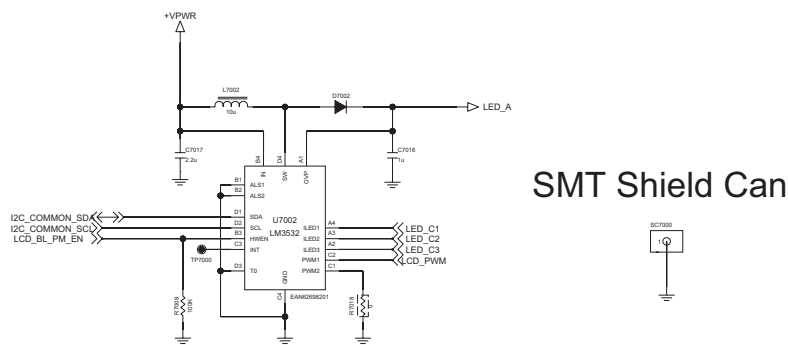


1.3M_VT_MIPI

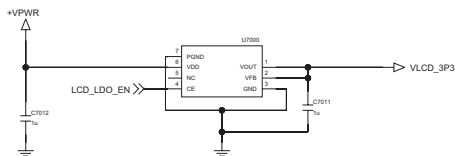
1.3M, Sony, 0 deg (EBP61881801)



LCD BACKLIGHT LED Boost



3.3V 500mA LCD POWER



VBAT1

VREG_S4_1P8

VBUS_USB_IN

VREF_BAT_THM

BATT_THERM

BATTERY CONNECTOR

RESENSE_P

RESENSE_M

Current Sensing

Main GND Line
Line Width must be over the 2mm.

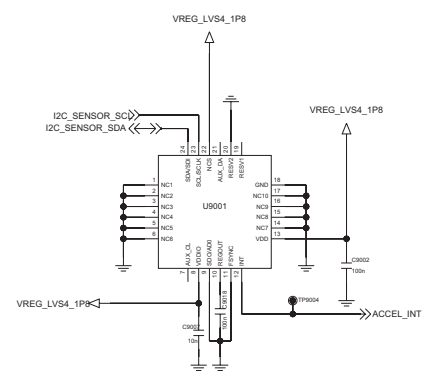
[illegible]

	S0,S1
LOW	A=B0
HIGH	A=B1

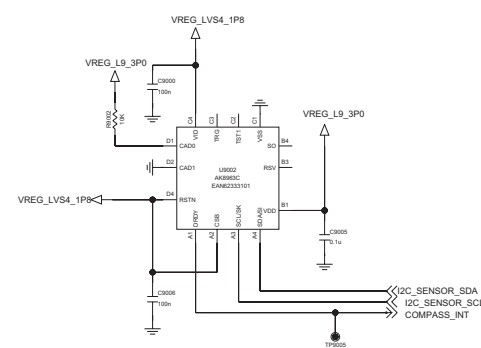
[illegible]

Sensor Block

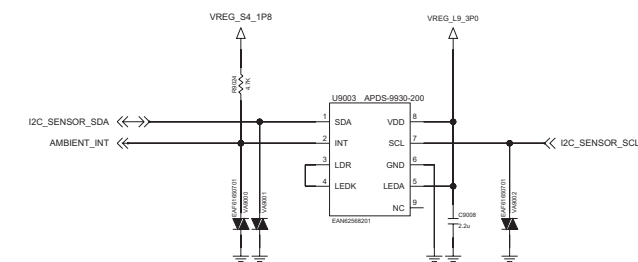
GYRO SENSOR



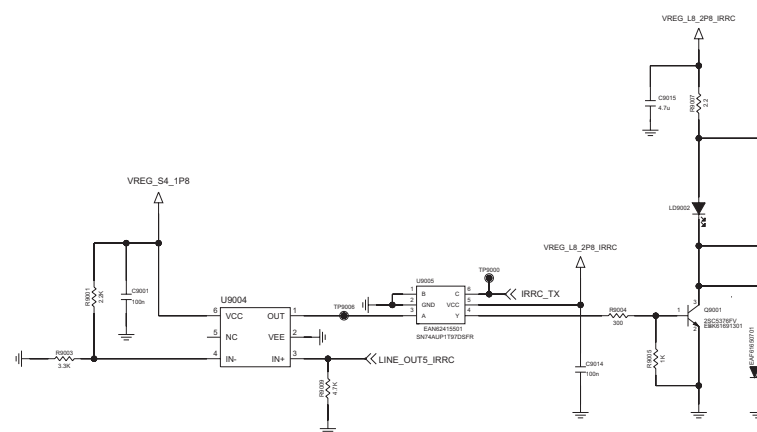
DIGITAL COMPASS



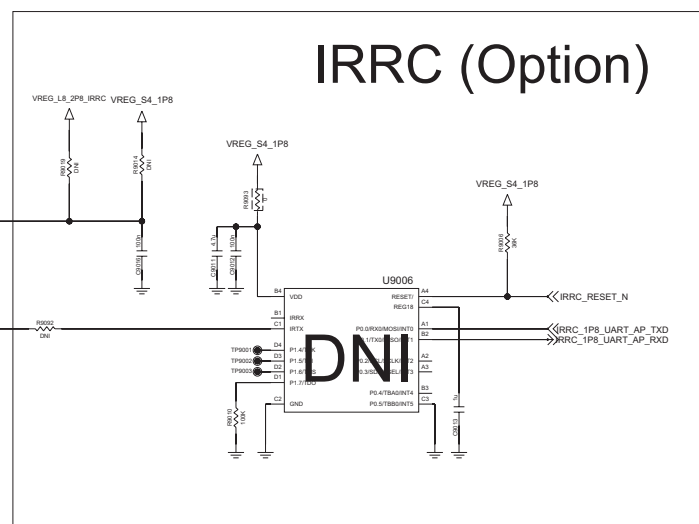
Light Sensor



IRRC controller



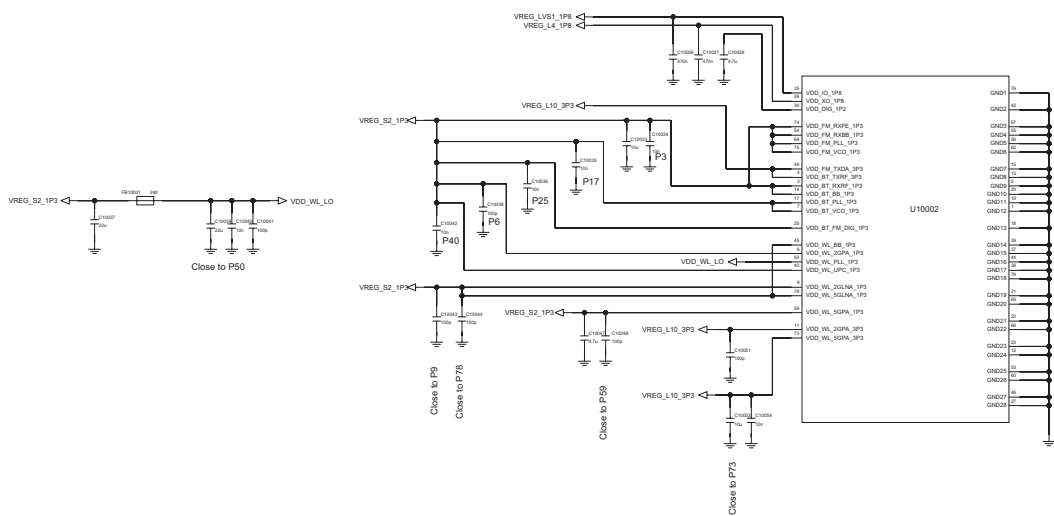
IRRC (Option)



IRRC(If use Maxim IC)
Connect : U9006,R9091,R9092
Disconnect : Q9001,R9014

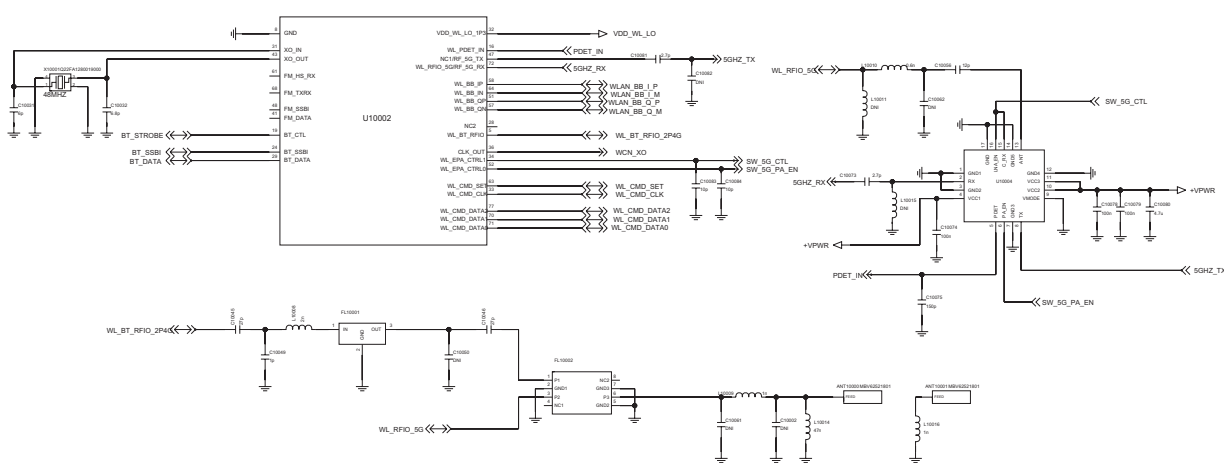
Connectivity

WCN 3660 PWR/GND

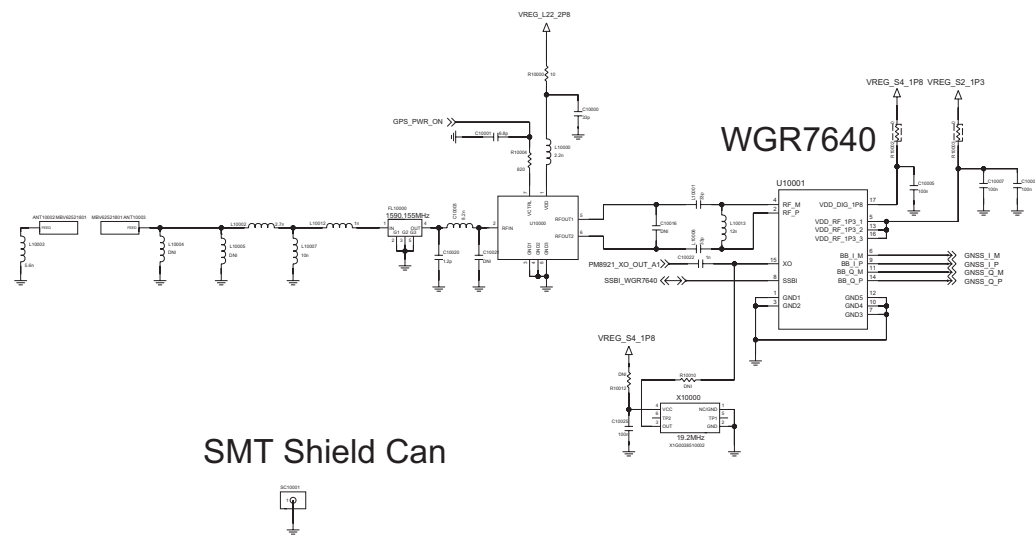


WCN 3660 RF/CONTROL

5G FEM(SPDT + PA + LNA)



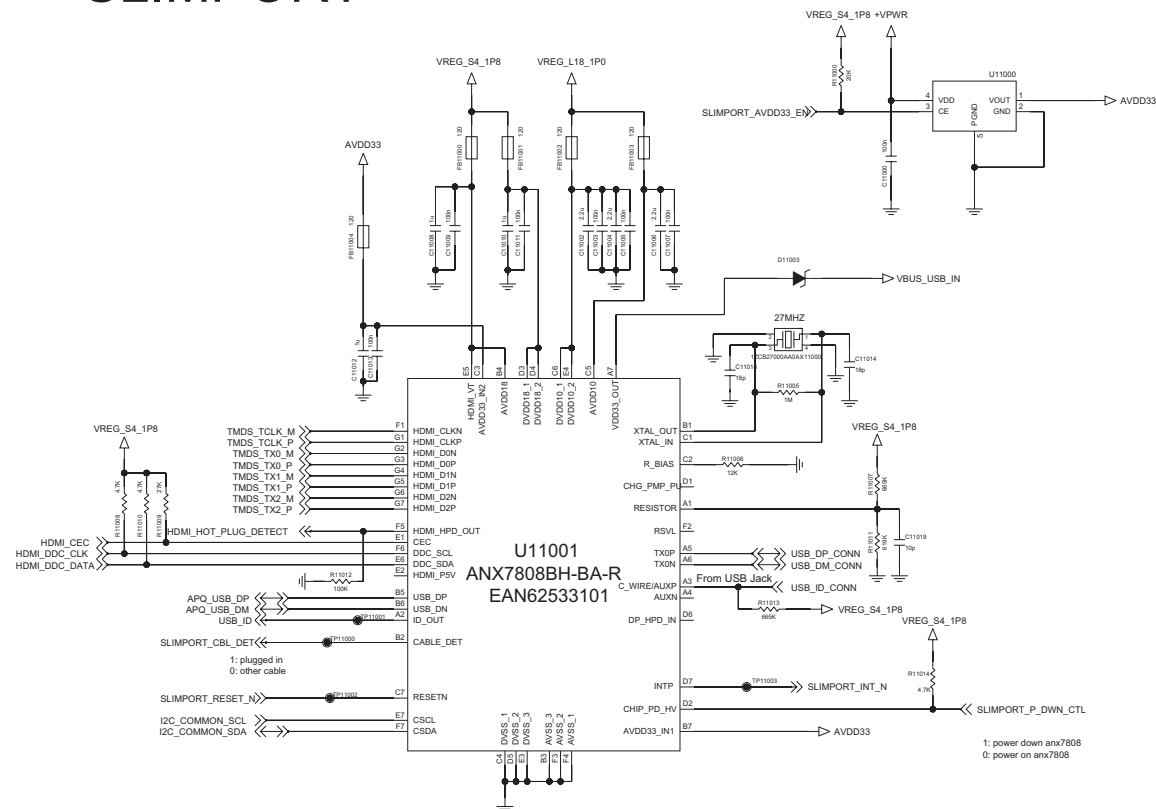
GPS(Qualcomm)



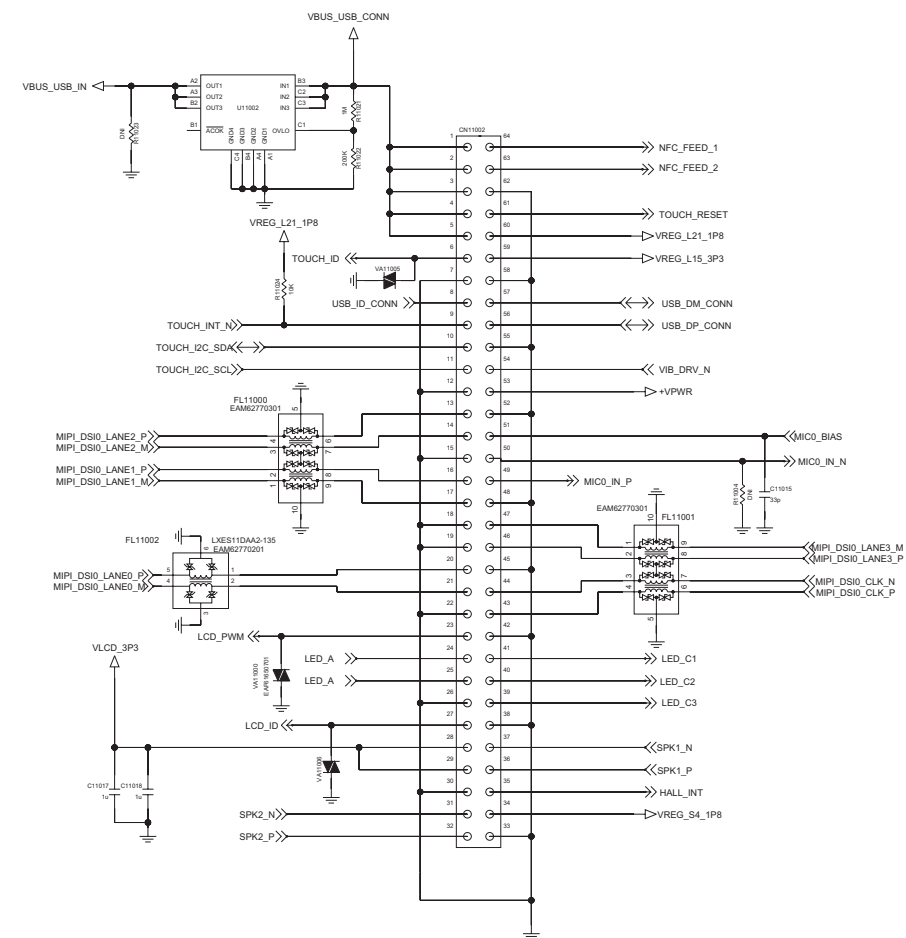
SMT Shield Can



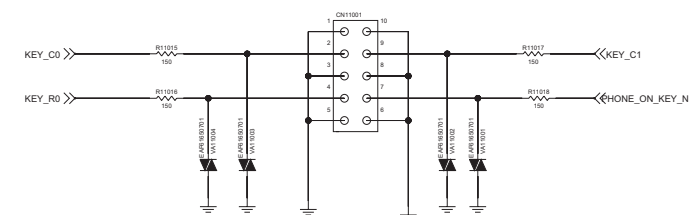
SLIMPORT



LOWER CONNECTOR

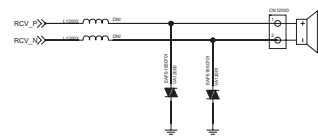


POWER & VOLUME KEY

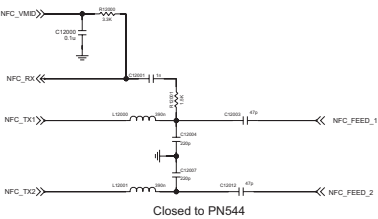
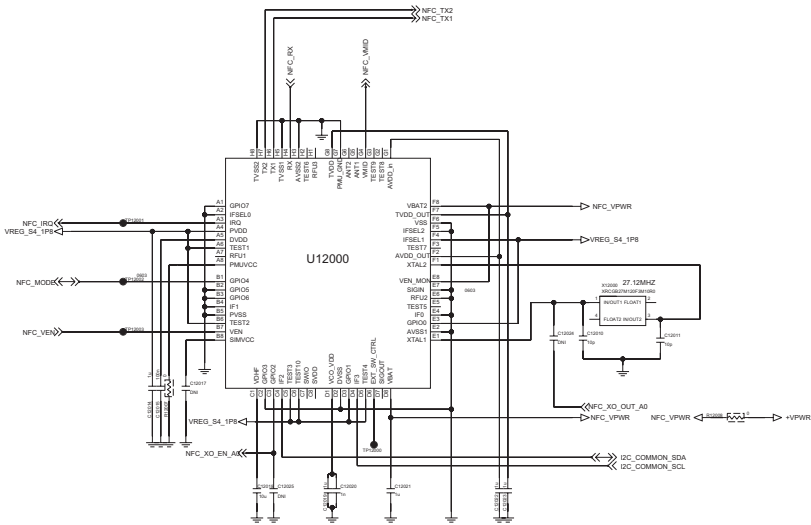


070Player

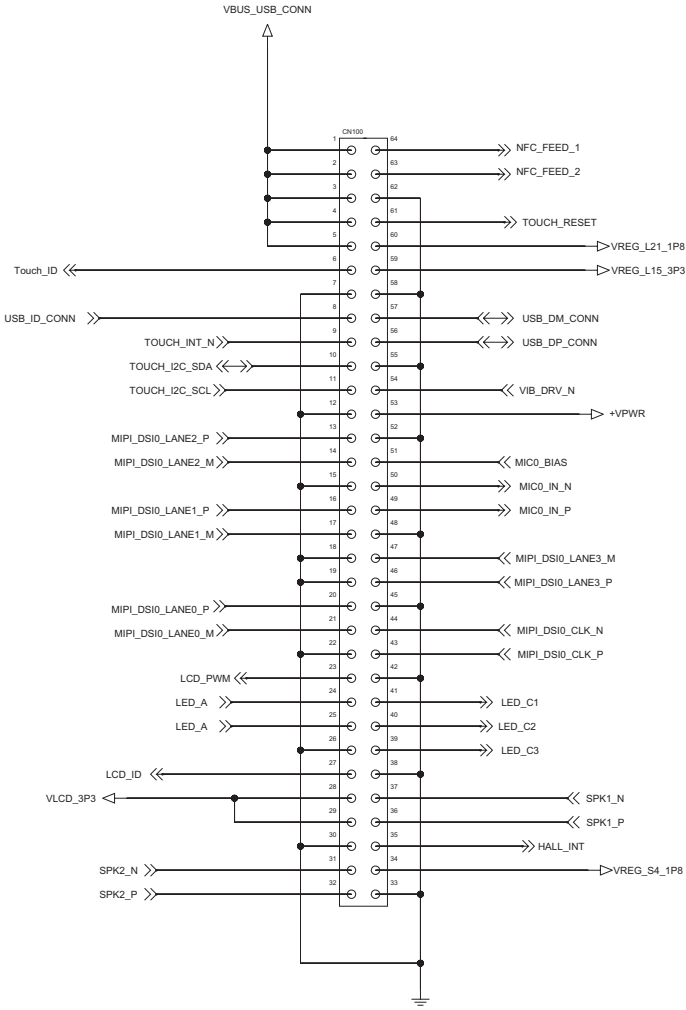
Receiver



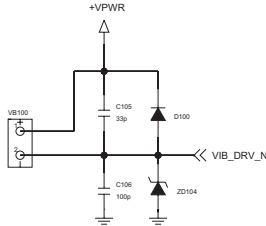
NFC



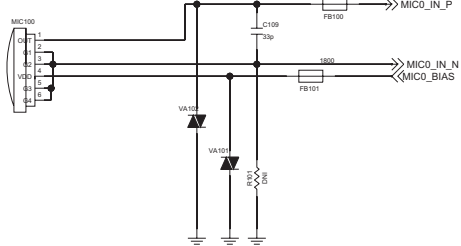
FPCB CONNECTOR BtoB



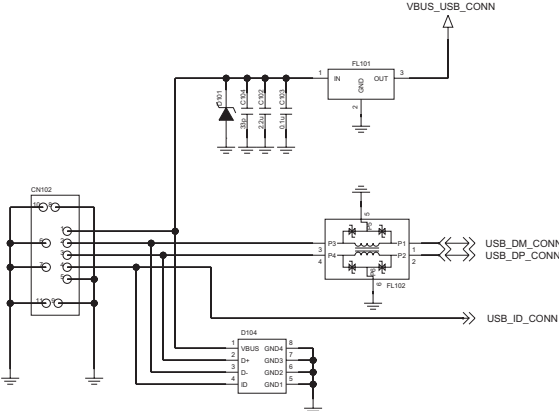
MOTOR



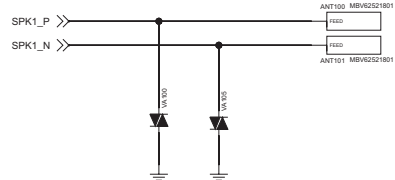
Main MIC



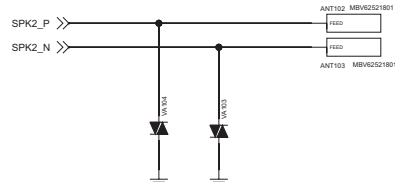
uUSB 5 pin connector



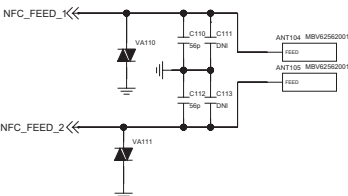
SPEAKER (Stereo)



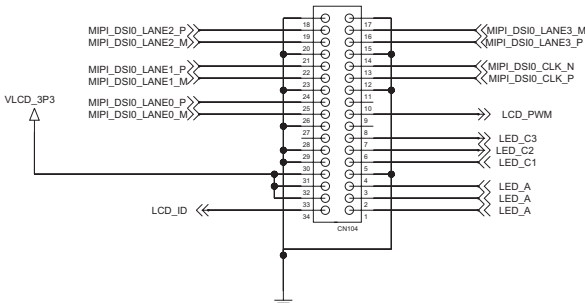
SPEAKER (Stereo)



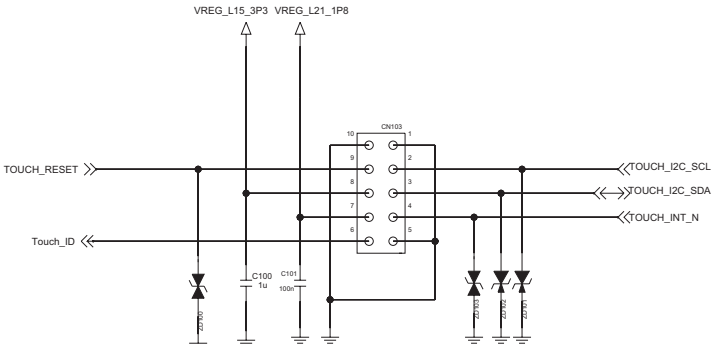
NFC



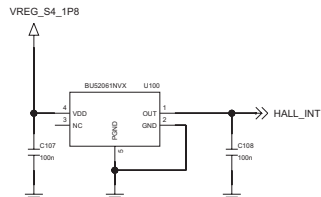
LCD CONNECTOR

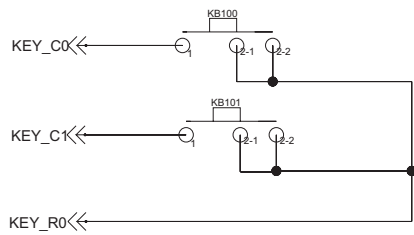
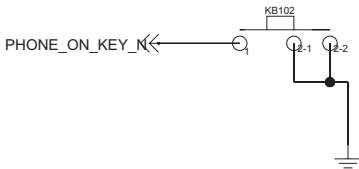
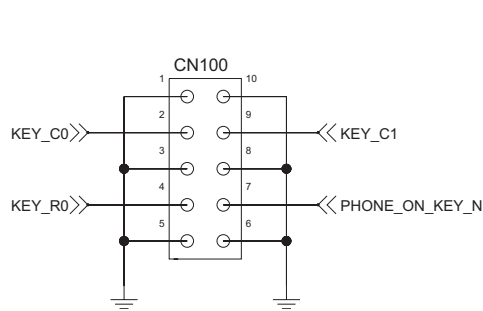


TOUCH CONNECTOR



Magnetic Sensor





	R0
C0	UP
C1	DOWN

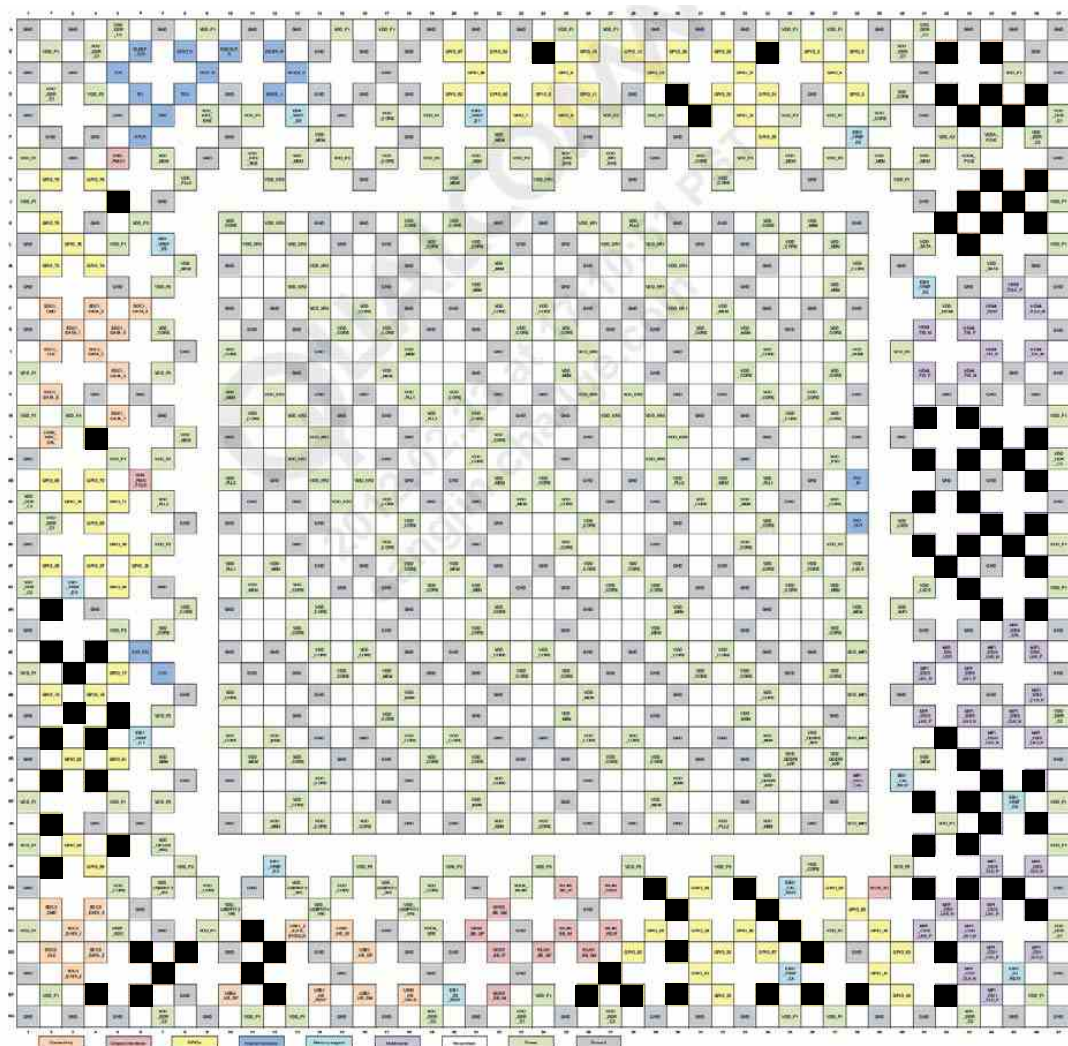
7. BGA PIN MAP

U1000 – APQ8064 top (Top View)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	
A																													A
B																													B
C																													C
D																													D
E																													E
F																													F
G																													G
H																													H
J																													J
K																													K
L																													L
M																													M
N																													N
P																													P
R																													R
T																													T
U																													U
V																													V
W																													W
Y																													Y
AA																													AA
AB																													AB
AC																													AC
AD																													AD
AE																													AE
AF																													AF
AG																													AG
AH																													AH
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	
	Connectivity	Chipset interface	GPUs	Internal functions	Memory support	Multimedia	No connect	Power	Ground																				

Not Used

U1000 – APQ8064 bottom (Top View)



Not Used

7. BGA PIN MAP

U3000 - PM8921 (Top View)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18		
A		XTAL_32K_IN	XTAL_32K_OUT	VREG_L25	VREG_L24	VDD_S1	VSW_S1	GND_S1	GND_S3	VSW_S3	VDD_S3	VDD_S2	VSW_S2	GND_S2	VDD_S5	VDD_S5	VSW_S5	VSW_S5	A	
B		VDD_L10_22	VREG_L22	VDD_L25	VDD_L24	VREG_S1	VSW_S1	RESIN_N	VREG_S3	VSW_S3	KYPD_PWR_N	PON_RESET_N	VSW_S2	VREG_S2	OPT_2	VREG_S5	VSW_S5	GND_S5	B	
C	VREG_L10	GND_REF															CBL_PWR0_N	CBL_PWR1_N	GND_S5	C
D	USB_ID	REF_BYP		VREG_L1	GPIO_1					GPIO_14	GPIO_19			MPP_01	MPP_04	GND_DRV			OVP_SNS	D
E	VOUT_SVS_OTG			VREG_L2	VREG_L18						GPIO_20				MPP_05			USB_IN	USB_IN	E
F	VREG_L17	VIN_SVS		VDD_L1_2_12_18	VREG_L12	GPIO_2								MPP_03	MPP_06	VIB_DRV_N		USB_OUT	USB_OUT	F
G	VDD_L3_15_17			VREG_L15	VREG_L3				GPIO_9	GPIO_15	GPIO_17	GPIO_18		PA_THERM	LED_DRV0_N			PHY_VBUS		G
H	VREG_L5			VREG_L21	VREG_XO		XO_OUT_A0	GND_XO	GND	GND	GND	BAT_ID						BMS_CSP	OPT_3	H
J	VREG_L8	VDD_L21_23_29		VREG_L29	VREG_RF_CLK		XO_OUT_A1	GND	GND	GND	GND			XO_OUT_D0_EN				VPRE_CAP	VREG_CHG	J
K	VDD_L5_8_16	XTAL_19M_IN		VREG_L23	VDD_L4			GND	GND	GND	GND			VCOIN	VREF_LPDDR2			VBAT	BAT_FET_N	K
L	VREG_L16			VREG_L11	VREG_L4		XO_OUT_D1	GND_XOBUF	GND	GND	GND_CHG			SSBI	BAT_THERM			BMS_CSM	VDD_CDRV	L
M	VREG_L9	VDD_L9_11		VREG_L7	VIN_LVS2		GPIO_26	GPIO_27	GPIO_31					VREF_BAT	VREG_L14			DC_IN	DC_IN	M
N	VREG_L6	VDD_L6_7		VIN_LVS4_5_7	VOUT_LVS2	XO_OUT_D0										MPP_12		VDRV_P	VSW_CHG	N
P	VREG_L26	VDD_L26		VOUT_LVS1	VOUT_LVS5	VOUT_LVS4							MPP_08					VDRV_N	VSW_CHG	P
R	VREG_L28	VDD_L28		VOUT_LVS3	VOUT_LVS7	GPIO_24		GPIO_28		GPIO_34	GPIO_37			MPP_07	MPP_10			GND_CHG_HP	GND_CHG_HP	R
T	GND_XOADC		VIN_LVS1_3_6													PS_HOLD		GND_CHG_DRV	GND_S6	T
U	VDD_NCP	GND_NCP		VOUT_LVS6	VDD_L27	VREG_S7	VSW_S7	PM_MDM_INT_N	PM_USR_INT_N			PM_SEC_INT_N	VSW_S4	VREG_S4	OPT_1	VREG_S6	VSW_S6	GND_S6	U	
V	GND		SLEEP_CLK0	VREG_L27	VDD_S7	VSW_S7	GND_S7	GND_S8			VDD_S8	VDD_S4	VSW_S4	GND_S4	VDD_S6	VDD_S6	VSW_S6	VSW_S6	V	
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18		
	Input Power Management	Output Power Management	General Housekeeping	User Interface	IC Interface	GPIO or MPP	NDR	Power	Ground											

Not Used

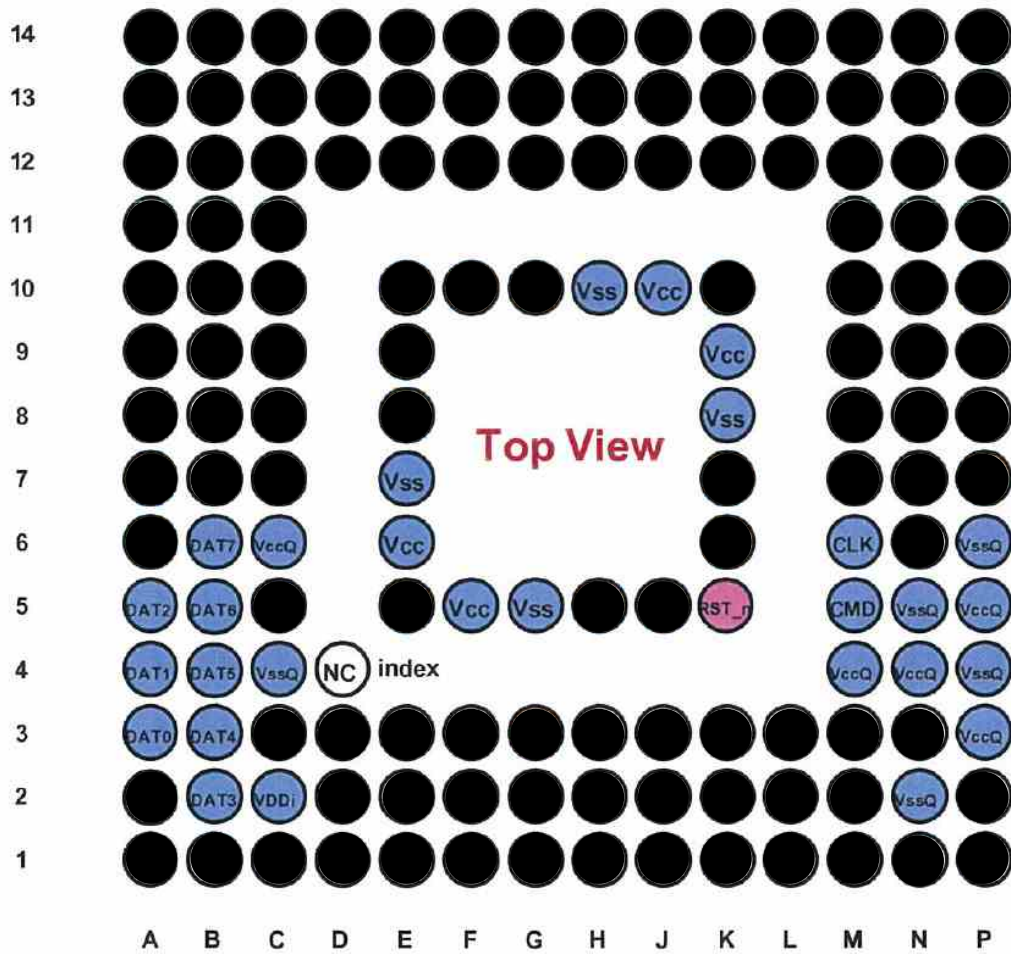
U4000 - PM8821 (Top View)

6 GND_S1	12 GND _DRV_S1				36 GND _DRV_S2	42 GND_S2
5 GND_S1	11 GND_S1	17 ALT _SSBI_CLK	23 GND		35 GND_S2	41 GND_S2
4 VSW_S1	10 VSW_S1	16 INT_N	22 GND	28 GND	34 VSW_S2	40 VSW_S2
3 VDD_S1	9 VDD_S1	15 VDD _DRV_S1	21 PON _RST_N	27 VDD _DRV_S2	33 VDD_S2	39 VDD_S2
2 PON	8 PS_HOLD	14 VREG_S1	20 VDD_DIG	26 VREG_S2	32 SSBI	38 SSBI_CLK
1 GND	7 GND_REF	13 REF_BYP	19 VPH_PWR	25 GND	31 VDD_MSM _IO	37 GND

Output power management	General housekeeping	IC interface	MPPs	Power	Ground
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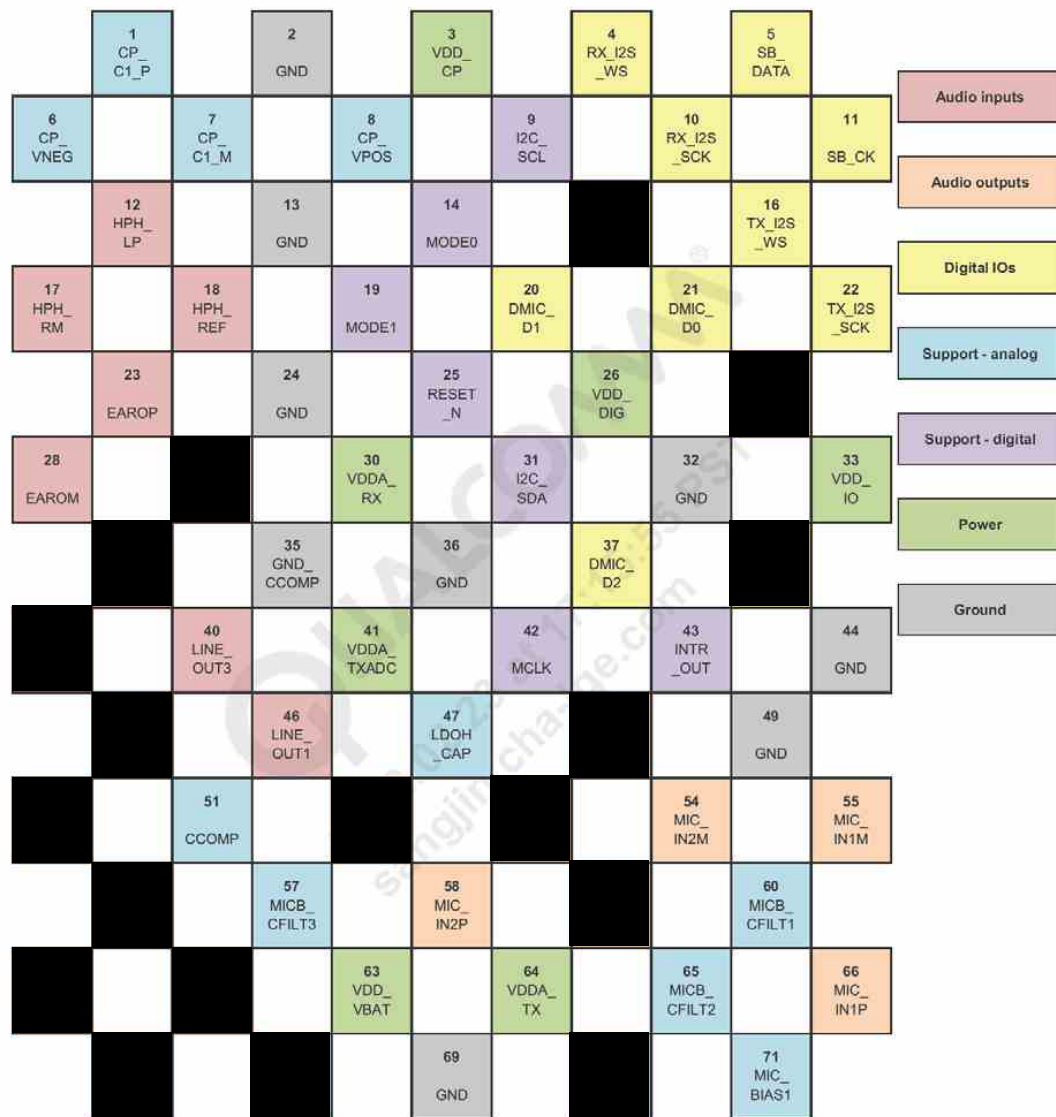
Not Used

U5001 - Memory MCP (Top View)



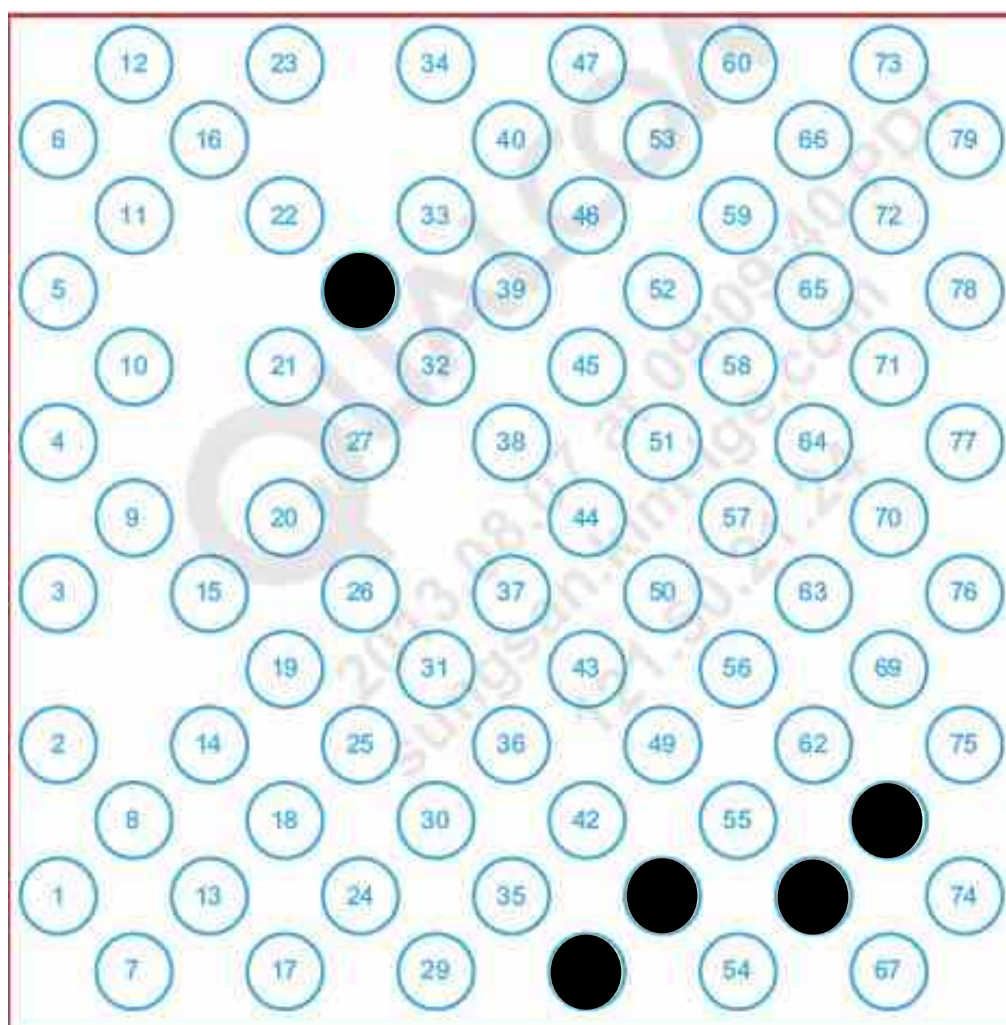
● Not Used

U6002-WCD9310 (Top View)



Not Used

U10002 – BT / WIFI WCN3660A (Top View)



● Not Used

U10001 – GPS WGR7640 (Top View)

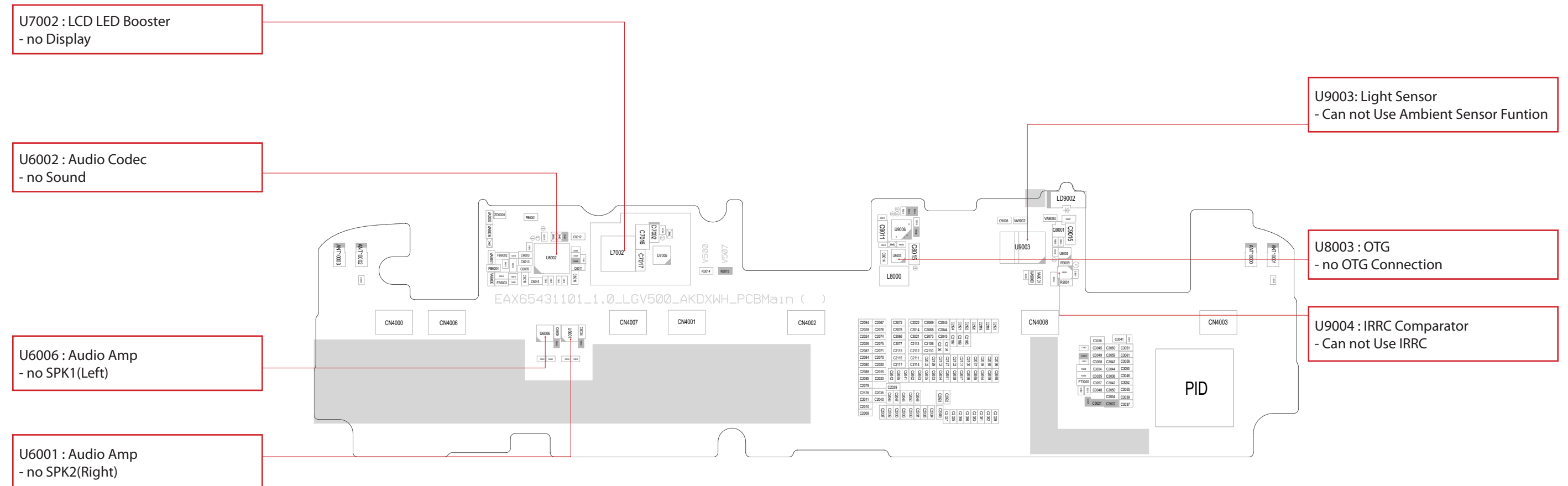
1 GND		2 RF_P		3 GND
	4 RF_M		5 VDD_ RF_1P3	
6 BB_I_M		7 GND		8 SSBI
	9 BB_I_P		10 GND	
11 BB_Q_M		12 GND		13 VDD_ RF_1P3
	14 BB_Q_P			
15 TCXO		16 VDD_ RF_1P3		17 VDD_ DIG_1P8

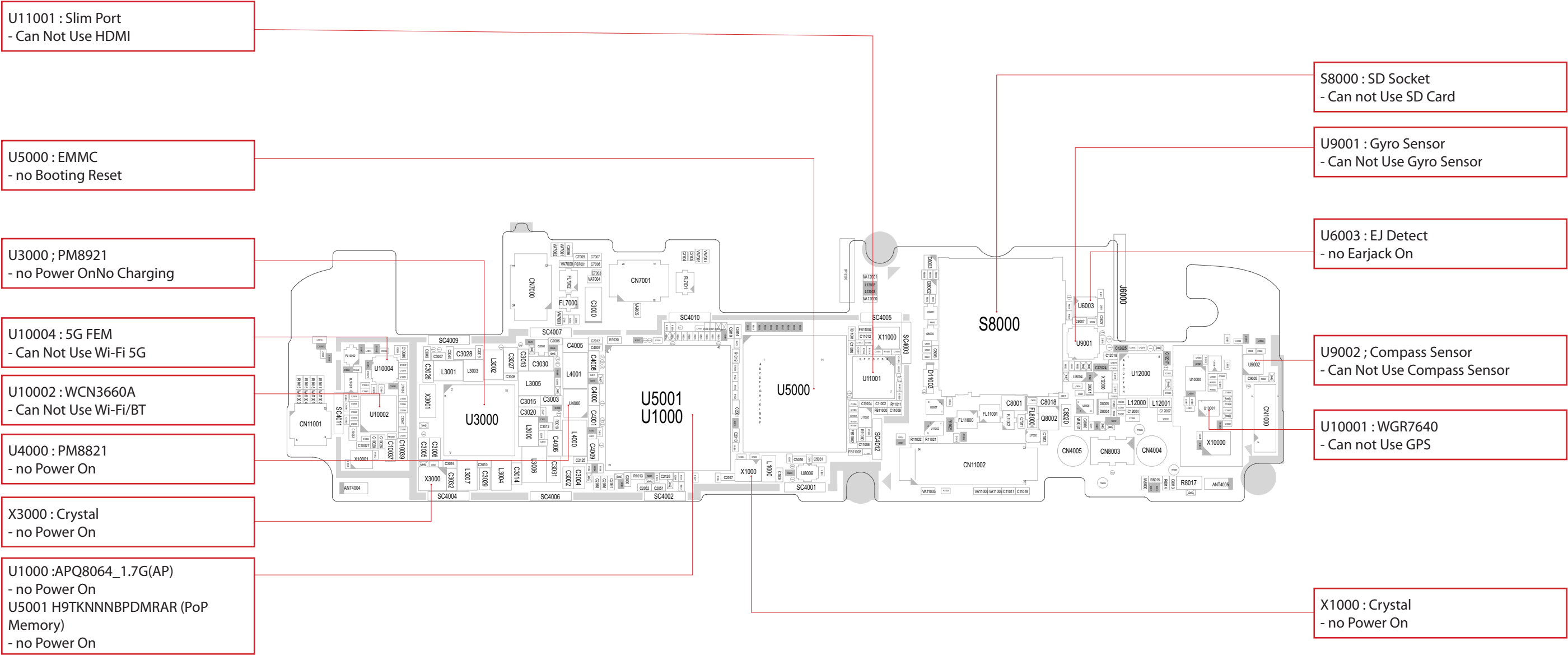
ALL USED

U11001 – Slim Port (Top View)

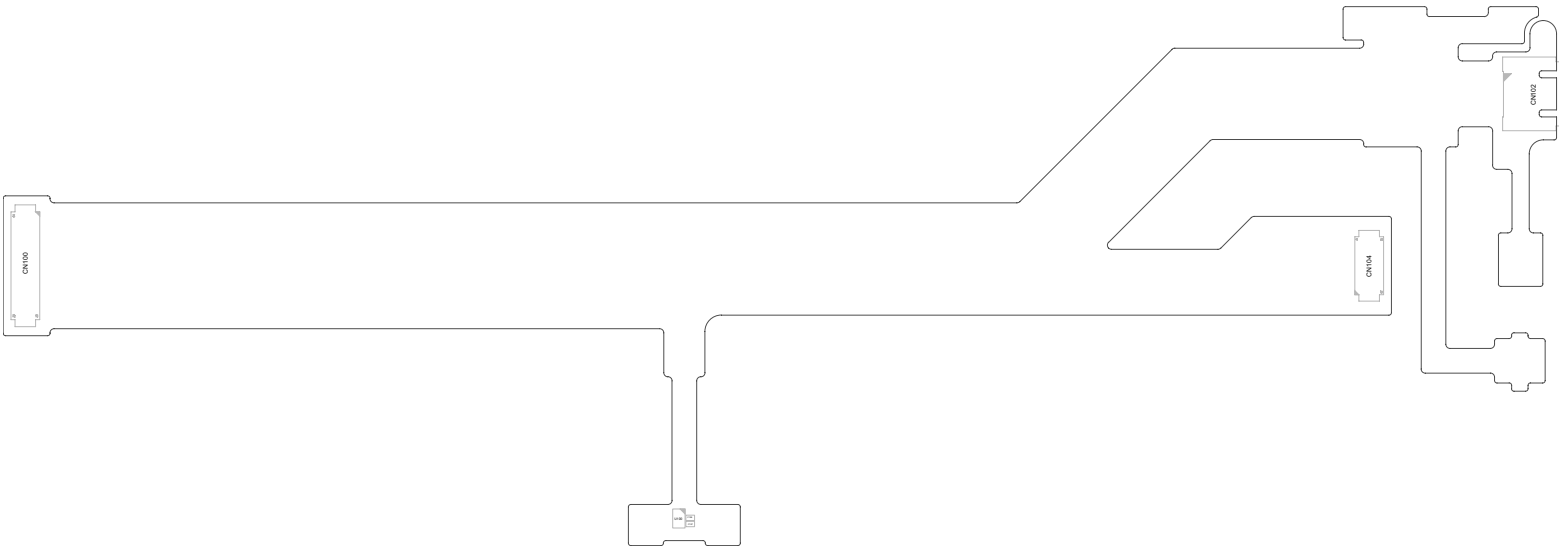
	1	2	3	4	5	6	7	
A	RESISTOR	ID_OUT	C-WIRE/ AUXP		TX0P	TX0N	VDD33_OUT	A
B	XTAL_OUT	CABLE_DET	AVSS	AVDD18	USB_DP	USB_DN	AVDD33_IN	B
C	XTAL_IN	R_BIAS	AVDD33_IN	DVSS	AVDD10	DVDD10	RESETN	C
D		CHIP_PD_HV	DVDD18	DVDD18	DVSS		INTP	D
E	CEC		DVSS	DVDD10	HDMI_VT	DDC_SDA	CSCL	E
F	HDMI_CLKN		AVSS	AVSS	HDMI_HPD_ OUT	DDC_SCL	CSDA	F
G	HDMI_CLKP	HDMI_D0N	HDMI_D0P	HDMI_D1N	HDMI_D1P	HDMI_D2N	HDMI_D2P	G
	1	2	3	4	5	6	7	

 Not Used

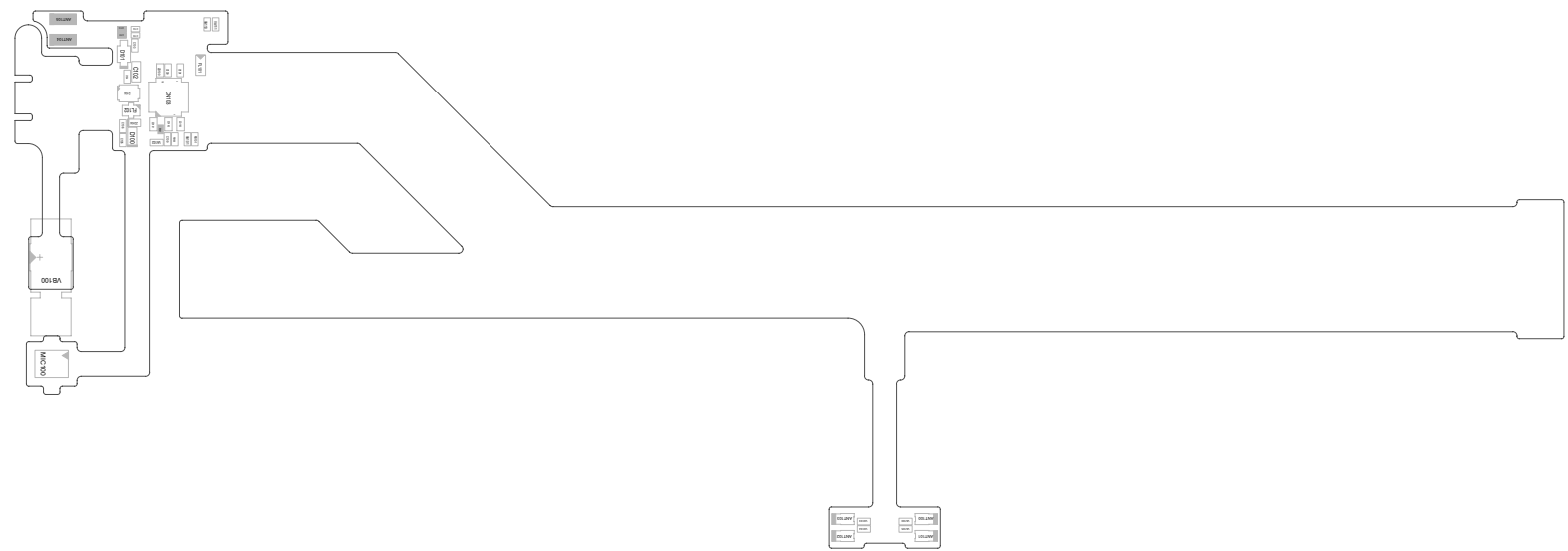




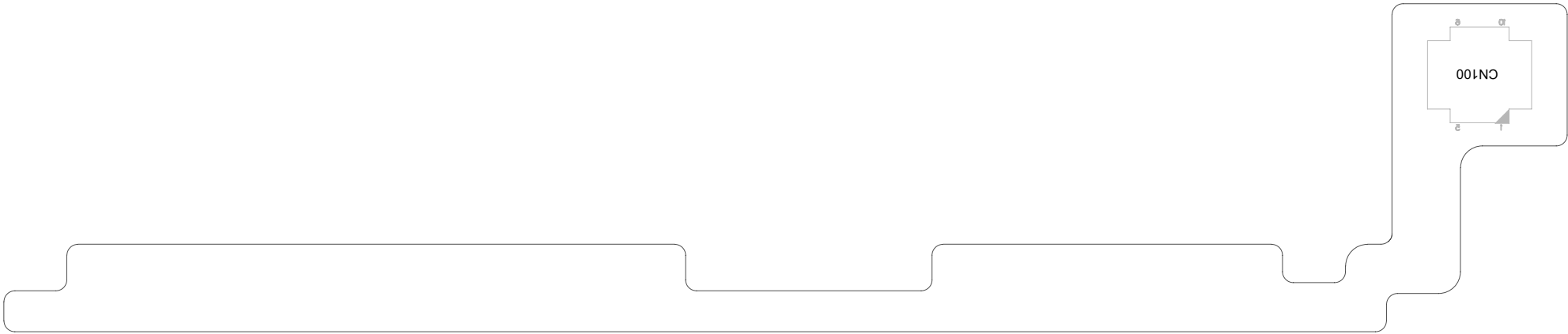
LG-V500_MAIN_EAX65431101_1.0_BOT



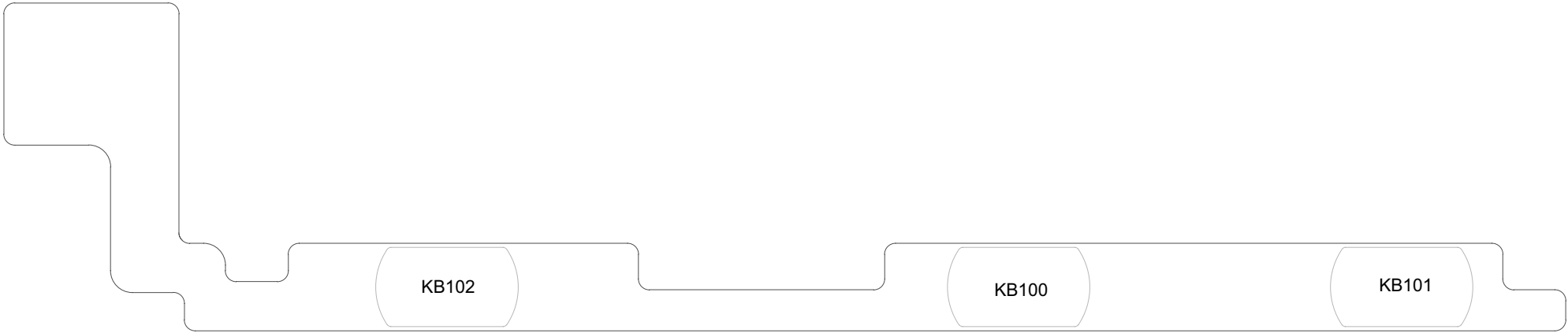
LG-V500_F_LOWE_EAX65430801_1.0_TOP



LG-V500_F_LOWE_EAX65430801_1.0_BOT

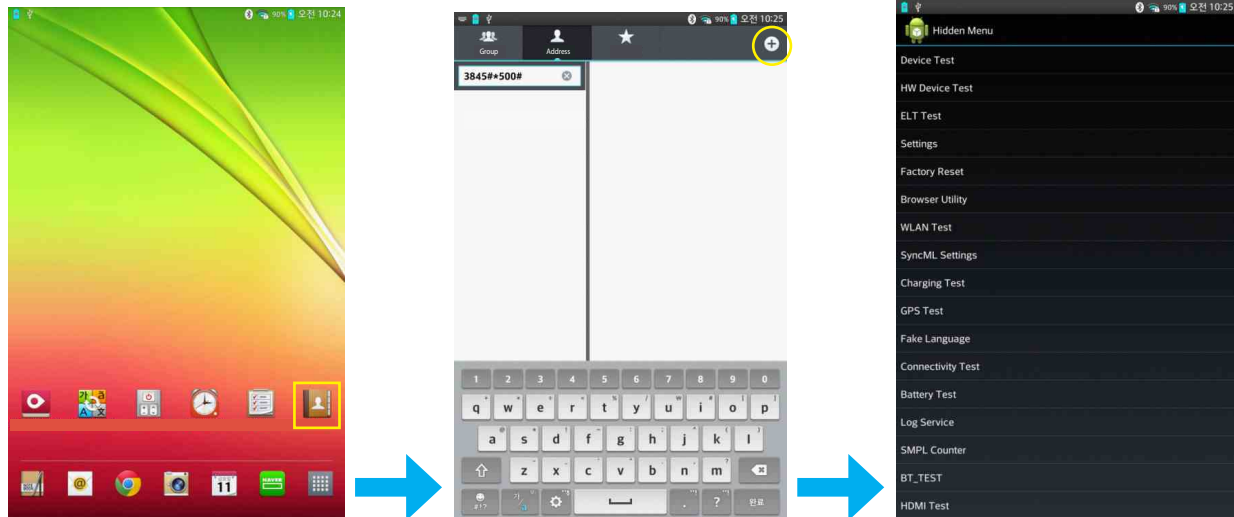


LG-V500_F_S_KEY_EAX65397001_1.0_TOP



LG-V500_F_S_KEY_EAX65397001_1.0_BOT

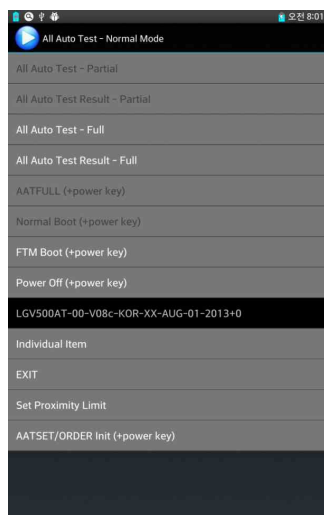
9. HIDDEN MENU



Press address book icon

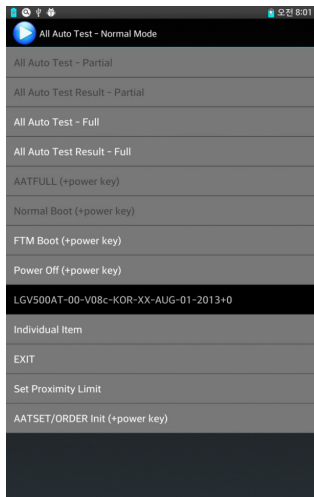
Input number 3845#*500#
Press + icon

Hidden menu Start

**[List]**

All Auto Test – Full
All Auto Test Result – Full
FTM Boot(+power key)
Power Off(+power key)
Individual Item
EXIT
Set Proximity Limit
AATSET/ORDER Init(+power key)

9. HIDDEN MENU

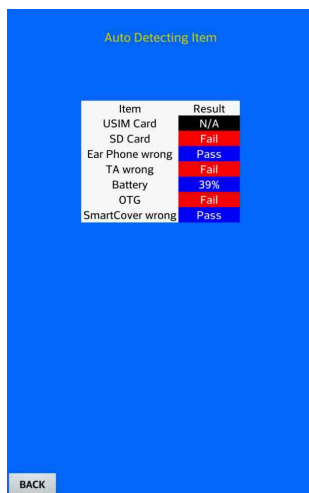


[List]

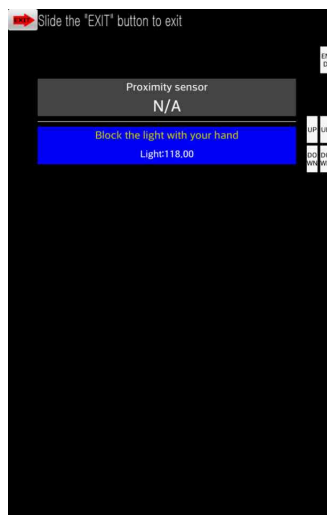
All Auto Test - Full :

-> All Auto Test - Full menu click

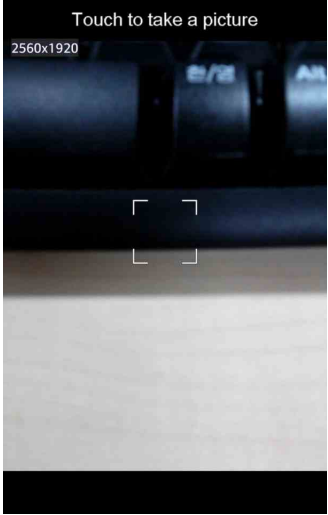
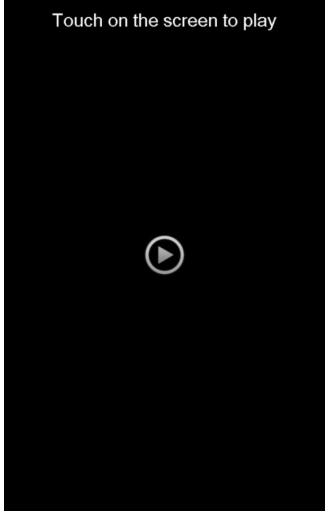
-> Continuous information on the menu, giving you ability test

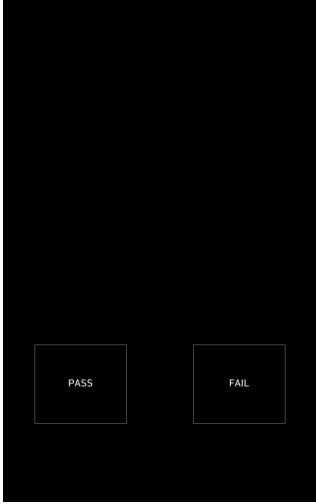
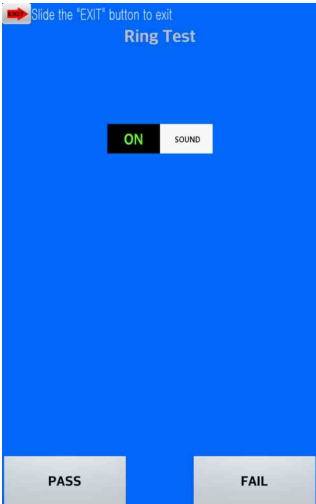


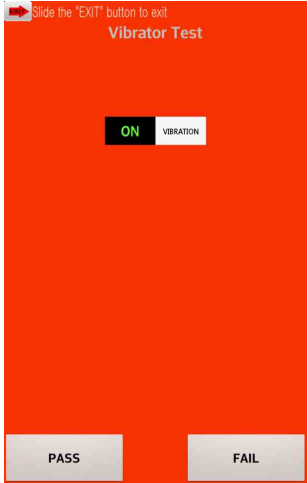
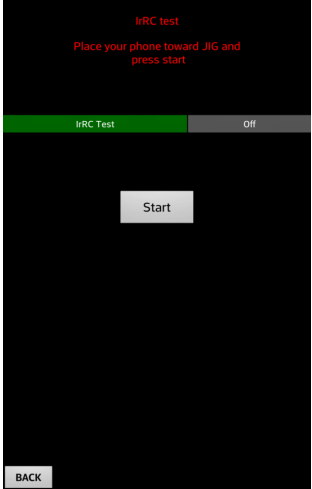
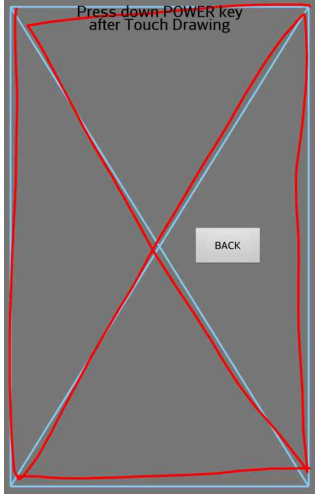
Auto Detecting Item



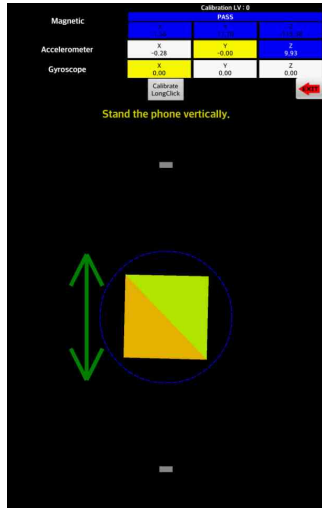
Key Press Test

	Camera(Main Test)
	Camcorder Test
	Camera(VT) Test

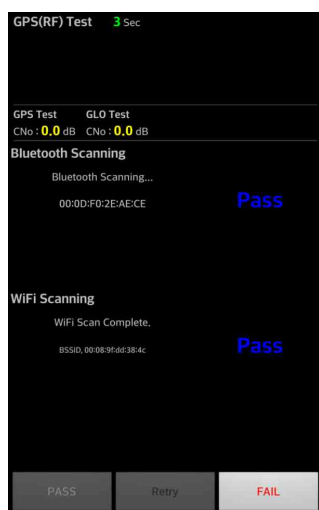
 A black screen with two white rectangular buttons at the bottom. The left button is labeled "PASS" and the right button is labeled "FAIL".	Display Check Test - Black Screen
 A white screen with two white rectangular buttons at the bottom. The left button is labeled "PASS" and the right button is labeled "FAIL".	Display Check Test - White Screen
 A blue screen with a red arrow pointing right and the text "Slide the 'EXIT' button to exit" at the top left. Below this is the text "Ring Test". In the center, there are two buttons: a green "ON" button and a white "SOUND" button. At the bottom, there are two white rectangular buttons labeled "PASS" and "FAIL".	Ring Test

	Vibrator Test
	IrRC Test
	Touch Draw Test - Manual

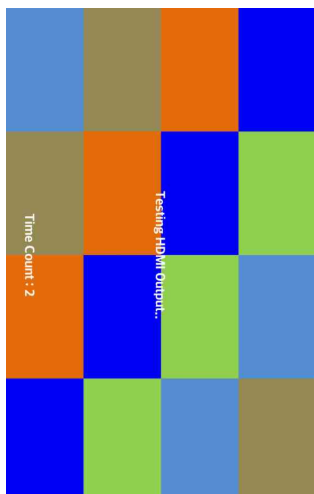
Motion Sensor Test

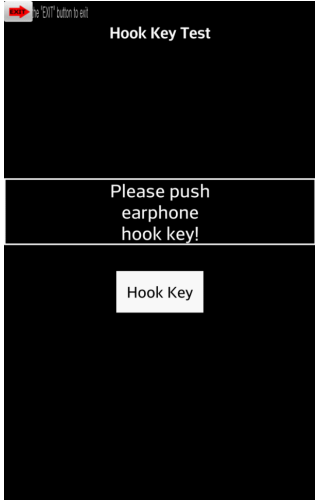
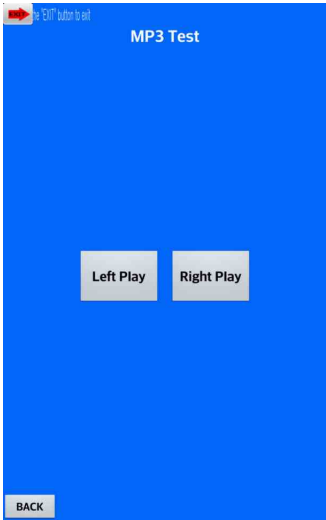
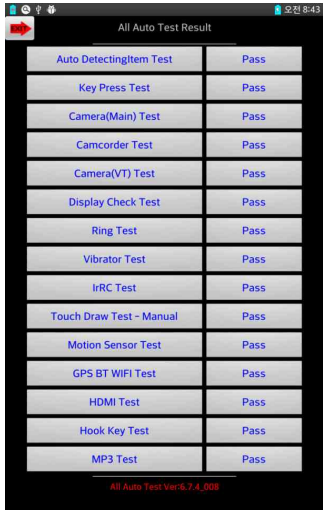


GPS BT WIFI Test



HDMI Test

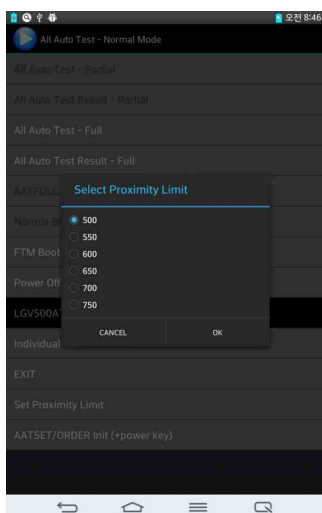


	Hook Key Test
	MP3 Test
	Device Test -> All Auto Test - Result

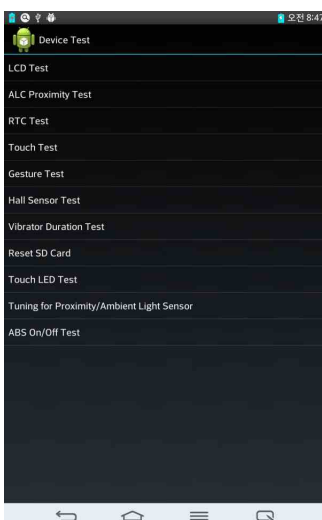


[List]

Auto DetectingItem Test, Key Press Test, Camera(Main) Test, Camcorder Test, Camera(VT) Test, Display Check Test, Ring Test, Vibrator Test, IrRC Test, Touch Draw Test – Manual, Motion Sensor Test, GPS BT WIFI Test, HDMI Test, Sensor Test, Touch Draw Test – Auto, Hook Key Test, MP3 Test, Back to Main

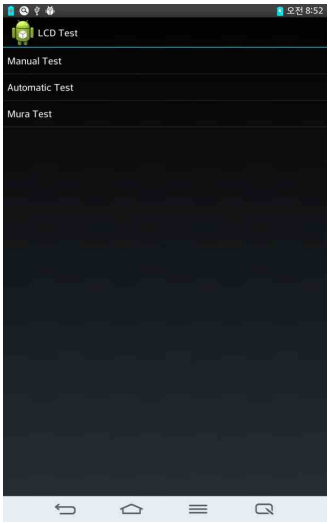
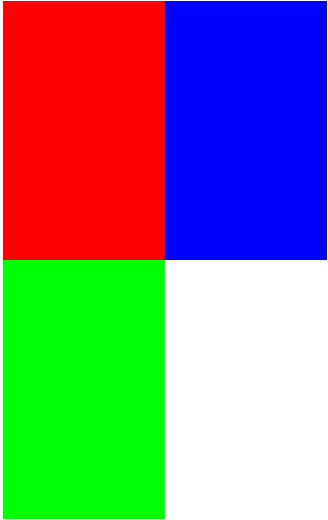
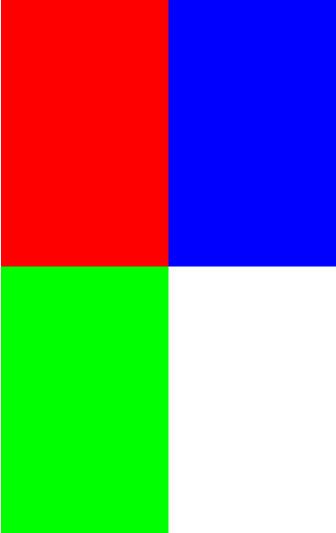


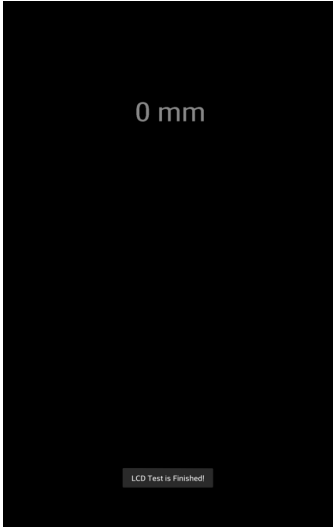
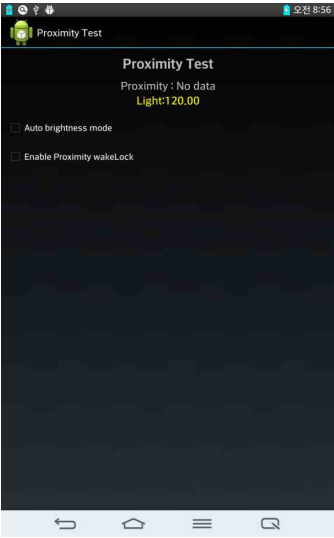
Device Test -> Set Proximity Limit



[List]

LCD Test, ALC Proximity Test, RTC Test, Touch Test, Gesture Test, Hall Sensor Test, Vibrator Duration Test, Reset SD Card, Touch LD Test, Tuning for Proximity/Ambient Light Sensor, ABS On/Off Test

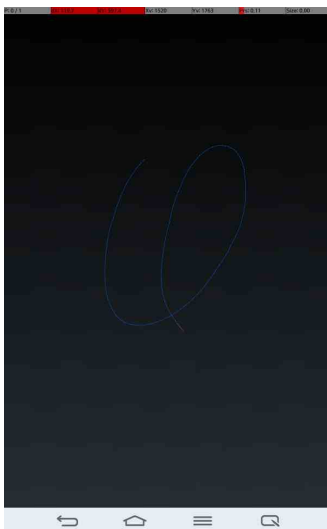
	[List] Manual Test, Automatic Test, Mura Test
	Manual Test
	Automatic Test

	Mura Test
	ALC Proximity Test
	RTC Test

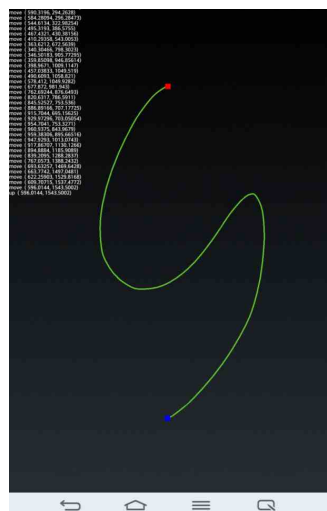


[List]

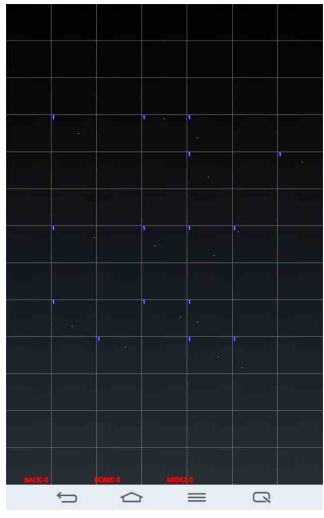
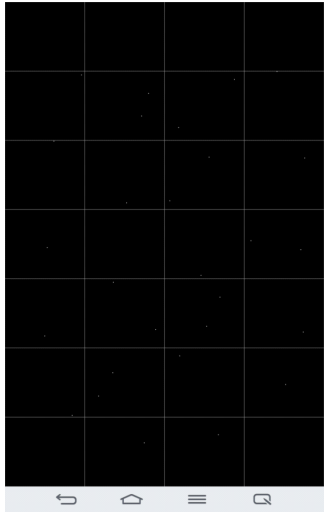
Multi Touch Test, Finger Paint,
Dot Finger Paint, Touch Sensing Test 7*13,
Key Touch Sensing Test 4*7,
Grid Touch Test 4*7, Touch Key Test,
Grid Touch Color Change, Block Touch Test,
Touch Firmware version,
Touch Firmware Upgrade,
Check Synaptics Touch Panel,
Check Synaptics Noise,
Touch Sensing Test 8*13,
Touch Sensing Test 9*14

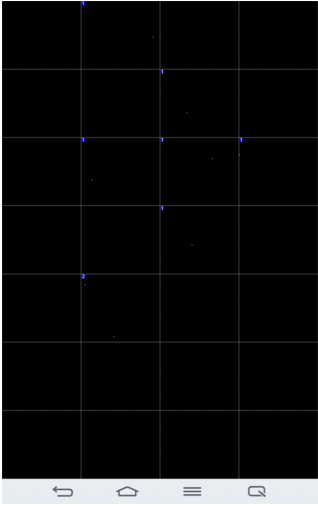
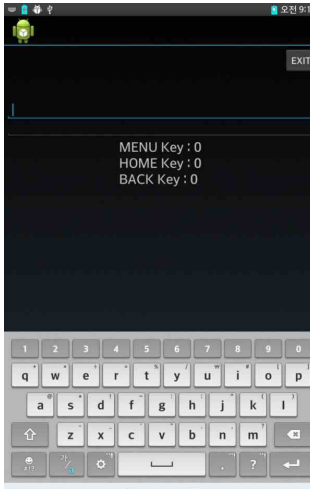
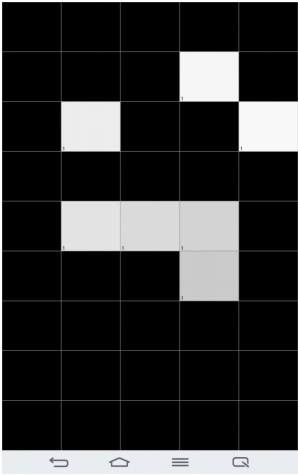


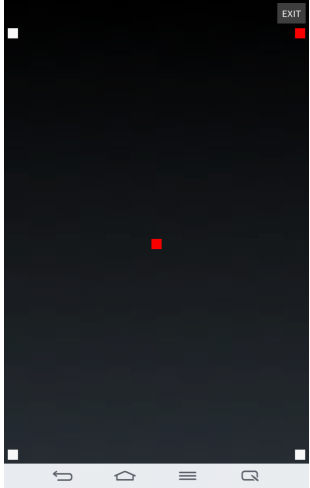
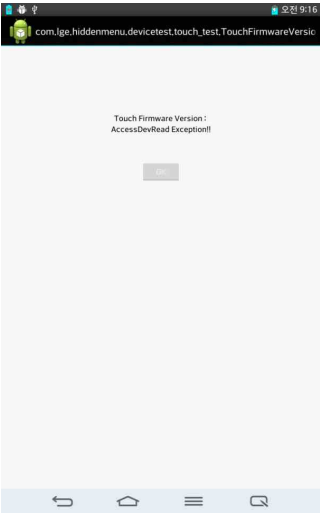
Multi Touch Test

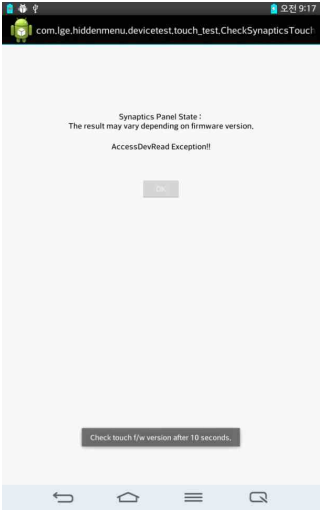
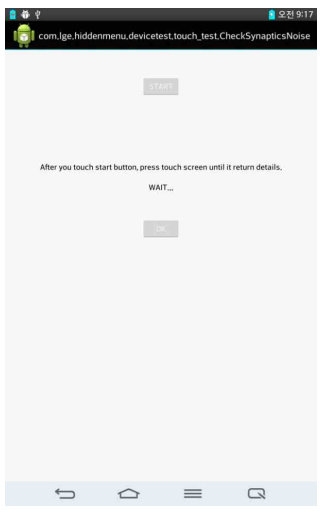
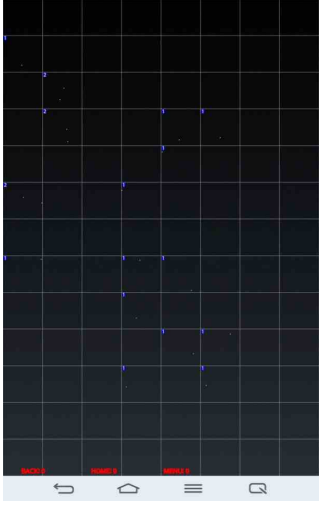


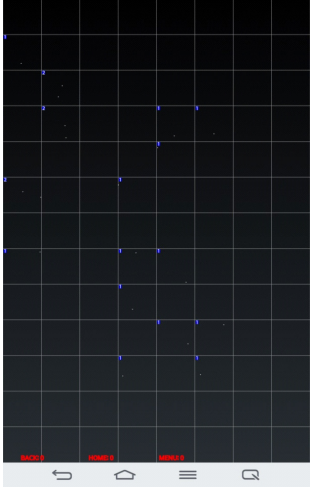
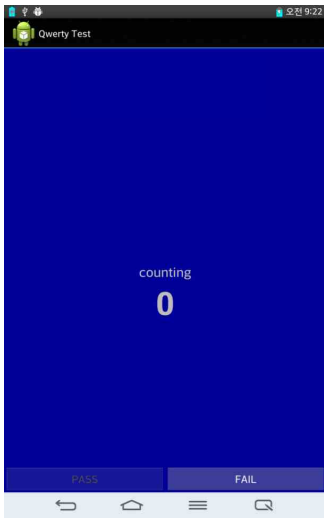
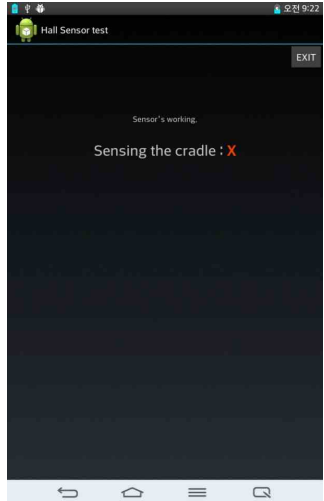
HW Device Test -> Touch Test Finger Paint

	Dot Finger Paint
	Touch Sensing Test 7*13
	Key Touch Sensing Test 4*7

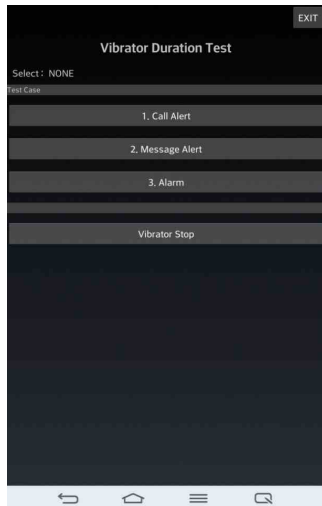
 A screenshot of a smartphone screen displaying a 4x7 grid of small blue dots on a black background. The Android navigation bar is visible at the bottom.	Grid Touch Test 4*7
 A screenshot of a smartphone screen showing a black interface with the text "MENU Key : 0", "HOME Key : 0", and "BACK Key : 0". An "EXIT" button is in the top right. A virtual keyboard is at the bottom. The status bar at the top shows the time 9:19.	Touch Key Test
 A screenshot of a smartphone screen displaying a 4x7 grid of squares. Most squares are black, but several are white or light gray, forming a pattern. The Android navigation bar is at the bottom.	Grid Touch Color Change

	Block Touch Test
	Touch Firmware Version
	Touch Firmware Upgrade

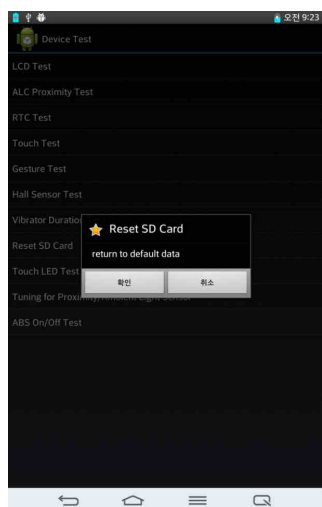
	Check Synaptics Touch Panel
	Check Synaptics Noise
	Touch Sensing Test 8*13

 A screenshot of a mobile application titled 'Touch Sensing Test 9*14'. The screen displays a dark grid with small blue dots and arrows indicating touch points. At the bottom, there are three red buttons labeled 'BACK', 'HOME', and 'POWER'. The Android navigation bar is visible at the very bottom.	Touch Sensing Test 9*14
 A screenshot of a mobile application titled 'Qwerty Test'. The screen is blue with the text 'counting' and a large '0' in the center. At the bottom, there are two buttons labeled 'PASS' and 'FAIL'. The Android navigation bar is visible at the very bottom.	HW Device Test -> Gesture Test
 A screenshot of a mobile application titled 'Hall Sensor test'. The screen is dark with the text 'Sensor's working.' and 'Sensing the cradle : X' in red. At the top right, there is an 'EXIT' button. The Android navigation bar is visible at the very bottom.	HW Device Test -> Hall Sensor Test

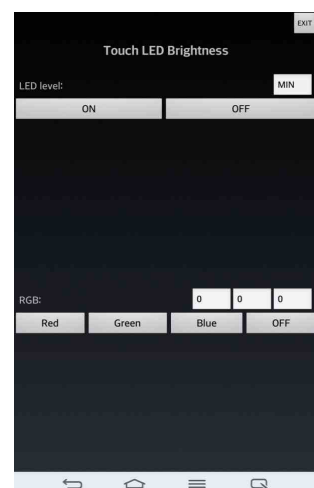
9. HIDDEN MENU



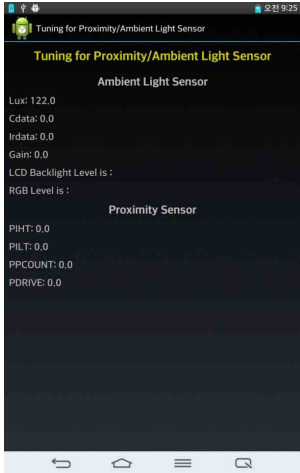
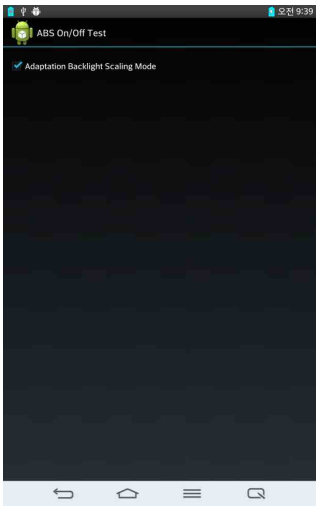
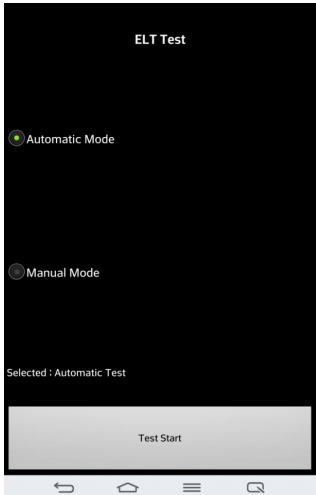
HW Device Test -> Vibrator Duration Test

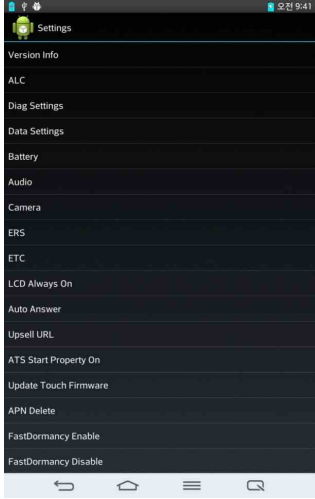
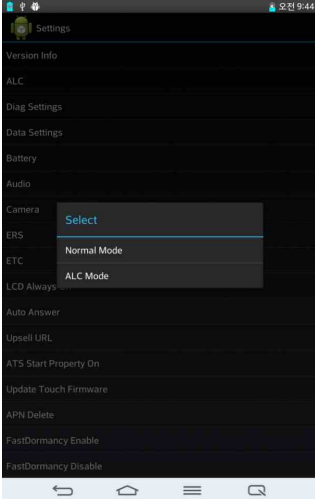


HW Device Test -> Reset SD Card

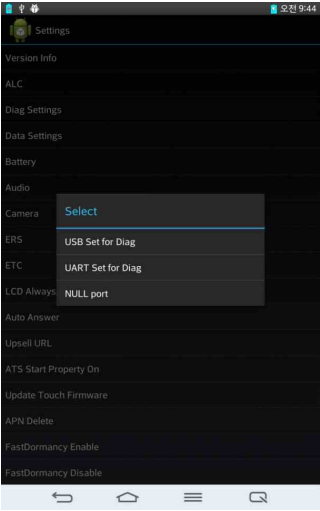
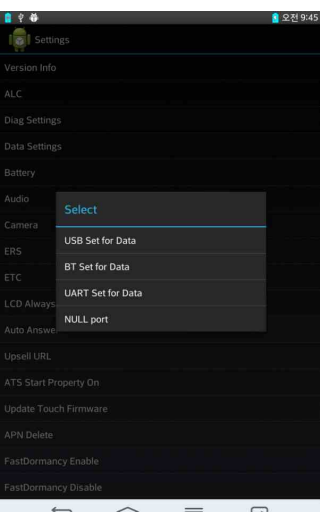
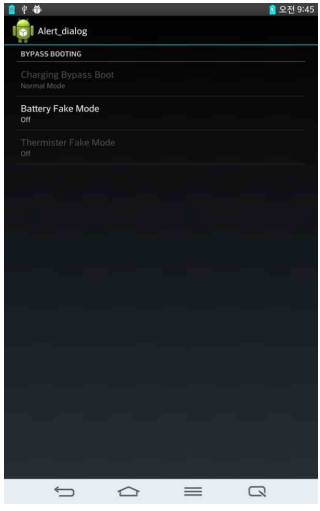


HW Device Test -> Touch LED Test

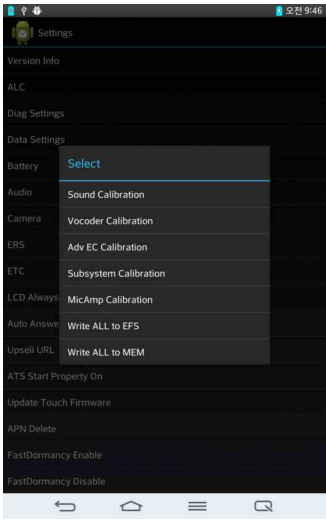
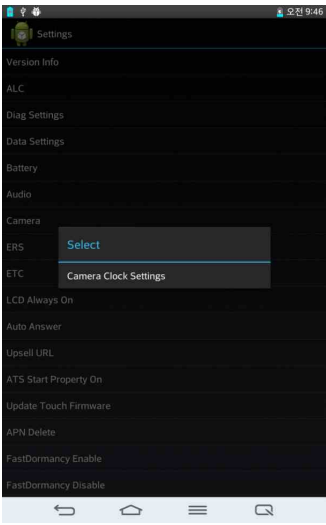
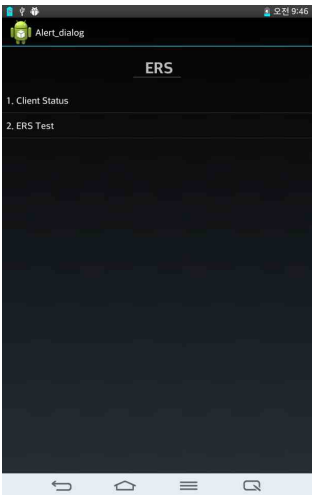
	HW Device Test -> Tuning for Proximity/Ambient Light Sensor
	HW Device Test -> ABS On/Off Test
	ELT Test

	Settings [List] Version Info, ALC, Diag Settings, Data Settings, Battery, Audio, Camera, ERS, ETC, LCD Always On, Auto Answer, Upsell URL, ATS Start Property On, Update Touch Firmware, APN Delete, FastDormancy Enable, FastDormancy Disable,
	Version Info
	ALC

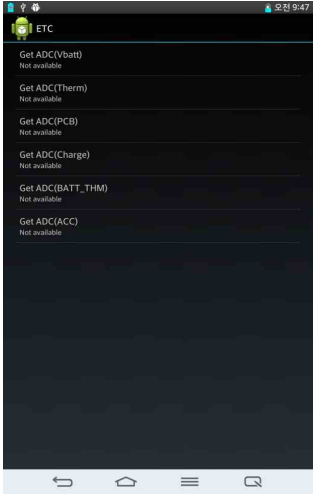
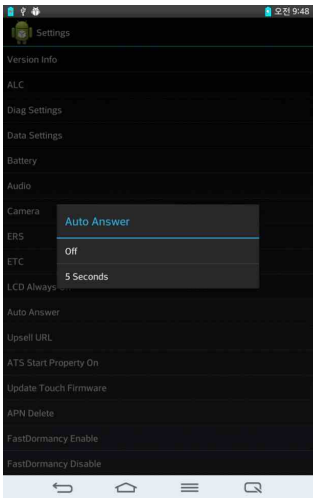
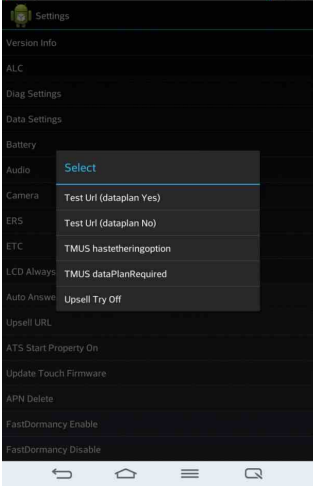
9. HIDDEN MENU

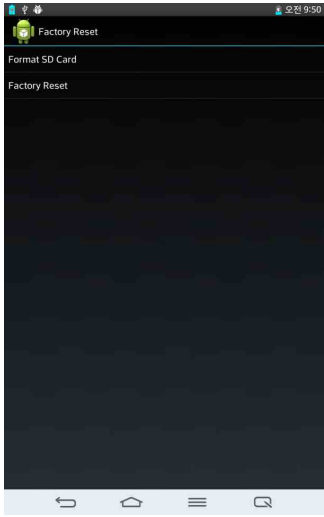
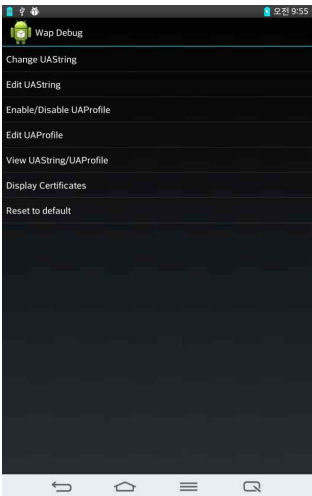
	Diag Settings
	Data Settings
	Battery

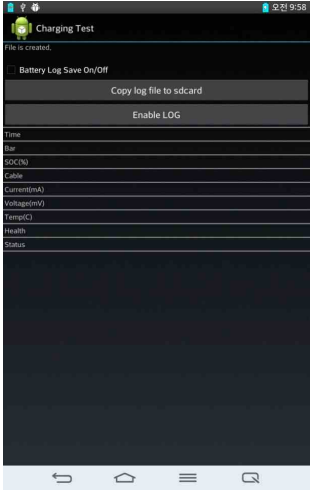
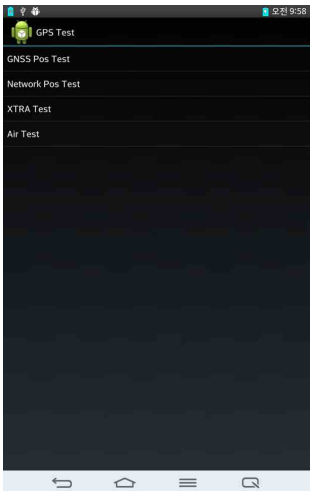
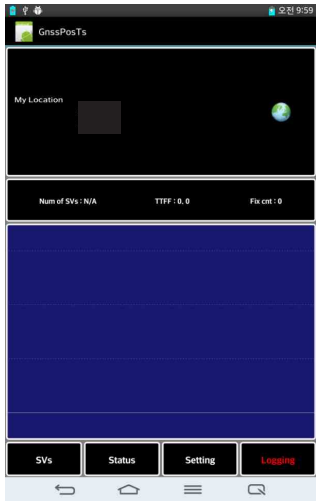
9. HIDDEN MENU

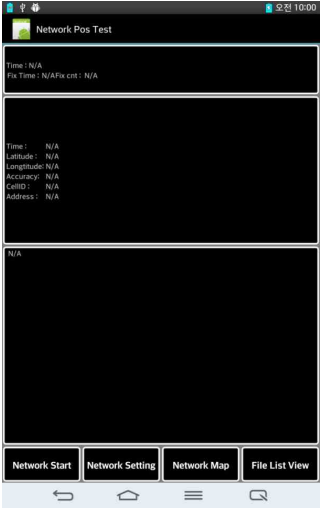
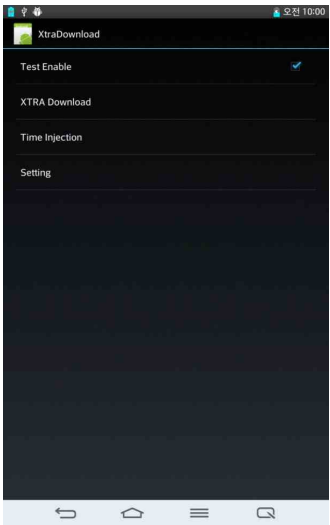
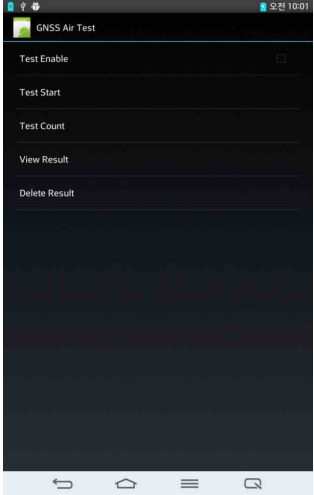
	Audio
	Camera
	ERS

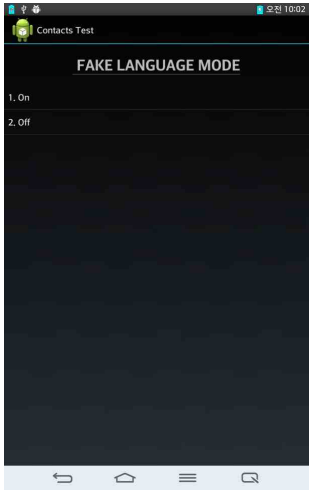
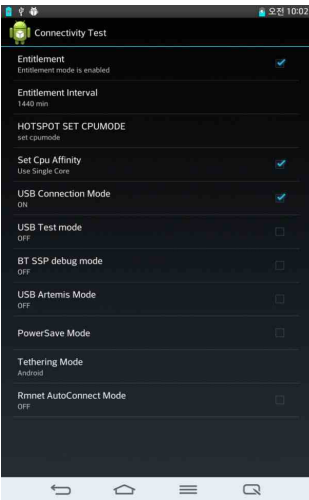
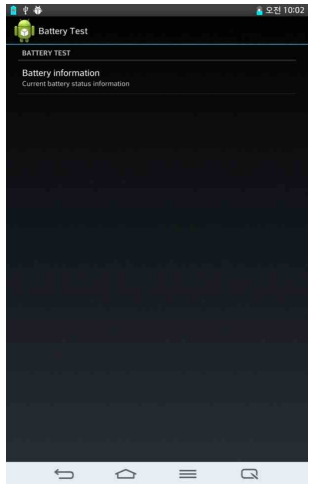
9. HIDDEN MENU

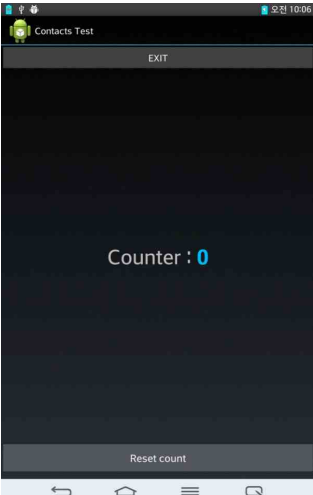
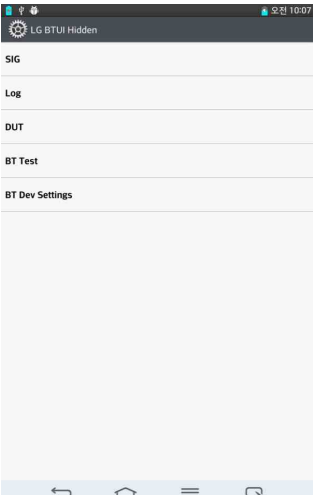
 ETC - Get ADC(Vbatt) Not available - Get ADC(Therm) Not available - Get ADC(PCB) Not available - Get ADC(Charge) Not available - Get ADC(BATT_THM) Not available - Get ADC(ACC) Not available	ETC
 Settings - Version Info - ALC - Diag Settings - Data Settings - Battery - Audio - Camera - ERS - ETC - LCD Always - Auto Answer - Upsell URL - ATS Start Property On - Update Touch Firmware - APN Delete - FastDormancy Enable - FastDormancy Disable Auto Answer - Off 5 Seconds	Auto Answer
 Settings - Version Info - ALC - Diag Settings - Data Settings - Battery - Audio - Camera - ERS - ETC - LCD Always - Auto Answer - Upsell URL - ATS Start Property On - Update Touch Firmware - APN Delete - FastDormancy Enable - FastDormancy Disable Select - Test Uri (dataplan Yes) - Test Uri (dataplan No) - TMUS hastetheringoption - TMUS dataPlanRequired - Upsell Try Off	Upsell URL

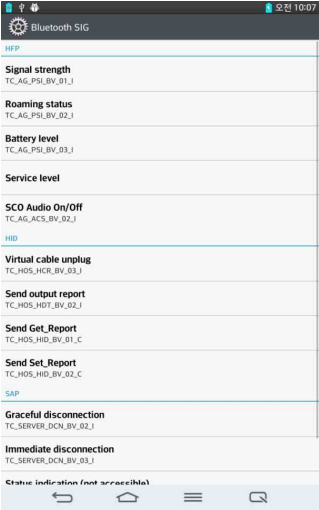
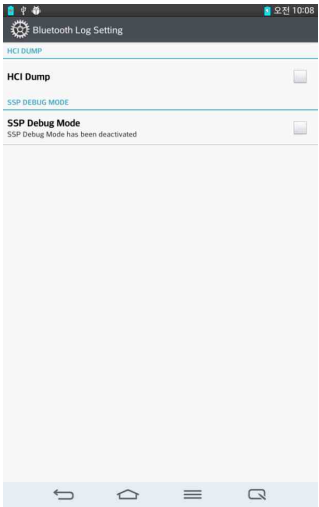
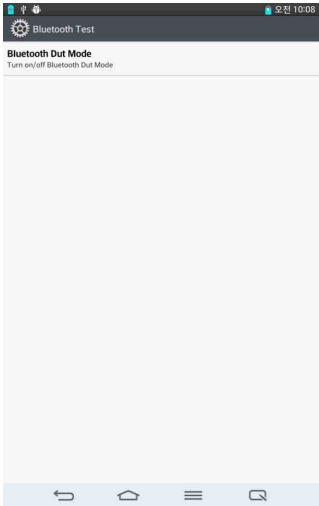
	Factory Reset
	Browser Utility [List] Change UAStrng, Edit UAStrng, Enable/Disable UAProfile, Edit UASProfile, View UAStrng/UAProfile, Display Certificates, Reset to default
	WLAN Test [List] Net Info, Available Net, Saved Ned, FTM Test, OTA Setting, One Way Test, Wi-Fi Debug Screen, Wi-Fi Ping Test/User Command

	Charging Test
	GPS Test [List] GNSS Pos Test, Network Pos Test, XTRA Test, Air Test
	GNSS Pos Test

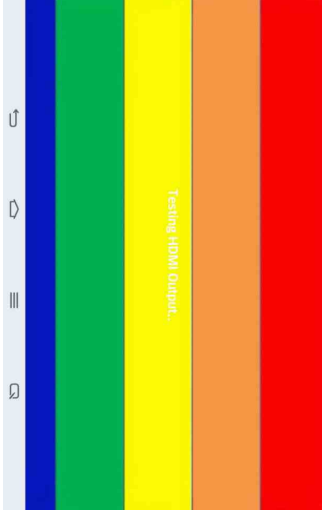
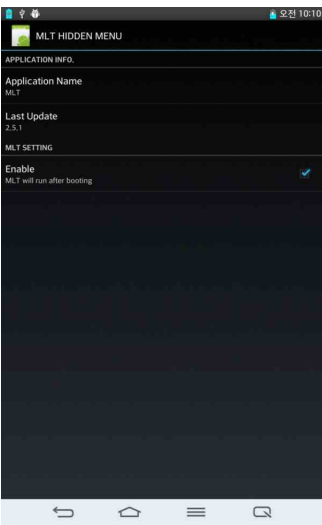
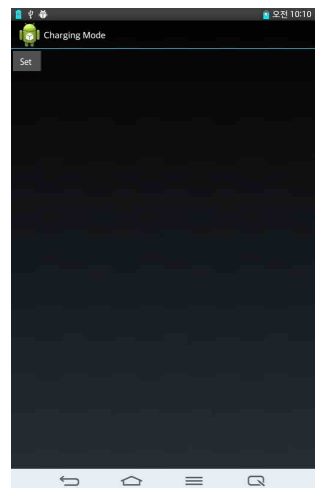
	GPS Test Network Pos Test
	GPS Test Xtra Test
	GPS Test Air Test

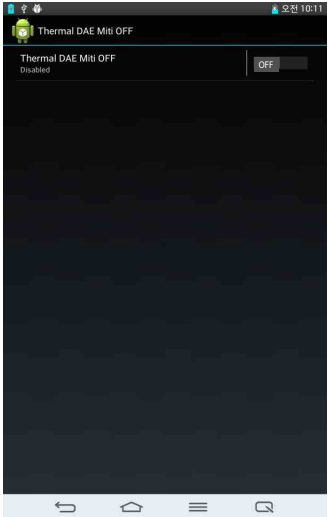
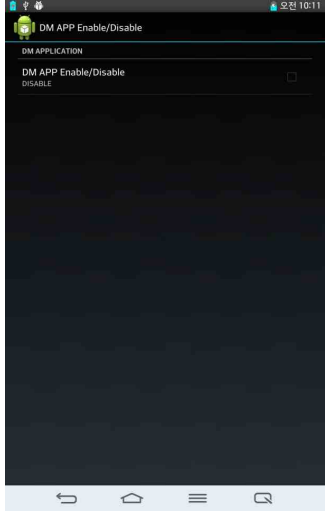
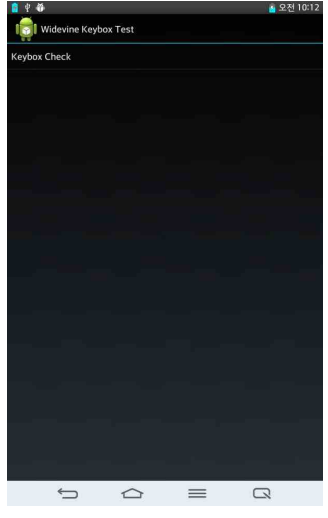
	Fake Language
	Connectivity Test
	Battery Test

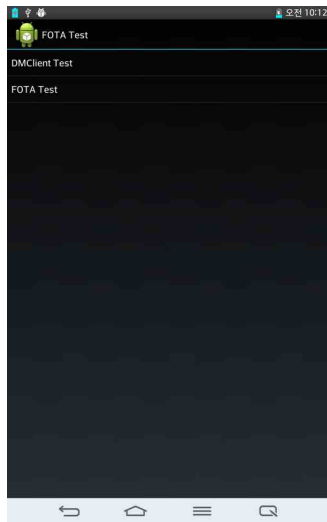
	Log Service
	SMPL Counter
	BT_Test [List] SIG, Log, Dut, BT Test, BT Dev Settings

	BT_Test SIG
	BT_Test Log
	BT_Test Dut

9. HIDDEN MENU

	HDMI Test
	MPT Test
	Charging Mode On/Off

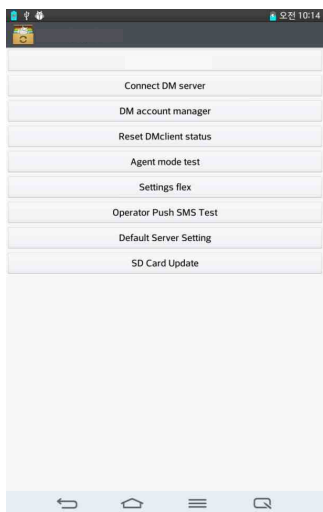
	Thermal Daemon Mitigation Off
	DM APP Enable/Disable
	WVDRM Keybox



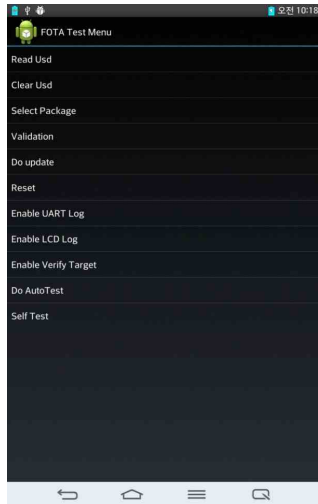
FOTA DM Test

[List]

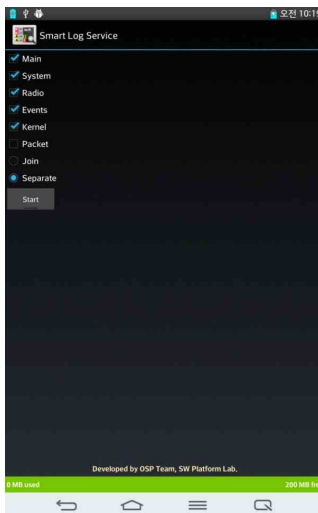
DMClient Test, FOTA Test



FOTA DM Test -> DMClient Test



FOTA DM Test -> FOTA Test



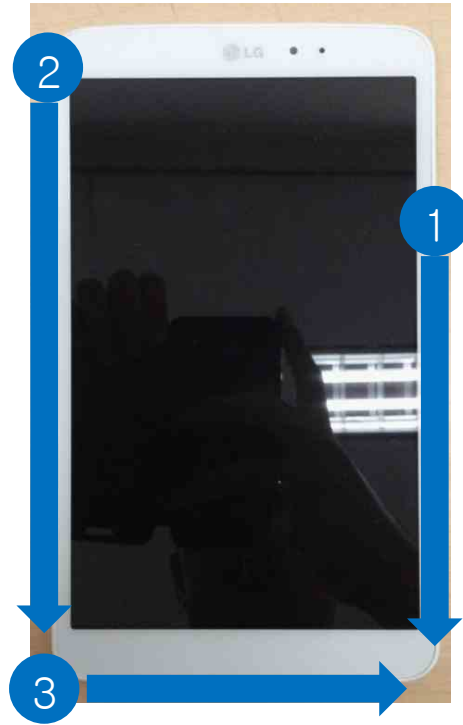
Smart Log Service

10. DISASSEMBLE GUIDE

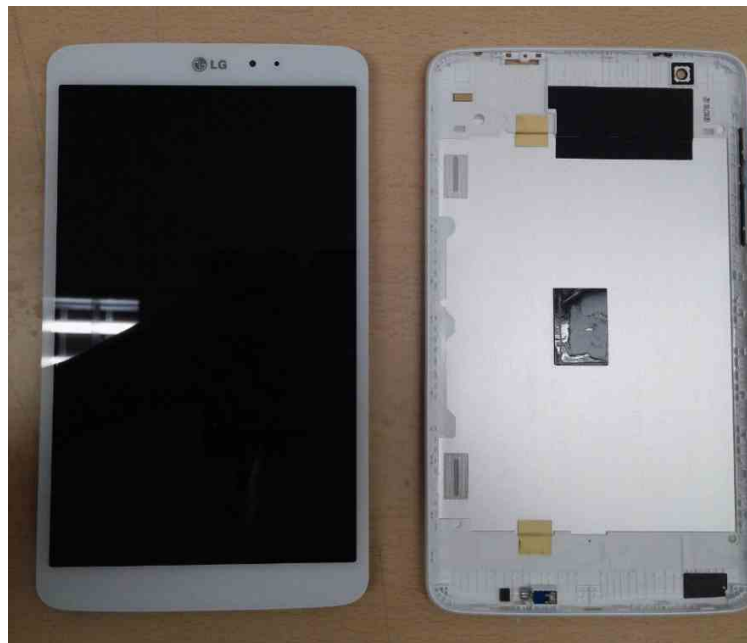
1. Disassemble Rear Cover



Disassemble 1 screw



Disassembling direction



2. Disassemble Speaker Module, I/O Bracket



Disassemble 17 screws



Disassemble speaker module



Disassemble I/O Bracket

3. Disassemble LCD FPCB



Disassemble 3 Connectors



Disassemble LCD FPCB



4. Disassemble Battery



Disassemble Battery Connector

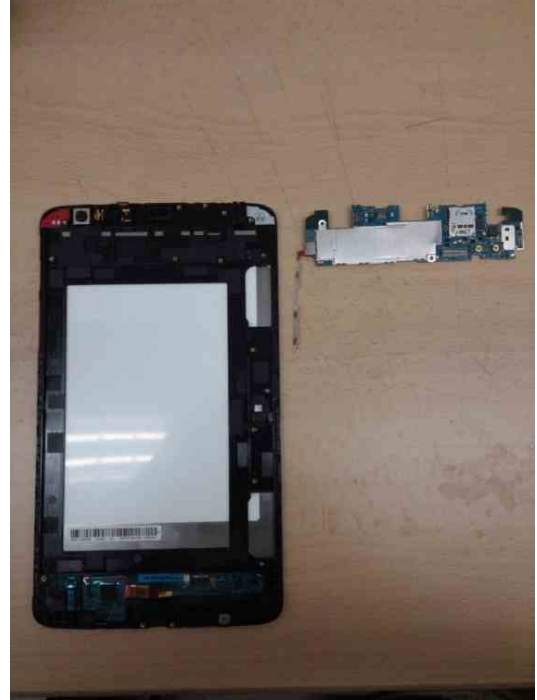


Disassemble Battery

5. Disassemble PCB, H/W Parts



Disassemble 2 Connectors



Disassemble PCB



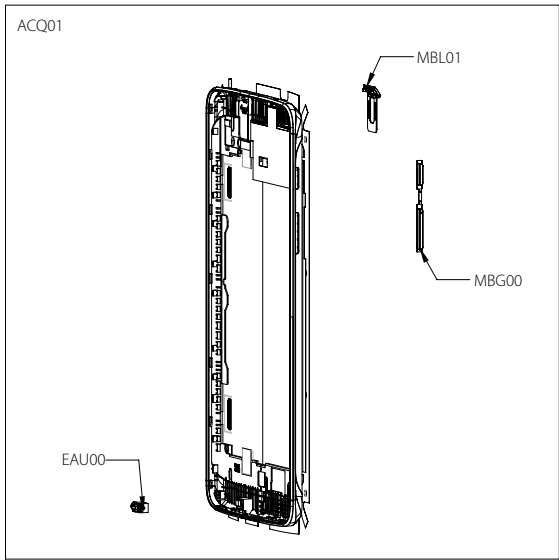
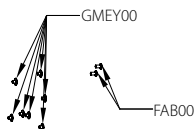
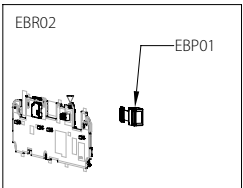
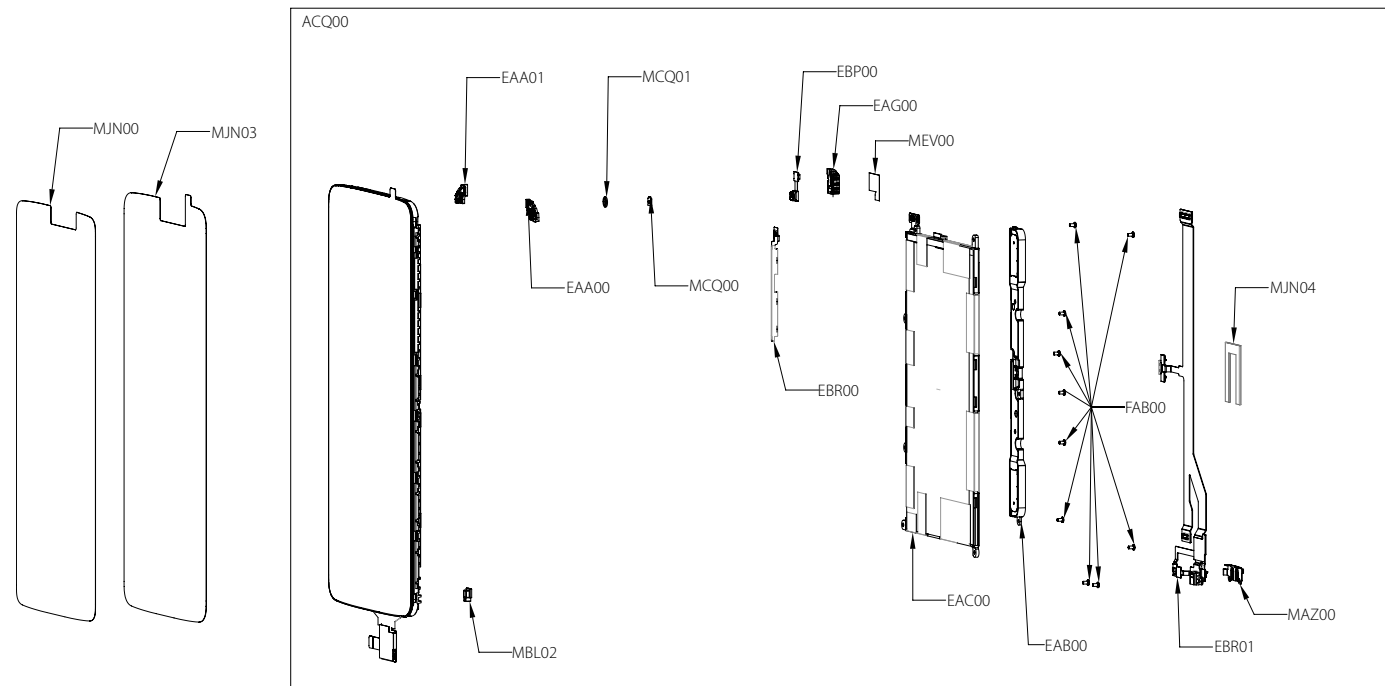
Disassemble H/W Parts

6. Complete disassembling V500



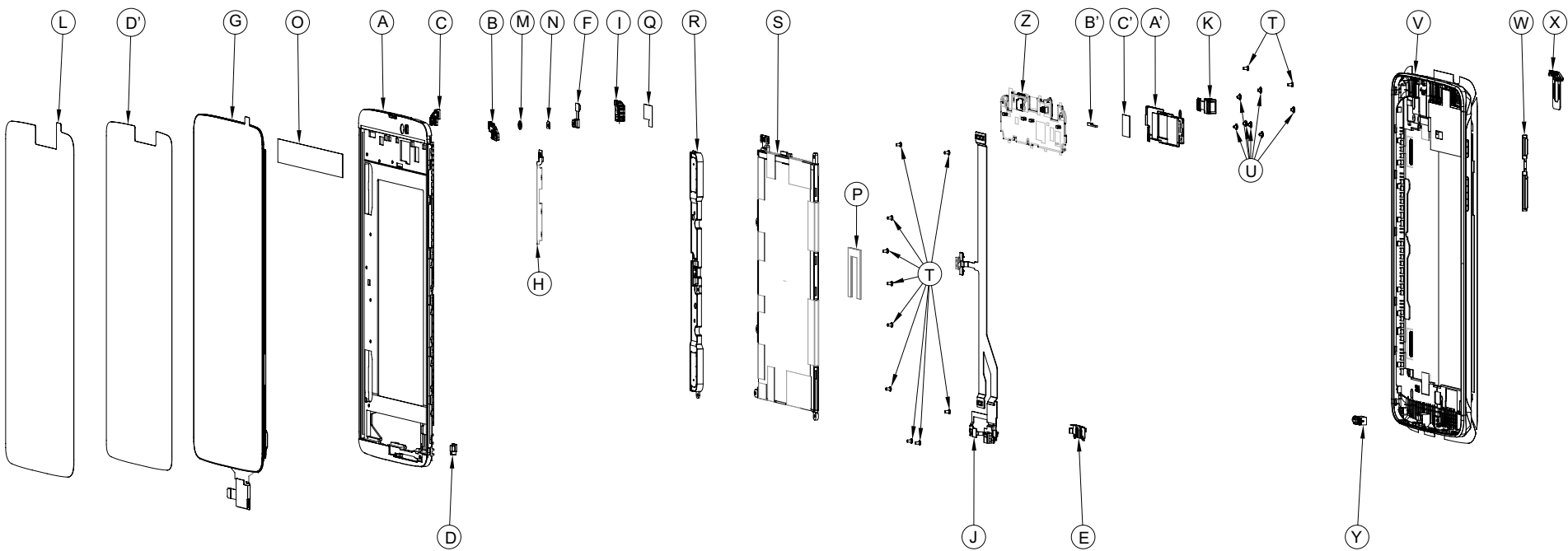
11. EXPLODED VIEW & REPLACEMENT PART LIST

11.1 EXPLODED VIEW(SBOM)



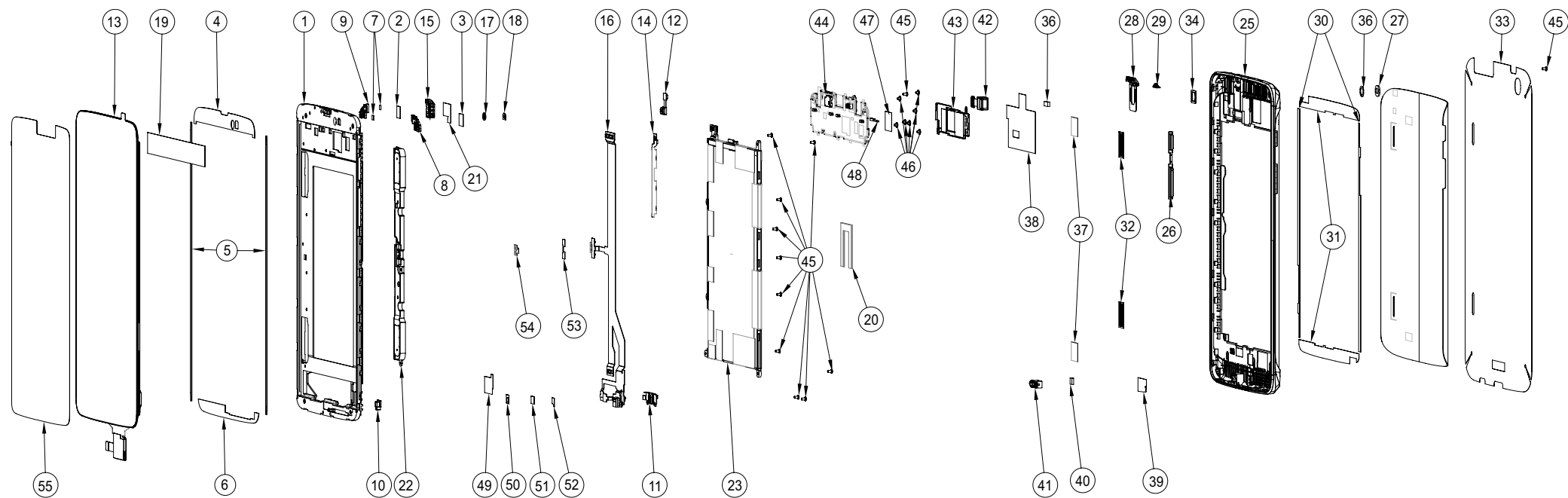
Location	Description
ACQ01	Cover Assembly,Rear
EAU00	Motor,DC
MBG00	Button,Side
MBL01	Cap,Receptacle
EAG00	Jack,Phone
EBP00	Camera Module
MAZ00	Bracket
EBR00	PCB Assembly,Flexible
EBR01	PCB Assembly,Flexible
EAC00	Rechargeable Battery,Lithium Polymer
EAB00	Speaker Module
FAB00	Screw,Machine
ACQ00	Cover Assembly,Bar(Sub)
EAA01	PIFA Antenna,GPS
EAA00	PIFA Antenna,WiFi
MBL02	Cap
MCQ00	Damper
MCQ01	Damper,Camera
MEV00	Insulator
MJN03	Tape,USP Film
MJN04	Tape,Window
MJN00	Tape,Protect
GMEY00	Screw,Machine
FAB00	Screw,Machine
EBR02	PCB Assembly,Main
EBP01	Camera Module

EXPLODED VIEW(ASSY)



No.	Part Name	Part Number
A	Cover Front Assembly	ACQ86607101
B	Wifi Carrier	EAA63086001
C	GPS Carrier	EAA63085901
D	Cap Mic (Rubber)	MBL65602501
E	Bracket I/O connector	MAZ63840801
F	VT Cam	EBP61881801
G	HYBRID LCD	EAT61993301
H	Side Key FPCB Assembly	EBR77721901
I	Earjack	EAG63292201
J	Main FPCB Assembly	EBR77739601
K	Main Camera	EBP61842101
L	USP Film	MJN68523301
M	Pad Cam VT	MCQ67393801
N	Pad Sensor	MCQ67358301
O	Thermal Sheet 1	MHK64033601
P	Tape Back Batt	MJN68560901
Q	Insulator Earjack	MEV64612601
R	Speaker Assy	EAB63028301
S	Battery	EAC62159101
T	Screw	FAB32039101
U	Screw	GMEY0018101
V	Cover Rear Assembly	ACQ86676401
W	Button Side	MBG64827301
X	Cap SD	MBL65546401
Y	Motor	EAU62006001
Z	Main PCB assembly	EBR77719602
A'	Can Shield Main Assy	ABM74057701
B'	Main PCB Absorber	MKU30623201
C'	Pad Batt Conn	MCQ67490601
D'	Tape Protection Bar	MJN68561001

EXPLODED VIEW(TOTAL)



No.	Part Name	Part Number	No.	Part Name	Part Number	No.	Part Name	Part Number
1	Cover Front(sub)	ACQ8669630I	21	Insulator Earjack	MEV6461260I	41	Motor	EAU6200600I
2	Tape Earjack	MJN6852290I	22	Speaker Assy	EAB6302830I	42	Main Camera	EBP6184210I
3	Tape Cam	MJN6852280I	23	Battery	EAC6215910I	43	Can Shield Main Assy	ABM7405770I
4	Tape Window Top	MJN6852310I	24	Cover Rear	MCK6773010I	44	Main PCB Assy SMT	EBR77739302
5	Tape window Side	MJN6856080I	25	Deco Rear	MCR6511360I	45	Screw	FAB3203910I
6	Tape Window Bottom	MJN6852320I	26	Button Side	MBG6482730I	46	Screw	GMEY001810I
7	LDI	MLAB000480I	27	Window Camera	MKC6462140I	47	Main PCB Absorber	MKU3062320I
8	Wifi Carrier	EAA6308600I	28	Cap SD	MBL6554640I	48	Pad Batt Conn	MCQ6749060I
9	GPS Carrier	EAA6308590I	29	Window IRDA	MFB6305430I	49	Tape FPCB LCD TOUCH	MDS6441030I
10	Cap Mic (Rubber)	MBL6560250I	30	Tape TBF 2	MJN6848270I	50	Tape FPCB LCD MIC	MJN6864690I
11	Bracket I/O connector	MAZ6384080I	31	Tape TBF 1	MJN6856130I	51	Tape FPCB LCD MOTOR	MJN6862700I
12	VT Cam	EBP6188180I	32	Spk mesh	MDJ6384500I	52	Tape FPCB LCD NFC	MJN6862710I
13	HYBRID LCD	EAT6199330I	33	Tape Protection Rear	MJN6848290I	53	Tape FPCB LCD HALL	MJN6862720I
14	Side Key FPCB Assembly	EBR7772190I	34	Pad Camera	MCQ6729830I	54	Tape Pull FPC	MJN6867080I
15	Earjack	EAG6329220I	35	Tape Window Camera	MJN6856140I	55	Tape Protection Bar	MJN6856100I
16	Main FPCB Assembly	EBR7773960I	36	Gasket Shield 1	MDS6427780I			
17	Pad Cam VT	MCQ6739380I	37	Gasket Shield 2	MDS6424160I			
18	Pad Sensor	MCQ6735830I	38	Thermal Sheet 2	MHK6411320I			
19	Thermal Sheet 1	MHK6403360I	39	Tape Light Block	MJN6866810I			
20	Tape Back Batt	MJN6856090I	40	Pad Mic	MCQ6742600I			

11. EXPLODED VIEW & REPLACEMENT PART LIST

11.2 Replacement Parts <Mechanic component>

Note: This Chapter is used for reference, Part order is ordered by SBOM standard on GCSC

Level	LocationNo.	Description	PartNumber	Spec	Remark
1	AGQ000000	Phone Assembly	AGQ87286303	LGV500.AUSABK BK:BLACK BLACK -	
2	ACQ100400	Cover Assembly,EMS	ACQ86750503	LGV500.AUSABK BK:BLACK BLACK -	
3	ACQ01	Cover Assembly,Rear	ACQ86676402	LGV500.AKDXWH ZZ:Without Color -	
4	ACQ105800	Cover Assembly Rear(SVC)	ACQ86676302	LGV500.AKDXWH ZZ:Without Color -	
5	MBG00	Button,Side	MBG64827302	MOLD PC LGV500.AKDXWH ZZ:Without Color V500 Button side	
5	MBL01	Cap,Receptacle	MBL65546402	MOLD PC LGV500.AKDXWH ZZ:Without Color V500 SD CAP	
5	MJN061100	Tape,Protect	MJN68482901	COMPLEX LGV500.AKDXWH ZZ:Without Color -	33
5	MKC009400	Window,Camera	MKC64621401	COMPLEX DEVPF-TEST.ADEUZY ZZ:Without Color v500 Camera window	27
5	MJN089300	Tape,Window	MJN68561401	COMPLEX LGV500.AKDXWH ZZ:Without Color -	35
5	MJN000001	Tape	MJN68561301	COMPLEX LGV500.AKDXWH ZZ:Without Color -	31
5	MJN000000	Tape	MJN68482701	COMPLEX LGV500.AKDXWH ZZ:Without Color -	30
5	MHK000001	Sheet	MHK64113201	COMPLEX LGV500.AKDXWH ZZ:Without Color -	38
5	MFB029600	Lens,Flash	MFB63054301	MOLD PC LEXAN141R DEVPF-TEST.ADEUZY ZZ:Without Color V500 IRLED	29
5	MDS000000	Gasket	MDS64277801	COMPLEX LGV500.AKDXWH ZZ:Without Color -	36
5	MDS000001	Gasket	MDS64241601	COMPLEX LGV500.AKDXWH ZZ:Without Color -	37

11. EXPLODED VIEW & REPLACEMENT PART LIST

Level	LocationNo.	Description	PartNumber	Spec	Remark
5	MDJ000000	Filter	MDJ63845002	PRESS STS 0.1 LGV500.AKDXWH ZZ:Without Color -	
5	MCR000000	Decor	MCR65113602	PRESS AL 0.6t LGV500.AKDXWH ZZ:Without Color V500 Deco Rear	
5	MCQ000000	Damper	MCQ67426001	COMPLEX LGV500.AKDXWH ZZ:Without Color -	40
5	MCQ009400	Damper,Camera	MCQ67298301	COMPLEX LGV500.AKDXWH ZZ:Without Color -	34
5	MCK063300	Cover,Rear	MCK67730102	MOLD PC LGV500.AKDXWH ZZ:Without Color V500 Cover Rear	
3	ACQ003400	Cover Assembly,Bar	ACQ86698802	LGV500.AKDXWH ZZ:Without Color -	
4	MAZ00	Bracket	MAZ63840801	PRESS STS 304 2B 0.3t DEVPF-TEST.ADEUZY ZZ:Without Color -	E,11
6	MJN000000	Tape	MJN68691201	COMPLEX LGV500.AKDXWH ZZ:Without Color -	
6	ADB048600	Dome Assembly,Metal	ADB74058601	LGV500.AKDXWH ZZ:Without Color -	
6	MJN000000	Tape	MJN68646901	COMPLEX LGV500.AKDXWH ZZ:Without Color -	50
6	MJN000004	Tape	MJN68670801	COMPLEX LGV500.AKDXWH ZZ:Without Color -	54
6	MJN000003	Tape	MJN68627201	COMPLEX LGV500.AKDXWH ZZ:Without Color -	53
6	MJN000002	Tape	MJN68627101	COMPLEX LGV500.AKDXWH ZZ:Without Color -	52
6	MDS000000	Gasket	MDS64410301	COMPLEX LGV500.AKDXWH ZZ:Without Color -	49
6	MJN000001	Tape	MJN68627001	COMPLEX LGV500.AKDXWH ZZ:Without Color -	51
7	ANT100 ANT101 ANT102 ANT103	Clip	MBV62521801	COMPLEX LGF320L.ALGTBK ZZ:Without Color -	
4	FAB00	Screw,Machine	FAB32039101	BH + 2.7M 3M SWRCH18A SILVER N N	45

11. EXPLODED VIEW & REPLACEMENT PART LIST

Level	LocationNo.	Description	PartNumber	Spec	Remark
4	ACQ00	Cover Assembly Bar(Sub)	ACQ87073002	LGV500.AKDXWH ZZ:Without Color -	
5	ACQ032700	Cover Assembly,Front	ACQ86607101	LGV500.AKDXWH ZZ:Without Color -	A
6	MJN000000	Tape	MJN68522901	COMPLEX LGV500.AKDXWH ZZ:Without Color -	2
6	ACQ033200	Cover Assembly Front(Sub)	ACQ86696301	DEVPF-TEST.ADEUZY ZZ:Without Color V500 Cover Front (sub)	1
7	MET099500	Insert,Nut	MICE0016901	COMPLEX MECH_COMMON ZZ:Without Color -	
7	MCK032700	Cover,Front	MCK67730001	MOLD PC DEVPF-TEST.ADEUZY ZZ:Without Color V500 Cover Front	
6	MEZ000900	Label,After Service	MLAB0004801	COMPLEX LG-LB3300 LGT ZZ:Without Color -	7
6	MJN009400	Tape,Camera	MJN68522801	COMPLEX LGV500.AKDXWH ZZ:Without Color -	3
6	MJN089300	Tape,Window	MJN68523101	COMPLEX LGV500.AKDXWH ZZ:Without Color -	4
6	MJN089302	Tape,Window	MJN68523201	COMPLEX LGV500.AKDXWH ZZ:Without Color -	6
6	MJN089301	Tape,Window	MJN68560801	COMPLEX LGV500.AKDXWH ZZ:Without Color -	5
6	MBL02	Cap	MBL65602501	MOLD SILICONE DEVPF-TEST.ADEUZY ZZ:Without Color -	D,10
5	MHK00	Sheet	MHK64033601	COMPLEX LGV500.AKDXWH ZZ:Without Color -	O,19
4	MCQ00	Damper	MCQ67358301	COMPLEX LGV500.AKDXWH ZZ:Without Color -	N,18
4	MCQ01	Damper,Camera	MCQ67393801	COMPLEX LGV500.AKDXWH ZZ:Without Color -	M,17
4	MEV00	Insulator	MEV64612601	COMPLEX LGV500.AKDXWH ZZ:Without Color V500 Earjack insulaotr	Q,21
4	MJN03	Tape,USP Film	MJN68523301	COMPLEX LGV500.AKDXWH ZZ:Without Color -	L

11. EXPLODED VIEW & REPLACEMENT PART LIST

Level	LocationNo.	Description	PartNumber	Spec	Remark
4	MJN04	Tape,Window	MJN68560901	COMPLEX LGV500.AKDXWH ZZ:Without Color -	P,20
4	MJN00	Tape,Protect	MJN68561001	COMPLEX LGV500.AKDXWH ZZ:Without Color -	D',55
4	MEZ000000	Label	MLAZ0038301	COMPLEX LG-VX6000 ZZ:Without Color PID Label 4 Array PRINTING,	
3	GMEY00	Screw,Machine	GMEY0018101	GMEY0018101 FH + 1.4mm 1.5mm SWCH FZB N - KUMGANG SCREW CO., LTD	U,46
3	MEZ002100	Label,Approval	MEZ65230101	COMPLEX LGVS980.AVRZBK BK:Black -	
3	FAB00	Screw,Machine	FAB32039101	BH + 2.7M 3M SWRCH18A SILVER N N	T
5	MKU101700	Absorber,Electromagnetic Wave	MKU30623201	COMPLEX LGV500.AKDXWH ZZ:Without Color -	B',47
5	ABM070300	Can Assembly,Shield	ABM74057701	LGV500.AKDXWH ZZ:Without Color -	A',43
6	MEV000000	Insulator	MEV64418601	COMPLEX LGV500.AKDXWH ZZ:Without Color -	
6	MBK070300	Can,Shield	MBK63517601	PRESS STS304 0.15 LGV500.AKDXWH ZZ:Without Color -	
5	MCQ000000	Damper	MCQ67490601	COMPLEX LGV500.AKDXWH ZZ:Without Color -	C',48
5	MEZ000000	Label	MEZ65049701	COMPLEX LGLS720.ASPRTS ZZ:Without Color PID label	
6	ANT100 ANT101 ANT102 ANT103	Clip	MBV62521801	COMPLEX LGF320L.ALGTBK ZZ:Without Color -	
6	SC7000	Can,Shield	MBK63517501	PRESS STS304 0.2 LGV500.AKDXWH ZZ:Without Color -	
6	SC10001	Can,Shield	MBK63517701	PRESS STS 304 0.2 LGV500.AKDXWH ZZ:Without Color -	

11. EXPLODED VIEW & REPLACEMENT PART LIST

Level	LocationNo.	Description	PartNumber	Spec	Remark
6	SC4001 SC4002 SC4003 SC4004 SC4005 SC4006 SC4007 SC4009 SC4010 SC4011 SC4012	Clip	MBV62321701	PRESS STS 301 0.15 LGLU6200.ALGTBK BK:Black -	
6	ANT4004 ANT4005	Clip	MBV62562101	PRESS CU 0.1 LGD800.AATTBK ZZ:Without Color -	
6	CN4004 CN4005	Insert	MET62537201	CUTTING STS SCPSD00.ASKTAV SV:Silver -	
1	AGF000000	Package Assembly	AGF77045902	LGV500.AUSAWH ZZ:Without Color V500(A PJT) USA_Package(USA UB/Global MB/Seal_UB1/Angle/480ea)	
2	MAY084000	Box,Unit	MAY66150804	BOX Paper 149 50 248 4 COLOR LGV500.AUSAWH BK:BLACK BLACK A Project(V500) USA Unit Box	
2	MAY047100	Box,Master	MAY66194502	COMPLEX LGV500.AUSAWH WH:WHITE WHITE LG-V500(A PJT) USA_Global(English) Master Box	
2	MEZ000000	Label	MLAZ0050901	COMPLEX KU990.AGBRBK ZZ:Without Color Battery Warning Label (Lithium ion Battery Label)	
2	MEZ047200	Label,Master Box	MLAJ0004402	PRINTING CG300 CGR DG ZZ:Without Color LABEL MASTER BOX(for CGR TDR 2VER. mbox_label) GSM standard_master box label	
2	AGJ000000	Pallet Assembly	AGJ73618302	LGV500.AUSAWH WH:WHITE WHITE LG-V500(A PJT) USA_Pallet Assy (Angle,Cap,Sleeve_1,219x1,016 Pallet/480ea)	
3	MCQ000000	Damper	MPBZ0001227	COMPLEX LG-TM910 VRZ DB ZZ:Without Color VX9000_Angle	
3	MAY115200	Box,Pallet Sleeve	MAY66194403	COMPLEX WH:WHITE WHITE LGV500.AUSAWH LG-V500(A PJT) USA_Sleeve For Dead Space (1,219x1,016)	
3	MGA000000	Pallet	MPCY0005202	COMPLEX LG-TM520 ZZ:Without Color -	

11. EXPLODED VIEW & REPLACEMENT PART LIST

Level	LocationNo.	Description	PartNumber	Spec	Remark
3	MBL007000	Cap,Box	MBL65718402	COMPLEX LGV500.AUSAWH WH:WHITE WHITE LG-V500(A PJT)_USA Cap (1,219x1,016 Pallet)	
2	MEZ084100	Label,Unit Box	MEZ65241801	COMPLEX LGV500.AUSABK BK:BLACK BLACK A Project(V500)_GSM Unit Box Label_90*35	
2	MAF086500	Bag,Vinyl	MAF63627401	COMPLEX LGV500.JKDXBK BK:BLACK BLACK -	
1	AAD000000	Addition Assembly	AAD86537703	LGV500.AUSABK BK:BLACK BLACK -	

11. EXPLODED VIEW & REPLACEMENT PART LIST

11.2 Replacement Parts <Main component>

Note: This Chapter is used for reference, Part order is ordered by SBOM standard on GCSC

Level	LocationNo.	Description	PartNumber	Spec	Remark
4	EAU00	Motor,DC	EAU62006001	WHSM-3408SA 3V 100mA 0A 12.5KRPM 0RPM 40mSEC 0GF.CM 22OHM WOOSUNG G&T CO.,LTD	Y,41
4	EAG00	Jack,Phone	EAG63292201	HG05-AB3550 5P 2P ANGLE R/TP 3.5M BLACK 5P Side contact HYUPJIN I&C CO.,LTD.	I,15
4	EBP00	Camera Module	EBP61881801	CW1353 CW1353 1.3M Sony 1/7.8" HD, 5x5x3.0t, FPCB 5mm, 0deg. CNT-Bottom COWELL ELECTRONICS CO.,LTD	F,12
4	EBR00	PCB Assembly Flexible	EBR77721901	LGV500.AKDXWH 1.0 Flexible	H,14
5	EBR070400	PCB Assembly Flexible,SMT	EBR77722001	LGV500.AKDXWH 1.0 Flexible	
6	EBR070300	PCB Assembly Flexible,SMT Top	EBR77740701	LGV500.AKDXWH 1.0 Flexible	
7	EAX010700	PCB,Flexible	EAX65397001	LGV500.AKDXWH 1.0 POLYI Double 2 0.18 Flexible	
7	CN100	Connector,BtoB	ENBY0050901	GB042-10P-H10-E3000 10P 0.4MM STRAIGHT MALE SMD R/TP 1M - LS Mtron Ltd.	
6	EBR070200	PCB Assembly Flexible SMT Bottom	EBR77740601	LGV500.AKDXWH 1.0 Flexible	
5	EBR070100	PCB Assembly Flexible,Insert	EBR78105801	LGV500.AKDXWH 1.0 Flexible	
4	EBR01	PCB Assembly Flexible	EBR77739601	LGV500.AKDXWH 1.0 Flexible	J,16
5	EBR070100	PCB Assembly Flexible,Insert	EBR78105901	LGV500.AKDXWH 1.0 Flexible	
6	RAA050100	Resin,PC	BRAH0001301	UF2040 or 3075BHF . . NONE	
5	EBR070400	PCB Assembly Flexible,SMT	EBR77720001	LGV500.AKDXWH 1.0 Flexible	

11. EXPLODED VIEW & REPLACEMENT PART LIST

Level	LocationNo.	Description	PartNumber	Spec	Remark
6	EBR070300	PCB Assembly Flexible,SMT Top	EBR77720101	LGV500.AKDXWH 1.2 Flexible	
7	EAX010700	PCB,Flexible	EAX65430801	LGV500.AKDXWH 1.0 POLYI Multi 4 0.32 Flexible	
7	CN102	Connector,I/O	EAG63149901	04-5161-005-101-868+ 5P 0.50MM ANGLE RECEPTACLE SMD R/TP Reverse type(New IO) KYOCERA ELCO KOREA SALES CO.,LTD.	
7	CN100	Connector,BtoB	ENBY0041701	GB042-64P-H10-E3000 - 0.40MM - MALE SMD BK - - LS Mtron Ltd.	
7	U100	IC,Hall Effect Switch	EAN62682901	BU52061NVX Hall IC 1.2X1.6 SSON R/TP 4P - ROHM Semiconductor KOREA CORPORATION	
7	CN104	Connector,BtoB	ENBY0040201	GB042-34P-H10-E3000 34P 0.4MM STRAIGHT PLUG SMD R/TP 1M - LS Mtron Ltd.	
7	C107 C108	Capacitor Ceramic,Chip	EAE62286801	CL03A104KP3NNNC 0.0000001F 10% 10V X5R - 55TO+85C 0603 R/TP 0.3 - SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	EBR070200	PCB Assembly Flexible SMT Bottom	EBR77739701	LGV500.AKDXWH 1.0 Flexible	
7	C100	Capacitor Ceramic,Chip	ECCH0004904	GRM155R60J105K 1uF 10% 6.3V X5R -55TO+85C 1005 R/TP - MURATA MANUFACTURING CO.,LTD.	
7	FL102	Filter,LCR	EAM62451101	ICMEF112P350MFR COMMON MODE NOISE FILTER 0HZ 0F 0H SMD R/TP Common mode Impedance at 100MHz : 35ohm+-30%, DC Resistance Max :3.0ohm INNOCHIPS TECHNOLOGY	
7	D100	Diode,Switching	EDSY0011901	SDB310Q 340mV 30V 200mA 1A 0SEC 150mW EMD2 R/TP 2P 1 AUK CORP	
7	VA100 VA101 VA102 VA103 VA104 VA105	Varistor	EAF61650701	LXES15AAA1-117 15V 25% 0.1pF 1.0X0.5X0.33 mm IEC61000-4-2 (ESD) level #4 SMD R/TP MURATA MANUFACTURING CO.,LTD.	
7	CN103	Connector,BtoB	ENBY0051001	GB042-10S-H10-E3000 10P 0.4MM STRAIGHT FEMALE SMD R/TP 1M - LS Mtron Ltd.	

11. EXPLODED VIEW & REPLACEMENT PART LIST

Level	LocationNo.	Description	PartNumber	Spec	Remark
7	D104	Diode,TVS	EAH62172601	RCLAMP1255P 12V 13.5V min. 25V 100A 2.5KW SLP2018P6 R/TP 8P 4 SEMTECH INTERNATIONAL AG	
7	C104 C105 C109	Capacitor Ceramic,Chip	ECZH0000830	C1005C0G1H330JT000F 33pF 5% 50V NP0 - 55TO+125C 1005 R/TP - TDK KOREA COOPERATION	
7	ZD101 ZD102	Diode,TVS	EAH61533301	RCLAMP0521PA.TCT 5V 6V min. / 9.3V typ. / 11V max. 15V 4A - SLP1006P2 R/TP 2P 1 SEMTECH INTERNATIONAL AG	
7	ZD100 ZD103	Diode,TVS	EDTY0012501	UCLAMP3311T.TCT 3.3V 3.5V min. 6.5V 5A - SLP1006P2T R/TP 2P 1 SEMTECH CORPORATION	
7	FB100 FB101	Filter,Bead	EAM62792501	BHS05S182MFR 1800 1.0x0.5x0.5 25% 1.9 0.2 SMD R/TP 2P - INNOCHIPS TECHNOLOGY	
7	D101	Diode,TVS	EAH61872601	PESD12VS1UA 12V 13.3V min. 19V 22.5A 360mW SOD323 R/TP 2P 1 NXP Semiconductors	
7	C103	Capacitor Ceramic,Chip	EAE62685201	C1005X5R1E104K 0.1uF 10% 25V X5R - 55TO+85C 1005 R/TP 0.55T max. TDK KOREA COOPERATION	
7	C106	Capacitor Ceramic,Chip	ECZH0000813	C1005C0G1H101JT 100pF 5% 50V C0G - 55TO+125C 1005 R/TP - TDK KOREA COOPERATION	
7	FL101	Filter,EMI/Power	EAM62790301	NFM18PC225B1A3D_ EMI - 0.0000022F 0H SMD R/TP size : 1.6x0.8x0.6 (3 terminal cap) MURATA MANUFACTURING CO.,LTD.	
7	ZD104	Diode,TVS	EDTY0009101	ESD9X5.0ST5G 5V 6.2 12.3V 8.7A 107W SOD528 R/TP 2P 1 ON SEMICONDUCTOR	
7	C101	Capacitor Ceramic,Chip	ECZH0004402	CL05F104ZO5NNNC 0.1uF -20TO+80% 16V Y5V - 30TO+85C 1005 R/TP - - SAMSUNG ELECTRO- MECHANICS CO., LTD.	
7	MIC100	Microphone Condenser	SUMY0010616	SUMY0010616 FPCB,dB,1.1TO10V, KNOWLES ACOUSTICS	
7	C102	Capacitor Ceramic,Chip	ECCH0042301	CL10A225KA5LNNC 0.0000022F 10% 25V X5R - 55TO+85C 1608 R/TP Max 0.6T L:1.6+-0.1 W:1.6+- 0.1 T:0.5+-0.1 SAMSUNG ELECTRO-MECHANICS CO., LTD.	
4	EAB00	Speaker Module	EAB63028301	1810-8T-42MP Nd-Fe-B 700mW 8OHM 81DB 1.5KHZ 151.50 x 13.30 x 3.95 FPCB KIRYN TELECOM CO., LTD	R,21

11. EXPLODED VIEW & REPLACEMENT PART LIST

Level	LocationNo.	Description	PartNumber	Spec	Remark
6	EAA01	PIFA Antenna GPS	EAA63085901	LS01-I-13072-A0 SINGLE -5DB 3.0 C+F - GPS Carrier LS Mtron Ltd.	C,9
6	EAA00	PIFA Antenna WiFi	EAA63086001	LS01-I-13073-A0 DUAL -5DB 3.0 C+F - BT Tape LS Mtron Ltd.	B,8
5	EAT130000	Module,Hybrid Touch LCD	EAT61993401	LD083WU1-SMA3 CAPACITIVE TOUCH GF2 Atmel MXT1188S 100 OCR 0.2 8.3 WUXGA B to B - LG Display Co., Ltd.	
3	EBR02	PCB Assembly Main	EBR77719602	LGV500.AUSAWH 1.0 Main	Z
4	EBR071500	PCB Assembly Main,Insert	EBR78012201	LGV500.AKDXWH 1.1 Main	
5	RAA050100	Resin,PC	BRAH0001301	UF2040 or 3075BHF . . NONE	
5	EBP01	Camera Module	EBP61842101	C5AA-A449B C5AA-A449B 5M AF, SS-LSI 1/4" BSI, 8.5x8.5x4.75t, MIPI, 90deg,FPCB LG INNOTEK CO., LTD	K,42
4	EBR071800	PCB Assembly Main,SMT	EBR77739302	LGV500.AUSAWH 1.0 Main	44
5	EBR071700	PCB Assembly Main,SMT Top	EBR77719801	LGV500.AKDXWH 1.2 Main	
6	EAX010000	PCB,Main	EAX65431101	LGV500.AKDXWH 1.0 FR-4 LX-BUMP 10 0.8 Main	
6	C100	Capacitor Ceramic,Chip	ECCH0004904	GRM155R60J105K 1uF 10% 6.3V X5R -55TO+85C 1005 R/TP - MURATA MANUFACTURING CO.,LTD.	
6	L7002	Inductor,Wire Wound,Chip	EAP62368101	VLP3612ST-100M-F 10UH 20% - 1A 1 1.2 0.39OHM - - SHIELD 3.6X3.6X1.2MM - R/TP TDK KOREA COOPERATION	
6	C6003 C6011 C6015 C6018	Capacitor Ceramic,Chip	EAE62962301	GRM155R61A225KE95 2.2uF -10TO+10% 10V X5R -55TO+85C 1005 R/TP 0.5 L:1.0+-0.05 W:0.5+-0.05 T:0.5+-0.05 MURATA MANUFACTURING CO.,LTD.	
6	R3013	Resistor,Chip	ERHY0009536	MCR006YZPF1003 100KOHM 1% 1/20W 0603 R/TP - ROHM.	

11. EXPLODED VIEW & REPLACEMENT PART LIST

Level	LocationNo.	Description	PartNumber	Spec	Remark
6	C2022 C2023 C2064 C2065 C2077 C2078 C2089 C2090 C2101 C2102 C2109 C2110	Capacitor Ceramic,Chip	ECCH0000198	CL05A225MQ5NSNC 2.2uF 20% 6.3V X5R - 55TO+85C 1005 R/TP . SAMSUNG ELECTRO- MECHANICS CO., LTD.	
6	C2116 C2117 C2123 C2124 C3045 C3049 C3053 C3059 C3060 C6012 C6013 C9008	Capacitor Ceramic,Chip	ECCH0000198	CL05A225MQ5NSNC 2.2uF 20% 6.3V X5R - 55TO+85C 1005 R/TP . SAMSUNG ELECTRO- MECHANICS CO., LTD.	
6	R6013 R6014 R9024	Resistor,Chip	ERHY0009526	MCR006YZPJ472 4.7KOHM 5% 1/20W 0603 R/TP - ROHM.	
6	L10003	Inductor Multilayer,Chip	EAP62227501	LG HK 0603 5N6S-T 5.6NH 0.3NH - 260mA - - 0.36OHM 4.6GHZ 5 SHIELD - 0.6X0.3X0.3MM R/TP TAIYO YUDEN CO.,LTD	
6	U6001 U6006	IC,Speaker Amplifier	EAN62206801	TPA2028D1 2.5~5.5V 999W WCSP R/TP 9P I2C I/F, ClassD TEXAS INSTRUMENTS KOREA LTD, HONGKONG BRANCH.	
6	C107 C108	Capacitor Ceramic,Chip	EAE62286801	CL03A104KP3NUNC 0.0000001F 10% 10V X5R - 55TO+85C 0603 R/TP 0.3 - SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	U9005	IC,Gate	EAN62415501	SN74AUP1T97DSFR 0~3.6V 0.000004 LEVEL SHIFTER SON-6 R/TP 6P Configurable Gate Logic TEXAS INSTRUMENTS KOREA LTD, HONGKONG BRANCH.	
6	FB100 FB101	Filter,Bead	EAM62792501	BHS05S182MFR 1800 1.0x0.5x0.5 25% 1.9 0.2 SMD R/TP 2P - INNOCHIPS TECHNOLOGY	
6	U6002	IC,Audio Codec	EAN62155601	WCD9310 1.14~4.5V 999W NSP R/TP 71P QCT CODEC QUALCOMM INCORPORATED.	

11. EXPLODED VIEW & REPLACEMENT PART LIST

Level	LocationNo.	Description	PartNumber	Spec	Remark
6	CN4000 CN4001 CN4002 CN4003 CN4008	Connector Terminal Block	ENZY0018801	25SMT-4442-01 0P A-LOGICS STRAIGHT SMD R/TP - W.L.GORE & ASSOCIATES INC	
6	U9004	IC,Comparator	EAN62116101	NCX2200GM 1.6~5.5V 9uA COMPARATOR SOT R/TP 6P SOT886, Push Pull Output STC CORP.	
6	R3002	Resistor,Chip	ERHZ0000204	MCR01MZP5F1003 100KOHM 1% 1/16W 1005 R/TP - ROHM.	
6	L8000	Inductor Wire Wound,Chip	ELCP0014301	1239AS-H-1R0N=P2 1UH 30% - 2.3A 3.0 2.3 0.059OHM - - SHIELD 2.5X2.0X1.2MM NONE R/TP TOKO, INC.	
6	C3001	Capacitor Ceramic,Chip	ECCH0009203	GRM033R60J333KE01D+A01 33nF 10% 6.3V X5R -55TO+85C 0603 R/TP - - MURATA MANUFACTURING CO.,LTD.	
6	C6009 C6019 C6034 C6039	Capacitor Ceramic,Chip	ECZH0001215	C1005X5R1A105KT000F 1uF 10% 10V X5R - 55TO+85C 1005 R/TP - TDK KOREA COOPERATION	
6	R6001 R6003 R6007 R7018	Wire Pad,Short	SAFP0000401	AX3100 ATL SV_SHIPBACK,MAIN,A	
6	C7016	Capacitor Ceramic,Chip	ECCH0017401	CL21B105KBFNNNE 1uF 10% 50V X7R - 55TO+125C 2012 R/TP 1.25MM - SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	R9001	Resistor,Chip	ERHZ0000243	MCR01MZP5F2201 2.2KOHM 1% 1/16W 1005 R/TP - ROHM.	
6	C6032 C6033 C6037 C6038	Capacitor Ceramic,Chip	ECCH0009106	C0603X7R1C103KT 10nF 10% 10V X7R - 55TO+125C 0603 R/TP - TDK CORPORATION	
6	R9005	Resistor,Chip	ERHY0009504	MCR006YZPJ102 1KOHM 5% 1/20W 0603 R/TP - ROHM.	
6	VA100 VA101 VA102 VA103 VA104 VA105	Varistor	EAF61650701	LXES15AAA1-117 15V 25% 0.1pF 1.0X0.5X0.33 mm IEC61000-4-2 (ESD) level #4 SMD R/TP MURATA MANUFACTURING CO.,LTD.	

11. EXPLODED VIEW & REPLACEMENT PART LIST

Level	LocationNo.	Description	PartNumber	Spec	Remark
6	R9007	Resistor,Chip	ERHZ0000456	MCR01MZP5J2R2 2.2OHM 5% 1/16W 1005 R/TP - ROHM.	
6	ZD6000	Diode,TVS	EDTY0012102	PESD5V0V1BL 5V 5.8V min. 12.5V 4.8A 45W SOD-882 R/TP 2P 1 STC CORP.	
6	R9003	Resistor,Chip	ERHZ0000267	MCR01MZP5F3301 3.3KOHM 1% 1/16W 1005 R/TP - ROHM.	
6	C6028 C6029 C6030 C6035 C6036 C6040 C6041	Capacitor Ceramic,Chip	ECZH0025916	GRM0335C1E330J 33pF 5% 25V NP0 - 55TO+125C 0603 R/TP - MURATA MANUFACTURING CO.,LTD.	
6	Q9001	TR,Bipolar	EBK61691301	2SC5376FV NPN 5V 15V 12V 400mA 100NA 300~600 150mW VESM R/TP 3P TOSHIBA ELECTRONICS KOREA CORPORATION	
6	R6015 R6016	Resistor,Chip	ERHZ0000402	MCR01MZP5J100 10OHM 5% 1/16W 1005 R/TP - ROHM.	
6	C3037 C3038 C3039 C3041 C3051	Capacitor Ceramic,Chip	ECCH0017601	CL05A475MQ5NRNC 4.7uF 20% 6.3V X5R - 55TO+85C 1005 R/TP 0.5MM - SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	R9009	Resistor,Chip	ERHZ0000286	MCR01MZP5F4701 4.7KOHM 1% 1/16W 1005 R/TP - ROHM.	
6	PT3000	Thermistor,NTC	SETY0007301	NCP15WF104F03RC NTC,100000 ohm,SMD,4250K+/-1%,100K,1%,0.1mA,4250K+/-1%,SMD,R/TP MURATA MANUFACTURING CO.,LTD.	
6	D7002	Diode,Schottky	EAH61992701	PMEG6002EB 600mV 60V 200mA 0SEC 20pF 0W SOD523 R/TP 2P 1 NXP Semiconductors	
6	U7002	IC,DC DC Converter	EAN62698201	LM3532 2.7V to 5.5V 40V max. 0W CSP R/TP 16P Asynchronous, Output current : 1A, 8bit, 10S 3P, 500KHz, CSP (1.745x1.845x0.6) TEXAS INSTRUMENTS KOREA LTD, HONGKONG BRANCH.	
6	R3014	Resistor,Chip	ERHZ0000405	MCR01MZP5J103 10KOHM 5% 1/16W 1005 R/TP - ROHM.	

11. EXPLODED VIEW & REPLACEMENT PART LIST

Level	LocationNo.	Description	PartNumber	Spec	Remark
6	C8015	Capacitor Ceramic,Chip	EAE62522101	CL10A226MP8NUNE 22uF 20% 10V X5R - 55TO+85C 1608 R/TP 1.05T max. - SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	LD9002	LED,Chip	EAV62092201	IR26-71C/L447/S63/TR8 CLEAR 1.15~1.6V 65mA 8.5 920~960nm 100mW 3012 R/TP 2P - EVERLIGHT ELECTRONICS CO., LTD.	
6	U8003	IC,DC,DC Converter	EAN62461401	TPS61256A V2.5 to V5.5 V5 800mA 0W WLP T/REEL 9P Small Package(1.2x1.3,0.4mm) TEXAS INSTRUMENTS KOREA LTD, HONGKONG BRANCH.	
6	R7008	Resistor,Chip	ERHY0009506	MCR006YZPJ104 100KOHM 5% 1/20W 0603 R/TP - ROHM.	
6	U9003	IC,Proximity	EAN62568201	APDS-9930-200 High T Proximity and Ambient Light Sensor 3.36X4.94X3.45 COB R/TP 8P High T Proximity and Ambient Light Sensor AVAGO TECHNOLOGIES INTERNATIONAL SALES PTE. LIMITED	
6	C9015	Capacitor Ceramic,Chip	ECCH0006201	C1608X5R0J475KT000N 4.7uF 10% 6.3V X5R - 55TO+85C 1608 R/TP - TDK CORPORATION	
6	R3003	Resistor,Chip	ERHY0000162	RC1005F2153CS 215KOHM 1% 1/16W 1005 R/TP - SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	C7017	Capacitor Ceramic,Chip	ECCH0005602	GRM188R61C225K 2.2uF 10% 16V X5R - 55TO+85C 1608 R/TP - MURATA MANUFACTURING CO.,LTD.	
6	R9004	Resistor,Chip	EBC62575901	RC0201FR-07300RL 300OHM 1% 1/20W 0603 R/TP - YAGEO CORPORATION	
6	C3055 C8014	Capacitor Ceramic,Chip	EAE62506501	CL05A475MP5NRNC 4.7uF 20% 10V X5R - 55TO+85C 1005 R/TP - - SAMSUNG ELECTRO- MECHANICS CO., LTD.	
6	L10016	Inductor Multilayer,Chip	EAP62226901	LG HK 0603 1N0S-T 1NH 0.3NH - 470mA - - 0.11OHM 10GHZ 4 SHIELD - 0.6X0.3X0.3MM R/TP TAIYO YUDEN CO.,LTD	
6	C3062	Capacitor Ceramic,Chip	ECZH0025920	GRM033R71C102K 1nF 10% 16V X7R - 55TO+125C 0603 R/TP - - MURATA MANUFACTURING CO.,LTD.	
5	EBR071600	PCB Assembly Main,SMT Bottom	EBR77719701	LGV500.AKDXWH 1.0 Main	

11. EXPLODED VIEW & REPLACEMENT PART LIST

Level	LocationNo.	Description	PartNumber	Spec	Remark
6	C10020	Capacitor (High Frequency) Ceramic,Chip	EAE62945901	GRM0335C1E1R2B_H 1.2pF 0.1PF 25V C0G - 55TO+125C 0603 R/TP 0.3+-0.03 L:0.6+-0.03 W:0.3+-0.03 T:0.3+-0.03 MURATA MANUFACTURING CO.,LTD.	
6	L10010	Inductor Multilayer,Chip	EAP62109001	LQP03TN0N6B02D 600PH 0.1NH - 850mA - - 0.07OHM 6GHZ 14 SHIELD - 0.6X0.3X0.3MM R/TP MURATA MANUFACTURING CO.,LTD.	
6	C10056	Capacitor Ceramic,Chip	ECZH0025911	GRM0335C1E120J 12pF 5% 25V C0G - 55TO+125C 0603 R/TP - MURATA MANUFACTURING CO.,LTD.	
6	R7008	Resistor,Chip	ERHY0009506	MCR006YZPJ104 100KOHM 5% 1/20W 0603 R/TP - ROHM.	
6	FB10001	Filter,Bead	EAM62471001	BLM03AX241SN1D 240 ohm 0.6X0.3X0.33 25% 0.38 ohm 0.35A SMD R/TP 2P 0 MURATA MANUFACTURING CO.,LTD.	
6	C10037 C10039 C3026 C3028 C3029 C3032	Capacitor Ceramic,Chip	ECCH0017501	CL10A226MQ8NRNE 22uF 20% 6.3V X5R - 55TO+85C 1608 R/TP 0.8MM SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	C3055 C8014	Capacitor Ceramic,Chip	EAE62506501	CL05A475MP5NRNC 4.7uF 20% 10V X5R - 55TO+85C 1005 R/TP - - SAMSUNG ELECTRO- MECHANICS CO., LTD.	
6	C11002 C11004 C11006 C8010	Capacitor Ceramic,Chip	ECCH0007804	CL05A225MP5NSNC 2.2uF 20% 10V X5R - 55TO+85C 1005 R/TP 0.5MM - SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	C100	Capacitor Ceramic,Chip	ECCH0004904	GRM155R60J105K 1uF 10% 6.3V X5R -55TO+85C 1005 R/TP - MURATA MANUFACTURING CO.,LTD.	
6	FL11002 FL7000	Filter,EMI/Power	EAM62770201	LXES11DAA2-135 COMMON MODE NOISE FILTER - 0.0000000000004F 0H SMD R/TP CM impedence : 60 ohm Leakage Current : 2uA MURATA MANUFACTURING CO.,LTD.	
6	C10031	Capacitor (High Frequency) Ceramic,Chip	EAE62946701	GRM0335C1E6R0D_H 6pF 0.5PF 25V C0G - 55TO+125C 0603 R/TP 0.3+-0.03 L:0.6+-0.03 W:0.3+-0.03 T:0.3+-0.03 MURATA MANUFACTURING CO.,LTD.	
6	C107 C108	Capacitor Ceramic,Chip	EAE62286801	CL03A104KP3NNNC 0.0000001F 10% 10V X5R - 55TO+85C 0603 R/TP 0.3 - SAMSUNG ELECTRO-MECHANICS CO., LTD.	

11. EXPLODED VIEW & REPLACEMENT PART LIST

Level	LocationNo.	Description	PartNumber	Spec	Remark
6	R1000 R1001 R1002 R1007 R1008 R1009	Resistor,Chip	ERHY0009522	MCR006YZPJ332 3.3KOHM 5% 1/20W 0603 R/TP - ROHM.	
6	R1030	Resistor,Chip	ERHZ0000519	MCR01MZIP5J912 9.1KOHM 5% 1/16W 1005 R/TP - ROHM.	
6	FB11000 FB11001 FB11002 FB11003 FB11004 FB7001	Filter,Bead	EAM62071101	BLM15PD121SN1D 120 ohm 1.0X0.5X0.5 25% 0.09 ohm 1.3A SMD R/TP 2P 0 MURATA MANUFACTURING CO.,LTD.	
6	R6001 R6003 R6007 R7018	Wire Pad,Short	SAFP0000401	AX3100 ATL SV_SHIPBACK,MAIN,A	
6	X3001	Crystal	EAW61843401	Q13FC1350000200 32.768KHZ 20PPM 0.000000000007F NONE SMD R/TP EPSON TOYOCOM CORP	
6	R1010 R1011 R1018 R11024 R5010 R8001 R8004 R9002	Resistor,Chip	ERHY0009505	MCR006YZPJ103 10KOHM 5% 1/20W 0603 R/TP - ROHM.	
6	L3005 L3006 L4000 L4001	Inductor Wire Wound,Chip	EAP62307001	1277AS-H-1R0M 1UH 20% - 3.7A 3.7 3.7 0.045OHM - - SHIELD 3.2X2.5MM - R/TP TOKO, INC.	
6	R1016	Resistor,Chip	EBC62581901	RC0201FR-0733RL 33OHM 1% 1/20W 0603 R/TP - YAGEO CORPORATION	
6	C8018 C8020	Capacitor Ceramic,Chip	EAE62969601	CL10A106MP8NNNC_ 10uF -20TO20% 10V X5R - 55TO+85C 1608 R/TP 0.8+/-0.1 L:1.6+-0.1 W:0.8+-0.1 T:0.8+-0.1 SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	R6004 R6017 R8000	Resistor,Chip	ERHY0009507	MCR006YZPJ105 1MOHM 5% 1/20W 0603 R/TP - ROHM.	

11. EXPLODED VIEW & REPLACEMENT PART LIST

Level	LocationNo.	Description	PartNumber	Spec	Remark
6	C3027 C3030 C3031 C4005 C4006	Capacitor Ceramic,Chip	EAE62767801	JDK212BBJ476MD 47uF -20TO20% 6.3V X5R - 55TO+85C 2012 R/TP 0.85T - TAIYO YUDEN CO.,LTD	
6	C10033 C10053 C3012	Capacitor Ceramic,Chip	EAE62502901	CL05A106MP5NUNC 10uF 20% 10V X5R - 55TO+85C 1005 R/TP 0.55T max. SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	FL11000 FL11001 FL7001 FL7002	Filter,EMI/Power	EAM62770301	LXES21DAA4-136 COMMON MODE NOISE FILTER - 0.00000000000004F 0H SMD R/TP 4ch-CMEF 2.0x1.25 60 ohm MURATA MANUFACTURING CO.,LTD.	
6	R1012	Resistor,Chip	ERHY0009303	MCR006YZPF1002 10KOHM 1% 1/20W 0603 R/TP - ROHM.	
6	C6009 C6019 C6034 C6039	Capacitor Ceramic,Chip	ECZH0001215	C1005X5R1A105KT000F 1uF 10% 10V X5R - 55TO+85C 1005 R/TP - TDK KOREA COOPERATION	
6	FL10002	Filter,Separator	EAM62511801	LFD182G45DCJD247 0.35 @2.4GHz / 0.5 @ 5GHz - - 2.4G(LPF)/5G(BPF) for QCT WiFi Chipset - Mirror type of LFD182G45DCHD277 for BRCM MURATA MANUFACTURING CO.,LTD.	
6	R6013 R6014 R9024	Resistor,Chip	ERHY0009526	MCR006YZPJ472 4.7KOHM 5% 1/20W 0603 R/TP - ROHM.	
6	C9005	Capacitor Ceramic,Chip	ECCH0002001	C1005JB0J104KT000F 0.1uF 10% 6.3V X5R - 30TO+85C 1005 R/TP - - TDK CORPORATION	
6	L3003	Inductor Wire Wound,Chip	EAP62045901	VLS2012ET-R47N 470NH 30% - 2A 2.05 2 0.059OHM - - SHIELD 2.0X2.0X1.2MM NONE R/TP TDK KOREA COOPERATION	
6	FL10000	Filter,Saw	SFSY0042901	SAFEA1G58KA0F00 1575.42M/1602M 1.4*1.1*0.5 SMD R/TP 5P MURATA MANUFACTURING CO.,LTD.	
6	C6028 C6029 C6030 C6035 C6036 C6040 C6041	Capacitor Ceramic,Chip	ECZH0025916	GRM0335C1E330J 33pF 5% 25V NP0 - 55TO+125C 0603 R/TP - MURATA MANUFACTURING CO.,LTD.	

11. EXPLODED VIEW & REPLACEMENT PART LIST

Level	LocationNo.	Description	PartNumber	Spec	Remark
6	C10038 C10041 C10043 C10044 C10048 C10051	Capacitor Ceramic,Chip	ECCH0009103	C0603C0G1H101JT00NN 100pF 5% 50V C0G - 55TO+125C 0603 R/TP - - TDK CORPORATION	
6	L10007	Inductor Multilayer,Chip	EAP62246201	LG HK 0603 10NJ-T 10NH 5% - 220mA - - 0.51OHM 2.9GHZ 5 SHIELD - 0.6X0.3X0.3MM R/TP TAIYO YUDEN CO.,LTD	
6	R1014	Resistor,Chip	ERHY0009311	MCR006YZPF51R0 51OHM 1% 1/20W 0603 R/TP - ROHM.	
6	R1020 R1021 R1022 R1023 R1024 R1025 R1026 R1027	Resistor,Chip	EBC61856201	RC0201FR-07240RL 240OHM 1% 1/20W 0603 R/TP - YAGEO CORPORATION	
6	R11022	Resistor,Chip	ERHZ0000439	MCR01MZP5J204 200KOHM 5% 1/16W 1005 R/TP - ROHM.	
6	C6032 C6033 C6037 C6038	Capacitor Ceramic,Chip	ECCH0009106	C0603X7R1C103KT 10nF 10% 10V X7R - 55TO+125C 0603 R/TP - TDK CORPORATION	
6	C10003	Inductor Multilayer,Chip	EAP62227701	LG HK 0603 8N2J-T 8.2NH 5% - 230mA - - 0.45OHM 3.4GHZ 5 SHIELD - 0.6X0.3X0.3MM R/TP TAIYO YUDEN CO.,LTD	
6	D8001 D8004 D8005	Diode,Schottky	EAH61794001	KDR720F 350mV 30V 100mA 5NSEC 12pF 100mW TFSC R/TP 2P 1 KEC CORPORAITION	
6	R11015 R11016 R11017 R11018	Resistor,Chip	ERHZ0000420	MCR01MZP5J151 150OHM 5% 1/16W 1005 R/TP - ROHM.	
6	C10083 C10084 C11019	Capacitor Ceramic,Chip	ECCH0009514	MCH032A(AN)100DK 10pF 0.5PF 25V X7R - 55TO+125C 0603 R/TP - ROHM.	
6	R8014	Resistor,Chip	EBC62259801	RC0402FR-07105KL 105KOHM 1% 1/16W 1005 R/TP 0.37T max. (mm) YAGEO CORPORATION	

11. EXPLODED VIEW & REPLACEMENT PART LIST

Level	LocationNo.	Description	PartNumber	Spec	Remark
6	Q8002	FET	EBK61711801	SSM6J502NU P-CHANNEL MOSFET -20V +-8 -6A 0.028OHM 1W DFN R/TP 6P TOSHIBA ELECTRONICS KOREA CORPORATION	
6	L8000	Inductor,Wire Wound,Chip	ELCP0014301	1239AS-H-1R0N=P2 1UH 30% - 2.3A 3.0 2.3 0.059OHM - - SHIELD 2.5X2.0X1.2MM NONE R/TP TOKO, INC.	
6	C2022 C2023 C2064 C2065 C2077 C2078 C2089 C2090 C2101 C2102 C2109 C2110	Capacitor Ceramic,Chip	ECCH0000198	CL05A225MQ5NSNC 2.2uF 20% 6.3V X5R - 55TO+85C 1005 R/TP . SAMSUNG ELECTRO- MECHANICS CO., LTD.	
6	C2116 C2117 C2123 C2124 C3045 C3049 C3053 C3059 C3060 C6012 C6013 C9008	Capacitor Ceramic,Chip	ECCH0000198	CL05A225MQ5NSNC 2.2uF 20% 6.3V X5R - 55TO+85C 1005 R/TP . SAMSUNG ELECTRO- MECHANICS CO., LTD.	
6	U9002	IC,Geomagnetic Sensor	EAN62333101	AK8963C 3 Axis Geomagnetic Sensor 1.6X1.6X0.5t CSP R/TP 14P - ASAHI KASEI CORPORATION	
6	X1000 X11000	Crystal	EAW61543401	1ZCB27000AA0A 27MHZ 10PPM 12F ; SMD R/TP DAISHINKU CORPORATION.	
6	U10004	Module, FEM (Front End Module)	EAT61775101	RFFM8502 0DBM,0DBM,0DBM 0DB,0DB,0DB 0%,0%,0% 0A,0A,0A 0A 0DB 0DBM 0DBM 16P 2.5x2.5x0.4MM WiFi FEM, 5GHz, SPDT/PA/LNA, 802.11a/n, for BCM4334 RF MICRO DEVICES INC	
6	R3010	Resistor,Chip	ERHY0003201	MCR01MZP5F1001 1KOHM 1% 1/16W 1005 R/TP - ROHM.	

11. EXPLODED VIEW & REPLACEMENT PART LIST

Level	LocationNo.	Description	PartNumber	Spec	Remark
6	VA100 VA101 VA102 VA103 VA104 VA105	Varistor	EA61650701	LXES15AAA1-117 15V 25% 0.1pF 1.0X0.5X0.33 mm IEC61000-4-2 (ESD) level #4 SMD R/TP MURATA MANUFACTURING CO.,LTD.	
6	C3013 C3014	Capacitor Ceramic,Chip	ECCH0007803	CL10A106MP8NNNC 10uF 20% 10V X5R - 55TO+85C 1608 R/TP 0.8MM SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	R11005	Resistor,Chip	ERHY0009553	MCR006YZPF1004 1MOHM 1% 1/20W 0603 R/TP - ROHM Semiconductor KOREA CORPORATION	
6	Q3000 Q8000	FET	EBK61892201	KTK5132V N-CHANNEL MOSFET 30V +-20 100mA 70HM 100mW VSM R/TP 3P KEC CORPORAITION	
6	U1000	IC,Digital Baseband Processor,4G	EAN62446802	APQ8064-1.7G Quad 1.7G Krait, 1080p,839 NSP PoP 14x14, Adreno320 BGA R/TP 839P QUALCOMM INCORPORATED.	
6	C1003 C1004 C11014 C11016	Capacitor Ceramic,Chip	ECCH0009504	MCH032A180JK 18pF 5% 25V NP0 -55TO+125C 0603 R/TP - ROHM.	
6	C3003 C3020	Capacitor Ceramic,Chip	ECCH0007802	CL10A475KP8NNNC 4.7uF 10% 10V X5R - 55TO+85C 1608 R/TP - - SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	C7018	Capacitor (High Frequency) Ceramic,Chip	EAE62947001	GRM0335C1E100J_H 10pF 5% 25V C0G - 55TO+125C 0603 R/TP 0.3+-0.03 :0.6+-0.03 W:0.3+-0.03 T:0.3+-0.03 MURATA MANUFACTURING CO.,LTD.	
6	C10045 C10046	Capacitor Ceramic,Chip	ECCH0009506	MCH032A270JK 27pF 5% 25V NP0 -55TO+125C 0603 R/TP - ROHM.	
6	C3037 C3038 C3039 C3041 C3051	Capacitor Ceramic,Chip	ECCH0017601	CL05A475MQ5NRNC 4.7uF 20% 6.3V X5R - 55TO+85C 1005 R/TP 0.5MM - SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	X10001	Crystal	EA61824301	Q22FA1280019000 48MHZ 10PPM 7F - SMD R/TP EPSON TOYOCOM CORP	
6	U5001	IC,Mobile SDRAM	EAN62461101	H9TKNNBPDMRAR-NGM 16GBIT LPDDR2 PoP 1.7VTO1.95V 533MHz 5500ns FBGA TR 220P 16Gb LPDDR2 PoP 220ball 533MHz (38nm 4Gb x 4) HYNIX SEMICONDOCTOR	

11. EXPLODED VIEW & REPLACEMENT PART LIST

Level	LocationNo.	Description	PartNumber	Spec	Remark
6	U4000	IC,PMIC	EAN62416801	PM8821 2.5 to 4.5V adj 0W WLCSP R/TP 42P - QUALCOMM INCORPORATED.	
6	C8002 C8006 C8019	Capacitor Ceramic,Chip	ECCH0009104	C0603C0G1H330JT00NN 33pF 5% 50V C0G - 55TO+125C 0603 R/TP - TDK CORPORATION	
6	C3062	Capacitor Ceramic,Chip	ECZH0025920	GRM033R71C102K 1nF 10% 16V X7R - 55TO+125C 0603 R/TP - - MURATA MANUFACTURING CO.,LTD.	
6	R10000 R5013 R7000 R7001	Resistor,Chip	ERHY0009502	MCR006YZPJ100 10OHM 5% 1/20W 0603 R/TP - ROHM.	
6	C8015	Capacitor Ceramic,Chip	EAE62522101	CL10A226MP8NUNE 22uF 20% 10V X5R - 55TO+85C 1608 R/TP 1.05T max. - SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	L10013	Inductor Multilayer,Chip	EAP62227801	LG HK 0603 12NJ-T 12NH 5% - 190mA - - 0.68OHM 2.7GHZ 5 SHIELD - 0.6X0.3X0.3MM R/TP TAIYO YUDEN CO.,LTD	
6	CN7000	Connector,BtoB	ENBY0034201	GB042-24S-H10-E3000 24P 0.40MM STRAIGHT SOCKET SMD R/TP 1M - LS Mtron Ltd.	
6	L10016	Inductor Multilayer,Chip	EAP62226901	LG HK 0603 1N0S-T 1NH 0.3NH - 470mA - - 0.11OHM 10GHZ 4 SHIELD - 0.6X0.3X0.3MM R/TP TAIYO YUDEN CO.,LTD	
6	C6027	Capacitor Ceramic,Chip	ECZH0001216	C1005X5R1A224KT000E 220nF 10% 10V X5R - 55TO+85C 1005 R/TP - TDK KOREA COOPERATION	
6	C5017 C5032	Capacitor Ceramic,Chip	ECCH0032801	GRM033R60J224M 0.00000022F 20% 6.3V X5R - 55TO+85C 0603 R/TP 0.3MM MURATA MANUFACTURING CO.,LTD.	
6	U10002	IC,WiFi	EAN62701201	WCN-3660A-0-79BWLNSP-TR-03 WiFi(11a/b/g/n, HT40)+BT4.0+FM Radio, 3.805x3.82x0.63 WLCSP R/TP 79P QUALCOMM INCORPORATED.	
6	U9001	IC,Gyro Sensor	EAN62574401	MPU-6500 Accelerometer embedded Gyro Sensor 3.0x3.0x0.9 QFN R/TP 24P One Chip Solution INVENSENSE	
6	R11007 R11013	Resistor,Chip	EBC62296301	RC0201FR-07665KL 665KOHM 1% 1/20W 0603 R/TP 0.26T max. YAGEO CORPORATION	

11. EXPLODED VIEW & REPLACEMENT PART LIST

Level	LocationNo.	Description	PartNumber	Spec	Remark
6	R1013	Resistor,Chip	ERHZ0000235	MCR01MZP5F2000 200OHM 1% 1/16W 1005 R/TP - ROHM.	
6	S8000	Card Socket	EAG63392501	SCHA4B0409 SD 8P STRAIGHT SMD T/REEL - ALPS ELECTRIC KOREA CO.,LTD.	
6	R8002 R8003 R8005 R8006 R8021	Resistor,Chip	ERHY0009538	MCR006YZPF1802 18KOHM 1% 1/20W 0603 R/TP - ROHM.	
6	C10049	Capacitor (High Frequency) Ceramic,Chip	EAE62945801	GRM0335C1E1R0C_H 1pF 0.25PF 25V C0G - 55TO+125C 0603 R/TP 0.3+-0.03 L:0.6+-0.03 W:0.3+-0.03 T:0.3+-0.03 MURATA MANUFACTURING CO.,LTD.	
6	FL8000	Filter,EMI/Power	SFEY0015301	NFM18PC104R1C3 ESD/EMI 0HZ 0.1uF 0H SMD R/TP - MURATA MANUFACTURING CO.,LTD.	
6	C10026 C10027	Capacitor Ceramic,Chip	ECZH0001217	GRM155R60J474K 470nF 10% 6.3V X5R - 25TO+70C 1005 BK-DUP - MURATA MANUFACTURING CO.,LTD.	
6	R11011	Resistor,Chip	EBC61959801	WR04X6193FTL 619KOHM 1% 1/16W 1005 R/TP WALSIN (Resistor Maker) - STC (WALSIN Resistor Supplier) STC CORP.	
6	R11009	Resistor,Chip	ERHY0009540	MCR006YZPF2702 27KOHM 1% 1/20W 0603 R/TP - ROHM.	
6	C8013	Capacitor Ceramic,Chip	ECCH0000161	MCH153CN333KK 33nF 10% 16V X7R - 55TO+125C 1005 R/TP -- ROHM Semiconductor KOREA CORPORATION	
6	CN11002	Connector,BtoB	ENBY0041801	GB042-64S-H10-E3000 64P 0.4MM STRAIGHT SOCKET SMD R/TP 1M - LS Mtron Ltd.	
6	D11003	Diode,Switching	EDSY0019602	PMEG3005AEA 430mV 30V 500mA 3.5A 0SEC 0W SOD323 R/TP 2P 1 NXP Semiconductors	
6	U11002	IC,Over Voltage Protection	EAN62768301	MAX14654EWC+T 2.3 to 28V adj 0W CSP R/TP 12P - MAXIM INTEGRATED PRODUCTS INC.	
6	C7003 C7004 C7005	Capacitor Ceramic,Chip	EAE62726601	CL03A225MQ3CRNC 2.2uF -20TO20% 6.3V X5R - 55TO+85C 0603 R/TP 0.3T - SAMSUNG ELECTRO-MECHANICS CO., LTD.	

11. EXPLODED VIEW & REPLACEMENT PART LIST

Level	LocationNo.	Description	PartNumber	Spec	Remark
6	U11000	IC,LDO Voltage Regulator	EUSY0367001	RP101K332D RP101K332D RP101K332D,DFN1612-4B ,4 ,R/TP ,300mA 3.3V LDO RICOH COMPANY, LTD. RICOH COMPANY, LTD.	
6	C9007	Capacitor Ceramic,Chip	EAE62685701	GRM155R71H103KA88D 10nF 10% 50V X7R - 55TO+125C 1005 R/TP 0.55T max. - MURATA MANUFACTURING CO.,LTD.	
6	U10000	IC,RF Amplifier	EAN62581401	SFMG2B0D001 1.5~3.1V 0 0 0W 0W 0 0 MCM R/TP 8P Gain : 17.2dB NF : 0.85dB WISOL.CO.,LTD	
6	C7000 C7001	Capacitor (High Frequency) Ceramic,Chip	EAE62946601	GRM0335C1E4R7C_H 4.7pF 0.25PF 25V C0G - 55TO+125C 0603 R/TP 0.3+-0.03 L:0.6+-0.03 W:0.3+-0.03 T:0.3+-0.03 MURATA MANUFACTURING CO.,LTD.	
6	U7000	IC,LDO Voltage Regulator	EAN62737501	RP111L331D-TR 1.4v to 5.25 3.3 0W DFN R/TP 6P - RICOH.	
6	L3002 L3004	Inductor Wire Wound,Chip	ELCP0014201	1239AS-H-2R2N=P2 2.2UH 30% - 1.6A 2.2 1.6 0.108OHM - - SHIELD 2.5X2.0X1.2MM NONE R/TP TOKO, INC.	
6	X3000	Oscillator,TCXO	EAW61843801	X1G003851000200 19.2MHZ 1.5PPM 1.8V 2.5x2.0x0.9MM none SMD R/TP EPSON TOYOCOM CORP	
6	C10075	Capacitor Ceramic,Chip	ECCH0009515	MCH032CN151KK 150pF 10% 25V X7R - 55TO+125C 0603 R/TP - ROHM.	
6	L10002	Inductor Multilayer,Chip	EAP62266401	LQP03TN2N7B02D 2.7NH 0.1NH - 500mA - - 0.2OHM 6GHZ 14 SHIELD - 0.6X0.3X0.3MM R/TP MURATA MANUFACTURING CO.,LTD.	
6	C8011	Capacitor Ceramic,Chip	ECCH0009228	GRM033R61A472K 4700pF 10% 10V X5R - 55TO+85C 0603 R/TP - - MURATA MANUFACTURING CO.,LTD.	
6	C10073 C10081	Capacitor Ceramic,Chip	ECCH0032901	GRM0335C1H2R7C 2.7pF 0.25PF 50V C0G - 55TO+125C 0603 R/TP 0.3MM MURATA MANUFACTURING CO.,LTD.	
6	C10001 C10032	Capacitor Ceramic,Chip	ECCH0042007	GRM0335C1E6R8BD01D 6.8 pF,25V,B,C0G,TC,0603,R/TP,0.0000000000068,0. 1PF,25V,C0G,-55TO+125C,0603,R/TP,0.3 mm MURATA MANUFACTURING CO.,LTD.	

11. EXPLODED VIEW & REPLACEMENT PART LIST

Level	LocationNo.	Description	PartNumber	Spec	Remark
6	U5000	IC,MCP,eMMC	EAN62512301	THGBM5G7A2JBAIR 16GBYTE 2.7VTO3.6V,1.7VTO1.95V 11.5x13.0x1.0 TR 153P MLC NAND FBGA 16GB eMMC v4.41 (19nm 64Gb MLC x 2, 11.5x13.0x1.0) TOSHIBA ELECTRONICS KOREA CORPORATION	
6	Q8001	FET	EBK61952101	KTJ6131V P-CHANNEL MOSFET -30V +-20 -0.05A 400HM 100mW VSM R/TP 3P KEC CORPORATION	
6	FL10001	Filter,Ceramic	EAM62250401	LFB212G45CG7D227 BPF 2.45KHZ 100Hz SMD R/TP 3P MURATA MANUFACTURING CO.,LTD.	
6	R8017	Resistor,Chip	ERHY0035601	PMR10EZPFU10L0 0.01OHM 1% 1/2W 2012 R/TP - ROHM.	
6	R11006	Resistor,Chip	ERHY0009555	MCR006YZPF1202 12KOHM 1% 1/20W 0603 R/TP - ROHM.	
6	R8015	Resistor,Chip	ERHZ0000224	MCR01MZP5F1602 16KOHM 1% 1/16W 1005 R/TP - ROHM.	
6	R6011	Resistor,Chip	ERHY0009516	MCR006YZPJ222 2.2KOHM 5% 1/20W 0603 R/TP - ROHM.	
6	D8002 D8003	Diode,TVS	EAH61794301	ESD5205P6T6G 5V 5.5V min. 10V 2UA 0W SOT- 963 R/TP 6P 5 ON SEMICONDUCTOR	
6	U6003	IC,Comparator	EAN62656801	MAX14628 3.1~5.5V 1.9~800uA COMPARATOR TDFN R/TP 8P 3button support headset detection IC MAXIM INTEGRATED PRODUCTS INC.	
6	L10008	Inductor Multilayer,Chip	EAP61866901	LQP03TN2N0C02D 2NH 0.2NH - 600mA - - 0.15OHM 6GHZ 14 SHIELD 0 0.6X0.3X0.3MM R/TP MURATA MANUFACTURING CO.,LTD.	
6	C3000	Capacitor TA,Conformal	ECTH0002703	TCTAL1A107M8R-V2 100uF 20% 10V 50UA - 55TO+125C 2.5OHM 3.2X1.6X1.1MM NONE SMD R/TP 1.2T max. ROHM CO.,LTD.	
6	L10000	Inductor Multilayer,Chip	EAP62227301	LG HK 0603 2N2S-T 2.2NH 0.3NH - 360mA - - 0.19OHM 8.8GHZ 4 SHIELD - 0.6X0.3X0.3MM R/TP TAIYO YUDEN CO.,LTD	
6	U8006	IC,Analog Switch	EUSY0186504	FSA2259UMX QFN ,8 ,R/TP ,Dual SPDT ,; ,IC,Analog Switch FAIRCHILD SEMICONDUCTOR	
6	CN8003	Connector Terminal Block	EAG63412101	BM22-4S- V 4P 1.78MM STRAIGHT SMD T/REEL - HIROSE KOREA CO.,LTD	

11. EXPLODED VIEW & REPLACEMENT PART LIST

Level	LocationNo.	Description	PartNumber	Spec	Remark
6	R11021	Resistor,Chip	ERHZ0000205	MCR01MZP5F1004 1MOHM 1% 1/16W 1005 R/TP - ROHM.	
6	L10014	Inductor Wire Wound,Chip	EAP61767901	LQW15AN47NJ00D 47NH 5% - 210mA 1.08OHM 2.9GHZ 25 SHIELD 1.0X0.5X0.5MM NONE R/TP MURATA MANUFACTURING CO.,LTD.	
6	L1000	Inductor Multilayer,Chip	EAP62366401	LQM21PN2R2MGHL 2.2UH 20% - 700mA 0.7 1.7 0.125OHM 0HZ - SHIELD - 2.0X1.25X1.0 MM R/TP MURATA MANUFACTURING CO.,LTD.	
6	R1019	Resistor,Chip	EBC61959901	WR04X4641FTL 4.64KOHM 1% 1/16W 1005 R/TP WALSIN (Resistor Maker) - STC (WALSIN Resistor Supplier) STC CORP.	
6	R10004	Resistor,Chip	EBC62236101	RC0603J821C 820OHM 5% 1/20W 0603 R/TP - SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	R8030	Resistor,Chip	ERHY0024401	RC0603F753CS 75KOHM 1% 1/20W 0603 R/TP - SAMSUNG ELECTRO-MECHANICS CO., LTD.	
6	U3000	IC,PMIC	EAN62095901	PM8921 2.5 to 4.5V adj 1.3W NSP R/TP 251P - QUALCOMM INCORPORATED.	
6	U11001	IC,Signal Bridge	EAN62533101	ANX7808BH-BA-R HDMI to DP tx, 4.5X4.5 VFBGA R/TP 49P ANALOGIX SEMICONDUCTOR	
6	C8001	Capacitor TA,Conformal	ECTH0002002	F981A336MSA 33uF 20% 10V 3.3UA -55TO+125C 6OHM 2.0X1.25X0.8MM NONE SMD R/TP 0.9T max. UNITELE ELECTRONICS CO. LTD.	
6	R11000	Resistor,Chip	ERHZ0000438	MCR01MZP5J203 20KOHM 5% 1/16W 1005 R/TP - ROHM.	
6	U8005	IC,Reset	EAN62688701	FTL11639UCX 1.2~5.5V 0~5.5V 410mW WLCSP R/TP 12P Reset Timer(11.5s) with Integrated Load Switch FAIRCHILD SEMICONDUCTOR HONG KONG LTD.	
6	U8007	IC I/O Support Chip	EAN62676701	TPD4S214YFFR USB OTG V I Protection IC CSP R/TP 12P TEXAS INSTRUMENTS KOREA LTD, HONGKONG BRANCH.	
6	CN7001	Connector,BtoB	ENBY0039601	GB042-20S-H10-E3000 20P 0.4MM STRAIGHT SOCKET SMD R/TP 1M - LS Mtron Ltd.	

11. EXPLODED VIEW & REPLACEMENT PART LIST

Level	LocationNo.	Description	PartNumber	Spec	Remark
6	CN103	Connector,BtoB	ENBY0051001	GB042-10S-H10-E3000 10P 0.4MM STRAIGHT FEMALE SMD R/TP 1M - LS Mtron Ltd.	
6	U10001	IC,GPS	EAN62649901	WGR7640 GNSS(GPS,GLONASS,COMPASS) RFIC, 2.07x1.51x0.63 WLCSP R/TP 17P QUALCOMM INCORPORATED.	
5	SAD010000	Software,Mobile	SAD34040301	Android JBMR1 Base V50010a - NORTH AMERICA TABLET(QCT) LGV500.AUSAWH	
2	EAD010000	Cable,Assembly	EAD62329302	OTB-LGE50DCMUA USB USB 1.2M 5 BLACK UL N CRESYN(H.K.)CO.,LTD.	

11. EXPLODED VIEW & REPLACEMENT PART LIST

11.3 Accessory

Note: This Chapter is used for reference, Part order is ordered by SBOM standard on GCSC

Level	LocationNo.	Description	PartNumber	Spec	Remark
4	EAC00	Rechargeable Battery, Lithium Polymer	EAC62159101	BL-T10_WW 3.8V 4.43AH 2AH 150x100x2.4 152.0x101.5x2.72 BLACK Embedded - LG Chem, LTD.	S, 23
2	EAY060000	Adapters	EAY63148602	MCS-04WR2 90Vac ~ 264Vac 5V 1.8A 50/60Hz UL, CSA NONE DC PIN PLUG - SUNLIN ELECTRONICS CO., LTD	
2	MFL053800	Manual, Operation	MFL67986001	PRINTING LGV500.AUSABK ZZ: Without Color United States ENGLISH+ SPANISH -	
2	MBM062600	Card, Quick Reference	MBM64079901	PRINTING LGV500.AUSABK ZZ: Without Color -	